HALO

Complete backend and frontend handoff

Prepared for Vibhav and Yogi

October 3, 2026

One searchable volume. Complete source documents. No feature summaries substituted for the implementation requirements.

45 planned feature groups / 24 source families / existing frontend integration reference

Build blueprint, not a claim of deployed functionality. Source, permission, product-decision and release gates remain explicit throughout.

Includes the original 83-page frontend specification as a historical appendix. Newer decisions and approval rules take precedence.

How to use this volume

Vibhav: start with Chapter 1, then use Chapters 2 and 3 for backend and data work. Chapters 4 through 6 define the full feature scope, frontend needs and acceptance requirements. Chapter 8 explains the already approved frontend integration surface.

Yogi: use the same shared plan, especially Chapters 4 through 6 and the preservation references. Do not create a separate edited copy that drifts from the backend requirements.

Preservation: every source document listed in the manifest is reproduced in full. Wide Markdown tables are reflowed into labelled records for legibility; no cells are summarized. Code is wrapped for reading. External URLs are printed as well as clickable. Original source bytes, including scripts and the source PDF, are also embedded as PDF attachments.

Precedence: the current product plans and explicit user decisions supersede conflicting historical specifications. Today retains its approved depth. PFAS belongs to Homeguard. Map and other new designs still require review. The proposed fourth-tab label remains a decision, not an approved visual.

Historical material: the onboarding/mobile/map references and original specification describe their own dated scope. Preserve historical status wording as evidence; do not treat old assignment, command, appearance or architecture suggestions as new instructions.

Repository dependency: this PDF contains the handoff documentation, not the entire application source, image library, test suite or immutable design archives. Vibhav still needs the halo-app repository to inspect and extend the existing implementation. Paths printed in the documents refer to that repository.

Validation boundary: public request success is not complete data validation. Keyed-provider checks, account rights, full imports, semantic review, migrations and integration tests remain the gates stated in the documents.

Reading: use the clickable contents and PDF bookmarks. The original specification retains its original page layout and numbering; use its appendix bookmark to open it. PDF attachments preserve exact source files, although some mobile viewers do not expose attachments.

Repository HEAD at compilation: 66a3246c5fde222754df100e1e5bb74784e7296d. Planning files are a local working-tree snapshot and are not implied to exist in this commit.

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docs/product/HALO_BACKEND_BUILD_HANDOFF.md

HALO Backend Build Handoff

Prepared for Vibhav on October 3, 2026. This is the entry point for implementing the added backend behind HALO's complete app plan. It connects all 45 planned feature groups to existing code, external sources, new owned records, jobs and acceptance gates. Yogi remains the frontend owner. No implementation, deployment, paid account changes or new visual approval is implied by these documents.

Use this handoff with the Backend and Data Plan (HALO_BACKEND_AND_DATA_PLAN.md), Source Acquisition Guide (HALO_SOURCE_ACQUISITION_GUIDE.md), Master Plan (HALO_MASTER_PLAN.md), Frontend Build Plan (HALO_FRONTEND_BUILD_PLAN.md) and Feature Acceptance (HALO_FEATURE_ACCEPTANCE.md). The backend plan contains the 22-route baseline, proposed field requirements, relationships, transactions, security, retention and rollout requirements. This handoff adds source verification and an ordered implementation checklist; it does not replace those requirements with a shorter feature list.

What can be built and what is still conditional

The application can be developed in connected stages using the existing Next.js and Supabase architecture. Shared actions, private tests, gardens, Plan, participation and reminders do not require a separate external data API for every screen. They require properly owned records and the evidence that actually applies to each record.

External-source research identified working public samples for Open-Meteo, NWS, US Drought Monitor, openFDA food recalls, water-boundary metadata, ECHO query initialization, NC fish-advisory records and NC conservation-status records. Verification details (verification/SOURCE_CHECKS_2026-10-03.md) are limited to those checks. Keyed providers, complete production imports, account rights and the deployed database still need the explicit gates below. No blanket promise of local food measurements, current advisories, full geographic coverage or production readiness is made.

Three especially important distinctions:

- PFAS in food includes sampled findings, applicable official notices, fish advisories and private reports. No verified source here supports an all-products or all-farms PFAS safety lookup.
- Fish and conservation ArcGIS services are technically accessible. Fish sample inconsistencies prevent automatic health-guidance publication; conservation status requires separate utility instructions.
- Backend completeness includes migrations, security, jobs, source permissions, fixtures, error states and operational ownership, not just a set of successful GET responses.

Rules that must survive the expansion

Keep Today as air, UV, pollen and mold. PFAS stays in Homeguard; lead is displayed but unscored. Preserve original Journal, summary, outreach, alerts, Learn and volunteer capabilities through relocation. Today and its existing factor detail pages are not to be substantially shortened. The immutable approved archives remain untouched. The proposed fourth tab is Plan, subject to the documented product decision; backend records must not depend on a tab label.

Household categories describe presence, not a named person, diagnosis or exact age. Food advisory audiences are selected using the official definitions, not guessed from HALO's booleans. No completion state improves an environmental score. No text or animation can disguise a proxy or missing measurement. No app-facing em dashes.

Feature to backend coverage matrix

Source IDs refer to the Source Acquisition Guide (HALO_SOURCE_ACQUISITION_GUIDE.md). A User record uses S24. A row lists the additional work, not a claim that it already exists. All rows also inherit ownership, validation, missing-data, accessibility-supporting errors, export/deletion and source-provenance requirements.

Table fields: Feature / Evidence and existing foundation / Added backend and required proof

G01 Entry and household setup

Evidence and existing foundation: S07 S23 S24; onboard/profile/household

Added backend and required proof: Atomic home-context transition, reliable household read/write, context-bound backfill, civic anonymous entry, recoverable request outcomes. Prove failed utility match cannot retain the previous home's utility.

T01 Today overview

Evidence and existing foundation: S01 S02 S03 S04 S05; daily/history

Added backend and required proof: Evidence times and scope, coverage, method version, compatible-change response, truthful score/contribution semantics, linked action summary. No invented factor score.

T02 Air quality

Evidence and existing foundation: S01 S02; daily/history/map

Added backend and required proof: Pollutant/area/station distinctions, permitted history, forecast intervals, source-basis comparison, Map and Plan evidence links. Test area maxima versus station data.

T03 UV

Evidence and existing foundation: S02 S05; UV window/history

Added backend and required proof: Hourly periods and gaps, actual valid times and peak-window basis, preparation actions. Test all-low, all-missing and incomplete windows separately.

T04 Pollen

Evidence and existing foundation: S04; three existing types

Added backend and required proof: Actual daily forecast horizon, type-specific series and permissions for retained history/exports. Resolve Google storage/display gate before expansion.

T05 Mold and moisture

Evidence and existing foundation: S03 S22 S24; proxy/Learn

Added backend and required proof: Reviewed proxy inputs/version, missing-value handling, private room humidity/leak records and follow-up. No spore or image-diagnosis claim.

T06 Today household guidance

Evidence and existing foundation: Conditions plus household/Learn

Added backend and required proof: Multiple applicable reviewed notes, evidence binding, deterministic priority and canonical action dedupe. Category changes cannot alter measurements.

H01 Homeguard overview

Evidence and existing foundation: S06 S07 S08 S09; home-guard

Added backend and required proof: Coverage explanation plus owned tests/actions/maintenance summaries. Private-water no result cannot become a healthy reading.

H02 Water and PFAS

Evidence and existing foundation: S06 S07 S22; water evidence

Added backend and required proof: Complete sample/provenance import, analyte aliases, tested/nondetect coverage, threshold versions, subscriptions, utility correction and private-well workflow.

H03 Radon

Evidence and existing foundation: S08 S22 S24; county zone

Added backend and required proof: Owned test timeline, collection duration/method/unit, confirmed transcription and reviewed follow-up; pending-test reminders respect real records.

H04 Lead and plumbing

Evidence and existing foundation: S09 S22 S24; year estimate

Added backend and required proof: Versioned inventory, property-specific document records, separate water sample evidence, home-year correction and outreach. No address material inferred from totals.

H05 Home household guidance

Evidence and existing foundation: Household/home/Learn

Added backend and required proof: Selected-category applicability with related owned tasks/tests; no named-person database. Removed-category links return general guidance without deleting history.

H06 Tests and records

Evidence and existing foundation: S18 S22 S24

Added backend and required proof: Structured tests/results, correction revisions, private attachments, quarantine/scan, interpretation gates and export/delete. Unknown units allow uninterpreted storage, not comparison.

H07 Maintenance and moisture

Evidence and existing foundation: S22 S24; action templates

Added backend and required proof: Assets, meter/room observations, recurrence mode, completion records, instructions and shared Plan actions. No inferred filter efficacy.

H08 Household summary

Evidence and existing foundation: Existing daily/home/household

Added backend and required proof: Authorized selected record composition, omissions manifest, local or short-lived private export, sensitive-field preview. No default public URL.

F01 Food and Garden hub

Evidence and existing foundation: S12 S13 S14 S16 S17 S18 S19

Added backend and required proof: Optional interest preferences, coverage-aware summary, saved records and links. General use works without a site or consumption log.

F02 Fish and waterways

Evidence and existing foundation: S13 S22 S24

Added backend and required proof: Reviewed full advisory rules, waterbody/species/audience lookup, overlapping rules, version watches; optional private catch/consumption CRUD. Quarantine conflicting raw fields.

F03 Garden workspace

Evidence and existing foundation: S18 S22 S24, applicable S17

Added backend and required proof: Private sites, independent irrigation context, tests/observations, known/unknown summary and advice preparation. Old samples keep their original site/source version.

F04 Food evidence and notices

Evidence and existing foundation: S12 S14 S22, applicable S13

Added backend and required proof: Separate study/sample/recall/local-notice records, scope matching, update/termination states, source-linked action and optional private report. No fabricated local food score.

F05 Grower support

Evidence and existing foundation: S18 S19 S22 S24

Added backend and required proof: Reviewed service/programme catalogue, eligibility and source revisions, inquiry/checklist/reference records, test and follow-up links. No grant or lab booking promise.

F06 Water stewardship

Evidence and existing foundation: S16 S17 S24

Added backend and required proof: Categorical county history, utility crosswalk/status and reviewed restriction text, subscriptions and optional meter/bill usage records. Compare compatible periods only.

A01 Act overview

Evidence and existing foundation: Existing volunteer/map/home/district

Added backend and required proof: Reviewed deterministic recommendations, relevant opportunities/projects and saved participation summary. Do not create a second task system.

A02 Organizations

Evidence and existing foundation: S20; volunteer directory

Added backend and required proof: Stable reviewed organizer IDs, service areas, reporting-outdated flow and archived state; distinguish query failure from no matching organizations.

A03 Opportunities

Evidence and existing foundation: S20 S24

Added backend and required proof: Versioned event catalogue, filters, cancellations, participation lifecycle, Plan link and reminder. Opening registration is not registration.

A04 Guided projects

Evidence and existing foundation: S19 S20 S21 S22 S24

Added backend and required proof: Reviewed project templates, prerequisites/safety, shared parent/child actions, progress and follow-up. No unmeasured cleanup or health benefit totals.

A05 Outreach

Evidence and existing foundation: Public evidence plus existing recipient modes

Added backend and required proof: Versioned source-bound drafts, independent recipient edits, optional private saved drafts with retention choice, user-reported sent/reply and Plan follow-up. No automatic send.

A06 Field observations

Evidence and existing foundation: S21 S24

Added backend and required proof: Private factual observations/photos, precise/coarse location choice, sanitized reporting preview/export and external handoff states. No public contamination feed.

M01 Map shell

Evidence and existing foundation: S07 S10 S11 S23; map route

Added backend and required proof: Production layer capability/coverage, query bounds, selection lookup, accessible list pagination, source dates and safe public cache. Preview geometry is not deployment proof.

M02 Water map

Evidence and existing foundation: S06 S07; water layer

Added backend and required proof: Sample quarter/detail/query contracts, non-detect/tested metadata, per-compound coverage, utility boundary/source distinction and follows. Density is not exposure.

M03 Other map layers

Evidence and existing foundation: S01 S08 S10 S11; optional S13 S15 S16

Added backend and required proof: Layer-specific evidence, CRS/geometry checks, facility labels, station dates and honest failure. Optional sample/advisory overlays remain gated.

M04 Public comparison

Evidence and existing foundation: Published normalized evidence

Added backend and required proof: Comparable analyte/unit/matrix/method/time rules and an explicit incompatibility result. Missing cannot rank as best.

M05 District evidence

Evidence and existing foundation: S06 S07 S10 S11; district view

Added backend and required proof: Versioned geometry/aggregates, defined mapped/tested denominators and partial-population flags, public export with no household fields.

P01 Your Week

Evidence and existing foundation: S24; shared actions

Added backend and required proof: Bounded agenda/unscheduled reads, occurrences, timezone handling, filters, one-way calendar export and canonical records. Cancellation updates every view.

P02 Shared actions

Evidence and existing foundation: S22 S24; current recommendations

Added backend and required proof: Action CRUD, recommendation dispositions, versioned evidence, idempotency, conflicts, lifecycle, parent steps, occurrence and reminder transactions.

P03 Outdoor planning

Evidence and existing foundation: S01 S02 S03 S04 S24

Added backend and required proof: Normalized per-factor forecast capability, planning location/context, chosen interval, preparation tasks, optional expiring change watch. Never auto-reschedule.

P04 Journal

Evidence and existing foundation: S24; journal/history/retrospective

Added backend and required proof: Real date/vocabulary validation, all-time first-use read, safe whole-entry revisions, create-if-absent batch and per-date outcomes, backfill status.

P05 Patterns and history

Evidence and existing foundation: Authorized Journal plus permitted history

Added backend and required proof: Versioned logged-day denominator/category analysis, acquisition-basis exclusions, counts and insufficient-evidence state. No causal or clinical claims.

P06 Seasonal review

Evidence and existing foundation: Existing summary plus S24

Added backend and required proof: Consistent analysis version and owned action/test/participation joins; season identity, dismissals and summary deep links. Separate activity from measured change.

X01 Learn and sources

Evidence and existing foundation: S22 plus selected evidence

Added backend and required proof: Reviewed topic index/search, locale/version, contextual source bindings, source detail and withdrawn/unavailable states. No arbitrary web search required.

X02 Luna

Evidence and existing foundation: Existing assistant/RAG plus S24

Added backend and required proof: Effective capabilities/consent, authorized selected-record context, bounded conversation, deletion and single-use confirmed action proposals. Verify corpus rights and citation support.

X03 Alerts and watches

Evidence and existing foundation: Existing alerts/push, applicable sources

Added backend and required proof: Entity subscriptions, channel-specific preference migration, durable source events/outbox, retries/quiet hours/expiry and correct unread pagination/count.

X04 Settings

Evidence and existing foundation: Existing profile/household/preferences

Added backend and required proof: Atomic revisioned edits, interests/new feature preferences, locale and home transition, truthful data/export/delete/recovery controls. No invisible loss of categories.

X05 Privacy and recovery

Evidence and existing foundation: S24; auth/account

Added backend and required proof: Supported optional recovery, file/row/job deletion orchestration, export manifests, retention and restore drills. Anonymous session replacement is not recovery.

X06 Accessibility and offline

Evidence and existing foundation: Existing frontend caches plus S24

Added backend and required proof: Identity-scoped permitted caches, explicit stale states, idempotent draft replay if enabled, bounded payloads and safe service-worker policy. No silent sensitive offline storage.

X07 Source operations

Evidence and existing foundation: All enabled external sources

Added backend and required proof: Operator roles, registry, acquisition/review/publication, translations, source health, bounded resumable jobs, spend controls, rollback and correction delivery.

Backend implementation order

These are build packages, not separate teams or additional owners. Each package delivers migrations, server schemas, pure logic tests, route tests, fixtures and operational notes. Keep the existing approved contracts compatible while adding fields or versioned routes.

B01 Baseline and private identity

Read the installed Next.js guides and repository instructions before implementation. Audit the actual deployed schema, migration ledger and RLS read-only with authorized access. Do not infer a reproducible baseline from migrations that reference pre-existing tables. Capture a schema-only baseline without private rows or secret values.

Harden date/coordinate/range validation, household transactionality, directory error states and authorization. Introduce home-context revisions and context-bound history/backfill. Decide optional recovery before valuable uploads. Extend retention/privacy disclosures before enabling sensitive input. Required test: two identities cannot read, mutate, link or export one another's nested records through either routes or direct authenticated database access.

B02 Evidence and source operations

Create the registry, source runs, public evidence/revisions, content reviews, published pointers, thresholds and source events. Define parser interfaces, allowlisted bounded network acquisition, snapshot-retention policy and rollback. Build the reviewed-import path first so a source without a stable API still has a legitimate operation.

A release has staging validation, completeness checks, units/geography/date validation, comparison with the prior release and atomic publication. Publish corrections/withdrawals as explicit revisions. Do not use a generic JSON blob as the sole schema for incompatible food, water, weather and private evidence.

B03 Shared actions Plan and durable delivery

Create actions, recommendation dispositions, parent/child relations, occurrences, reminders, subscriptions, source events and outbox/delivery attempts. Saves return an authoritative record; retried requests resolve to the original write. Revision conflicts preserve the user's draft. Complete, skip, cancel, snooze and reopen have different effects.

Implement a durable scheduler/worker compatible with the chosen hosting plan. The existing daily 11:00 UTC cron cannot satisfy a 15-minute reminder target. Jobs lease, retry, resume and reconcile; they recheck deletion/consent/context before private work. Fixed calendar recurrence and completion-relative recurrence need separate algorithms and DST tests. Push permission is not required for Plan or the inbox.

B04 Existing evidence enrichment

Add source metadata to daily/history/home/map and score explanations without inventing factor `/100` values. Fix or justify the null-precipitation assumption. Acquire and reconcile a complete UCMR release, retaining qualifiers and sampling context. Audit static radon, lead, boundary and district versions. Confirm Google Pollen and Mapbox storage rights before expanded persistence. Each provider can fail independently without generating healthy defaults.

B05 Tests maintenance gardens and attachments

Add owned tests/results/revisions, assets, moisture observations and garden sites. Support source matrix, original units, qualifiers and confirmed transcription. Garden irrigation and household tap contexts are independent. Add optional interest preferences without changing required onboarding.

Select private object storage, scanning and temporary-upload cleanup. Use upload intents, quarantine, finalize-to-owned-record, signed retrieval and complete deletion. Extraction is optional, never auto-confirmed. A failure can preserve an uninterpreted user record while blocking an unsafe derived comparison. Tie follow-up to B03 rather than inventing per-feature task stores.

B06 Food advisories and grower resources

Implement separate FDA study, food recall, local notice and reviewed fish-rule record types. Start official NC fish ingestion as reviewed candidate records: retain raw discrepancies, full official source and review decision. A point or five-mile buffer is not the advisory extent. Resolve source reuse and exact audience/combined-limit semantics before personalized matching.

Add advisory lookup/detail/version follow, private fish notes, reviewed lab/Extension/NRCS catalogue and grower inquiry records. Retain terminated recalls and withdrawn advisories as dated historical records. Do not turn a keyword

search, empty response or local producer directory into a safety verdict. Every published card must support an actual applicability explanation and useful next action, not only an article link.

B07 Water stewardship

Add USDM release ingestion and categorical regional history, NC utility status acquisition, PWSID crosswalk validation, separately reviewed utility instructions and subscriptions. Store optional user meter/bill readings with period, unit, context and method. Test cumulative-meter reset/rollover, gaps and unlike periods before displaying usage differences. Do not claim attributed water savings or prescribe irrigation from county drought.

B08 Act and reporting workflows

Extend organizations, opportunities, event revisions, user participation and project templates. Add outreach draft snapshots and private observations, report export and official destination directory. Existing letter modes and edited drafts are preserved. No booking, message delivery, public incident map or agency write access is implied. Explicit user action is required to transmit content to an external recipient.

B09 Forecasts and regional exploration

Implement per-factor forecast capability and period arrays, issue/valid times, source-specific maximum horizon and missing periods. Store an outdoor plan's selected interval and source references; watches flag material change without moving it. Air category-only forecasts and daily pollen stay at their native resolution.

Extend map detail/filter/compare contracts and geographic release metadata, preserving unlocated records and last-good permitted data. The WQP sampling layer is optional and cannot block unrelated Map features. Source-backed area charts and accessible lists share the same records and counts as their visual map.

B10 Journal analysis and seasonal summaries

Preserve existing entry workflows while adding revisions, protected retrospective batch saves, all-time first-use state and backfill readiness. Recompute descriptive findings only from eligible logged dates and comparable environmental series, with category-specific scope and transparent exclusions. Carry source/method and permitted retention into historical records. Align seasonal summaries with the same versioned analysis.

B11 Learn Luna and cross feature integration

Version and review guidance/translations once, reuse it in factor pages, tasks, food/garden and Luna. Expose effective server capabilities and model-context consent before any private context leaves the app. Authorize selected records on the server, exclude Journal free text by default, and distinguish provider retention from local transcript storage.

Luna proposals are structured previews; confirmation revalidates ownership, current source/applicability and proposal expiry, then commits once. Test malicious retrieved text, invented citations, source contradictions, prompt leakage, rate limits and duplicate confirmation. A model must not invent missing laboratory values or resolve disputed fish rules.

B12 Release operations and handoff

Finish full export/deletion, storage cleanup, reauthentication/recovery, backup/restore, source outage drills, observability and cost limits. Record code/schema/source/guidance versions and enabled features. Test end-to-end journeys in isolated staging with disposable identities, not shared production household records. Give Yogi the actual contract/fixtures and integration status before connecting each frontend stage.

Contracts the frontend needs

Use the field and relationship tables in the backend plan as the baseline. Before implementing each package, check in a versioned OpenAPI or equivalent executable server schema and derive frontend types/test fixtures from it. Names below are proposed paths; if Vibhav chooses different names, update the shared contract before frontend integration. Do not leave two competing undocumented contracts.

Table fields: Surface / Proposed endpoints and minimum response additions

Capabilities and coverage

Proposed endpoints and minimum response additions: `GET /api/capabilities`: contract version, enabled operations, per-domain/source coverage and horizon, effective locale and privacy choices, unavailable reasons. Do not expose operator configuration or secrets.

Context and preferences

Proposed endpoints and minimum response additions: Home-context list/detail/transition; interests/preferences read and revisioned update. Return actual saved profile/context, match method and backfill state.

Evidence and source detail

Proposed endpoints and minimum response additions: `GET /api/evidence/:id`, source list/detail and reviewed topic search/detail; return evidence revision, original/native values, scope, source dates, freshness, attribution and allowed uses.

Conditions

Proposed endpoints and minimum response additions: Additive daily/history/home fields; `GET /api/forecast` and `/api/changes`. Return per-factor availability, supported granularity, source issue/valid dates, source basis and comparability reasons.

Actions and recommendations

Proposed endpoints and minimum response additions: `/api/actions`, `/api/actions/:id`, occurrence completion/reopen/skip, recommendation dispositions and reminder operations. Return stable IDs, revision, canonical related IDs, allowed transitions and duplicate match.

Plan and outdoor plans

Proposed endpoints and minimum response additions: `GET /api/plan`, outdoor-plan CRUD and watch operations. Keep unscheduled/all-day/timed items distinct, return date/timezone and per-source forecast limits; calendar export is one-way.

Tests and maintenance

Proposed endpoints and minimum response additions: Test/result/correction, asset/maintenance and attachment intent/finalize/read/delete. Return original and normalized data, qualifier, workflow state, interpretation blocked reason and authorized signed download.

Food and gardens

Proposed endpoints and minimum response additions: Advisory list/detail/match, food-report list/detail, garden CRUD, fish-note CRUD, grower resource/inquiry operations. Preserve kind, geography/audience/product scope, official action, current revision and coverage.

Stewardship

Proposed endpoints and minimum response additions: Drought history and utility-conservation reads; water-use CRUD/comparison. Separate regional, official-rule and private measurement records.

Act

Proposed endpoints and minimum response additions: Organizer/opportunity/project reads, participation CRUD, outreach draft/follow-up, private observation/report preview/export. Source status and user-reported status remain separate.

Map

Proposed endpoints and minimum response additions: Extend existing layers/detail/quarter/compare/district reads; server-backed public record search and list pagination. Respond with the same record set, filtering definition and source versions used by the map.

Journal

Proposed endpoints and minimum response additions: Extend existing entry/findings/retrospective/summary contracts; authoritative first-use/backfill status and per-date batch outcomes. Return analysis version and included/excluded denominators.

Luna

Proposed endpoints and minimum response additions: Capabilities, questions with authorized context refs, bounded conversations/delete, proposal and explicit confirmation. Response citations bind claims to reviewed evidence.

Alerts and subscriptions

Proposed endpoints and minimum response additions: Existing alert/preferences routes plus entity subscriptions, cursor pagination and exact unread total. Return effective channels, watch status and source event identity.

Data controls

Proposed endpoints and minimum response additions: Owned export job/status/download/cancel, selective deletion, full deletion job/receipt and supported recovery/device flow. Return true completion/partial states, not optimistic success.

Operators

Proposed endpoints and minimum response additions: Source acquisition/review/publication/withdrawal and job monitoring behind dedicated roles. Never expose service-role operations merely through a hidden frontend route.

Required data envelopes

For an evidence-backed value, the frontend needs the following fields or explicitly named equivalents:

```
id, revision, kind, source_id, source_record_id
native_value, native_unit, qualifier, detection_limit, reporting_limit
normalized_value, normalized_unit, conversion_version
sample_matrix, analyte_id, method, basis
observed_at OR sample_date, issued_at, valid_from, valid_to
published_at, retrieved_at, reviewed_at
scope_type, geometry_ref, geography_version, association_method
coverage, missing_reason, freshness, interpretation_version
source_locator, attribution, permitted_uses
```

Fields not applicable to a domain are absent or explicitly null under its typed schema. They are never guessed to fill this template. Use separate typed envelopes for observations, forecasts, advisory rules, notices, guidance and private records. Evidence revisions are not mutable user tasks.

List responses need `items`, stable pagination, applied filters, coverage, source versions and true empty/unavailable distinction. Mutations need an authoritative saved record and revision. Errors need a stable code, safe message, field errors, retryable flag and request ID; add `Retry-After` where applicable. Use 409 or a documented precondition response for stale revisions. Do not leak existence across owners when returning authorization failures.

Each new form must have complete create/read/edit/delete/export rules, a retained draft on error, optional-field semantics, linked-record validation and a maximum payload size. A task with no date, a test with unknown units, or an advisory search with unknown species can be valid states; none authorizes a fabricated result.

Source and schema mapping examples

These are proposed mappings from inspected fields, not complete production parsers.

Table fields: Input / HALO normalized output / Required guard

USDM `mapDate`, `fips`, `none`, `d0` **to** `d4`, `statisticFormatID=2`

HALO normalized output: Regional release and mutually exclusive county area percentages

Required guard: Preserve FIPS as string and categorical mode; do not use county maximum as a home class.

NC conservation `systemPWSID`, `conserveStatus`, `conserveReason`, `conserveDateEffective`

HALO normalized output: Utility status evidence plus raw ID and validated federal-ID crosswalk

Required guard: No fabricated restriction wording or matching solely by utility name.

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openFDA `recall_number`**, product/code/distribution fields, lifecycle dates/status**

HALO normalized output: Food recall revision with specific applicability

Required guard: Firm state is not distribution; terminated historical record cannot trigger a current recall alarm.

NC fish `Popultn`, `MlsAllw`, `Advisory`, `Site`

HALO normalized output: Unpublished candidate requiring full-document review

Required guard: Contradiction, truncation or missing extent blocks automatic personalized advice.

User lab report and entered value

HALO normalized output: Owned report plus separately verified transcription revision

Required guard: Verification confirms transcription, not laboratory authenticity; uncertain matrix/unit blocks comparison.

Current environmental source with missing date

HALO normalized output: Record with unknown observation time and real retrieval time

Required guard: Do not label as observed just now or merge into a dated trend without a reviewed basis.

Completion tests for the added backend

Build tests alongside each package, not as an end-of-project manual check. Required scenario groups are:

1. **Identity and authorization:** no foreign ID linking, no public/private cache mixing, no service-role bypass in nested reads, invalid/expired sessions preserve drafts and do not create replacement identities silently during a write.
2. **Source acquisition:** missing key, 401/403, quota/429, timeout, malformed 200, partial pagination, duplicate row, source schema change, revised release, empty-but-unexpected result and revoked source rights.
3. **Scientific meaning:** unknown versus zero/nondetect, differing units/matrices, unavailable method, forecast versus observation, model versus monitor, advisory conflict, date-only records and explicitly incomparable history.
4. **Concurrency:** repeated create key, same key with different payload, two tabs editing, out-of-order refresh, two recurrence workers, interrupted publication/fanout, stale Luna proposal and duplicate confirmation.
5. **Calendar and reminders:** DST gaps/folds, user timezone change, all-day dates, late completion, skipped occurrence, cancelled event, expired forecast watch, quiet hours and delayed push. Test with a controlled clock rather than real waiting.
6. **Attachments:** claimed versus actual type, oversized upload, malware/scan failure, abandoned upload, foreign attachment link, expired URL, extraction disagreement, metadata leakage and original/derivative deletion.
7. **Lifecycle:** source correction/withdrawal, advisory supersession, event reschedule/cancellation, house move, category deselection, test correction, record deletion and consent revocation while jobs are queued.
8. **Food and agriculture:** product/lot mismatch, broad distribution, ambiguous fish species, overlapping advisory rules, conflicting raw audience/limit, missing effective status, unrelated garden water source and unsupported crop-safety request.
9. **Operational recovery:** worker crash/retry, cache loss, source unavailable, quota ceiling, failed export, partial delete, restore with deletion replay and source-parser rollback.
10. **Full connected journeys:** save guidance to Plan; schedule test and record result; follow fish advisory revision; inspect applicable food recall and prepare a question; create garden site and lab inquiry; compare drought releases and record usage; save event and handle cancellation; prepare official observation report; Journal edit recomputes eligible analysis; Luna confirmed proposal saves exactly once.

No external registration, message send, agency submission, shared-production delete or paid-provider stress test belongs in automated tests without a specifically authorized isolated setup. Stress tests use fixtures and isolated local/staging services first.

Open decisions with explicit closure evidence

Table fields: Gate / Required closure before dependent release

Google Pollen retention and map rights

Required closure before dependent release: Confirm applicable agreement and enforce distinct expiry classes. Current non-EEA terms allow today's forecast values for 365 days, other forecast/heatmap values for 24 hours; see S04. Review exports/derived/model uses and map restrictions. Audit existing data without destructive planning changes.

Mapbox persistence

Required closure before dependent release: Confirm the actual account/version storage entitlement; use an appropriate permanent-storage request/plan if required. Rounding alone is not a resolution.

Keyed provider configuration

Required closure before dependent release: Redacted account-level success and quota/billing checks for AirNow, Google Pollen and optional OpenUV; no secrets in fixtures.

Source imports

Required closure before dependent release: Complete UCMR release/dictionary mapping and count reconciliation; reviewed radon/lead/geography versions; no arbitrary sample treated as complete coverage.

Fish rules

Required closure before dependent release: Resolve contradictory/truncated sample fields against full official source, verify current status/extent and source reuse, then test all supported audience/species combinations.

Food studies and local notices

Required closure before dependent release: Real reviewed study rows and notice scope, exact source/table locators and a defined supported region. No suitable local evidence returns unavailable, not safe.

Utility conservation

Required closure before dependent release: Confirm acquisition rights/cadence, PWSID crosswalk and official local restriction instructions. Status alone cannot generate detailed rules.

WQP extension

Required closure before dependent release: Select/version supported profile, acquire a real filtered result fixture and prove required matrix/method/limit fields. Optional layer stays conditional otherwise.

Scores and interpretation

Required closure before dependent release: Review factor `/100` meaning, contribution semantics, proxy handling, threshold effective status and expanded health guidance. Native measures remain the truthful fallback.

Platform and budget

Required closure before dependent release: Choose and verify scheduler, private storage/scanning, recovery mechanism, retention periods, source review capacity and actual cost limits.

Deployment baseline

Required closure before dependent release: Authorized schema/RLS/config audit and isolated migration/restore tests. Do not apply speculative migrations to unknown production state.

New visuals

Required closure before dependent release: Yogi's review remains necessary for new pages. Backend implementation does not grant permission to replace approved frontend geometry or archives.

These gates are work to complete, not reasons to delete planned functionality or claims that permission has already been granted. If a gate cannot be resolved, document the precise affected promise and obtain a product decision about its fallback before marking that feature complete.

What Vibhav should return for frontend integration

For every completed build package, provide:

- Migration identifiers and safe rollout/backfill/rollback instructions.
- Versioned executable API schema, generated or validated frontend types and documented compatibility changes.
- Representative success, partial, empty, unavailable, stale, conflict and unauthenticated fixtures with no real private data.
- Source manifest with verified request/coverage/rights, parser tests, reviewed interpretation and actual refresh policy.
- Endpoint and job test results, RLS isolation results, cost limits and source freshness monitoring.
- Enabled feature/capability list and any remaining source/operation blockers.
- A staging demonstration of the complete connected journey, including reload persistence and failure recovery.

The frontend can then integrate against explicit contracts instead of guessing source semantics. Calling the package ready for integration does not mean every data source or feature in the entire app has shipped.

Repository status of this handoff

This documentation and its verification tools were saved locally in the existing repository. This task does not itself create a PR, push a branch, contact Vibhav, alter approved archives or implement backend routes. Preserve the unrelated map work in the current working tree. The source checks can be rerun explicitly; results and permissions must be revalidated at implementation and release time.

02 / Backend and data requirements

docs/product/HALO_BACKEND_AND_DATA_PLAN.md

HALO Backend and Data Plan

This document defines the proposed backend and data target for HALO's full product plan. It is written for Yogi and Vibhav so each interface has a credible source, a complete action workflow, and an explicit implementation boundary. It complements the HALO Master Plan (HALO_MASTER_PLAN.md), Frontend Build Plan (HALO_FRONTEND_BUILD_PLAN.md), and HALO Feature Acceptance (HALO_FEATURE_ACCEPTANCE.md).

Version 1.2, October 3, 2026. This is the proposed full-app backend target and implementation specification, not implemented functionality, verified deployment readiness, or approval of new visuals. Existing capability statements below describe inspected repository code, not an assertion that every provider, migration, credential, or scheduled job is configured in production. No existing feature is removed by this plan. Endpoint and table names introduced here are proposals; implementation may consolidate them without losing their behavior. The route inventory and hardening list make explicit what can be reused and what must change before the richer product can depend on it.

Start implementation with the Backend Build Handoff (HALO_BACKEND_BUILD_HANDOFF.md), which maps all 45 feature groups to backend work and completion gates. The Source Acquisition Guide (HALO_SOURCE_ACQUISITION_GUIDE.md) adds exact acquisition paths, inspected field mappings, permissions and unresolved source-specific constraints. Its verification report (verification/SOURCE_CHECKS_2026-10-03.md) records fourteen public sample requests, not production validation. These companions supersede earlier statements that no NC fish or conservation endpoint had been located; actual public services were found, with important interpretation and operational gates still open.

Architecture decisions

Extend the existing Next.js application and Supabase database rather than introducing a service for each feature. Shared evidence, saved actions, source references, and reminder delivery should connect the product. Public source ingestion is separate from household-owned records. A new food advisory, household test, conservation event, and daily air reading are different evidence types, not interchangeable inputs to a universal safety score.

The central connection is an immutable reference to evidence plus a mutable household action. A user can save a recommendation from Today, Homeguard, Act, Food and Garden, Map, or Luna and manage that same action in Plan. Completing it changes task state, not the environmental measurement or medical interpretation.

Public data may be cached and reused across nearby households where the source's resolution permits it. Private notes, household composition, appointments, test attachments, and check-ins never enter public map layers or shared provider caches.

Existing backend baseline

Table fields: Capability / Inspected implementation / Implication for the target

Identity and household

Inspected implementation: Anonymous Supabase sessions, verified API identity, seven presence categories, profile home details

Implication for the target: Reuse ownership and low-friction onboarding. Recovery of a rejected session currently creates a new anonymous identity; it does not recover the old household's records.

Today

Inspected implementation: Daily readings, source-specific failure handling, score, factor severities, household wording, saved history

Implication for the target: Extend observation metadata and forecasts. Do not rebuild the current daily pipeline as a separate feature.

Homeguard

Inspected implementation: Water contaminant detail, county radon estimate, lead building-era estimate, private-water testing guidance, deterministic action recommendations

Implication for the target: Add saved records and follow-through while retaining all existing source distinctions.

Journal

Inspected implementation: Entries by date and household category, history, retrospective entries, findings, season summaries

Implication for the target: Preserve these capabilities and correct comparison semantics before presenting stronger patterns.

Map

Inspected implementation: Cached water, radon, air monitor, facilities, and district layers

Implication for the target: Reuse data and geography. Map is not currently a continuous contamination measurement surface.

Act

Inspected implementation: Public organization directory, cause and county matching

Implication for the target: Extend into opportunities and participation. An organization entry is not an event feed.

Luna

Inspected implementation: Curated-source retrieval, cited answers, household context, diagnostic refusal, rate limits

Implication for the target: Add explicit privacy capabilities, selected-record context, conversation handling, and confirmed action proposals.

Notifications

Inspected implementation: Alert inbox, read/dismiss handling, five preference groups, browser push, deduplication

Implication for the target: Extend entity subscriptions and scheduled reminders instead of creating a second notification stack.

Account deletion

Inspected implementation: Verified-session deletion through the auth user with database cascades

Implication for the target: Extend coverage to every new table, attachment, conversation record, and queued job.

The current migrations assume some original tables already exist, including `profiles`, `daily_scores`, `home_risks`, and `ucmr5_utilities`. Before new migrations, capture and review the actual database schema and policies. The files in `supabase/migrations` alone must not be assumed to constitute a complete fresh-install baseline.

Current route inventory

This inventory covers all 22 route files currently under `app/api`. Every household route calls the existing verified-bearer-token helper. Public-reference responses still require a session at the route even where database read policies allow public reference access. The daily job instead requires its scheduler secret. Table entries describe current behavior, including limitations, rather than the desired replacement contract.

Table fields: Existing route and methods / Current inputs and response purpose / Ownership and compatibility requirement

`/api/onboard` POST

Current inputs and response purpose: `address`, optional `water_source`, `home_year`, `request_id`; geocodes, matches service area, stores approximate location, returns location/utility and request marker, starts 28-day history backfill

Ownership and compatibility requirement: Session supplies owner; legacy `profile_id` must match. Preserve request-outcome reconciliation. Current `request_id` is a stored marker, not a full idempotency transaction.

`/api/profile` GET and PATCH

Current inputs and response purpose: Reads onboarding flags, home/profile and seven household booleans; PATCH accepts `water_source` and/or `home_year`

Ownership and compatibility requirement: Keep existing field names and null behavior. Renter and locale writes currently belong to household PUT, not profile PATCH.

`/api/household` GET and PUT

Current inputs and response purpose: Reads composition, renter mode and locale; PUT replaces the seven booleans and optionally writes `renter_mode`, `locale`

Ownership and compatibility requirement: Omitted booleans become false. Preserve full replacement or add an explicitly separate partial-update contract. Current bands/preferences writes are not one transaction.

`/api/daily-score` GET

Current inputs and response purpose: Optional `lat`, `lng`, `fresh=1`, legacy matching `profile_id`; returns `air`, `uv`, `pollen`, `mold`, `score`, `retrieved_at`, `cached`

Ownership and compatibility requirement: Other-point lookup does not write home history. Current own-location cache lasts one hour and provider refresh is limited per household. Extend source times without renaming current fields.

`/api/home-guard` GET

Current inputs and response purpose: Profile defaults with county, utility, water-source and home-year overrides; returns `radon`, `water`, `score`, `lead`, `action_plan`, `breakdown`, `assembled_at`

Ownership and compatibility requirement: Overrides allow public context exploration; do not expose another household's private tests through them. Keep private-water/no-match/lookup-failed distinctions.

`/api/history` GET

Current inputs and response purpose: `days` defaults to 90 and is capped at 365 when used, or `from`, `to`; newest stored row per household date becomes a history point

Ownership and compatibility requirement: Owner scoped. Explicit ranges currently need stronger validation/bounds. Add retained source/basis metadata before richer comparisons.

`/api/journal` GET, POST and DELETE

Current inputs and response purpose: GET date range; POST whole date/category record: `entry_date`, `band`, `symptoms`, `severity`, `note`, `retrospective`, `possibly_illness`; DELETE `id` or `all=true`

Ownership and compatibility requirement: Unique date/category upsert. Preserve all fields during edit or Undo. DELETE returns actual row count. New revision checking must coexist with legacy upsert.

`/api/journal/findings` GET

Current inputs and response purpose: Current household-level 90-day analysis and readiness/counts

Ownership and compatibility requirement: No current category-filter contract. New category analysis must compute scoped findings, not relabel existing results.

`/api/journal/retrospective` POST

Current inputs and response purpose: `dates` array, at most 62 accepted dates; returns environmental comparison across the applicable months

Ownership and compatibility requirement: This comparison call does not save Journal entries. Separate entry saves need partial-batch reporting and retry reconciliation.

`/api/journal/summary` GET

Current inputs and response purpose: Optional `season`, `year`, otherwise last ended season; returns seasonal descriptive summary

Ownership and compatibility requirement: Preserve winter-year definition. Extend with action/test participation records without changing old analysis meaning silently.

`/api/map` GET

Current inputs and response purpose: `layer` accepts water, radon, facilities or air; optional water `quarter`; cached layers or live rebuild

Ownership and compatibility requirement: Public-reference layer cached without household fields. Quarter response uses samples in that quarter, not a cumulative plume.

`/api/district` GET

Current inputs and response purpose: NC08 aggregate plus the authenticated household's county context

Ownership and compatibility requirement: Shared aggregate and per-household county enrichment must remain separate cache scopes. Geography vintage requires release verification.

`/api/volunteer` GET

Current inputs and response purpose: County and comma-separated causes; defaults county from profile; returns matched organization rows

Ownership and compatibility requirement: Reuse directory and matching. A returned empty list with an unavailable note currently differs from a proven no-match result.

`/api/learn` GET

Current inputs and response purpose: `topic`, `locale`, plus factor-specific context query fields; returns definition, reading-specific explanation, one applicable household note, protective steps and sources

Ownership and compatibility requirement: Existing seven topics and English fallback remain. New multiple-category guidance, glossary search and additional domains require additive contracts.

`/api/assistant` GET and POST

Current inputs and response purpose: GET page suggestions; POST `question` up to 1,000 characters and optional allowlisted `page`; one cited answer or refusal/unavailable result

Ownership and compatibility requirement: No current conversation, upload, selected-record or mutation-tool contract. Household context is assembled server-side. Do not attach private text to GET query parameters.

`/api/alerts` GET and POST

Current inputs and response purpose: GET latest 100 undismissed alerts and unread count in that returned set; POST one read, all read, or one dismissed boolean

Ownership and compatibility requirement: Owner scoped and updated count returned. New global unread count/cursor pagination is an explicit addition. Dismissal is not deletion or task completion.

`/api/notifications` GET and PUT

Current inputs and response purpose: Five booleans: `air_quality_change`, `weather_advisory`, `new_water_results`, `radon_season`, `season_summary`; PUT accepts a partial preferences object

Ownership and compatibility requirement: Existing false preference suppresses both inbox creation and push. Channel-specific controls need defined migration, not a frontend-only reinterpretation.

`/api/push/subscribe` POST and DELETE

Current inputs and response purpose: Browser subscription JSON, or DELETE `{endpoint}`

Ownership and compatibility requirement: Endpoint unique; subscription upsert currently reassigns an existing endpoint to the current verified user. Rebinding and multi-device tests must prevent old-household notifications on a reused device.

`/api/sources` GET

Current inputs and response purpose: Source list and last retrieved values composed from household history and reference datasets

Ownership and compatibility requirement: Retrieval metadata is not observation date. Add registry/source-revision detail without breaking the existing list.

`/api/account` DELETE

Current inputs and response purpose: Literal `{confirm: "DELETE"}` then auth-admin user deletion

Ownership and compatibility requirement: Current cascade-based delete has no file/job/export lifecycle. Extend deletion before enabling those domains.

`/api/health` GET

Current inputs and response purpose: Authenticated fixed-location probes of providers, database and Redis

Ownership and compatibility requirement: Does not prove valid schemas, source freshness, corpus quality, migrations, deployed scheduler, or successful end-to-end workflows. Bound and restrict probe cost before operational use.

`/api/cron/daily` GET and POST

Current inputs and response purpose: Header or legacy query scheduler secret; geography/layers, water-change detection, daily readings, alerts, push

Ownership and compatibility requirement: Keep one authorized scheduler entry point while moving expensive work into resumable batches. Never invoke against production as a harmless health check.

The seven household fields remain `has_toddler`, `has_child`, `has_teen`, `has_adult`, `has_senior`, `has_pregnant`, and `has_respiratory`. Presentational category labels do not change these stored keys. Dates currently follow `America/New_York`; UTC instants and local calendar dates must not be interchanged during expansion.

Access is classified by data, not by whether a bottom tab is visible. Public regional Map, district evidence, organization and reviewed-source browsing should remain useful without a completed household profile. The proposed civic entry can reuse verified anonymous bootstrap without asking for an address. Home pin, household rankings, personalized guidance and all private records remain absent until their own authorized context exists. A truly unauthenticated public route would require a separately reviewed read-only cache, abuse and quota policy; this plan does not silently remove existing route authentication.

Existing schema and service reuse

Table fields: Existing storage or module / Keep and extend / Boundary that must remain visible

`auth.users`, `profiles`, `household_bands`

Keep and extend: Same anonymous identity and household category model; additive home contexts and consent records

Boundary that must remain visible: `profiles.id` matches the auth identity today. New recovery or membership cannot casually rekey every row.

`daily_scores`, `home_risks`

Keep and extend: Preserve owned history/cascades and any deployed dependencies; inspect actual use before migration

Boundary that must remain visible: Daily scores allow multiple writes per date, with newest chosen for history. Do not add date-only uniqueness that loses legitimate revisions.

`ucmr5_utilities`, geography cache, reference JSON

Keep and extend: Retain import and presentation adapters; add sample provenance and qualified evidence revisions

Boundary that must remain visible: Existing detections and boundary matches do not prove complete tested denominators or household exposure.

`symptom_logs`

Keep and extend: Same original Journal records and uniqueness

Boundary that must remain visible: Preserve note, illness, retrospective, category and dates through migration and export.

`volunteer_orgs`, `learn_content`

Keep and extend: Extend existing curated catalogs with explicit publication/review/version metadata

Boundary that must remain visible: Directory name uniqueness is not a permanent source identity. Published guidance and translations need revision history.

`alerts`, `notification_prefs`, `push_subscriptions`, `water_snapshots`

Keep and extend: Existing inbox and push delivery form the one notification subsystem

Boundary that must remain visible: Add subscriptions, delivery attempts and outbox rather than a second inbox; migrate old preference semantics deliberately.

`map_layers`, `lib/mapBuild.js`, `lib/districtView.js`

Keep and extend: Reuse canonical spatial assembly and district calculations

Boundary that must remain visible: Current stored layers age out after 36 hours and public layer queries are capped. Real source freshness differs by layer.

`assistant_corpus`, `match_assistant_corpus`, `lib/assistantRag.js`

Keep and extend: Same curated retrieval foundation, source citations and refusal path

Boundary that must remain visible: Migration 0013 changes embeddings to 2,048 dimensions. Model, dimensions, stored vectors and RPC must match; adding food/garden material needs reviewed source ingestion.

`lib/actionPlan.js`, `lib/wellTestData.js`, `lib/explain.js`

Keep and extend: Seed versioned action templates and explanations from reviewed existing logic

Boundary that must remain visible: These generate recommendations, not persistent user actions. P02 adds follow-through, not a second competing recommendation engine.

`lib/serverAuth.js`, `lib/authResult.js`, `lib/ratelimit.js`

Keep and extend: Verified identity, 401 versus503 handling, rate/spend guards

Boundary that must remain visible: Service-role access bypasses RLS. Every query and nested record must still be authorized in the server.

Baseline hardening before expansion

These are implementation requirements discovered in current code, not fixes performed by this planning document:

1. **Historical comparability:** `lib/backfill.js` retains `backfilled`, daily-peak modelled AQI and daily-max UV basis, but `/api/history` currently omits `details` and `aqi_source`. Return basis/time/quality before comparisons, and keep a current snapshot distinct from a daily peak. New-source history must not rewrite a past measurement as if it were observed then.

2. **Date and request validation:** Journal currently accepts a date-shaped string and arbitrary symptom strings capped at 20, while history explicit ranges are not equivalently bounded. Add real calendar-date validation, permitted vocabulary, finite coordinate bounds, range ordering and payload limits; retain an explicit read-only legacy path for historical unsupported values rather than deleting them.

3. **Atomic household edits:** the existing household read can normalize absent results when some database reads fail, and PUT writes bands before profile preferences. New edits need reliable read-error distinction, transactionality and a revision so partial or stale saves cannot silently erase category selection. Keep the current frontend's identity preflight until a tested replacement exists.

4. **Address transitions:** geocoding can update coordinates while retaining a previous utility after service-area lookup failure. New home contexts must atomically isolate that new location, and queued history backfills must be bound to the old/new context correctly. A request marker alone is not an atomic move.

5. **Durable source-change delivery:** current utility fingerprints are advanced before household alert fanout, and push sends only for newly inserted alerts. A crash can consume a change or lose push retries. Store source-event/outbox intent transactionally and resume each fanout/delivery independently.

6. **Last good map data:** current reads reject a stored layer older than 36 hours and may return 503 after live rebuild failure even though a previous copy remains stored. Serve a clearly dated stale copy under a validated domain policy, suppress unsupported interpretation, and refresh in the background; retaining a row alone is not offline/failure support.

7. **Meaningful freshness:** daily cache retrieval time, map assembly time and source-list fetch time cannot replace provider observation/publication times. Add provider issue/valid times, measurement basis and missing-metadata reason to canonical adapters.

8. **Alerts and reminders:** correct unread pagination/count semantics, add read/dismiss Undo tests, and migrate preference/channel behavior. Existing seasonal radon evaluation hardcodes no home test; use qualified H06 test state instead of continuing to claim none exists.

9. **Explicit degradations:** distinguish missing optional migration, failed query and genuine empty data in Learn, directory, profile and map responses. `/api/volunteer` currently returns 200 with an empty list and unavailable note on any directory database error; add an explicit directory-unavailable state so empty matches are not mistaken for working coverage. Audit legacy raw error forwarding and source/provider copy before new screens depend on it. Keep unknown provenance unknown.

10. **Operational honesty:** `/api/health` is reachability only. Scheduler completion, row counts, actual source lag, deployed RLS, source rights, model context behavior, and device push delivery require separate evidence. No production configuration or secret values were inspected for this plan.

11. **Provider rights and retained history:** the October 3 audit identified specific Google Pollen storage/display restrictions and Mapbox temporary/permanent storage distinctions. Resolve applicable account/version permissions before expanded persistence, exports, model context or map reuse. Do not assume existing implementation proves permission, and do not delete existing data as part of this planning task.

12. **Weather proxy null handling:** `moldFromNws` currently maps null precipitation to zero. The audit did not establish an authoritative basis for that rule. Correct or justify it with endpoint-specific evidence and tests; return missing when

required inputs are unknown. Select the applicable time interval rather than assuming array position alone establishes current conditions.

Common evidence contract

Every evidence-backed screen and recommendation needs the same meaning for provenance and missing information. Additive adapters can provide this contract while older endpoints remain compatible.

Table fields: Field group / Required meaning

Identity

Required meaning: Stable evidence ID, revision, domain, source record ID, and any superseded record ID

Value

Required meaning: Original value and unit, normalized value and unit when valid, parameter or analyte, and explicit reporting qualifier

Evidence type

Required meaning: Observed, modelled, forecast, proxy, official advisory, user reported, laboratory document, or reviewed guidance

Time

Required meaning: Observation or sample time, publication time, provider issue time when relevant, ingestion time, and validity interval. These must not be substituted for each other.

Geography

Required meaning: Point, monitor area, utility service area, county, waterbody segment, district, site, or unknown. Include geometry version and association method.

Coverage

Required meaning: Parameters available and missing, reporting limits, sampling coverage, and known exclusions

Provenance

Required meaning: Source ID, exact supporting URL or document locator, method version, transformation version, and retrieval record

Interpretation

Required meaning: Severity where a defensible rule exists, threshold or guidance reference, effective date, and applicability conditions

Freshness

Required meaning: Source-specific current, stale, superseded, expired, unavailable, or unknown state with a reason

An absent value must carry a reason such as no coverage, never collected by HALO, provider failure, not applicable, or insufficient information. Do not encode these states as zero, a healthy status, or an empty chart that resembles an all-clear.

For non-detect laboratory results, retain the qualifier and detection or reporting limit. A result reported below a limit is not a measured zero. Preserve the source's statement rather than coercing it to an exact concentration.

Staleness is domain-specific. An hourly air observation, a county radon map, a dated utility sample, and an event registration notice cannot share one expiration rule. The user should see the actual source date even if the ingestion process ran a minute ago.

Source registry and acquisition gates

Maintain a small reviewed source registry rather than embedding untracked source assumptions in components. Each entry needs an owner, official landing page, access method, geographic coverage, update cadence, usage terms, attribution, cost or quota, parser version, sample payload, failure policy, and last successful verification. A reviewed manual import is valid when no reliable API exists. It must be described as a reviewed import, not live monitoring.

Table fields: Domain / Repository foundation or candidate source family / Proposed acquisition and feasibility gate

Air

Repository foundation or candidate source family: Existing AirNow lookup with Open-Meteo model fallback

Proposed acquisition and feasibility gate: Preserve measured versus modelled. Verify observation timestamps, reporting area, parameter coverage, licensing, and permitted forecast access before increasing polling. AirNow documentation (https://docs.airnowapi.org/faq) describes variable forecast availability and category-only cases; do not manufacture numeric forecasts.

UV and weather

Repository foundation or candidate source family: Existing Open-Meteo current UV and one-day hourly lookup; NWS hourly data; guarded OpenUV fallback

Proposed acquisition and feasibility gate: Add issue time and valid time. Verify forecast horizon and uncertainty per provider. The NWS API (https://www.weather.gov/documentation/services-web-api) supports weather forecasts, not an automatic seven-day forecast for every HALO factor.

Pollen

Repository foundation or candidate source family: Existing Google Pollen request for one day

Proposed acquisition and feasibility gate: Source S04 documents the one-to-five-day request range and current non-EEA retention exceptions: today's values 365 days, other forecast/heatmap values 24 hours. Verify account applicability, geography, attribution, quota, cost and exports/model/map rights. Preserve categories and retention classes separately; do not promise seven days.

Mold

Repository foundation or candidate source family: Existing proxy from NWS humidity and precipitation conditions

Proposed acquisition and feasibility gate: Version and review the heuristic. It is not a measurement of outdoor spores or indoor mold. No background illustration or wording may imply detection.

Utility water

Repository foundation or candidate source family: Existing UCMR utility data and geography

Proposed acquisition and feasibility gate: Extend import records to retain tested analytes, non-detect qualifiers, dates, sampling context, and source releases where available. Verify current import coverage before claiming a denominator.

Radon and lead

Repository foundation or candidate source family: Existing county radon map, building-era lead estimate, utility inventory context

Proposed acquisition and feasibility gate: Retain geography and estimate basis. Add household test evidence separately. Verify dataset versions and any source-specific limits.

Official alerts

Repository foundation or candidate source family: Existing NWS weather advisory parsing and stored water-result change detection

Proposed acquisition and feasibility gate: Add authoritative utility or agency advisories only through a verified feed or reviewed publication workflow. A NWS warning is not a boil-water notice.

Fish advisories

Repository foundation or candidate source family: State health agency public ArcGIS records plus reviewed official documents

Proposed acquisition and feasibility gate: October 3 samples verified two public services behind the NC advisory landing page (https://epi.dph.ncdhhs.gov/oee/fish/arcg.html). A sampled row has truncated advice and an apparent audience/limit conflict. Use the exact endpoints and quarantine/review rules in source S13 of the acquisition guide. Complete text, extent, audience, effective status and reuse still require validation.

Food information

Repository foundation or candidate source family: Official food studies, openFDA food enforcement notices and reviewed local records

Proposed acquisition and feasibility gate: Source S14 supplies a sampled openFDA recall API, including historical PFAS-related records. Keep lifecycle/product/lot scope. FDA sampling publications (https://www.fda.gov/food/process-contaminants-food/testing-food-pfas-and-assessing-dietary-exposure) remain separate study imports, not a local-product safety API. Do not label unsampled nearby products or farms.

Gardens and growers

Repository foundation or candidate source family: Proposed Extension, agriculture agency, certified testing, and conservation program resources

Proposed acquisition and feasibility gate: Routine NC soil testing (https://www.ncagr.gov/Divisions/Agronomic-Services/soil-testing) serves fertility needs, not automatic contaminant clearance. NC drinking-water laboratory information (https://www.deq.nc.gov/about/divisions/water-resources/drinking-water/drinking-water-laboratories) requires checking certification scope. Store program eligibility and review dates. Guidance must specify when professional advice or testing is required.

Environmental water samples

Repository foundation or candidate source family: Optional Water Quality Portal services (https://www.waterqualitydata.us/webservices_documentation/)

Proposed acquisition and feasibility gate: Services expose stations, results, and detection-limit information. Gate ingestion by parameter, matrix, method, location, and date. These are contextual environmental samples, not household tap or food clearance.

Water stewardship

Repository foundation or candidate source family: US Drought Monitor and the public service behind NC utility conservation status

Proposed acquisition and feasibility gate: October 3 samples verified a categorical county drought response and the official NC map's conservation service. Sources S16 and S17 in the acquisition guide give exact endpoints and fields. Confirm reuse/cadence, utility ID crosswalk and full official restriction instructions; status alone cannot generate local watering rules.

Conservation

Repository foundation or candidate source family: Existing reviewed-organization structure; proposed organizer event pages and program feeds

Proposed acquisition and feasibility gate: Verify organizers, service areas, source permission, event changes, and registration links. Begin with curated coverage rather than claiming universal listings.

Field observations

Repository foundation or candidate source family: Proposed private observations and official environmental reporting destinations

Proposed acquisition and feasibility gate: NC Stream Watch (https://www.deq.nc.gov/about/divisions/water-resources/water-resources-public-information/water-education-programs/water-education-and-outreach/nc-stream-watch) offers an official participation/reporting path; no submission API is established here. Verify the appropriate destination, jurisdiction, and form for each report. No agency receipt is assumed.

Map context

Repository foundation or candidate source family: Existing EPA ECHO, AirNow, public-water geography, and NC district data

Proposed acquisition and feasibility gate: Preserve facility selection rules, source dates, district version, and limits of centroid-based association. A facility record is not proof of household exposure.

Every new source must pass a feasibility review before a live UI promises its output. Required proof is a real sample, parsing tests, coverage assessment, attribution and access review, sustainable refresh cost, and an honest missing-data state. Sources without a usable machine-readable interface can support reviewed content and manual updates without an invented realtime claim.

For grower assistance, the NRCS conservation evaluation and planning directory (https://www.nrcs.usda.gov/programs-initiatives/eqip-environmental-quality-incentives/cpas-dias-and-cemas) includes CEMA 209 for PFAS assessment and CEMA 207 for site contamination assessment. These are possible official pathways, not an assurance of eligibility, local availability, funding, or approval. The Extension county-center directory (https://www.ces.ncsu.edu/local-county-center/) supports contact discovery, not an assumed appointment-booking integration.

Score and interpretation safeguards

Preserve Today as air, UV, pollen, and mold. PFAS remains in Homeguard. Lead remains shown but unscored under the approved contract. Household categories change relevant guidance and priority, not environmental measurements.

Retain the current canonical severity vocabulary and the documented `no_data` fallback. `lead.level` and the water-status union are existing specialized fields, not errors to rename globally. New types may add explicit domain states but must not break Part D clients by replacing existing keys. Preserve dates as ISO fields additively while retaining original source strings where needed. Historical severity comparisons require the methodology version used at the time; `/api/history` currently derives severity with the current code and is not a versioned interpretation archive.

Existing air-change notifications trigger earlier for sensitive categories under `lib/alertRules.js`. The actual cron marks respiratory, toddler and senior presence as sensitive through `hasSensitiveGroup`. Preserve that behavior under a versioned alert policy; do not broaden it to all household categories or revise thresholds without a reviewed rationale. It changes notification relevance, not a reading or score.

The daily API provides risk inputs and a composite display score, but not an independently defined display score for each factor. Decide and validate the server-side meaning of any factor `/100` score before making it a live requirement. Otherwise show the native measure and severity, retaining the approved card structure with honest data states.

Current contribution bars normalize available risk inputs for visual geometry. They are not automatically percentages of the composite equation or shares of a person's health risk. A proposed explanation endpoint should return the actual formula version, included inputs, exclusions, and contribution semantics. Do not derive an unsupported explanation from an attractive ring.

Keep confidence, data coverage, score, and action completion separate. A user installing a filter or completing a test does not establish a measured improvement. Household results can support a separate documented history, but incorporation into a summary assessment requires a reviewed methodology with sample comparability, time, location, and quality rules.

Regulatory limits, health guidance, and HALO display thresholds are different objects. Store their jurisdiction, legal or advisory status, effective date, source, and review state. Verify current sources before changing them; comments in `lib/scoring.js` are not a continuing legal review.

Qualified domain review is a release dependency for expanded health interpretation, symptom findings, audience-specific guidance, test comparisons and product claims. A disclaimer alone does not establish clinical validation, regulatory status or an exemption. This document specifies the review requirement rather than making a legal determination about the finished product.

A single result above a comparison benchmark is not automatically a legal violation. Compliance may depend on averaging, sampling rules, effective dates, and jurisdiction. Separate a reported concentration comparison from an official compliance determination, and cite the specific record supporting either claim.

Proposed domain records

Use domain tables or clearly validated typed records where fields and ownership differ. Do not create a generic unvalidated JSON record that permits private evidence, public facts, and model-generated suggestions to become interchangeable.

Evidence and reviewed content

Proposed groups are `sources`, `source_runs`, `evidence_records`, `evidence_revisions`, `guidance_versions`, and `content_reviews`. Public ingestion uses service credentials only. Publication requires validation and, for curated content, a review state. Keep raw source snapshots only when terms and privacy allow; retain a hash and retrieval metadata otherwise.

A recommendation references the evidence revision and guidance version that produced it. Updated evidence may mark a saved recommendation as needing review. It must not silently rewrite what the user originally acted on.

The proposed Learn glossary/topic index uses these reviewed guidance records with topic IDs, related factors, synonyms, language and content version. A bounded text search over published topics is sufficient initially. Search results retain source/review metadata and distinguish no match from a failed search. Do not introduce arbitrary web retrieval or a second unreviewed content store for this index.

Home contexts and changes of address

Add proposed `home_contexts` with an owner, stable context ID, effective start/end, approximate coordinates, geography and timezone, water-source type, utility association and matching method, verification state, and current/archived status. Relevant daily history, home estimates, tests, garden sites, observations, actions and subscriptions reference the appropriate context. A garden or outdoor activity may instead have its own optional location context; it does not change the home.

An address change must be a validated idempotent transition: resolve or explicitly fail the new association, create the new context, switch the current reference, and retain the old context with its records. A genuinely new home cannot inherit the old utility merely because the new lookup failed. Preserve the entered draft and offer retry or an honest unmatched-utility state. Reconfirm applicable subscriptions and let the user deliberately carry portable generic tasks forward. Do not relabel old tests or daily history as belonging to the new property.

The current onboarding handler updates the profile in place, can retain its previous utility after lookup failure, and refreshes history keyed to household/date. The new context design is an explicit backend change. Preserve legacy history with an unknown context where its original location cannot be established; do not guess from the present address. Define timezone migration for dates, since current date logic assumes America/New_York. Historical calendar dates keep their original interpretation.

Actions and Plan records

Proposed `household_actions` fields include household owner, action type, template and version, origin page and record, evidence references, selected household categories, title, editable note, priority rationale, optional due time, time zone, state, completion time, and concurrency version. A recommendation key supports duplicate detection without preventing a deliberate repeat test or recurring task.

Proposed `plan_occurrences`, `reminders`, and `action_events` store recurring instances, reminder delivery, and a bounded history of meaningful changes. Recurrence rules are validated and edited as this occurrence, future occurrences, or the whole series. Store calendar-anchored versus completion-relative mode explicitly. A late completion or skipped fixed occurrence must not move the calendar series; completion-relative intervals start from the confirmed completion under a disclosed rule. Scheduler and completion-triggered generation share an occurrence dedupe key. Existing occurrences must not shift unexpectedly when daylight saving changes.

Add proposed `recommendation_preferences` or an equivalent typed owned record for Not relevant, Already doing this, Save for later, and Stop reminder. Key preferences to the recommendation/template version, applicable home/site context, optionally evidence revision, disposition and validity/review date. These choices are reversible and distinct from task completion. Already doing this may offer to link an existing action, but never inserts an unverified completion

or improves a score. Stop reminder cancels the chosen reminder scope; it does not hide a relevant official advisory. A materially changed source or context can request review under a disclosed deterministic policy, not silently discard the preference. Recommendation ranking uses reviewed applicability and priority rules, not model-generated health-risk rankings.

Saved, scheduled, in progress, waiting, completed, paused, and cancelled are distinct states. Overdue is normally derived from a due time, not a terminal state. Completion and reopening require an explicit user action. A walkthrough's Next button is navigation, not completion.

Proposed `outdoor_plans` store owner, shared action ID, activity label/type, optional approximate planning location, timezone, duration, preferred/candidate periods, chosen interval, relevant household categories, preparation tasks and referenced forecast revisions. The source issue time and period resolution are retained. A plan can exist before forecasts cover its date. An opt-in `forecast_watch` binds the plan, applicable factors, supported area/period, expiration and reviewed material-change rule; it cannot depend on arbitrary continuously precise tracking. Refreshes mark the plan for review and offer a deliberate reschedule, never silently change the chosen time. Create/edit/cancel/delete and export obey the same ownership, revision, retention and idempotency rules as other private records.

Tests and maintenance

Proposed `household_tests`, `test_results`, `home_assets`, and `maintenance_records` cover radon, drinking water, relevant soil or garden testing, and user-chosen maintenance. Store test purpose, collection context, planned and sampled dates, lab or provider, analytes, original and normalized units, qualifiers, reporting limits, result document reference, and user verification state.

A result moves through draft, awaiting result, received, needs clarification, user verified, and superseded states as applicable. User verification means the user confirmed transcription, not scientific certification. Preserve original results when corrected. Prevent invalid unit conversions, ambiguous dates, incompatible sample contexts, and missing qualifiers from generating confident interpretation.

Retain the original document and extracted or entered transcription as distinct records for as long as the user's retention choice permits. A normalized value must record its conversion and source value. Units depend on sample matrix: water concentration cannot be compared directly with soil or tissue concentration, and soil results may use dry-weight or wet-weight bases. Any unsupported conversion or missing basis blocks automated comparison rather than silently selecting a default.

Maintenance intervals need an explicit basis such as a manufacturer instruction, reviewed guidance, or user choice. Calendar elapsed time alone cannot establish filter performance or remaining contaminant removal capacity. A completed replacement is a household record, not a verified water improvement.

Food and Garden

Proposed public groups are `advisories`, `advisory_rules`, `waterbodies`, `food_reports`, `grower_resources`, and `conservation_programs`. Advisory rules retain official species, geographic extent, audience definitions, preparation guidance where present, frequency or restriction text, issue date, and revocation state. Do not infer that HALO's seven household categories exactly match official advisory groups.

Matching also needs species groups and synonyms, waterbody segments, serving definitions, time periods, and any combined limits across species or contaminants. When statewide and local rules overlap, the reviewed source interpretation must decide applicability. Do not add meal allowances from separate advisories or infer a remaining allowance without an explicitly supported method.

Optional private `fish_records` support catch or consumption notes as a distinct user-entered record kind, with owner, observed date, optional waterbody/location context, species or unresolved label, original quantity/unit, linked advisory

revision and optional note. Creating, reading, editing with revision history, deleting and exporting are explicit owned operations. These records are optional for advisory lookup and never update official evidence, public map layers, inferred dose or remaining meal allowances. They need the same minimization, retention and complete deletion coverage as Journal records before release.

Proposed private `garden_sites` and `site_observations` are optional. Collect only information used by a feature: irrigation source, approximate site context, user-entered soil or water tests, chosen practices, and relevant task records. Exact property polygons, crop yields, fertility schedules, and inferred contamination are not necessary defaults. Additional grower fields require a concrete user workflow and clear provenance.

The feature may monitor updates to official advisories and saved evidence, test follow-up, and chosen garden activities. It does not monitor the chemistry of crops without a relevant measurement. Water, soil, and food are separate media with separate methods and units.

Water stewardship

Use typed evidence kinds for regional drought classification, utility conservation notices or restrictions, and optional household or garden water-use records. County statistics retain the category distribution across county area; they are not a point classification at a home. A point lookup requires a valid dated source geometry and intersection method, never the county's maximum class. A weekly regional drought classification is not a plot's soil moisture, a household meter reading, or a utility restriction. Keep issued dates, effective periods, geographic scope, and any superseding notice visible.

Optional `water_use_records` retain date or period, original quantity and unit, whether measured from a meter or reported by the user, applicable site, and notes. Compare only compatible periods and methods, displaying gaps and interval length. Describe this as measured usage change, not savings caused by HALO or a particular task; weather, occupancy and other conditions may differ. Attributed savings need a separately validated method. Do not infer an unmeasured consumption total, assign drought responsibility to a household, or calculate irrigation prescriptions from a regional drought category.

Utility restrictions can lead to reviewed guidance, a saved action, and a chosen subscription. Regional drought updates can provide context and stewardship resources. A saved reminder must defer to the actual applicable restriction and source, not present a generic watering schedule as permission.

Conservation and participation

Extend `volunteer_orgs` with reviewed service areas, accessibility information when verified, contacts, and status. Proposed `opportunities`, `opportunity_revisions`, and `participation_records` add an organizer, cause, activity description, location, dates, time zone, age restrictions, access requirements, materials, cost, cancellation state, source, and last checked date.

Distinguish saved, external registration opened, user reports registered, attended, cancelled, and follow-up states. An outbound link is not proof of registration. Hours or participation outcomes are user reported unless independently confirmed. Never translate attendance into estimated pollutant removal or health benefit without a defensible measurement method.

Private field observations and reporting handoff

Proposed `field_observations` contain an owner, observed time, optional approximate location, user-selected observation category, description, optional private image, and attachment metadata. Categories describe what the person saw, such as unusual appearance or odor, rather than asserting a chemical or cause. Images and notes remain unverified reports and never populate a public contamination layer.

The workflow supports draft, saved privately, prepared for reporting, external reporting opened, and user-reports-submitted states. HALO may prepare an export or copyable description and link to a verified official reporting destination. Exact coordinates or images are included only after a user preview. Agency acknowledgement is shown only if an actual receipt is available; an opened browser link cannot establish it.

Provide source-reviewed safety and reporting instructions where applicable. Do not encourage entering hazardous sites, collecting unsafe samples, or approaching wildlife or suspected contamination to improve a report. Public sharing, automated agency submission, or community incident aggregation would require a separately reviewed privacy, moderation, safety, and receipt contract.

Outreach and summaries

Proposed `outreach_drafts` and `outreach_followups` retain recipient type, cited evidence revisions, template version, editable content, and user-reported sent or reply dates. Draft creation does not send a message. Any later delivery integration requires a preview, exact recipients, explicit confirmation, delivery status, and abuse controls.

Household summaries are generated from a user-selected set of records with source dates and caveats. Prefer on-demand private export. A public or bearer-link share is a separate opt-in capability with expiry and revocation, never the default. The generated record must exclude location, household health categories, Journal notes, or attachments unless explicitly selected.

Subscriptions and alerts

Proposed `subscriptions` connect a household to a supported public entity or a chosen reminder. Include subscription type, entity ID, selected parameters, notification channels, quiet-time preference, and cancellation state. Extend existing alerts with a stable target, source revision, event time, reason, and delivery state.

Deduplicate by meaningful event identity and revision, not merely text. Keep advisories, source updates, reminders, and organizer changes visually and structurally distinct. Suppressing push does not necessarily remove a relevant in-app record. Make those preference semantics explicit.

Journal and patterns

Reuse `symptom_logs` and history. Add explicit check-in state if needed to separate felt fine, symptoms, and incomplete entry, with a validated symptom vocabulary and real calendar-date validation. Preserve retrospective and possible-illness flags, edited timestamps, and selected category.

Add an owned all-time `has_any_entries` or exact entry-count read for first-use routing. An empty 90-day or 365-day list cannot prove the household has never used Journal. Expose history-backfill state as pending/running/complete/partial/failed, including coverage dates and source quality; a pending backfill with no rows is not a genuine no-history conclusion. Preserve the existing retrospective comparison maximum of 62 dates, default 90-day entry window, 12-week grid and seven recent entries as presentation/query requirements without inventing an API result for an unimplemented filter.

Retrospective batch entry creation must detect existing date/category records before overwriting them. Add create-if-absent or expected-revision semantics so a recalled rough-day retry cannot erase a richer current entry written by another tab. Return per-date saved/already-present/conflict/failed outcomes with identifiers, and retry only genuinely failed keys. An analysis request alone does not save entries. A corrected/deleted entry invalidates private findings and seasonal summaries using the same analysis version and logging denominator.

The current algorithm's denominator includes environmental-history days without a check-in. Future wellness comparisons must use explicit logged days, expose included and excluded counts, and label retrospective entries.

Missing logging is not evidence of feeling well. Category filtering needs category-specific calculation rather than relabelled household findings. A category is not a uniquely identified person.

Version the analysis and validate minimum evidence rules. Show descriptive associations only, with no diagnosis, causal inference, or medical probability. Do not create engagement incentives to report symptoms or imply a finding will appear after a promised number of entries.

Luna context and proposals

Add a capabilities response that discloses whether household context may be sent, what categories are eligible, provider-processing policy reference, retention setting, and available actions. The existing environment flag enables household context unless exactly `false`; `uses_household_data` in an answer does not prove what was already sent to the provider.

Proposed conversation handling stores minimal context or uses a short-lived server session. The user should be able to start a new conversation and delete stored content. Selected record IDs are authorized and loaded server-side. User-supplied snippets, retrieved documents, and attachment text are data, never instructions to execute tool actions.

Structured proposals contain a proposed action, citations, fields requiring confirmation, and a short expiry. The confirm operation revalidates ownership, evidence, and freshness before saving. Luna does not directly write tests, register for events, send letters, change subscriptions, or schedule tasks without that confirmation. Free-text Journal notes remain excluded by default.

Recovery and device access

Keep anonymous entry. Optional recovery or account linking can protect long-lived records, but requires a specific authenticated flow, collision handling, device/session management, and an explanation of what is recoverable. Never use possession of a household ID or a public URL as authorization. Current anonymous-session replacement is not a recovery feature.

Shared household accounts, if added, require membership roles, explicit invitations, revocation, and a decision about which health records other members may see. They must not be introduced as an incidental consequence of export or device linking.

Proposed endpoint and mutation contracts

All paths below are target proposals, not routes that already exist. Existing endpoints should be extended additively or adapted, not broken because navigation changes.

Table fields: Operation group / Proposed surface / Contract and state requirements

Home context

Proposed surface: Atomic current-home transition and archived-context reads

Contract and state requirements: Verified owner, idempotent transition, explicit new utility match/failure, retained historical associations, timezone handling, no silent transfer of old tests or utility.

Learn index

Proposed surface: Proposed reviewed-topic list, search and detail reads

Contract and state requirements: Published content only, bounded query, topic/language/version, no-match versus failure, source metadata, stable links back to originating context.

Evidence

Proposed surface: `GET /api/evidence/:id` and source detail

Contract and state requirements: Return authorized record, revision, provenance, scope, freshness, and applicability. Removed or withdrawn records return a legible status rather than disappearing from saved actions.

Conditions

Proposed surface: Extend `/api/daily-score`; add `/api/forecast` and `/api/changes`

Contract and state requirements: Separate observations from forecast issue/valid times. Change reasons identify environmental change, new coverage, source correction, or methodology/profile change.

Actions

Proposed surface: `GET/POST /api/actions`, `PATCH/DELETE /api/actions/:id`

Contract and state requirements: Owner derived from session, idempotent create, validated state transitions, duplicate suggestion detection, revision conflict detection, explicit delete or archive semantics.

Plan

Proposed surface: `GET /api/plan`, occurrence and reminder operations

Contract and state requirements: Date range plus time zone, bounded range, recurrence edit scope, save without schedule, explicit reminders, and cancelled-instance handling.

Outdoor plans

Proposed surface: Proposed plan CRUD, preparation and forecast-watch operations

Contract and state requirements: Authorized optional location, duration/timezone, saved evidence revisions, per-factor horizon/resolution, explicit material-change rules, no automatic rescheduling.

Tests

Proposed surface: `GET/POST /api/tests`, result and correction operations

Contract and state requirements: Typed measurements, qualifiers, original units, evidence distinction, validation messages, versioned corrections, and no score change merely from task completion.

Maintenance

Proposed surface: Assets and maintenance operations

Contract and state requirements: User choice or cited interval basis, repeated events, pause/resume, replacement history, no unsupported efficacy estimate.

Food and Garden

Proposed surface: Advisory search/detail, food reports, grower resources, garden sites

Contract and state requirements: Public records scoped by source geography and actual coverage. Private observations isolated from public evidence. No source match is not a clean result.

Fish notes

Proposed surface: Optional catch/consumption record CRUD and export

Contract and state requirements: Private owner, original quantity/unit and source context, optional advisory link, revision conflict handling, full deletion, no dose or remaining-allowance calculation.

Water stewardship

Proposed surface: Drought and utility notice reads, private water-use record operations

Contract and state requirements: Separate regional context, utility rules, and actual user measurements. Date and unit validation, compatible-period comparison, and source-aware subscriptions.

Conservation

Proposed surface: Opportunities, organizer details, participation operations

Contract and state requirements: Verified source fields, expiry/cancellation, filtering without invented metadata, and explicit user-reported external status.

Field observations

Proposed surface: Private observation and attachment operations, reporting preview/export

Contract and state requirements: Distinguish observed description from cause. Explicit location/image inclusion, verified destination, no automatic external submission or false agency receipt.

Outreach

Proposed surface: Draft preview/save, export, user-reported follow-up

Contract and state requirements: Cite actual evidence, preserve edits, validate recipients if delivery is later introduced, never auto-send.

Subscriptions

Proposed surface: `GET/POST /api/subscriptions`, update/cancel

Contract and state requirements: Supported entities only, opt-in channels, duplicate handling, unsubscribe confirmation, retained history where appropriate.

Journal

Proposed surface: Extend current entry, findings, retrospective, summary routes

Contract and state requirements: Real date validation, logged-day denominators, category scope, analysis version, corrected-entry recomputation.

Luna

Proposed surface: Extend assistant capabilities/questions; proposal/confirm

Contract and state requirements: Privacy disclosure before sending, authorized context references, bounded conversation, source-grounded output, confirmed side effects.

Export and recovery

Proposed surface: Data export job/status/download; separate recovery operations

Contract and state requirements: Reauthentication where appropriate, short-lived private download, ownership checks, lifecycle audit, and deletion handling.

Common request behavior

Every mutation must validate an explicit schema on the server. The verified session supplies the owner; clients cannot select another `profile_id`. Database row-level security is required even where service-role route handlers currently perform the operation. Service-role access must be narrowly scoped and tested against cross-household IDs.

Creates that can be retried accept an idempotency key bound to owner, operation, and request hash. Reusing a key with different content returns a conflict. Updates use a revision or `If-Match` precondition. Deleting an already-deleted owned resource has documented idempotent behavior without revealing another household's resource.

Return a consistent error envelope with code, safe message, field errors where relevant, retryability, and request ID. Distinguish validation errors, authentication required, forbidden or inaccessible resource, missing resource, revision conflict, rate limiting, and temporary provider/service failure. Never surface service credentials, raw SQL, private provider payloads, or other users' data in errors.

An offline draft is not a confirmed server save. Show pending sync and retry explicitly. Replayable writes need idempotency and conflict handling; a stored delete or external-send operation must not execute silently after a long offline period. Do not require the frontend to infer success from a closed dialog.

Typed field requirements for new flows

This table supplies the minimum proposed contract fields beyond the existing endpoints. Optional means the user or source may genuinely omit the field, not that its meaning can be guessed. Use strict server schemas and generate TypeScript clients/fixtures from the agreed schema. Naming can change during implementation; loss of the listed behavior cannot. Do not introduce all these endpoints before the corresponding complete journey is ready.

Table fields: Contract and feature IDs / Request and record fields / Required response and behavior

Current-home transition G01 X04

Request and record fields: Expected profile revision, address lookup reference/request key, confirmed water source/year, prior context ID, carry-forward task choices

Required response and behavior: New context ID, match status/method, timezone, old-context archive, affected subscriptions, retained records and authoritative current profile. Resolve lookup before atomic transition; no partial cross-context save.

Conditions and changes T01 to T06

Request and record fields: Location/context, requested observation or valid period, optional comparison revision

Required response and behavior: Per-factor native value/unit, severity, evidence kind, source revision/times, coverage, explanation version; changed field and old/new evidence; `comparable` plus reason. No change percent when baseline is null/zero or method incompatible.

Forecast and outdoor plan P03

Request and record fields: Supported place reference or bounded optional coordinates, interval and timezone, duration, chosen factors/categories; chosen period and source references on save

Required response and behavior: Actual period resolution and available horizon per factor, issued/valid times, expired/gap reasons, plan revision and optional watch status. No invented intervals. Rescheduling requires a separate confirmed mutation.

Action and project P02 A04 H07

Request and record fields: Action type/template version, origin reference, optional parent, home/site/category applicability, title/note, evidence revision IDs, optional schedule and recurrence

Required response and behavior: Stable action ID/revision, source snapshot, current state, permitted transitions, duplicate match, linked occurrences and follow-up. Reuse the existing action when selected; intentional repeat gets a distinct occurrence.

Plan read P01

Request and record fields: From/to, timezone, bounded filters and cursor

Required response and behavior: Items with all-day local date or timed instant, item timezone, type, canonical linked record, state and next occurrence. Return unscheduled records separately; one underlying item is never counted once per screen.

Test and result H06 H03 H04 F03 F05

Request and record fields: Test type/matrix, purpose, context/site, collection start/end, provider/lab, method, sample label, optional report reference; result analyte, original text/value/unit, qualifier/limits and basis

Required response and behavior: Saved workflow state, result revisions, validated normalization or blocked reason, transcription verification status, related action, optional reviewed interpretation and its method/source. Permit uninterpreted records when parsing is ambiguous.

Home asset and maintenance H07

Request and record fields: Asset type/user label, installation date, optional model/instruction evidence, calendar or completion-relative interval, service action and optional meter/room observation

Required response and behavior: Last confirmed completion, next due under explicit policy, archived/replaced state and history. Unsupported product/model certification stays unknown.

Garden F01 F03 F05

Request and record fields: Site label/type, coarse context, irrigation source independent of tap source, optional soil history, linked test/action/resource IDs

Required response and behavior: Site revision, evidence summary, relevant source notices, open questions and owned follow-up. Source changes version site context rather than relabelling old samples.

Advisory matching F02

Request and record fields: Waterbody/segment reference, species/group reference or unresolved text, exact official audience code, optional valid date

Required response and behavior: Matching advisory revisions, rule applicability and extent, overlap/conflict explanation, official serving/frequency language, warnings and no-match reason. Never auto-map broad HALO age categories into exact advisory groups.

Fish note F02

Request and record fields: Note kind catch/consumption, date/timezone, optional waterbody, species or unresolved label, original quantity/unit, optional advisory reference/note

Required response and behavior: Owned record and revision, unknown labels preserved, selected export/deletion support. No health dose or allowance computation.

Food evidence F04

Request and record fields: Region/food type/search/cursor within actual coverage; detail reference

Required response and behavior: Notice/study/sample kind, issuer, commodity/lot/site scope, analyte/unit/qualifier, dates, source extent, official action, current/superseded status. Evidence match is not a farm or brand safety certificate.

Grower inquiry F05

Request and record fields: Reviewed resource/program ID, optional garden/test references, chosen document checklist, user-reported inquiry/application state

Required response and behavior: Requirements snapshot, official contact/handoff, optional reference number and follow-up action. User-reported application status never becomes agency approval.

Water stewardship F06

Request and record fields: County/dated geometry or verified utility reference; optional usage period, original volume/unit, meter/bill/source, site

Required response and behavior: Regional category distribution and separately scoped utility notice; private usage series with interval and comparability metadata; update event references. No automatic irrigation prescription.

Opportunities A02 A03

Request and record fields: Cause/place/date filters, organizer/opportunity ID; explicit participation-state change and optional hours/note

Required response and behavior: Source verification, dated event revision, availability/cancellation, required preparation, linked Plan action and user-reported/official status provenance. Age/accessibility unknown is distinct from unrestricted.

Outreach A05

Request and record fields: Recipient class, purpose, selected evidence revisions, template revision, optional private draft only under retention consent

Required response and behavior: Independent draft per recipient, citations, outdated-evidence warning, export/copyable text, user-reported sent/reply state. No recipient email or actual message delivery implied.

Observation A06

Request and record fields: Observed time/timezone, factual category/text, optional chosen location precision, attachment IDs, destination record

Required response and behavior: Private record revision; reporting preview of exact included fields; official destination/review date; state and genuine receipt reference only if supplied. Metadata removal happens before sharing preview.

Household summary H08

Request and record fields: Chosen date range/context, included sections/records, allowed category/note/location/attachment choices

Required response and behavior: Preview manifest with source times and omissions, then generated artifact with scope/hash/time/expiry. One failed optional section returns an omission, not invented content.

Learn and guidance X01 T06 H05

Request and record fields: Topic or bounded text query, locale, selected evidence and category context

Required response and behavior: Reviewed content version, delivered/requested locale, valid contextual notes with evidence bindings, sources and no-match/unavailable state. Persist reviewed explanations once; do not duplicate a new corpus per page.

Subscriptions X03

Request and record fields: Supported entity type/ID, event kinds, channels, quiet hours/timezone, optional expiry and parameters

Required response and behavior: Subscription revision, effective preferences, last evaluation/source state, opt-in status, cancel confirmation and existing-match detection. No arbitrary client URL polling.

Luna X02

Request and record fields: Question, approved page/context references, optional bounded conversation ID, privacy choice/version

Required response and behavior: Capabilities reflecting effective server behavior, answer/refusal/source status, citation IDs, selected-context disclosure, optional structured proposal with expiry. Confirmation supplies proposal revision and idempotency key; reauthorize before write.

Export X05

Request and record fields: Explicit scopes, contexts/date ranges, format and attachment inclusion

Required response and behavior: Export job ID/status or local artifact manifest, counts/omissions, safe private download with expiry, cancellation/deletion state. Never embed a bearer session in a share URL.

Recovery X05

Request and record fields: Approved authentication challenge and verified destination, separate link/recover intent, current session where applicable

Required response and behavior: Verified identity continuity or explicit collision/recovery failure; session/device revocation support and records available. Household ID alone never proves ownership.

Source review X07

Request and record fields: Source/run/revision ID, reviewer role, validated publish/withdraw decision and rationale

Required response and behavior: Atomic published pointer, superseded history, affected-entity event and audit record. An editor cannot publish unvalidated raw import text as advice by merely checking a box.

Proposed relationships and database invariants

Keep private ownership as `profile_id` compatible with the current model unless an independently approved household-membership migration changes it. Every private child references both its parent and the same owner using a composite foreign key or equivalently enforced database constraint. A route's ownership check alone does not prevent accidentally linking another household's attachment or test.

Table fields: Record family / Relationships and uniqueness / Access and deletion rule

Source and ingestion run

Relationships and uniqueness: Source has many runs; run keyed by source/release/parser request; external record stable within its source namespace

Access and deletion rule: Service/operator writes. Public reads expose approved metadata only, never credentials, fetch headers or raw private data.

Evidence and revisions

Relationships and uniqueness: Evidence identity unique by source and stable source record key; revision keyed by canonical content hash plus source revision semantics; current pointer references validated revision

Access and deletion rule: Published public records are read-only to ordinary users. Private measurements remain in owned domains, not a public evidence table row with a forgotten flag.

Guidance and project templates

Relationships and uniqueness: Topic/template, locale and version; published version has sources and review record

Access and deletion rule: Ordinary users read published only. Withdrawal preserves referenced history while blocking new recommendations.

Home context

Relationships and uniqueness: Many per profile, at most one current context; effective dates cannot overlap for current-home interpretation

Access and deletion rule: Ownership enforced; archiving is not deletion. Full deletion includes all historical contexts.

Household action

Relationships and uniqueness: Many per profile; optional parent/project must share owner; optional evidence references public reviewed revisions or owned private record

Access and deletion rule: Parent completion does not silently complete children. Delete/archive policy handles children and linked tests explicitly.

Recommendation preference

Relationships and uniqueness: Unique owner/recommendation/context/version scope; disposition plus validity/review date and optional linked canonical action

Access and deletion rule: Reversible owner control. Suppression and Already doing this never write a completion or hide an official notice; full deletion includes these records.

Occurrence and reminder

Relationships and uniqueness: Unique action/series-version/local occurrence key; unique occurrence/channel/reminder kind; store original timezone/fold decision

Access and deletion rule: Server generates, owner changes authorized schedule; cancellation immediately invalidates future delivery intent.

Outdoor plan and forecast watch

Relationships and uniqueness: Owned plan has one canonical action; watch unique per plan/factor/policy and references supported forecast area/time

Access and deletion rule: Server may flag review, not change selected interval. Removing plan cancels watches and reminder jobs.

Test and results

Relationships and uniqueness: Test belongs to context or garden site; many analytes and corrected revisions; result sequence unique within test

Access and deletion rule: No one-result-per-analyte constraint if a report has multiple sample points. Corrections supersede but do not overwrite original evidence.

Assets and maintenance

Relationships and uniqueness: Owned asset, service records and linked shared actions; replacement can archive one asset and start another

Access and deletion rule: Cascades/explicit unlink must not erase an unrelated laboratory result. Cleanup handles attachments and reminders.

Garden and site context

Relationships and uniqueness: Many optional sites per profile; irrigation/source context versioned; observations/tests retain original context

Access and deletion rule: Do not require unique site labels or use labels as IDs. Delete offers clear linked-record scope.

Advisory and audience rules

Relationships and uniqueness: Issuer/advisory source ID plus version; many species groups, extents and audience/combined-limit rules

Access and deletion rule: Public reviewed reads; no household consumption data in rules. Withdrawal creates a change event.

Fish and water-use records

Relationships and uniqueness: Many private observations; source/advisory links optional; revision checked

Access and deletion rule: Do not prohibit legitimate same-day records with a day-only uniqueness key. Delete/export includes notes and owned files.

Opportunity and participation

Relationships and uniqueness: Public opportunity has organizer and revisions; participation unique by household/opportunity except distinct sessions explicitly modelled

Access and deletion rule: Attendance not inferred from opening a link. Cancelled source retains saved participation context.

Outreach and observations

Relationships and uniqueness: Owner, optional context, selected evidence revisions, snapshots and shared follow-up actions

Access and deletion rule: External-sent snapshot immutable to edits; draft and later amendments separate. Default remains private.

Attachment

Relationships and uniqueness: Owner, private object key, checksum, MIME, byte count, scan state, sanitized derivative references and typed domain binding

Access and deletion rule: Foreign-owner linking blocked. Deletion considers originals, thumbnails, extraction output and temporary upload objects.

Subscription and event

Relationships and uniqueness: Subscription unique by owner/entity/type/parameter set; source event unique by entity/revision/event kind

Access and deletion rule: Viewing is never subscription. All jobs recheck cancelled status and current preferences before delivery.

Outbox and delivery attempt

Relationships and uniqueness: Outbox intent keyed by target household/event; channel/device attempts keyed by outbox/channel/device revision

Access and deletion rule: Service-only leasing; repeated execution safe. Public logs contain IDs and error codes, not private notification bodies.

Luna session and proposal

Relationships and uniqueness: Owner-scoped conversation; proposal references authorized context and expires; confirmation nonce single-use

Access and deletion rule: Provider processing and local/server storage are distinct choices. Private free-text Journal notes excluded by default.

Export and deletion job

Relationships and uniqueness: Owner/request hash/idempotency scope; export manifest and expiry; deletion retains minimal non-content completion marker

Access and deletion rule: No private job result accessible using guessable ID. Deletion status needs a narrow authenticated or scoped receipt flow after auth deletion.

Index private lists by owner, current status and date/due time; public catalogs by source/status/updated time plus actual filter dimensions; queue polling by status, next-attempt time and lease expiry. Use cursor pagination with a stable secondary ID. Geospatial indexing is warranted only for actual supported intersection queries, not approximate rankings based on invented precision. Validate index plans and payload sizes on representative datasets before publishing load claims.

Database constraints complement RLS: finite validated values and permitted units, real date types, valid state enums, nonnegative durations/quantities where appropriate, start before end, required revision numbers, allowed relation types and ownership-consistent foreign keys. Chemical concentration with a reporting qualifier retains the original text even when a numeric representation is absent. Migration tests must cover both direct authenticated database access and server service-role queries.

Transaction and concurrency boundaries

Use a database transaction or a tightly scoped RPC for operations spanning records. Existing separate route calls cannot be made atomic by disabling the Save button.

- Saving an action with its initial occurrence, reminder and audit event commits once. A timed-out response is reconciled by the same idempotency key, not a second task.
- Completing an occurrence, recording its completion event, cancelling obsolete reminders and generating a completion-relative next occurrence commits once. Two workers or two tabs cannot generate two next occurrences.
- Publishing a source revision, updating the current pointer and writing a durable source-change event commits once. Fanout checkpoints advance only after their own durable acceptance, not before processing all households.
- Confirming a test transcription creates a result revision and updates linked workflow state together. It never mutates the public regional reading. Attaching a file requires an authorized completed upload, not a client-provided storage path.
- Moving home atomically changes current context and invalidates old-context future matching. Existing backfill jobs carry their original context and cannot write current-home readings after the move.
- Participation state, linked Plan scheduling and event-change notices share canonical references. Source cancellation flags the item without silently claiming the household cancelled or deleting its history.
- A Luna confirmation validates the proposal, owner, latest evidence and field limits again. The consumed proposal key and saved action are atomic. A retry returns the original outcome.
- Delete-everything first blocks new private writes and cancels future processing, then coordinates database/object/export cleanup before removing the identity. Crash recovery resumes without recreating deleted content.

Concurrent PATCH uses an expected revision and returns conflict plus the current authorized version. The client can compare and reapply explicitly; last-write-wins is not acceptable for private test transcriptions, household categories, or schedules. Public caches use source/version keys and single-flight refresh locks to prevent a cold layer request from triggering many parallel provider builds.

Ingestion and monitoring operations

The existing daily cron performs geography refresh, map construction, history recording, alert evaluation, and push. It has a 60-second maximum and selects up to 5,000 profiles and 2,000 utility rows. Adding every new feature to that handler would risk incomplete coverage and runtime failure.

Keep the public trigger small. Use persisted cursors and bounded job batches for source fetches, validation, canonical publication, household matching, and notification delivery. Begin with the existing deployment's available job mechanism; a durable database-backed work queue is sufficient if it can safely lease, retry, and resume jobs. No new distributed architecture is required merely for appearance.

Each run needs a source, cursor, start and finish time, record counts, rejected records, last successful watermark, parser version, and error classification. Jobs use leases, retry backoff with jitter, maximum attempts, and a review queue for persistent failures. Publishing a partial fetch must not delete records that were simply not reached.

Source changes publish only after validation. Preserve the last successful snapshot with its original dates when an upstream provider fails. Alerting uses a durable outbox so data publication and intent to notify are committed together. Delivery retries reuse a dedupe key. Expired push endpoints are removed as they are today.

Scheduler and event execution

Current `vercel.json` schedules `/api/cron/daily` at 11:00 UTC, so local wall-clock time changes with daylight saving. A daily trigger cannot promise 15-minute reminders or timely official-advisory updates. New refresh targets require a verified scheduler capability, bounded job runtime and durable queue before those promises appear in the UI. Deployment configuration presence is not proof that the host executes it.

Table fields: Worker responsibility / Trigger and durable key / Execution and failure requirements

Source acquisition

Trigger and durable key: Source-defined next check or explicit operator run; source/release/request key

Execution and failure requirements: Use permitted conditional requests, explicit timeout and redirect policy. Store the run and raw/hashed acquisition evidence before parser publication. A partial page never becomes a complete dataset.

Validation and publication

Trigger and durable key: Completed acquisition run plus parser/method version

Execution and failure requirements: Check required fields, count changes, units, geography and dates; quarantine suspect runs; publish canonical revision and change event transactionally.

Public derived views

Trigger and durable key: Published compatible source versions; layer or aggregate/version key

Execution and failure requirements: Rebuild Map/district/forecast indexes in bounded batches. Persist assembled-at and original source times independently; keep previous compatible version until success.

Household history and matching

Trigger and durable key: Regional observation/source event plus stable home-context ID

Execution and failure requirements: Batch by eligible context and permitted shared spatial resolution; persist cursor; preserve actual capture basis; match subscriptions before fanout. Recheck context when executing.

Reminder generation

Trigger and durable key: Scheduled scan and completion-relative event; action/series/occurrence key

Execution and failure requirements: Materialize a bounded horizon, preserve all-day versus timed semantics, respect timezone and quiet hours, and cancel superseded occurrences. Never replay an obsolete safety-related reminder as timely.

Alert fanout

Trigger and durable key: Published source event plus matching policy version; household/event key

Execution and failure requirements: Recheck ownership, watch state, current consent/preferences and supported scope. Fanout resumes per checkpoint. Event retained until all intended batches reconcile.

Push delivery

Trigger and durable key: Committed outbox item and allowed device/channel

Execution and failure requirements: Record attempts, retry transient provider failures, retire 404/410 endpoints, respect expiry and quiet hours. Accepted push is not proof the person read it.

Content and link review

Trigger and durable key: Source/template next review date or manual report

Execution and failure requirements: Queue a curator review; HTTP 200 is only link reachability. Source changes do not auto-publish new interpretation or translations.

Export and deletion cleanup

Trigger and durable key: User-confirmed job or retention expiry

Execution and failure requirements: Verify owner and current status at every step; remove temporary files; bounded retries and operator-visible failure. No sensitive payload in queue logs.

Classify errors as retryable transport, HTTP 429 or 5xx, unauthorized provider, invalid schema, unsupported coverage, or valid no-record response. Retry temporary failures with jitter and a bounded attempt policy; invalid schemas and permission failures require review rather than endless paid retries. A lease expiry permits another worker to resume, not to repeat an external side effect without checking its key. All jobs recheck household deletion state before private reads/writes.

Change detection must be semantic. Reordered JSON, renewed retrieval time or a source wording correction must not create a worsening-condition alert. Retain old/new values, event kind, effective period, applicable scope and source correction flag. A withdrawn advisory, cancelled event or invalidated interpretation creates a distinct update. New subscriptions establish a baseline instead of flooding the inbox with historical changes; the user can still inspect history explicitly.

Forecast watches stop at activity end/cancellation and consider only the periods, factors and region actually supported by the selected provider. Evaluate a source's genuine category change or reviewed material difference, not arbitrary visual rounding. Task snooze affects the selected reminder or occurrence and does not alter the underlying scientific record or official event time.

Proposed refresh classes

These are planning targets, not source guarantees. Confirm them against provider cadence, terms, quotas, user coverage, and cost before implementation.

Table fields: Data class / Proposed approach / User-facing meaning

Current outdoor conditions

Proposed approach: Shared cache refreshed within an appropriate 15 to 60 minute window while used, with measured provider cadence respected

User-facing meaning: Latest available observation or model output, with its actual time

Forecasts

Proposed approach: Refresh near provider releases or on stale reads, not once per household request

User-facing meaning: Forecast issued at a stated time, valid for stated periods

Urgent official advisories

Proposed approach: Frequent polling or supported feed delivery only where operationally sustainable

User-facing meaning: Last checked time and link to official instructions; HALO is not an emergency dispatch service

Utility laboratory results

Proposed approach: Check source releases or scheduled imports; notify after validated new records

User-facing meaning: New published evidence, not continuous tap monitoring

Fish and food advisories

Proposed approach: Check official sources on a reviewed schedule, with human review when automated interpretation is unsafe

User-facing meaning: Current known advisory and last verification, not continuous food testing

Drought and conservation notices

Proposed approach: Ingest drought on its actual weekly publication cycle; check utility notices on a separately verified schedule

User-facing meaning: Regional drought and utility restrictions are distinct, with their own dates and scope

Conservation events

Proposed approach: Recheck approaching events and source changes more often than distant records

User-facing meaning: Organizer information can change; registration is confirmed by the organizer

Tests and maintenance

Proposed approach: Monitor household-record states and chosen reminders

User-facing meaning: HALO tracks the process and reported result, not an absent physical sensor

Gardens

Proposed approach: Track official updates, entered observations, recorded tests, and chosen tasks

User-facing meaning: No claim to continuously detect soil or crop chemistry

No job may infer a normal condition from a failed fetch. Distinguish provider outage, unsupported coverage, missing record, stale record, and withdrawn record. A successful HTTP response with malformed or unexpectedly empty content is an ingestion failure until validated.

Reliability and cost targets

The following are proposed acceptance budgets to validate in staging, not claims about current performance:

- Cached reads should normally respond within 1 second at the 95th percentile, and ordinary private writes within 2 seconds under the agreed test load. Report provider-dependent operations separately.
- A cached screen must remain usable during source failure, with honest stale labels. Writes should acknowledge only after durable persistence, and jobs only after durable acceptance.
- Paginate lists and job scans. Start with a maximum page size of 100 records and explicit limits for large history queries and exports.
- Set per-provider call and spending budgets, share public source requests by permitted spatial/time resolution, and degrade the affected feature when a budget is reached. Preserve existing assistant global and per-user limits unless deliberately revised.
- Proposed attachment start: allowlisted PDF and common image formats, at most 10 MB per file and a clearly disclosed household storage budget. Review these limits with real test documents before launch. Do not accept executable formats or trust file extensions alone.
- Proposed routine reminders: dispatch within 15 minutes of the selected time under normal operation, with visible delay when the scheduler is impaired. Do not market this as a safety-critical alarm guarantee.
- Monitor request failures, latency, source staleness, queue age, failed job count, missed scheduled jobs, RLS denials, attachment failures, and notification delivery. Page the operator for persistent actionable failures, not every transient retry.
- Establish backup and restore coverage for both database rows and object storage before enabling valuable user records. Define and test recovery objectives against the selected hosting plan rather than inventing a promise.

Privacy and security requirements

Maintain the existing minimization: approximate home coordinates and category presence without names, birthdates, or required child accounts. Additional details are optional and requested at the point of use. A garden, outing, or event should not require collecting a home address when approximate context is enough.

All private tables require ownership policies, tested positive and negative access cases, and deletion relationships. Public imports have no client write policies. Editorial roles are separate from ordinary users and audited. Logs use request IDs and source/job metadata, not raw Journal notes, precise locations, health summaries, or complete model prompts.

Attachments stay in private buckets. Upload authorization, content-type validation, file scanning where feasible, signed URLs with short expiry, download authorization, quotas, and deletion are required. Extracted text is untrusted input and needs explicit correction before becoming a verified result. Stale preview URLs and object copies must be considered in deletion and retention design.

Deletion cancels future jobs and subscriptions, revokes active shares, removes object storage, clears application caches, and deletes private records. Database cascades alone do not remove blobs or third-party processing history. The privacy explanation must accurately distinguish local deletion, server deletion, backups, and provider retention.

Household categories and symptoms are sensitive even without a name. Consent for notifications, recovery, model context, export, and third-party sharing must remain separate. Product evaluation uses consented usability sessions and nonsensitive research notes by default. Production behavioral analytics are not assumed; adding them requires a separate explicit privacy decision consistent with existing expectations. Operational reliability metrics must not contain free-text notes, source attachments or private prompts.

Attachment and document lifecycle

Use one private attachment service for tests, site records, instructions, observations and optional exports. Do not independently implement uploads in each feature. Proposed flow is create upload intent, validate owner/domain/quota, upload into quarantine, verify actual type/size/checksum, scan, remove metadata in display/share derivatives, then finalize its authorized record binding. Unscanned, rejected or expired uploads cannot be publicly viewed or passed to Luna. Enforce quotas on both intent and finalization so concurrent uploads cannot bypass them.

Before the first valuable private record or lab upload, explain anonymous-session loss and the actual available recovery/export options. Do not promise recovery that is only on the roadmap, and do not defer this warning until after accepting an irreplaceable file. Anonymous use can continue after an informed choice; new recovery must remain optional and real.

If document extraction is enabled, store its parser version, source-page locator, uncertain fields and original text. The user compares the original and confirms a transcription. Extraction never silently supplies a missing unit, lab, analyte, sample date or limit. Render documents in a restricted viewer; do not execute active HTML/SVG/macros from an upload. Source parsers and external URL fetchers need allowlists, private-network/redirect protection and bounded download/decompression sizes. Files are untrusted content even when uploaded by their owner.

Signed URLs are short-lived and authorized on issuance; revocation stops future issuance but cannot recall an already downloaded copy. Original image metadata may be important to a private record but must not escape through thumbnails or report exports unintentionally. Record the user's chosen exact/coarse/no-location export scope and test the generated file metadata, not only the visual preview.

Retention and export coverage

Publish actual configured retention periods before collecting new private data. The following target policy identifies required choices without claiming an unimplemented legal or hosting guarantee:

Table fields: Data class / Proposed retention and user control / Required release proof

Household profile and category records

Proposed retention and user control: Retained while the household is active; editable; removed through full deletion

Required release proof: Current identity, old home contexts and linked records cannot survive invisibly outside documented retention.

Journal and private home records

Proposed retention and user control: Retained until user deletion or clearly disclosed inactivity policy; distinct Clear Journal scope

Required release proof: Export and deletion list actual rows, derived findings and attachments. Inactivity purge cannot surprise a user with valuable records.

Chat

Proposed retention and user control: Default no durable transcript beyond the selected session unless history is explicitly enabled

Required release proof: UI states actual server/browser/provider retention separately. Provider policy is reviewed, not inferred from no local table.

Local drafts

Proposed retention and user control: Session-only by default; optional device persistence requires disclosure and identity binding

Required release proof: Clear/logout/delete removes owned drafts and sensitive caches. Shared devices do not reveal a former identity's draft.

Temporary uploads and export artifacts

Proposed retention and user control: Short configured expiry and scheduled cleanup, with cancel/delete available

Required release proof: Interrupted uploads, derivative files and expired downloads are actually removed; signed URL expiry alone is not deletion.

Notifications and action audit

Proposed retention and user control: Bounded private history with explicit archive/delete policy

Required release proof: Read/dismiss/expiry semantics do not leave hidden sensitive payloads in a separate log forever.

Public evidence and guidance revisions

Proposed retention and user control: Retained as needed for provenance and referenced history within source terms

Required release proof: Licensing/retraction rules are obeyed and withdrawn content is not reused as current guidance.

Operational logs and job receipts

Proposed retention and user control: Short, minimized retention; identifiers/error codes rather than private content

Required release proof: Access limited to operators, retention configured and reviewed, diagnostic output does not leak provider secrets.

Backups

Proposed retention and user control: Provider-specific window documented separately from active-record deletion

Required release proof: Restore procedure reapplies deletions/tombstones before restored data becomes accessible; recovery window and file coverage tested.

Full export returns a manifest of requested scopes, record counts, schema version, generation time, warnings and omitted unavailable files. Include original values, qualifiers and source links in machine-readable records; a screenshot-only export cannot preserve a laboratory result or Journal history. The human-readable H08 summary is a separately selected subset. Exports are owned artifacts with expiry, not public app routes.

Full deletion needs an orchestration record outside the to-be-deleted private graph containing only the minimum non-content receipt needed to reconcile progress. Stop new jobs and revoke sessions/shares as designed, cancel

outbox/reminders, delete blobs/derivatives and private rows, then finalize identity removal. An interrupted multi-step operation must resume and report partial or pending status honestly. The design of a status receipt after identity deletion must not create a household lookup or reusable recovery credential. Do not retain health/location/content snapshots in a deletion audit. Selective deletion must not delete an unrelated garden/test because a shared action points to it.

Recovery and consent implementation gate

Choose one supported recovery method before advertising recoverability: verified account linking or a carefully designed device-transfer/recovery credential flow. Prefer the existing identity provider's supported capabilities over custom password/session storage. Define verification, replay expiry, account-collision resolution, device revocation, lost-device behavior and rollback. Never merge two identities' records solely because they share an address, email-looking string, device label or household ID. Export does not constitute account recovery.

Consent records need purpose, disclosed policy/version, effective choice/time and revocation. Model-context permission, transcript storage, browser push, reminder channels, private-file processing and external export are separate purposes. A declined purpose must not block unrelated core use. If children can use the product directly rather than an adult managing household-category information, complete a separate audience/privacy review before expanding age collection, accounts or sharing; category selection alone does not authorize collecting children's identities.

Deployment configuration and operational readiness

Configuration below is identified from source code by variable name only. No secret values, provider account settings, deployed environment, migration ledger, or actual production job execution were inspected. A variable named in source is not evidence that it is correctly configured. Do not add service-role keys, model keys, private VAPID keys or scheduler secrets to `NEXT_PUBLIC_` variables or committed fixtures.

Table fields: Configuration group / Current source names or implementation / Verification before release

Supabase

Current source names or implementation: `NEXT_PUBLIC_SUPABASE_URL`, `NEXT_PUBLIC_SUPABASE_ANON_KEY`, `SUPABASE_SERVICE_ROLE_KEY`; anonymous bootstrap and `requireUser`

Verification before release: Separate development/staging/production projects or demonstrably isolated test identities; anonymous auth/CAPTCHA policy; RLS on all private tables; key rotation and no client secret bundle.

Human verification

Current source names or implementation: `NEXT_PUBLIC_TURNSTILE_SITE_KEY` and provider-side auth configuration

Verification before release: Domain allowlist and challenge verification actually work; inaccessible challenge has a supported recovery path, not silent bypass.

Geocoding and map

Current source names or implementation: `MAPBOX_TOKEN`, ArcGIS adapter; production renderer chosen separately

Verification before release: Server geocoding credential permissions/quotas, public rendering token if needed restricted by origin/scope, attribution, allowed cache and data terms. No reuse of a private server token in a client renderer.

Environmental providers

Current source names or implementation: `AIRNOW_API_KEY`, `GOOGLE_POLLEN_API_KEY`, existing Open-Meteo/NWS/OpenUV adapter paths

Verification before release: Verify keys/access, API shape, supported geography, polling limits, declared units/source times and fallback behavior for each adapter. Review any hardcoded provider configuration without exposing it in logs.

Redis and budget guards

Current source names or implementation: `UPSTASH_REDIS_REST_URL`, `UPSTASH_REDIS_REST_TOKEN`, current limiter policies

Verification before release: Connectivity and scopes, shared cache persistence/TTL, circuit breakers and fail-open versus fail-closed spending consequences. Anonymous identity creation is not itself an abuse limit.

Luna provider

Current source names or implementation: `ASSISTANT_MODEL_KEY` or `OPENAI_API_KEY`; `ASSISTANT_CHAT_URL`, `ASSISTANT_CHAT_MODEL`, `ASSISTANT_CHAT_FALLBACK_MODELS`, `ASSISTANT_EMBED_URL`, `ASSISTANT_EMBED_MODEL`, `ASSISTANT_EMBED_DIM`

Verification before release: Approved provider processing policy and permitted domains, model/token costs, timeouts and citation tests; model/dimension/corpus/RPC compatibility and retained provenance. Do not switch providers as an undisclosed privacy fallback.

Luna privacy and spending

Current source names or implementation: `ASSISTANT_SEND_HOUSEHOLD_CONTEXT`, `ASSISTANT_GLOBAL_DAILY_LIMIT`

Verification before release: Explicit effective configuration returned to clients; missing context flag currently enables context. Proposed user consent must be enforced server-side. Global cap retains fail-closed behavior when budget service fails.

Push

Current source names or implementation: `VAPID_PUBLIC_KEY`, `VAPID_PRIVATE_KEY`, `VAPID_SUBJECT`; browser uses the corresponding public application-server key

Verification before release: Public/private pair, secure service-worker scope, allowlisted notification destinations, subscription lifecycle, browser/platform support and physical-device delivery tests. Backend sender existing does not prove every frontend registration path exists.

Scheduler

Current source names or implementation: `CRON_SECRET`, `vercel.json`, current 60-second maximum routes

Verification before release: Host actually dispatches authorized jobs on the supported schedule. Prefer header secret, retire logged query-secret use carefully. New reminder cadence needs real scheduler capacity.

Previews and isolated tests

Current source names or implementation: `HALO_ENABLE_FRONTEND_PREVIEW`, `HALO_STAGE_TEST`, fixture routes and isolated build output

Verification before release: No fixture payload on live routes, no test writes to shared production, previews cannot become an authentication bypass or leak private state.

Proposed platform additions

Current source names or implementation: Private storage buckets, queue/lease tables, source flags, retention settings, content-review roles

Verification before release: Configure only when the corresponding feature is enabled; stage secret rotation, backup, dead-letter review and rollback before taking user records. These are not existing deployed capabilities.

The current rate-limit module enables provider-side limiter analytics. Inventory what identifiers/metadata that service records and align actual retention with the privacy explanation. Operational metrics are not automatically anonymous because they exclude names. Do not introduce behavioral analytics as an incidental dependency of the new pages.

Recommended release gates are feature flags per complete capability, not many permanent per-component flags. At minimum distinguish approved core, actions/Plan, private records/uploads, food-advisory sources, opportunities/observations, forecast watches, recovery and expanded Luna. Flags must be enforced server-side where they change writes, provider spending or data processing. Disable unavailable capabilities with an honest explanation; do not expose an enabled-looking control that can never succeed.

Maintain an environment-specific readiness report recording deployed commit, schema version, content/source versions, enabled features, scheduler success/watermarks, provider test outcomes, current budgets, storage/deletion drill, and test evidence. Health probes must be bounded/rate-limited and restricted appropriately; do not have normal page loads repeatedly call all external providers. Public status reveals high-level availability only, not internal addresses, stack traces, credentials or household identifiers.

Backup recovery and rollback runbook

1. Record the current deployed code/schema/content versions, valid provider credentials without copying values, queue watermarks, storage inventory and latest successful backup.
2. Restore database and private objects in an isolated environment with outbound reminders, model calls, emails and other side effects disabled.
3. Reapply completed deletion records, verify ownership/RLS, validate file hashes and links, and reconcile source/job checkpoints before enabling reads.
4. Test representative household histories, a pending result, recurring action, retained evidence revision and expired share. Demonstrate that restored queues cannot resend old notifications blindly.
5. Switch traffic only under an authorized operational release; observe error rate, latency, stale source count and job backlog. Preserve a reversible prior compatible code version.
6. If a new interpretation is wrong, disable its producer/UI claim and publish correction state while retaining original evidence. If a schema migration fails, forward-fix or disable the dependent feature without destroying user records. Test both source rollback and private-write rollback in staging.

Recovery time, acceptable data-loss window and on-call response targets depend on the chosen hosting tier and budget. Vibhav must record measurable accepted values before release; this plan does not promise a backup feature or recovery objective the deployment has not demonstrated.

Migration and rollout plan

1. Audit the current deployed schema, migration ledger, policies, scheduled jobs, environment configuration, source permissions, and backup coverage using read-only checks before a migration is prepared.
2. Add versioned, additive tables and fields for shared evidence and actions first. Write policies and cross-household tests with the migration, not after release.
3. Introduce source adapters and feature flags. Shadow-validate new interpretations against stored fixtures without changing existing displayed scores or sending notifications.
4. Backfill in bounded resumable jobs, retaining original source dates and tagging derivations. Quarantine ambiguous units, qualifiers, geography, and legacy records rather than guessing.
5. Expose new contracts to isolated preview and staging fixtures. Keep sample records explicitly separate from live household data.
6. Roll out one complete connected journey at a time after user review, source validation, contract tests, and negative-path testing. A staged delivery order does not remove later planned features.
7. Monitor freshness, job lag, writes, access isolation, and notification volume before increasing coverage.

Rollback should disable the new feature or producer while preserving user-entered records and the older compatible contract. Avoid destructive down-migrations once a feature has user data. If an interpretation is wrong, withdraw that version, label affected records, and recompute from preserved evidence after review. Do not rewrite history as though the incorrect version never existed.

Backend acceptance requirements

The detailed screen checks live in HALO Feature Acceptance (HALO_FEATURE_ACCEPTANCE.md). Backend completion additionally requires:

- Every promised live feature has a validated source or a clearly labelled user-input workflow, not only a successful empty response.
- Every mutation is idempotent where retried, ownership-checked, schema-validated, and safe under concurrent edits.
- Unknown, true zero, non-detect, partial coverage, stale, invalid, and withdrawn records have distinct tested outcomes.
- Time zones, daylight saving changes, corrected source records, duplicate imports, and delayed notifications do not corrupt schedules or comparisons.
- Journal denominators, category scope, retrospective status, and illness exclusions are tested against known fixtures.
- Household tests retain original units, qualifiers, and sample context; ambiguous input cannot silently drive a score.
- Source outages, quota exhaustion, dropped connections, job restarts, and attachment failures preserve honest states and allow recovery.
- Export and deletion cover rows, files, derived reports, shares, reminders, and queues, with the provider-retention limitation stated accurately.
- No public layer, cache key, assistant context, signed URL, log, or export leaks another household's data.
- A complete restore drill and a rollback drill succeed in an isolated environment before valuable records are accepted in production.

Repository evidence

The following inspected files support the existing-capability statements and identified limitations. Relative links refer to this repository, not a verified production deployment.

- Workflow and preservation (../WIREFRAME_WORKFLOW.md) and approved stage handoff (../foundation/STAGE_FRONTEND_HANDOFF.md).
- Daily source acquisition (../../lib/dailyReadings.js), daily response contract (../../lib/dailyPayload.js), and scoring (../../lib/scoring.js).
- Current profile (../../app/api/profile/route.js), household replacement and preferences (../../app/api/household/route.js), and onboarding transition (../../app/api/onboard/route.js).
- Historical backfill basis (../../lib/backfill.js) and history response (../../app/api/history/route.js).
- Homeguard route (../../app/api/home-guard/route.js), action recommendation logic (../../lib/actionPlan.js), and household categories (../../lib/household.js).
- Journal route (../../app/api/journal/route.js), comparison algorithm (../../lib/journalAnalysis.js), and Journal schema (../../supabase/migrations/0003_symptom_logs.sql).
- Map sources (../../lib/mapLayers.js) and map cache schema (../../supabase/migrations/0011_map_layers.sql).
- Organization route (../../app/api/volunteer/route.js) and directory review metadata (../../supabase/migrations/0004_volunteer_orgs.sql).
- Daily scheduled work (../../app/api/cron/daily/route.js), notification preferences (../../app/api/notifications/route.js), and alert and push schema (../../supabase/migrations/0010_alerts_notifications.sql).
- Scheduler configuration (../../vercel.json), current alert rules (../../lib/alertRules.js), alert inbox (../../app/api/alerts/route.js), and push delivery (../../lib/push.js).
- Assistant route and context flag (../../app/api/assistant/route.js), household context collection (../../lib/assistantContextData.js), and client identity handling (../../lib/auth.js).
- Embedding migration (../../supabase/migrations/0013_assistant_embed_2048.sql), verified server identity (../../lib/serverAuth.js), budget guards (../../lib/ratelimit.js), and reachability probes (../../app/api/health/route.js).
- Anonymous identity and cascades (../../supabase/migrations/0001_anon_auth.sql), household ownership (../../supabase/migrations/0002_household_and_profile.sql), location minimization (../../supabase/migrations/0009_daily_details_and_state.sql), and account deletion (../../app/api/account/route.js).

03 / Source acquisition guide

docs/product/HALO_SOURCE_ACQUISITION_GUIDE.md

HALO Source Acquisition Guide

Prepared for Vibhav and Yogi on October 3, 2026. This is the source-level companion to the Backend and Data Plan (HALO_BACKEND_AND_DATA_PLAN.md) and Backend Build Handoff (HALO_BACKEND_BUILD_HANDOFF.md). It identifies where the full planned product obtains its information, what must be stored, and which interpretations cannot ship yet. It is a development specification, not a claim that the integrations are implemented or that production credentials and permissions have been checked.

The audit covers all 45 feature groups in the master plan. Existing Today detail pages retain their approved depth. No feature is removed because its source needs review. Where automated acquisition cannot support a claim, the implementation uses reviewed official records, an honest coverage state, or an explicitly private user record. It must not substitute fabricated measurements.

Verification and limitations

Four evidence levels are used below:

- **Sample checked:** a bounded public request returned a payload whose relevant fields were inspected. This does not establish complete coverage, permission to retain data, or reliability.
- **Documentation checked:** the official documentation or publication was read. A production credential, complete import, or parser has not been tested.
- **Reviewed content:** the source supports a human-reviewed record and external handoff. Do not imply a supported transactional API.
- **User record:** the household supplies the information; HALO supplies storage, validation, provenance and follow-through, not independent measurement.

The verification report (verification/SOURCE_CHECKS_2026-10-03.md) records actual requests, selected response fields, hashes and failures. The public probe script (verification/probe-public-sources.mjs) is explicit-run, read-only and does not read environment files, credentials or household information. No HALO production route, cron, mutation, provider billing account or deployed Supabase schema was tested. Do not mistake this documentation audit for a deployment audit.

Findings that change implementation

1. **Food notices have a real API option.** openFDA food enforcement returned PFAS-related historical recall records. Use this for scoped notices, not laboratory concentrations or a universal food checker. The sampled recall was terminated, so the status must survive every transformation.
2. **NC fish data is accessible but not ready for automatic interpretation.** Public services behind the official dashboard were located and sampled. One row has truncated advisory text and an apparent audience/meal-limit conflict. The full official advisory must resolve it before publication.
3. **NC utility conservation status has a public service.** Its sampled fields support dated utility status. They do not contain the complete local restrictions or prove that the endpoint has a support or reuse agreement.

4. **Google Pollen needs explicit retention classes.** Existing code stores pollen in daily history. The current non-EEA service-specific terms allow today's forecast values for 365 days and other forecast/heatmap values for 24 hours, with deletion afterward. Confirm account applicability and implement the correct retention classes rather than assuming a blanket storage ban or indefinite retention. Map display, exports and model-context rights still need review. Do not delete existing user history as part of this planning task.
5. **Geocoding storage needs a licence check.** Current onboarding uses Mapbox v5 and stores rounded coordinates. Rounding is not evidence of storage permission. Check the actual account/version entitlement before expanding home contexts.
6. **A source can be technically reachable but semantically unusable.** NWS null precipitation is not established as zero by this audit; the current proxy parser's assumption needs correction or authoritative justification. EPA regulatory comparisons need versioned interpretation and cannot become automatic violation claims.
7. **Dataset versions matter.** The WQP legacy service has a documented recent-USGS-data limitation. District boundaries, service boundaries, regulatory references and static lead inventories must carry release versions, not just download dates.

Registry required for every external source

Implement one source registry. A source may have several endpoints, parsers and coverage classes. Fields below are proposed HALO fields, not assertions that providers return them.

Table fields: Record / Required fields and invariants

Identity and access

Required fields and invariants: `source_id`, issuer, official landing URL, endpoint allowlist, access mode, secret reference name only, source owner/contact, operational owner, enabled domains, supported geography. No credentials in records or client bundles.

Permission

Required fields and invariants: Terms URL, checked date, applicable account agreement, attribution text/assets, retention rules, redisplay/export/model-context rights, map restrictions, approval state and next review. Unknown permission blocks the affected use, not every unrelated page.

Cost and cadence

Required fields and invariants: Quota by interval, account billing tier, request weights, maximum daily/monthly spend, expected source release cadence, HALO check interval, stale policy, timeout and retry policy. Values must be verified for the actual account.

Acquisition run

Required fields and invariants: Requested URL with secrets removed, request parameters excluding private data, HTTP status, provider request ID if supplied, start/end, bytes, content hash, ETag/Last-Modified if supplied, parser version, expected and received counts, pagination completion, rejects and failure class.

Publication

Required fields and invariants: Source record identity, raw snapshot reference only where permitted, normalized revision, exact source locator, scope, source dates, review status, reviewer, publication/withdrawal event, superseded revision. Never treat `retrieved_at` as sample date.

Coverage

Required fields and invariants: Parameter/matrix/geography/time coverage, unsupported queries, known exclusions, dataset complete/partial status, available history/forecast horizon and actual resolution. Missing data is not zero.

Every source adapter returns `records`, `coverage`, `source_run_id`, `complete`, `next_cursor`, `warnings` and classified errors. Publication is a separate operation. A successful first page cannot publish an apparently complete dataset or withdraw all missing records.

Outdoor conditions and forecasts

S01 AirNow

Status: existing adapter; documentation checked, authenticated calls not made. Read the service catalogue (https://docs.airnowapi.org/webservices) and FAQ (https://docs.airnowapi.org/faq). Observation latitude/longitude queries return reporting-area data, not necessarily an individual nearby monitor. Forecasts can be category-only and have variable coverage. The FAQ recommends appropriate caching and file products for bulk database acquisition instead of looping over locations.

Acquire: existing `https://www.airnowapi.org/aq/observation/latLong/current/` with bounded latitude, longitude, distance, JSON format and server-side `AIRNOW_API_KEY`. Existing station layer uses `https://www.airnowapi.org/aq/data/` with dates, bounding box and parameters. For forecasts, obtain the exact current reporting-area request and sample from the authenticated service documentation before implementing it; an unverified forecast URL is not specified here as working.

HALO mapping: preserve `DateObserved`, `HourObserved`, `LocalTimeZone`, `ReportingArea`, `StateCode`, coordinates, `ParameterName`, `AQI` and category when supplied. Validate provider sentinel values and time-zone conversion. Keep station identity/time separate from area maxima. Dominant pollutant calculation must retain all available inputs.

Operations: share permitted requests by actual reporting area, not arbitrary home rounding; initial target hourly observations and source-cadence forecasts. Test category-only forecasts, missing pollutant, empty coverage, stale dates, rate limit and changing area association. Current per-household cron acquisition needs a source-budget review before scale-up.

S02 Open Meteo

Status: current UV and air payloads sample checked. Forecast documentation (https://open-meteo.com/en/docs), air quality documentation (https://open-meteo.com/en/docs/air-quality-api) and terms (https://open-meteo.com/en/terms) were checked. Free-service eligibility and commercial use differ; verify HALO's actual use and account. Retain required attribution, including the air model provider where required.

Acquire: `https://api.open-meteo.com/v1/forecast` for supported UV/weather fields and `https://air-quality-api.open-meteo.com/v1/air-quality` for modelled AQI. The probe includes exact successful query examples. Current free requests need no key; a paid deployment may require a customer endpoint and credential configured separately.

HALO mapping: latitude/longitude of returned grid, time zone/offset, `current.time`, `current.interval`, current units, hourly timestamps/units and values. Label modelled/forecast values accurately. `generationtime_ms` is processing duration, not model issue time. If a model run time is unavailable, return unknown rather than substituting retrieval time.

Build: per-factor forecast periods, UV windows that preserve gaps, history basis metadata and source-specific null handling. Preserve time offsets through DST. Existing backfill uses daily maxima and must not be merged invisibly with current snapshots. Validate the current `past_days` and requested daily variables with real historical fixtures before claiming backfill readiness. One successful present-day probe does not prove historical endpoints.

S03 NWS weather alerts and mold inputs

Status: points and hourly forecast sample checked; official API documentation (https://www.weather.gov/documentation/services-web-api) checked. No key is currently required by the documented workflow; use an identifying User-Agent with a maintained contact and the actual provider's limits.

Acquire: `https://api.weather.gov/points/{lat},{lon}`, then its returned `properties.forecastHourly`; alerts through `https://api.weather.gov/alerts/active?point={lat},{lon}`. Do not hardcode a grid forever. Sampled grid was `GSP/127,74`, which is a test location, not a production default.

HALO mapping: forecast generation/update times, time zone, each period's start/end, native units, relative humidity, precipitation probability, temperature and wind. Alert records need official ID, event, status, message type, sent/effective/onset/expires/ends, severity/certainty/urgency, affected area, description, instructions and update/cancel references when returned.

Critical gate: `moldFromNws` currently converts null precipitation to zero under a code comment. Treat missing input as unknown unless an endpoint-specific documented exception is proven. Select the interval applicable to now, not blindly the first array entry. Version any changed proxy and test before updating historical interpretations. This remains a weather-based indicator, not a spore measurement.

S04 Google Pollen

Status: adapter exists; request schema (https://developers.google.com/maps/documentation/pollen/reference/rest/v1/forecast/lookup), billing (https://developers.google.com/maps/documentation/pollen/usage-and-billing) and policy (https://developers.google.com/maps/documentation/pollen/policies) checked. No paid credential call was made. The schema supports one to five daily records, subject to actual coverage, not hourly pollen.

Acquire: `GET https://pollen.googleapis.com/v1/forecast:lookup`, location, `days`, optional language and paging. Server key `GOOGLE_POLLEN_API_KEY`, enabled billing and restricted key scopes. Map `regionCode`, `dailyInfo.date`, pollen type code, `inSeason`, `indexInfo` value/category/code and missing types; keep the provider index separate from concentration.

Retention implementation: the current non-EEA service-specific terms, section 16 (https://cloud.google.com/maps-platform/terms/maps-service-terms) explicitly permit caching today's forecast values for 365 consecutive days and other forecast/heatmap values for 24 hours, followed by deletion. These exceptions are more specific than the policy page's general summary. Confirm HALO's account jurisdiction/agreement; classify each acquired value at acquisition and store a rights-based expiry. Do not retain a multi-day forecast indefinitely because its valid date later becomes today. Review treatment of derived values, exports, backups and Luna context separately.

Display gate: the policy prohibits displaying pollen results on non-Google maps and specifies attribution. Ordinary cards do not require a map. Do not add a pollen layer to HALO's non-Google map incidentally. A properly permitted one-year today-value history may support the planned history window; longer retention needs separate rights. Test expiry and incomplete older series without deleting unrelated Journal entries or task records. No alternative provider is required merely because these retention classes exist.

S05 OpenUV fallback

Status: current adapter inspected; provider site (https://www.openuv.io/) was reachable but exposed no useful text in the research tool. Current plan entitlement and quota remain unverified.

Acquire: existing `https://api.openuv.io/api/v1/uv?lat={lat}&lng={lon}` with server `OPENUV_API_KEY` in `x-access-token`. The existing fallback guard is reused. Confirm `result.uv`, source time, units, error shape, retention and attribution using the real account before enabling it. Do not rely on the code comment about a free daily allowance as a current entitlement.

Failure: fallback unavailable means missing UV, not zero. This source does not automatically supply every hourly forecast period required by outdoor planning.

Water home risks and geography

S06 UCMR occurrence data

Status: official download index checked; complete current release not downloaded or parsed in this audit. EPA's occurrence page (https://www.epa.gov/dwucmr/occurrence-data-unregulated-contaminant-monitoring-rule) links nationwide and state text archives. The index currently identifies a July 2026 summary. The year in an archive URL is not the data's release date.

Acquire: `https://www.epa.gov/system/files/other-files/2023-08/ucmr5-occurrence-data-by-state.zip` or the corresponding `ucmr5-occurrence-data.zip` linked from that index. Public download, no account key identified. Use a bounded scheduled import, not a browser request or a 60-second page handler. Retain a release manifest/checksum and reviewed dictionary; bound compressed and expanded sizes.

HALO target fields: source system ID/PWSID, system name/state/type where supplied, sample/event/point IDs, collection date, analyte identity and source label, reported result text, numeric value only if valid, units, qualifier, reporting limit, method, water/sample context and release revision. Exact upstream column mapping must be completed against the downloaded dictionary before ingestion. Preserve non-detect and tested-but-not-detected rows. Existing utility aggregates alone cannot recover discarded sampling metadata.

Build: deterministic import and reconciliation, stable sample identities, corrections/deletions after complete-release validation, legacy aggregate compatibility, `tested_count` with a valid denominator, new-publication events and map quarter queries. No reproducible UCMR acquisition script was found in the inspected `scripts` directory. Obtain the existing import procedure and reconcile it before replacing its output.

S07 Water service boundaries

Status: existing layer metadata sample checked; EPA source page (https://www.epa.gov/ground-water-and-drinking-water/public-water-system-service-areas) links the same service. It describes mixed authoritative and modelled boundaries and version updates. Association is not proof that a particular property is served.

Acquire: `https://services.arcgis.com/cJ9YHowT8TU7DUyn/arcgis/rest/services/Water_System_Boundaries/FeatureServer/0/query`. Use point intersection for association, PWSID for record retrieval, explicit spatial reference and paginated/ID-batched queries. Metadata returned a 2,000-record maximum.

HALO mapping: `PWSID`, `PWS_Name`, `Population_Served_Count`, `Original_Data_Provider`, `Data_Source`, `Model_Method`, `Verification_Status`, `Original_Data_Created_Date`, geometry and source revision. Existing code currently requests fewer fields. Store candidate matches and association method. Overlaps need disambiguation; no match does not establish private-well use. Never relabel a modelled polygon match as a measured household result.

Operations: review on source releases, bounded refresh, geometry checks, coordinate reference tests and index validation. Keep the prior context on a failed address transition without assigning its utility to the new location. Verify reuse terms and attribution at the dataset/item level; an empty copyright field is not a permission grant.

S08 Radon context and testing guidance

Status: existing static county lookup; EPA zone source (https://www.epa.gov/radon/epa-map-radon-zones) checked. County zones are regional context and cannot decide a home's test result.

Acquire: reviewed county zone release and documented source link; no new live sensor API. Preserve state plus county FIPS, zone, release/version and matching method. Audit the existing NC lookup against the official release and store unknown outside coverage. Reviewed testing content supplies method/duration/unit interpretation requirements. Household reports use private tests, not replacement county values.

Gate: map every current static county value to a source revision; verify test guidance before shipping interpretation or reminders. An operator review updates public guidance, not past private test values.

S09 Lead inventory and plumbing

Status: repo static July 2025 inventory source inspected; NC DEQ inventory page (https://www.deq.nc.gov/about/divisions/water-resources/drinking-water/lead-service-line-inventory) checked. Repo data references `https://www.deq.nc.gov/water-resources/pws/lead-service-line-inventory-july-2025/download?attachment`. The current complete download and column mapping were not re-imported.

HALO mapping: PWSID, inventory reporting date, source release, lead/galvanized/non-lead/unknown counts and any source definitions. Keep home-year estimate separate from utility totals and private plumbing evidence. A stored note saying no address-level data exists must not be generalized into a claim that no utility has published a newer local inventory.

Build: reviewed release importer, null-aware totals and utility correction pathway. A specific service-line result requires an actual utility/property record and user-confirmed applicability. Lead remains shown but excluded from the approved composite. No new source is required merely to let a household save a question or test.

S10 ECHO facilities

Status: a bounded current query returned a QueryID and 11 rows for the test county; result-page ingestion was not validated end to end. ECHO services (https://echo.epa.gov/tools/web-services) and all-program search (https://echo.epa.gov/tools/web-services/facility-search-all-data) are the official references.

Acquire: existing `https://echodata.epa.gov/echo/echo_rest_services.get_facilities` followed by `.get_qid`, retaining the documented columns and all pages. Use the existing active-major/repeat-noncompliance selection deliberately; it is not a complete census of every local facility. QueryID is transient, not a stable facility identity.

HALO mapping: registry/program IDs, name, coordinates, program scope, compliance categories and time basis, quarters in noncompliance, inspection/penalty dates and amounts where supplied, official detail URL. Program counts are not distinct violation counts. Public regulatory evidence does not identify the cause of household exposure.

Operations: use bulk downloads if scale warrants, source-appropriate refresh, last-good dated cache, explicit partial coverage, schema drift checks and coordinate-null list entries. No public layer may include a private home pin in shared exports.

S11 County and district geometry

Status: Census TIGER files (https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html) and NC legislative redistricting source (https://www.ncleg.gov/Redistricting) checked. The latter distinguishes the enacted 2025 plan for the 2026 elections. Do not confuse election boundaries with the currently serving district at another effective date.

Acquire: versioned official boundary download with state/county/district identifiers, coordinate reference and effective use dates. Validate the chosen product against `lib/data/nc08District.json`, `lib/data/ncCounties.json` and the map preview assets. Record a checksum and the exact download URL in the import manifest before production.

Build: boundary-version-aware aggregation, point/polygon attribution, partial-county flags, explicit denominator and missing-population counts. Geographic assignment by centroid is an approximation and cannot allocate all utility customers precisely to districts. A changing geometry version requires aggregate recomputation and separately labelled history, not silent rewriting. Representative contact links need a current official directory review before automatic addressing.

Food fish gardens and stewardship

S12 FDA food sampling publications

Status: FDA PFAS testing publication (https://www.fda.gov/food/process-contaminants-food/testing-food-pfas-and-assessing-dietary-exposure) checked. It describes limited food sampling and targeted investigations. It does not establish a universal local-product lookup or a continuous farm monitoring service. Study-table extraction remains a reviewed-import task.

Acquire: follow the publication's study/result links, retain their original locator and revision, and manually validate the selected table extraction. Store study/sample identity, commodity, tested matrix, analyte, concentration/unit/basis, qualifier/limit, method, collection dates and geographic/product scope only where actually supplied. Unknown brand, lot or farm stays unknown.

HALO behavior: broader research findings are visibly separate from an applicable notice. A food category result must not be assigned to every item in that category. Guidance and questions can lead to Plan; a user-held laboratory report stays private. Water thresholds cannot interpret food or soil concentrations.

Gate: reviewed real rows and units, a permission/attribution review, publication-date parsing and checks against the original table. No API, local contamination rate, ingestion frequency or product coverage is invented to fill this gap.

S13 NC fish advisory records

Status: public service schemas and sample rows checked. Discovery chain: official landing (https://epi.dph.ncdhhs.gov/oee/fish/arcg.html), official StoryMap (https://storymaps.arcgis.com/stories/eab23dd84e8b403b95dc1223703aa487), embedded dashboard (https://experience.arcgis.com/experience/470cb0bb993a47969f065e6087282286/). Its public configuration exposes the following services:

- `https://services.arcgis.com/iFBq2AW9XO0jYYF7/arcgis/rest/services/FA_Lakes_Rivers_Points2/FeatureServer/32`
- `https://services.arcgis.com/iFBq2AW9XO0jYYF7/arcgis/rest/services/FA_Lakes_Rivers_Points_Complete/FeatureServer/38`

Acquire: metadata `?f=json`; bounded `/query` with selected fields, pagination, `returnGeometry` and explicit `outSR=4326` when requesting geometry. Public samples needed no key. This establishes technical access, not a supported feed

agreement. Official item metadata had no explicit licence; confirm permitted retention/redisplay and maintain a reviewed-content fallback.

Mapping: `Wtr_Bdy`, `Site`, `Fsh_Spc`, `Polltnt/Pollutnt`, `Popultn`, `MlsAllw`, `Advisry/Advisory`, source item/layer ID and row key. Layer 32 supplies `FAURL` for source material. Do not treat FID alone as permanent across source republication.

Critical sample findings: FID 1 paired a sensitive-population label and a one-per-month value with text indicating that population should not eat the specified fish. Both advice text fields are limited to 254 characters and the sample ends mid-sentence. `Lat` and `Long` contained projected coordinates, not valid latitude/longitude. The layer geometry is a point, not a waterbody advisory boundary. The dashboard also references a five-mile buffer layer, which must never become the official advisory extent. Some linked fish images are third-party stock imagery; do not copy them as government-owned assets.

Build: quarantine automatic advice; join the complete official document, exact audience, species grouping, serving definition, combined limits, effective/withdrawn status and actual extent under review. Publish a reviewed rule revision, not the raw `MlsAllw` field. Test against complete official examples such as the Cape Fear announcement (https://www.ncdhhs.gov/news/press-releases/2023/07/13/ncdhhs-recommends-limiting-fish-consumption-middle-and-lower-cape-fear-river-due-contamination). Do not assume that a historical announcement establishes today's status.

S14 FDA food recall notices

Status: public JSON sample checked, including a PFAS keyword query. API description (https://open.fda.gov/apis/food/enforcement/), fields (https://open.fda.gov/apis/food/enforcement/searchable-fields/), authentication (https://open.fda.gov/apis/authentication/), licence (https://open.fda.gov/license/) and terms (https://open.fda.gov/terms/) checked. This is a useful new acquisition path for F04, not a general recall dashboard expansion.

Acquire: `https://api.fda.gov/food/enforcement.json`. The probe's `search=reason_for_recall:PFAS&limit=1` returned two matches and a historical terminated recall. That count only describes this search at that time, not all PFAS incidents. Use reviewed contaminant synonyms and periodic complete reconciliation of the selected domain; a keyword query alone cannot guarantee recall coverage. Do not filter by `state` as if it were consumer distribution: the firm location and distribution text are different fields.

Mapping: source key `recall_number` with `event_id`, `status`, `classification`, product/lot descriptions, `code_info`, `reason_for_recall`, `distribution_pattern`, initiation/report/termination dates, dataset `meta.last_updated` and official notice link when verified. No concentration field is assumed. Preserve source lifecycle without turning a terminated recall into either a current alert or a product safety certificate.

Operations: source documentation says weekly updates; HALO can check daily conditionally, but must not advertise immediate recall alerts. Obtain a free server key for deployment; recheck its actual quota. Applicable openFDA data is generally CC0 subject to stated exceptions and service terms. Credit FDA without implying endorsement. Cache public revisions, preserve changes, and link to the issuing notice for exact consumer instructions.

S15 Water Quality Portal

Status: service guide (https://www.waterqualitydata.us/webservices_documentation/) and user guide (https://www.waterqualitydata.us/portal_userguide/) checked; an actual NC PFAS result fixture was not acquired. Optional environmental sampling extension, not a dependency for the initial food hub.

Acquire: documented `/data/Station/search`, `/data/Result/search` and domain lookup services with bounded state/site/characteristic/date filters and supported output. Pin the API/profile version: the guide warns that legacy WQX 2.2 excludes USGS data added after March 11, 2024 and points to WQX 3.0 beta. Do not assume a legacy query is complete recent coverage.

HALO mapping: organization, project/station/activity/result identifiers; sample medium; characteristic/analyte; value/unit; detection condition and limits; analytical method; sample date/time/depth/basis; coordinates and location method. Verify exact field names against the chosen profile. No reviewed concentration interpolation, household tap equivalence or crop safety inference is authorized.

Gate: real filtered fixtures, completeness assessment, provider-specific attribution, detection-limit parsing, station/result joins, retained method/matrix and no-result state. Leave this optional layer disabled until those checks pass; food notices and private garden records can work independently.

S16 US Drought Monitor

Status: documented REST and a county sample checked. Service documentation (https://droughtmonitor.unl.edu/DmData/DataDownload/WebServiceInfo.aspx) defines area, date range and statistics format; permission page (https://droughtmonitor.unl.edu/About/Permission.aspx) gives map attribution requirements.

Acquire: `https://usdmdataservices.unl.edu/api/CountyStatistics/GetDroughtSeverityStatisticsByAreaPercent?aoi={FIPS}&startdate={M/D/YYYY}&enddate={M/D/YYYY}&statisticsType=2`, `Accept: application/json`. Public sample required no key. Store `mapDate`, string FIPS, county/state, `none`, `d0` to `d4`, `validStart`, `validEnd`, `statisticFormatID`. Format 2 is categorical; do not sum cumulative format 1 as if mutually exclusive.

Build: weekly regional history, compatible-release differences, area-category distribution, explicit source and release dates. Validate percentages and identity. If a home-point map is later needed, acquire the matching dated geometry; county maximum is not a home measurement. Follow source map attribution instructions when reproducing its maps; separate source attribution from HALO branding.

S17 NC water conservation status

Status: official table (https://www.ncwater.org/WUDC/conservation-status), linked map configuration, schema and one record checked. The official map exposes:

`https://services2.arcgis.com/kCu40SDxsCGcuUWO/arcgis/rest/services/NC_Drought_Monitor_Water_System_Conservation_Status/FeatureServer/0/query`

Acquire: bounded ArcGIS queries with pagination and selected fields. No key was needed for the sample. Verify automated reuse and update expectations with the source policy or maintainer before enabling ingestion. A public endpoint is not a service-level agreement.

Mapping: `systemPWSID`, `systemName`, `countyName`, `conserveStatus`, `conserveReason`, `conserveDateEffective`; keep `countyDroughtStatus`/`countyDroughtDate` separate. Raw IDs such as `03-63-020` require a validated crosswalk to the federal PWSID namespace, not a name-only join. Preserve both IDs and reject ambiguous matches.

Build: source revisions, utility watches and separately reviewed official restriction instructions/contact URLs. The service's status alone does not specify allowed watering days, exemptions or enforcement. Keep old effective dates visible; a new fetch is not a new notice. Failed reads retain dated prior evidence, not Not Restricted. A proposed daily check needs approved access and workload limits.

S18 Labs Extension and garden preparation

Status: reviewed-content acquisition. NC lab directory guidance (https://www.deq.nc.gov/about/divisions/water-resources/drinking-water/drinking-water-laboratories), certified laboratory search (https://slphreporting.dph.ncdhhs.gov/Certification/CertifiedLaboratory.asp), Extension contacts (https://www.ces.ncsu.edu/local-county-center/), soil testing (https://www.ncagr.gov/Divisions/Agronomic-Services/soil-testing) and PFAS guide (https://content.ces.ncsu.edu/Guide-to-Understanding-and-Addressing-PFAS-in-our-communities) were checked.

Acquire: curated provider/resource entries with exact source URL, service region, supported matrix/analyte/method, certification identifier and checked date where actually verified, contact method, preparation requirements and last review. No supported appointment, payment, laboratory-result or universal certification API was established. Routine fertility tests are not contaminant clearance.

Build: searchable reviewed resource catalogue plus private site/test/inquiry records. Show unknown price or availability honestly; version quoted estimates with date, region and basis. Explain lab-specific sampling requirements through the official provider instead of inventing a universal PFAS collection protocol. User-entered results are not independently lab-verified.

S19 NRCS conservation assistance

Status: reviewed official programme content. The national activity list (https://www.nrcs.usda.gov/programs-initiatives/eqip-environmental-quality-incentives/cpas-dias-and-cemas) includes CEMA 207 and 209; NC EQIP information (https://www.nrcs.usda.gov/programs-initiatives/environmental-quality-incentives-program/north-carolina/environment-quality) supplies the state entry point. This audit did not validate eligibility for an individual applicant or a live application API.

Acquire: programme/activity ID, jurisdiction, fiscal year, scope, official eligibility/checklist/deadline/contact references, review date and replaced version. Read the selected current activity requirements before publishing detailed instructions. A national directory listing is not proof the activity is funded locally this year.

Build: preparation checklist, chosen site/test references, official handoff, optional user-reported application/reference/response state, linked follow-up task. Do not upload reports to an agency or promise assistance approval without a separately authorized integration.

Action content and platform dependencies

S20 Organizations and opportunities

Status: existing directory plus curated expansion. Yadkin Riverkeeper's official events (https://www.yadkinriverkeeper.org/events) was checked and includes dated event pages and calendar links. It is an example of an organizer source, not an established universal feed or partnership.

Acquire: reviewed organizer/service areas and event-specific source URL, stable organizer/event IDs, cause, start/end/time zone, location, age/supervision/accessibility requirements, cost/materials, registration link, cancelled/updated status and checked date. Missing metadata stays unknown. Calendar links can aid verified import but do not prove registration or attendance.

Build: source-specific approved adapters or curator entry, version diff/review, event expiry, user participation and Plan linkage. Recheck approaching events daily if access permits; review broader directories monthly as a proposed starting policy. No new booking/payment/organizer messaging service is required. All original directory causes remain available.

S21 Official reporting destinations

Status: NC Stream Watch (https://www.deq.nc.gov/about/divisions/water-resources/water-resources-public-information/water-education-programs/water-education-and-outreach/nc-stream-watch) and NC DEQ algal bloom reporting (https://www.deq.nc.gov/about/divisions/water-resources/water-sciences-section/algal-blooms) checked. These have different purposes. Stream Watch invites publicly shared observations; it is not a substitute for incident/emergency reporting.

Acquire: reviewed destination type, jurisdiction, official URL, appropriate observation classes, privacy/publication warning, urgency instructions, required user-supplied fields and review date. No automated submission/receipt integration was verified.

Build: private factual observations, sanitized attachments, report preview, explicit external handoff and user-reported submission/reference state. Alert users before leaving HALO if the receiving programme publishes observations. Do not transmit photos, locations or notes just because an official form exists. Refresh destination ownership/contact information, particularly around programme changes.

S22 Guidance thresholds translations and certification

Status: existing Learn/corpus plus new editorial work. Existing source URLs live in `lib/learnContent.js` and `scripts/ingest-corpus.mjs`. EPA PFAS rule information (https://www.epa.gov/sdwa/and-polyfluoroalkyl-substances-pfas) currently distinguishes the 2024 rule and 2026 proposals. NSF's contaminant-claim directory (https://www.nsf.org/consumer-resources/articles/contaminant-reduction-claims-guide) provides a model-specific verification path, not a general filter performance API.

Acquire: topic/claim ID, authoritative source, exact supporting section, publication/effective/review dates, domain/audience/matrix/units, original and reviewed translated text, legal/advisory status and method version. Supplement with qualified review for health interpretation. A translation must preserve audience and numerical meaning; no automatic synonym substitution for official rules.

Build: source-reviewed templates, glossary, contextual explanations, source changes and withdrawals, score-method registry and model-specific certification links. Separate legal limit, health benchmark and HALO display threshold. Existing scoring comments do not establish present compliance status. No single sample automatically proves a violation, and no standard number alone proves a product reduces every PFAS. Retrieval corpus publication follows content review and respects source rights.

S23 Geocoding basemap and search

Status: existing Mapbox v5 adapter; current Mapbox geocoding documentation (https://docs.mapbox.com/api/search/geocoding/) checked. It distinguishes temporary and permanent storage. Account rights and exact legacy-version terms were not inspected.

Acquire: existing `/geocoding/v5/mapbox.places/{query}.json` with server `MAPBOX_TOKEN`; proposed migration only after validating the chosen endpoint/response. Keep address queries out of logs. Current returned center/context parsing is not automatically compatible with v6.

Build: exact account/storage decision, restricted rendering token if needed, basemap style/tile attribution and budget, response-version adapter, candidate disambiguation, location bounds and rounded storage under the privacy policy. Do not add POI search, navigation or continuous GPS as hidden requirements. An accessible public-record search can use HALO's own published catalogue without an external search API.

S24 Household records and platform services

Status: user records and existing platform dependencies, not missing environmental feeds. Actions, Plan, tests, garden observations, fish notes, water-use records, participation, Journal, outreach and summaries can use owned Supabase records plus reviewed content and the sources above.

Build dependencies: actual database schema/RLS audit; private storage with quarantine and a chosen scanning mechanism; durable job execution; identity recovery via a supported Supabase flow; browser Web Push subscription/service worker; secret management and current provider budgets. Luna keeps its existing configured chat/embedding provider until deliberately reviewed. No additional external AI provider is required by this plan.

Unresolved operational choices: storage/scanning provider, scheduler frequency and hosting tier, recovery flow, retention durations, account costs and deployment credentials must be recorded by the implementation owner before launch. These are explicit tasks, not assumed free capabilities. No external calendar API is necessary for one-way ICS

export; bidirectional sync is outside the established target. Public official data must never absorb household notes or lab attachments.

Shared normalization and matching requirements

Table fields: Problem / Required implementation and test

Chemical identity

Required implementation and test: Maintain source label plus reviewed analyte identifier/aliases. PFAS is a family, not one interchangeable measured substance. Preserve mixtures and method scope.

Units and sample matrix

Required implementation and test: Retain raw text/unit/qualifier. Normalization includes conversion version and matrix. Water mass-per-volume, soil mass-per-mass and food tissue results cannot share one automatic threshold. Wet/dry basis and nondetect limits remain explicit.

Time

Required implementation and test: Distinguish sample/observation, issue, valid, publication, retrieval and review times. Preserve date-only records as dates and user schedules with IANA time zone. Never invent midnight UTC observation time.

Identity

Required implementation and test: Namespace PWSIDs, station IDs, facility registry IDs, advisory IDs, programme IDs and source row keys. Preserve raw values and reviewed crosswalks. Fuzzy names can suggest candidates, not confirm identity.

Geography

Required implementation and test: Store point/area/segment plus CRS, geometry release and matching method. Source point, centroid and buffer are not exposure boundaries. Keep approximate user context separate from public source geography.

Completeness

Required implementation and test: Know which pages/files/parameters were fetched. Distinguish not tested, nondetect, no coverage, fetch failure, stale, withdrawn and valid zero. No record count becomes a testing denominator without its population definition.

Corrections

Required implementation and test: A new revision can correct metadata without indicating worsening conditions. Saved tasks retain evidence snapshots and flag material changes. Withdrawal invalidates current guidance but preserves provenance.

Scope

Required implementation and test: No household location inference from event attendance, no pregnancy/age inference from general categories, no food inference from water, no agency receipt from an opened link.

Source release checklist

Before enabling a source-dependent feature, attach the following evidence to its implementation PR:

1. Exact source/request and applicable terms review, with no key values.
2. Successful representative raw fixture or permitted minimal derivative, source checksum, parser version and expected canonical output.
3. Coverage tests for in-region, out-of-region, ambiguous, empty, stale and unsupported inputs.
4. Unit/qualifier/date/geometry tests and complete pagination reconciliation.
5. Source-to-screen field mapping, including which source fields are unavailable rather than silently derived.
6. Approved refresh interval, quota/cost ceiling, timeout/retry limits and last-good-data behavior within storage rights.
7. Content review and actual official instructions for every personalized or actionable interpretation.
8. Versioned source-change matching, withdrawal handling, dedupe and notification tests.
9. Attribution in cards, maps, exports and model context as applicable.
10. Feature flag, stale/error UI contract, operational owner and rollback procedure.

The source remains conditional until its applicable checklist is complete. A page can ship with honest unavailable sections only after that fallback is explicitly accepted; that must not be reported as full live feature completion.

04 / Complete app master plan

docs/product/HALO_MASTER_PLAN.md

HALO Complete App Plan

Version 1.1, October 2, 2026. Prepared for Yogi and Vibhav from the current repository, approved work, original specification, and the product decisions in this conversation. This revision adds the detailed frontend build guide, backend implementation inventory and cross-feature integration requirements.

This is the consolidated target product plan, not a claim that the target is already built, scientifically validated, deployed, or visually approved. It preserves established capabilities and gives the new features concrete jobs, data requirements, interactions, and completion criteria. New layouts still require Yogi's wireframe review. New backend work is planned here, not performed by creating this document.

The recommended navigation is **Today, Homeguard, Act, Plan**. Plan is the recommended fourth tab and remains a product decision for Yogi to confirm. Moving Map and Journal into this structure preserves their capabilities. Implementation order is not a cut list.

Read this document for product behavior, Frontend Build Plan (HALO_FRONTEND_BUILD_PLAN.md) for screens, controls, state and reuse, Backend and Data Plan (HALO_BACKEND_AND_DATA_PLAN.md) for implementation contracts, and Feature Acceptance (HALO_FEATURE_ACCEPTANCE.md) for preservation and release tests. Together they form the plan. None replaces the original reference or an immutable approved archive. The 45 feature identifiers describe capability groups, not a limit of 45 screens; each group can include multiple detail pages, forms, dialogs and states.

October 3 source audit: the Backend Build Handoff (HALO_BACKEND_BUILD_HANDOFF.md) maps every feature group to implementation work. The Source Acquisition Guide (HALO_SOURCE_ACQUISITION_GUIDE.md) records exact source paths, inspected payloads, storage/display constraints and remaining gates. Public NC fish and conservation services and an FDA food recall API were located and sampled. Their availability does not establish complete local food coverage or validate raw advisory interpretation.

Purpose and intended outcomes

HALO helps a household understand environmental conditions, distinguish evidence from uncertainty, act on what matters, and participate in local environmental improvement. Its organizing sequence is **evidence, meaning for this household, practical choice, action, and follow-up**.

The app connects three timescales without mixing their meaning:

- Today's outdoor conditions and available forecasts.
- Longer-lived home, water, soil, and official advisory information.
- The household's own plans, observations, records, and participation over time.

Success means a person can answer a useful question and complete an appropriate next step. Examples include understanding a water result, arranging a test, preparing a sourced question for a utility, checking a fish advisory, choosing between forecast periods, following a conservation opportunity, or recording a maintenance task. A completed task is not proof of lower exposure, better health, or pollution removed.

HALO remains useful with no Journal entries, no notification permission, no linked account, and no purchased hardware. It is not a medical diagnostic system, emergency alert replacement, laboratory, farm certification service, or a declaration that a property or food is safe.

People the design must work for

Table fields: Situation / Product need / Design consequence

Busy household

Product need: Understand the immediate situation without reading an essay

Design consequence: Short landing pages, deeper detail on demand

Renter

Product need: Options they can actually control

Design consequence: Renter wording, information requests, portable actions, landlord follow-ups

Private well or spring user

Product need: Useful help even with no utility sample

Design consequence: Testing journey instead of invented results

Household with a sensitive category

Product need: Relevant explanations without medical profiling

Design consequence: Selected category guidance, no named-person risk predictions

Older or low-vision user

Product need: Legibility and predictable controls

Design consequence: Larger text, visible labels, low motion, no gesture-only actions

Gardener, fisher, or grower

Product need: Specific local evidence and practical resources

Design consequence: Optional focused branches, source-aware guidance, test and advisory records

Community participant

Product need: A credible way to help

Design consequence: Verified opportunities, clear participation requirements, official reporting paths

Weak connection or limited device

Product need: Core information remains usable

Design consequence: Small initial load, cached evidence with dates, text map alternative

Reviewer or professional

Product need: Understand how a result was obtained

Design consequence: Sources, methods, limitations, and a private export chosen by the user

Established decisions and authority

The latest direct user decisions and repository instructions take precedence over older reference wording. The original PDF is a functional inventory, not an instruction to restore old names, ownership, layout, or deployment commands.

Preserve these decisions:

- Yogi is the sole frontend person. Vibhav handles backend integration. This plan does not reintroduce the PDF's old assignments.
- Today is the page name, not Personal Hero. Its scored factors are air, UV, pollen, and mold. PFAS belongs to Homeguard and never enters the Today score.
- October 3, 2026 clarification: Today and its air, UV, pollen and mold detail pages keep their current content depth, sections, visuals and overall layout for now. Do not substantially shorten them under the broader simplification plan. Small refinements and reviewed additive connections remain possible; substantial cuts or restructuring need a separate user decision.
- Homeguard retains water and radon scoring. Lead remains shown but unscored. Excluded water compounds remain visible with explanations.
- Keep the approved logo, onboarding handwriting, centered Welcome and to HALO treatment, and immediate access under reduced motion.
- Keep the approved blue and teal identity, rounded navigation, factor tints, contextual scenes, selected household figures, Luna name and symbol, and UV rising-sun treatment. No deer.
- Factor cards navigate to dedicated pages. They do not revert to expandable overview cards.
- Keep the step-card deck with previous and next arrows. Browsing a step never marks it completed.
- Household icons appear only for selected categories. Their count is not the number of people in the home. Respiratory Issue(s) replaces the old Breathing label.
- Keep all seven category routes already preserved in the approved stage. Child, teen, and adult guidance can share general content where there is no meaningful distinction; no forced duplicate advice.
- Appearance follows the device. Atmospheric time follows the clock and is not a weather observation. Accessibility preferences remain available.
- No em dashes in app-facing text, including sourced or generated content and accessibility labels.
- Approved Settings use explicit Save and retained drafts, not the older PDF's automatic saves.
- Never replace the sealed onboarding or October 1 foundation archives. New approval receives a separate baseline.

Existing production-stage scope is recorded in the handoff (../foundation/STAGE_FRONTEND_HANDOFF.md). The current Map work (../map/README.md) is an unapproved fixture-driven review, not a connected production map. New backend willingness broadens this plan's possibilities; it is not authorization to mutate the production database or deploy unfinished features.

Build on the existing application

This is an extension of the current HALO application, not a replacement project. Repository code is evidence of an implementation, not proof that the same revision and every credential or job are configured in production. The historical context describes earlier versions and must not override inspected code or newer user decisions.

Table fields: Area / Existing foundation to retain / Change required for the target

Entry and identity

Existing foundation to retain: Approved onboarding, anonymous Supabase identity, authoritative profile gate, request recovery

Change required for the target: Add optional durable recovery and home-context history deliberately, without adding mandatory registration

Live shared shell

Existing foundation to retain: StageApp, StageSession, StageRuntime, protected transport and safe return routing

Change required for the target: Extend the same shell and route allowlists for reviewed new pages and four-tab navigation

Today and factors

Existing foundation to retain: Approved components, environmental scenes, daily/history/Learn adapters

Change required for the target: Add supported change explanations, saved actions and forecast entry; do not invent missing factor scores

Homeguard and family

Existing foundation to retain: Existing home response, evidence panels, household flags, action-plan rules and guidance routes

Change required for the target: Add owned tests, maintenance, garden contexts and follow-up; keep regional and property evidence separate

Luna

Existing foundation to retain: Approved UI and current question/page transport, retrieval and source links

Change required for the target: Add truthful context capabilities, bounded conversation and confirmed action proposals

Alerts

Existing foundation to retain: Inbox, read/dismiss, preferences and browser-push server routes

Change required for the target: Finish the preference UI, service-worker/device delivery integration, subscriptions and durable reminder delivery

Journal

Existing foundation to retain: Entry, retrospective, findings and summary server routes

Change required for the target: Build the reviewed frontend and correct interpretation semantics without erasing recording capabilities

Map

Existing foundation to retain: Server map/district datasets and an unapproved interactive preview

Change required for the target: Review and implement the production renderer, source/date semantics, query adapters and accessible equivalent

Act

Existing foundation to retain: Organization lookup and existing home/context data

Change required for the target: Build civic pages, letter/export flows, then opportunities, projects and reporting records

Full Settings

Existing foundation to retain: Essential live forms and device accessibility preferences

Change required for the target: Complete language, notification, privacy, export, deletion, source/method and recovery surfaces

New persistent domains

Existing foundation to retain: Current identity, database, API and source pipeline

Change required for the target: Add shared actions, Plan, evidence revisions, tests, food/advisory records and participation within the existing stack

The current production route set does not yet include working `/map`, `/journal` or `/act` page implementations. Their original planned URLs and notification destinations should guide stable future routes. A backend API, a preview route and a finished live page are three different completion states.

Platform remains a responsive web app with a mobile-first layout and a usable wider-screen presentation. Native iOS/Android packaging, hardware procurement, continuous background location and shared family accounts are not implied. Browser-install/offline capabilities and Web Push require their actual supported browser and service-worker paths to work; a manifest or Capacitor configuration alone does not establish them.

Keep current inline SVG chart patterns and the established component/icon system. A new package needs a concrete functional or maintenance reason, size/accessibility review and approval of any visible changes. Do not replace the existing auth client, create a second global household store, or add a new charting framework simply because a new page needs one chart.

Navigation and page ownership

Every feature has one primary home and may have several contextual entrances. A contextual entrance reuses the same underlying record and action, not another copy of the feature.

Table fields: Primary home / Pages and subpages / Main cross links

Today

Pages and subpages: Overview, air, UV, pollen, mold, daily household guidance, meaningful changes

Main cross links: Relevant map record, Journal check-in, outdoor plan, saved action

Homeguard

Pages and subpages: Overview, selected household guides, water, radon, lead, tests, maintenance, household summary, Food and Garden

Main cross links: Plan follow-up, Act outreach, sources, selected utility on Map

Act

Pages and subpages: Overview, organizations, opportunities, stewardship projects, outreach, field observations, Explore Your Area and district view

Main cross links: Plan participation, relevant home evidence, official registration or reporting

Plan

Pages and subpages: Your Week, saved actions, action detail, outdoor plans, Journal and Patterns, seasonal recap

Main cross links: Original reading, test, organization, event, or advisory

Shared

Pages and subpages: Luna, Learn, alerts, source detail, Settings, privacy, accessibility, export preview

Main cross links: Return to the exact originating context

Recommended route organization preserves delivered destinations and the original planned URL contract:

Table fields: Route family / Treatment

`/onboarding`, `/today`, `/home`, `/factors/[factor]`, `/family`, `/household/[member]`, `/settings`

Treatment: Preserve existing routes and safe return behavior

`/map`

Treatment: Build the planned canonical map route; discoverable under Act, not a bottom tab. Its current wireframe is at a separate preview route.

`/journal`

Treatment: Build the planned canonical Journal route; discoverable under Plan, not a bottom tab. Current Journal server routes are not a finished page.

`/act`, `/act/organizations/[id]`, `/act/opportunities/[id]`, `/act/projects/[id]`

Treatment: Recommended civic discovery and detail routes

`/act/outreach`, `/act/observations`

Treatment: Recommended outreach and private observation workflows

`/plan`, `/plan/actions/[id]`, `/plan/outdoors/[id]`

Treatment: Recommended commitments and planning routes

`/home/tests`, `/home/tests/[id]`, `/home/maintenance`, `/home/summary`

Treatment: Recommended home record routes

`/home/food-garden`, `/home/food-garden/fish`, `/home/food-garden/garden`, `/home/food-garden/food`, `/home/food-garden/growers`, `/home/food-garden/water`

Treatment: Recommended Food and Garden branches

Subpages use a clear back control. Opening a sheet adds one history entry; replacing its content does not stack sheets. Filters normally replace the URL state. Private notes, household details, precise locations, and draft letters never enter query strings. Links to public map selections can carry public IDs. Active navigation uses the owning tab; a contextual back link returns to the originating feature without losing its scroll, filters, or unsaved draft. All return paths are allowlisted.

Public civic exploration must not require a household address. Support a reviewed entry into public Map/district evidence, organizations and available public opportunities through the existing anonymous bootstrap or a separately reviewed public-read contract. The current APIs still require an authenticated anonymous token; this is not a promise that they are unauthenticated public endpoints. Before household setup, omit the home pin, personal ranking and household-matched wording. Saving private actions can explain its record/identity requirements without preventing someone from reading public information. Private pages retain their stronger profile and ownership gates. Any new Browse your area entry beside onboarding needs visual review and must not alter the approved Welcome presentation without approval.

Shared feature requirements

The following apply to every feature below, so they are not repeated as a long checklist on every screen.

Evidence and monitoring

Every displayed observation or advisory identifies its kind, geographic applicability, source, observation or sample time, publication time when relevant, retrieval time, units, quality, and missing-data reason. Not every source supplies every field; unavailable metadata stays unavailable. Retrieval today cannot turn an old sample into today's measurement.

Measured, modelled, proxy, forecast, official advisory, self-reported, and manually entered laboratory result are different evidence types. No automatic promotion between them. A regional reading is not a property measurement. Non-detect, measured zero, not tested, no record, failed source, and outside coverage are distinct states.

Monitoring means checking a named source or a user's explicitly recorded information at a stated cadence, detecting a defined change, and producing a traceable update. It does not imply an installed sensor, continuous observation, or access to unpublished information. Suggested operating cadences are targets to validate against provider rules and budgets, not current guarantees:

Table fields: Information / Proposed monitoring behavior / User meaning

Air and forecast periods

Proposed monitoring behavior: Scheduled regional refresh plus cached reads; provider-specific freshness

User meaning: Latest available observation or forecast, not second-by-second exposure

Official weather notices

Proposed monitoring behavior: More frequent batched polling with source identifiers, updates, expiry and cancellation

User meaning: Supplemental information, not a replacement for official emergency channels

Utility samples and lead inventories

Proposed monitoring behavior: Check publication feeds or imports on their actual release schedule

User meaning: A new published record, not a new sample taken by HALO

Radon zones

Proposed monitoring behavior: Versioned reference dataset

User meaning: Background context, not live home monitoring

Fish and local advisories

Proposed monitoring behavior: Approved feed checks or documented human review

User meaning: Last verified advisory status, with stale-source warnings

Drought

Proposed monitoring behavior: Source's weekly release

User meaning: Regional weekly conditions

Utility conservation notices

Proposed monitoring behavior: Source-specific checks with original update dates

User meaning: Published utility status, not an inference from drought

Events and organizations

Proposed monitoring behavior: Source checks, expiry rules, and human verification

User meaning: Known upcoming opportunities, not guaranteed availability

Home tests, routines, observations

Proposed monitoring behavior: Updates when the user records them; reminders when chosen

User meaning: Household record tracking, not automatic sensing

Action and interaction contract

Every actionable recommendation has a reason, applicability, source, suggested next step, and a definition of completion. Save and Schedule are separate. A recommendation is not automatically a commitment. User-created tasks can exist without an environmental claim.

The shared action state supports saved, scheduled, in progress, waiting, completed, paused, and cancelled. Only relevant choices appear in a given task. Completion can be reopened; deleting and cancelling are distinct. Original evidence is retained as a dated snapshot while the latest evidence can be viewed separately. Changed evidence flags review rather than silently rewriting a promise or the user's text.

Mutations show pending, confirmed success, failure with retained input, and conflict when another device changed the record. Retrying must not create duplicate tasks, entries, reminders, or exports. External links never imply a registration, purchase, report, or message was completed.

Relevant guidance without a competing recommendation feed

Keep severity, relevance and urgency separate. Existing Homeguard recommendations already use severity, household context and renter/owner wording; extend that deterministic, reviewed approach rather than asking an AI to rank health risks. Cross-feature suggestions add applicability, source freshness, whether the user already saved an equivalent action, current task state and a clear reason. No hidden personalized health-risk percentage is needed.

Present an official applicable notice distinctly from a household task. Then show the most useful eligible next step, with other suggestions available below it or in detail. A missing result may make obtaining information useful, but must not be ranked as a measured severe condition. An overdue administrative task is not an environmental emergency.

Users can choose Not relevant, Already doing this, Save for later or Stop this reminder where appropriate. Record these as recommendation preferences or task changes, not edits to environmental evidence. Show why a suggestion appeared and how to reverse a dismissal. Changed sources can request review of a dismissal only under a defined rule; they must not create repeated nagging.

What completion actually records

Table fields: Action type / Required user confirmation or record / Follow-up behavior

Read an explanation

Required user confirmation or record: No completion required to use the app

Follow-up behavior: Offer a relevant practical choice without an artificial achievement

Arrange or perform a test

Required user confirmation or record: Arranged, sample collected and result received are separate events

Follow-up behavior: Await the result, then review the confirmed record and any next step

Replace or maintain equipment

Required user confirmation or record: Completion date and optional user note

Follow-up behavior: Schedule the next occurrence using the selected recurrence basis

Attend an opportunity

Required user confirmation or record: User-reported registration and attendance remain distinct

Follow-up behavior: Retain the organizer source and optional personal participation record

Send a letter or report

Required user confirmation or record: User reports external submission, or a genuine integration supplies a receipt

Follow-up behavior: Track a chosen follow-up and any user-entered reference or reply

Complete a project

Required user confirmation or record: Required steps resolved and a user-confirmed outcome

Follow-up behavior: Keep observations separate from verified environmental measurements

Make an outdoor plan

Required user confirmation or record: A selected time and preparation choices

Follow-up behavior: Review a relevant forecast change; do not assume the activity happened

The shared action detail supports these typed outcomes with a small form relevant to the task. It does not show every possible status field to every user. Completion never updates a source reading, household score, symptom record or public participation total by inference.

Cross feature changes

Table fields: Change / Views that update / Information that must remain unchanged

Household category edited

Views that update: Figure selection, eligible guidance and future category-sensitive alerts

Information that must remain unchanged: Historical records and raw environmental values

New home confirmed

Views that update: Current environmental context, utility association, relevant subscriptions and reminders after review

Information that must remain unchanged: Prior-home tests, source snapshots and original historical dates

Utility result corrected

Views that update: Water detail, Map record, relevant watch notice and saved-action review flag

Information that must remain unchanged: Original source revision and the user's earlier draft or action reason

Test result corrected

Views that update: Current test display, compatible comparisons and affected follow-up

Information that must remain unchanged: Original transcription/document and its correction history

Event cancelled or moved

Views that update: Opportunity page, linked Plan item and applicable alert

Information that must remain unchanged: The user's original choice, participation history and source revision

Journal entry edited or deleted

Views that update: Calendar, day detail, findings and season summary

Information that must remain unchanged: Unrelated home records and public history

Advisory withdrawn

Views that update: Advisory detail, watch notice, relevant saved-item review state

Information that must remain unchanged: Historical evidence and optional catch notes

Session expires or identity changes

Views that update: Authorization gates and current private surfaces

Information that must remain unchanged: No private record may be reassigned to a replacement identity by guesswork

Server mutations return the affected stable IDs and versions or trigger a bounded refetch. Frontend invalidation must cover these dependents, not only the originating card. Polling, focus refresh and push cannot independently create three copies of the same event.

Common states and access

All features support loading, populated, genuine empty, partial evidence, source failure, stale data, outside coverage, offline cached, offline unavailable, and permission or identity failure where applicable. Forms also handle validation, duplicate submission, save failure, unsaved exit, expired referenced records, and deletion confirmation. No realistic fixture appears as live fallback.

Use semantic controls, keyboard paths, focus restoration, text alternatives, visible severity words, large tap targets, and reduced motion. The app must work without hover, swiping, animation, precise dragging, or color perception. Privacy-sensitive offline persistence requires explicit design; do not casually cache Journal notes, letters, photographs, or tests alongside public environmental data.

Entry and onboarding

G01 Entry and household setup

Job: reach a useful household view with minimal data and without mandatory account creation.

Keep: the approved Welcome animation and logo, address flow, selected household categories, home details, result reveal, anonymous session, progress/back behavior, failure recovery, and change-address pathway. Do not add interests, a farm questionnaire, notification permission, or AI consent as mandatory setup steps.

Data and actions: the existing onboarding and profile interfaces remain authoritative. Typed addresses are used for lookup and discarded under the existing policy. Household booleans are presence categories, not diagnoses or person counts. Users can skip optional information, correct it in Settings, and inspect how it changes guidance. Unknown water source and unmatched utility must lead to useful correction or testing choices.

Enhancement: ask feature-specific optional questions only when opening Food and Garden, outdoor planning, tests, or subscriptions. Offer optional recovery after the person creates something worth keeping, without making it a prerequisite. An address change must not attach old home test results to the new home: archive the old context, explain the transition, and let users deliberately carry appropriate generic tasks forward. This needs new home-context versioning.

Allow public civic browsing without requiring address setup as described in Navigation and page ownership. This is a separate route/access-mode enhancement, not permission to show household-specific mock data to an incomplete profile.

Done: first use, returning session, unavailable profile, expired session, denied location, no utility match, outside coverage, interrupted save, and changed-home flows are all exercised without losing or crossing household data.

Today

T01 Current conditions and meaningful changes

Job: answer what matters now and what the household might do differently.

Surface: centered title, approved ambient score hero, plain interpretation, source freshness, compact contribution explanation, fixed four-factor order, household guidance below the cards, then a small relevant planned-action or check-in entrance. Do not stack several recommendation feeds. Expand detail rather than repeating introductory paragraphs.

Interactions: tap a factor or ring segment to its detail; inspect score basis and missing inputs; compare a valid previous period; refresh with existing rate-limit feedback; open relevant alerts; save an offered action; resume today's saved action. A notification does not mark a task done. A second refresh does not remove retained readings.

Real monitoring: daily observations and history, with a proposed backend change summary distinguishing environmental change, changed source coverage, new publication, and method change. Compare only compatible periods and methods. Suppress unsupported score comparisons.

Backend: reuse daily/history/profile/alerts. Add per-source observation metadata and structured change reasons. Define any new factor display scores server-side with reviewed semantics. The existing API does not supply individual daily `/100` scores, so the live fallback remains Not scored, not fabricated values. Explain the ring's visual shares honestly; they are not automatically causal contribution percentages or formula weights.

T02 Air quality

Job: interpret the current outdoor air reading and help choose a practical response.

Content: AQI and severity, dominant pollutant, measured or modelled basis, monitor or region when known, a dated trend with gaps, and source-linked household guidance. Keep raw AQI distinct from HALO's higher-is-better score.

Interactions: inspect individual trend points and accessible table; open pollutant explanation; view relevant monitor in Map; compare available forecast periods through P03; save an applicable action; open Journal for the same date without implying the air caused symptoms.

Backend: existing AQI, source, pollutant and history; add precise source observation metadata and supported forecast periods. Preserve area-versus-station distinctions. A distant monitor does not measure a user's route, classroom, or indoor air. Revalidate provider endpoints, licence terms, budget, and attribution before integration.

History must retain acquisition method and aggregation. The current backfill stores modelled daily-peak air values, while ordinary daily records can reflect a different observation context. A chart must expose this distinction or split the series; connecting them as comparable measurements can create a false apparent change. Extend the history response to carry the stored basis instead of losing it at the API boundary.

Visual: detailed sky with slow atmospheric movement and limited particles; chart and action areas remain still and legible. Pollution imagery must not falsely encode an unmeasured concentration.

T03 UV

Job: understand available UV conditions and plan protection or timing.

Content: current index, rising half-sun window, available daily maximum, historical trend, protective guidance, and source date. Explicitly distinguish the forecast's local time from decorative sky lighting.

Interactions: inspect the returned protection window, open an explanation, compare times when forecast resolution permits, save a preparation step or activity plan, and inspect prior readings. A null window does not assert all-day low UV. Do not invent an hourly curve from two window endpoints.

Backend: reuse current UV/window/history; new forecast intervals must retain issue time, valid time, location and missing periods. User reminders are chosen tasks, not medical dosing or automated assurances that going outside is safe.

T04 Pollen

Job: show which category matters and make changes understandable.

Content: tree, grass, and weed readings with their own severity; dominant category; genuine history; source-reviewed regional seasonal context; household guidance.

Interactions: select a category without moving the page, inspect its history, compare available forecast days, save an action, and view related Journal history. Do not treat a category change as improvement in the same series. General seasonal text is never presented as a current local measurement.

Backend: existing categories and dominant values; new multi-day forecasts depend on paid-provider coverage, terms, actual returned horizon, and caching permission. No invented hourly pollen timing from a daily forecast.

Visual: warm pale-yellow botanical accents and subtle plant movement; not the same scene as UV.

T05 Mold conditions and indoor moisture context

Job: explain the weather-based indicator and help users address an actual moisture concern without pretending to detect mold.

Content: estimate label, humidity/rainfall basis, what the indicator does and does not represent, source-reviewed practical steps, and link to home moisture records. Do not add a fabricated mold trend when the source has none.

Interactions: inspect inputs; save a moisture check; record a user-observed leak, condensation, or meter reading in H07; track repair and follow-up; create a landlord information request when appropriate.

Backend: reuse proxy data and Learn; add private observation/maintenance records, not a new sensor claim. User-entered relative humidity includes unit, time, room label and measurement source. It remains separate from the outdoor estimate. No image-based mold identification or diagnosis.

Visual: restrained botanical/moisture scene with slow drifting detail. Decorative mushrooms are not presented as evidence of the indoor condition.

T06 Daily household guidance

Job: compile what the selected household should consider today without duplicating all factor pages.

Content: the selected categories, most relevant current factors, concise source-backed interpretation, practical choices and any already-saved related task. General guidance works with no selected category.

Interactions: switch a relevant category or Outdoors/At home view; open the factor; save a step; edit household selection; open a relevant planned activity. Reuse the canonical action if another category suggests the same thing.

Backend: existing household and Learn are the starting point. A new compiled guidance response can return multiple applicable notes with evidence IDs, provenance and deduplication, rather than only one priority sentence. Categories affect wording and action priority, not environmental measurements or a person's inferred medical risk.

Homeguard

H01 Home overview and evidence coverage

Job: understand the home assessment, people it considers, and what needs follow-through.

Surface: existing hero and evidence basis, selected-household illustration, water/radon/lead cards, current ranked home actions, then compact entrances to tests, maintenance, Food and Garden, and the household summary. All branches are discoverable without making them giant dashboards on the landing page.

Interactions: open a factor or selected household figure, inspect assessment coverage, change incorrect home information, start a relevant test, save an action, or resume a pending follow-up.

Backend: reuse home/profile/household/learn. Integrate task and record summaries by stable IDs. Do not treat private-water missing data as a healthy result or re-score the house because a task was checked off. A clear coverage explanation accompanies a partial score.

H02 Drinking water and PFAS

Job: understand the household's water evidence and take a feasible next step.

Content: utility identity, association basis, compound results and dates, source-defined comparisons, historical samples, coverage, unregulated detections, excluded substances, and regulatory/benchmark distinctions. PFAS remains prominent without concealing other existing water evidence.

Interactions: choose a compound, inspect a threshold row or historical sample, show exact units and methodology, open the utility in Map, follow new results, view certification guidance, save a practical action, draft a sourced utility or landlord question, or arrange a test. Confirm or challenge a utility association through a reviewed correction flow; do not let arbitrary client IDs silently replace verified service-area matching.

Private water: show the existing prioritized test plan, reasons, provider guidance, cadence and labelled estimated costs. No utility data is assigned to a well. Say HALO has no result rather than asserting nobody has ever tested the well. Distinguish well and spring context.

Backend: retain current home response. Add complete sample provenance, explicit tested/non-detect coverage when available, consistent analyte identities, threshold versions, publication changes, and links to home test records. Detections-only ingestion cannot establish complete testing denominators. Never copy drinking-water thresholds to food or soil results.

H03 Radon

Job: move from regional context to a valid home testing and follow-up process.

Content: county zone as regional context; test options and approved source guidance; any household-entered test as a separate dated record; next step based on reviewed interpretation rules where supported.

Interactions: understand the zone, save a testing task, record test setup/start/end, track an expected result, enter or attach the report, review confirmed data, record a professional follow-up, and set an appropriate chosen review reminder.

Backend: reuse zone and action content; add test lifecycle and source-backed interpretation. Test location, duration, method, unit and report date matter. A user report does not overwrite the county dataset. Stop irrelevant pending-test reminders when a relevant record is confirmed, without inventing a universal retesting interval.

H04 Lead and plumbing

Job: explain the existing estimate and help obtain better property-specific evidence.

Content: building-era estimate, unknown-year state, utility-wide service-line inventory where returned, shown-not-scored label, renter/owner options and testing guidance.

Interactions: correct the home year; inspect the utility inventory; ask about the specific service line; plan a test; add a report or professional record; track a landlord/utility follow-up. Retain context for well households through the appropriate testing plan.

Backend: existing estimate and inventory remain. Proposed records separate plumbing/service-line evidence from water sample results. Utility totals never establish a particular address's material. Lead remains excluded from the

composite even when a record is added unless Yogi explicitly revisits that established decision through scientific and product review.

H05 Household guidance pages

Job: compile relevant home guidance for a selected category.

Content: relevant evidence, why it matters, feasible actions, existing tasks/tests, and linked sources. The house illustration uses selected categories only and labels their meaning. General advice is shared for categories without a meaningful difference.

Interactions: open a household category; inspect the linked factor or test; save a nonduplicate action; return to the same house state. A category deselected in Settings disappears from the house, but does not silently delete historical Journal records or completed tasks. A removed-category deep link offers general guidance and an explanation instead of manufacturing a profile.

Backend: seven boolean categories remain default. No names, birthdates, or diagnoses are needed. Named household members, multi-user collaboration, or person-specific analyses are separate future decisions, not requirements hidden in this plan.

H06 Tests and home records

Job: close the gap between advice to test and a useful record with a next step.

Pages: record list, test detail, provider/resource selection, sample/result entry, attachment review, and follow-up. The detail page shows a short timeline, not an elaborate project-management board.

Inputs: home-context ID; test medium and purpose; planned/sample/result dates; sampling location label; laboratory/provider; method if supplied; analyte; original value and unit; qualifier such as less-than or non-detect; detection/reporting limit if supplied; report reference; optional private attachment. Missing fields are explicit, never guessed.

Actions: start from an existing test recommendation; schedule; mark collected; await result; enter; validate; confirm; edit with revision history; compare compatible records; prepare questions; create follow-up; export or delete. Retain raw reported values alongside normalized display values. Unknown units or medium prevent automated interpretation, not saving an uninterpreted record.

States: planned, collected, awaiting result, draft result, confirmed record, needs clarification, follow-up open, completed/cancelled. These are record workflow states, not safety verdicts. A manually entered laboratory result is user-reported until independent verification exists.

Backend: owned structured records, encrypted transport, secure attachment storage, access policies, size/type limits, malware handling and metadata removal, idempotent saves, revisions, export and full deletion. Automatic extraction is optional assistance with human confirmation, not an authoritative result parser.

H07 Maintenance and moisture follow through

Job: help the household carry out and maintain an improvement.

Examples: filter replacement, a planned plumbing question, moisture repair follow-up, or a user-chosen periodic home check. These are not claims of automatic equipment monitoring.

Inputs: equipment/task label, relevant manufacturer or professional instruction, installation/completion date, chosen cadence, applicable factor, optional meter reading or observation. No sensitive serial number or exact room geometry is required.

Actions: create from guidance, choose a due date, mark completed, record what changed, pause, replace a product, adjust a recurrence, attach instructions, and open a related test. Recurrence explicitly distinguishes fixed calendar dates from intervals after completion. A late completion or skipped occurrence does not shift a calendar-anchored series. The scheduler or completion event creates each next occurrence at most once. A skipped occurrence is not silently completed.

Backend: reuse the shared Plan task engine and home record store. Certification claims require product/model-specific verified evidence. A generic standard or completed filter task cannot certify current tap quality. User measurements are tagged by source; reminders use documented instructions or the user's chosen interval.

H08 Household summary and sharing

Job: give the household a concise, credible record to discuss with a clinician, landlord, relative, or adviser.

Keep: daily and home information, category table, evidence dates, household notes, top actions or well tests, source list, image export and print. Provide a short default summary and optional detail appendix so additional records do not destroy readability.

Interactions: choose purpose and included sections, preview, remove sensitive details, choose date range, export image or file, print, use native sharing where supported. Share cancellation is not a failure. A failed source section does not blank everything else. Clipboard refusal has a selectable-text fallback where relevant.

Privacy: no public household URL by default. Exports are deliberate snapshots with generation time and underlying evidence dates. Distinguish user entries, proxy results and official records. Once downloaded or shared externally, HALO cannot revoke the recipient's copy; say so before enabling any later cloud-share feature.

Backend: much of the original summary can be composed from existing responses. New test/action content uses the new owned records. Generate sensitive exports locally where feasible; server generation, if necessary, uses short-lived authenticated retrieval, explicit retention, and no public storage.

Food and Garden

F01 Food and Garden overview

Job: connect environmental evidence to food, growing, fishing, and water stewardship without turning HALO into an unrelated diet or farm-management product.

Surface: a small relevant-updates area and five clear destinations: Fish and waterways, My garden, Food information, Grower resources, and Water stewardship. Show saved advisories and outstanding records only when they exist. A first visit explains what is available and asks optional interests, not an extensive questionnaire.

Interactions: choose an interest; select a waterbody, garden context or county; resume a saved item; review coverage; manage followed sources. General guidance remains available without supplying a growing site or personal consumption history.

Backend: optional interest preferences, source registry, advisory records, garden contexts, subscriptions and shared Plan links. A region without verified local food information gets an honest coverage state and useful source/contact paths, not a false all-clear. New backend capacity cannot manufacture measurements that no source has collected.

F02 Fish and waterway advisories

Job: make an official advisory understandable and usable for a specific waterbody and fish selection.

Data: authoritative advisory ID and version, exact waterbody segment or published extent, species or group, contaminant, official audience categories, serving definition, frequency/time window, combined-species conditions, effective/update dates, source and supersession status. Preserve overlapping statewide and waterbody-specific advice. Do not reduce several rules to a single badge.

Interactions: search/select waterbody; select species with plain labels and reference imagery if useful; choose the official audience category without requiring date of birth or reproductive history; see the matched advisory and its geographic extent; inspect why it applies; save/share a selected advisory; follow revisions; ask Luna about the cited wording; add a planned source check or contact task.

Deeper records: optional private catch or consumption notes can record date, species, location context and user-entered quantity. Users can create, correct, inspect, export and delete these records. They remain separate from advisory evidence and optional for using the feature. They do not become an allowance-remaining meter until serving definitions, grouped species, overlapping advisories, and time-window semantics are validated. This keeps the record feature without presenting an unsupported dose calculation.

Failure and edge cases: ambiguous species, waterbody with similar names, outside published extent, unspecified category, no advisory record, conflicting notices, withdrawn advisory, old cached version, and unsupported source language. No advisory found does not mean fish are safe to eat.

Backend: a curated/versioned advisory ingestion service and matching rules, tested with official examples. The October 3 audit located and sampled public services behind the official NC fish advisory resources (https://epi.dph.ncdhhs.gov/oee/fish/arcg.html). A sampled record has truncated text and an apparent audience/meal-limit conflict, so the full official document must govern reviewed interpretation. Source S13 in the acquisition guide gives exact paths and guards. Verify reuse, current status and actual advisory extent before publication. Monitoring reports a published advisory change, not real-time contamination in a fish.

F03 Home garden workspace

Job: organize what is known about a garden and turn uncertainty into a practical testing, advice, or stewardship plan.

Optional inputs: garden/site label; container, raised bed or in-ground context; approximate location or county; irrigation source, explicitly separate from drinking-water source; known soil source/history; user concerns; dated test records; optional observed moisture or rainfall. Avoid exact plot geometry unless a validated use needs it. Use Unknown rather than forcing guesses.

Surface: known information, relevant source notices, open questions, and a short action list. A site detail connects water source, soil records, consultation notes and planned follow-ups. Multiple sites are supported through the same reusable record model if the user needs them, without placing farm-management controls on a normal household page.

Interactions: add/edit a site; associate an existing water source only after confirmation; choose the correct test class; prepare questions for Extension/laboratory staff; save sampling preparation; record results; track a chosen stewardship task; export a site brief. Changing irrigation source keeps older results tied to their original context.

Backend: owned garden contexts and records linked to H06/H07, reviewed recommendation rules, advisory matching and source links. Routine fertility testing and contaminant testing are distinct services. Do not calculate edible-crop safety from county data, nearby facilities, rainfall, or an unverified water-to-plant transfer model. NC State Extension's PFAS guide (https://content.ces.ncsu.edu/Guide-to-Understanding-and-Addressing-PFAS-in-our-communities) supports this evidence-and-consultation approach; NCDA soil testing (https://www.ncagr.gov/Divisions/Agronomic-Services/soil-testing) describes a fertility service, not a general contaminant screen.

F04 Local food evidence and notices

Job: let a family distinguish a real food notice or sampled finding from a general concern and act on information that actually applies.

Content: verified local notices where available, broader food monitoring as a separately labelled category, sampled food/analyte/date/geography, evidence limitations, and the issuing organization's instructions. If a notice identifies a product or lot, preserve that specificity. A nearby farm is not included merely because its coordinates are near a water result.

Interactions: filter by place or food category when supported; inspect source and sample scope; follow a notice; save a question for the issuer or producer; prepare a source-linked information request; attach a user-held test report as a private record; export the selected notice. Any local-produce directory is a discovery resource, not a safety ranking or certification.

Backend: reviewed food notice/study records with stable identifiers, version history, matching scope, withdrawn status and update handling. Current FDA PFAS food monitoring (https://www.fda.gov/food/process-contaminants-food/testing-food-pfas-and-assessing-dietary-exposure) is a source of studies and findings, not a complete local-product lookup API. The feature must declare where no suitable local dataset exists. Do not create a universal PFAS food score, infer brand-level results from a few samples, or imply missing records mean no contamination.

The October 3 audit also sampled openFDA food enforcement records with PFAS-related recall reasons. Source S14 in the acquisition guide defines this concrete notice path, including product/lot scope, distribution and terminated status. These notices complement sampled findings; they do not supply chemical concentrations or cover every local food.

Depth test: the feature passes only when a user can identify applicability, inspect evidence, and take the appropriate source-defined step. A static article catalogue alone does not fulfil this feature.

F05 Grower sampling and conservation support

Job: help a grower prepare a useful test, consultation, or conservation-program inquiry rather than guessing at agronomic decisions.

Flow: concern and site context, correct test or service class, provider requirements, preparation checklist, contact/application handoff, result or response record, professional follow-up. Preserve agricultural depth without requiring a separate full farm-management system.

Data: official program/resource, jurisdiction and service area, applicable matrices/analytes, eligibility description, required documents, dates/deadlines, estimated or verified costs, provider qualifications and last review. Laboratory certification is method/analyte/matrix-specific. A general lab listing is not sufficient evidence that it can run the requested test.

Interactions: filter resources by concern and region; compare service scope; save questions/documents needed; open the official application or contact route; mark user-reported progress; record reference number or response; plan a follow-up. HALO does not promise approval, funding, free testing, a booking, or professional certification.

Sources and backend: use reviewed county Extension contacts (https://www.ces.ncsu.edu/local-county-center/), NC drinking-water laboratory guidance (https://www.deq.nc.gov/about/divisions/water-resources/drinking-water/drinking-water-laboratories), and appropriate NRCS conservation activity resources (https://www.nrcs.usda.gov/programs-initiatives/eqip-environmental-quality-incentives/cpas-dias-and-cemas). Verify current state applicability and eligibility before surfacing a named program. Add a curated resource catalogue and preparation records; an external portal is not an integration API.

F06 Water stewardship watch

Job: monitor actual regional drought and published utility conservation information, then support appropriate household or garden follow-through.

Three separate signals: regional weekly drought; the user's verified utility's published conservation status and rules; optional user-measured household/garden water records. Do not blend them into a new health score. A drought category is not a binding utility restriction, and neither measures a garden's soil moisture.

Content: current source release, what changed since a comparable release, utility notice with its own date, known restrictions and official detail link, relevant stewardship choices, and optional private measurement history. County drought statistics show the distribution of county area by category, not a single condition at the home. A point-specific classification requires the appropriate dated source geometry; do not use the county maximum as the household's class. Unmatched utility, absent notice and stale notice are distinct.

Interactions: inspect the regional history; open the precise utility rule; follow material updates; save a water-use check, leak investigation or stewardship task; enter dated meter/usage readings with units; compare compatible intervals; correct a reading; export a chosen record. A forecast can supply context but cannot alone determine irrigation quantity.

Backend: US Drought Monitor REST services (https://droughtmonitor.unl.edu/DmData/DataDownload/WebServiceInfo.aspx) provide documented regional statistics and were sampled on October 3. The official map linked by the NC conservation-status table (https://www.ncwater.org/WUDC/conservation-status) also exposes a sampled public service. Sources S16 and S17 in the acquisition guide define exact requests and mappings. Reuse/cadence, utility-ID crosswalk and full local restriction instructions remain release gates. Add source-version changes, watch subscriptions and optional measurement records. Show measured consumption change between comparable periods, including interval length and known limitations. Weather and occupancy can also change usage. Attribute savings to a particular intervention only under a separately validated method, never from a completed checklist.

Act

A01 Conservation and civic action overview

Job: help the household act beyond its own immediate protection while connecting to local evidence.

Surface: one relevant issue or opportunity, then Get involved, Explore your area, Speak up, Ways to help, and a small continuation link to saved participation. Preserve original volunteer matching, contextual help, all letter recipients, and household-summary access. No social feed is needed to accomplish these jobs.

Interactions: browse by cause or area, see why a recommendation was shown, open evidence, find an organization, save an opportunity/project, prepare outreach, or document an observation. Recommendations never prevent browsing other causes. Nothing elevated should still produce useful community choices.

Backend: existing volunteer/home/map/district data plus opportunities, projects and source-reviewed recommendations. Keep issue labels accurate: a reported facility compliance status does not establish the cause of a user's symptoms or a specific pollutant at their property.

A02 Organizations and trustworthy resources

Job: identify who works on the issue and how to engage them.

Content: mission and relevant activities, service geography, causes, contact/official site, verified opportunities, accessibility information if supplied, last link check and separate human verification. Explain the match to a household finding without exposing private details externally.

Interactions: filter, inspect an organization, follow available updates, save a contact task, open official website, report an outdated listing to the private review process. Unknown access information stays unknown rather than defaulting to accessible. No badge implies a partnership.

Backend: extend the existing county/cause directory with stable IDs, verification source/date, archived status and scoped contact fields. Expired/dead listings leave clear history for saved items. Keep unverified records labelled; a successful HTTP response is not human verification.

Correct the current route's database-error behavior before integration: a failed directory query currently falls back to a successful empty organization list. Return a distinguishable unavailable result so the frontend can retain prior dated listings and offer retry. No matches and inability to query must remain different.

A03 Opportunities and participation

Job: help a person find and actually participate in an appropriate event or opportunity.

Content: work involved, cause, organizer, location/service area, date/time/timezone, duration, cost, age restrictions, supervision, accessibility, preparation/equipment, registration URL, capacity/deadline only if known, and source verification. Filters hide unsupported fields rather than inventing defaults.

Interactions: inspect; save without a date if ongoing; add to Plan; open external registration; record user-reported registration; set an optional reminder; see cancellation/time changes; mark attended/not attended; record optional personal hours or notes; reopen. Saving is not signing up, opening a form is not registration, and a checked attendance box is self-reported.

Backend: opportunities linked to organizations, curator/feed provenance, expiry/cancellation detection, user participation records and shared action scheduling. No attendee list, child account, organizer messaging, or booking platform is required. Officially confirmed participation is a separate status possible only with a real integration.

A04 Guided stewardship projects

Job: turn a relevant conservation goal into a practical sequence with a follow-up.

Examples of project categories: water stewardship, a source-reviewed garden/habitat activity, preparation for a supervised cleanup, or participation in an established monitoring program. Exact instructions require source and suitability review before publication.

Anatomy: purpose, who it suits, prerequisites/permissions, materials, approximate effort/cost with basis, steps, safety constraints, completion evidence chosen by the user, and follow-up. Do not show the same generic environmental tips under every category.

Interactions: choose a project, review suitability, save selected steps or the whole project, schedule, record progress, pause, and document an outcome. Reuse P02 tasks instead of a second checklist store. Small-step cards retain the approved interaction with an accessible full-list option.

Backend: versioned project templates, source-backed instructions, applicability rules and action relationships. Participation hours or completed steps are valid self-reported metrics. Avoid unmeasured carbon, pollution-removal, water-quality or health-improvement claims. Hazardous collection, entering waterways, and chemical testing require an appropriate official program and safeguards, not an unsupervised app challenge.

A05 Evidence based outreach

Job: enable a useful, informed question to a utility, landlord, representative, or relevant official source.

Keep: all three original recipient modes, independent edited drafts per recipient, reset current draft, Copy, native Share and fallback, and transparent evidence selection. Add information-request templates for cases without a measured exceedance so missing data is not a dead end.

Inputs: selected public evidence and its date, recipient class, purpose, editable questions, optional local-only greeting/signature. Do not automatically persist names, addresses or landlord details in the existing anonymous profile. Optional saved sensitive drafts require a clear separate consent and retention decision.

Interactions: choose purpose and recipient; inspect citations; edit; preview; copy/share/print; optionally save a nonsensitive follow-up task; record user-reported sending and any reference/response; revisit if the evidence changes. A new sample flags an old draft for review rather than changing the user's text. When there is no appropriate measurement, generate a request for information, not an accusation.

Backend: source-aware templates and evidence snapshot binding can build on existing home/map/district data. Optional persistent outreach records need owned storage and privacy rules. A sample above a comparison value is not automatically a formally determined legal violation. Recipient directories and current district boundaries need verification before automatic addressing. No automatic external sending in the target workflow.

A06 Private field observations and official reporting

Job: help a user make a clear factual record and reach the appropriate official reporting channel.

Inputs: observation type, date/time, plain description, optional approximate or precise location chosen for the purpose, waterbody/site reference, and optional photograph. Avoid asking users to enter water, approach facilities, touch suspect material, or diagnose a bloom from appearance. For immediate danger, show relevant official emergency guidance rather than positioning HALO as a response service.

Interactions: create a private draft, review sensitive details, choose the official destination, copy/export the report, open the official form, record user-reported submission/reference number, and schedule a follow-up. User notes remain editable; a sent snapshot stays distinguishable from later changes.

States: draft, prepared, official form opened, user-reported submitted, acknowledged only with genuine confirmation, follow-up, archived. HALO cannot invent agency investigation status. Observations are not automatically public or added to a contamination heatmap.

Backend and source: private records and attachments, export, location/EXIF controls and a reviewed official destination directory. NC Stream Watch (https://www.deq.nc.gov/about/divisions/water-resources/water-resources-public-information/water-education-programs/water-education-and-outreach/nc-stream-watch) links observation/reporting forms; public form availability does not establish a submission API or authorization to transmit user data. Use explicit external handoff until a real integration is approved.

Regional exploration

M01 Map shell and discovery

Job: place a household concern in regional context and let users ask their own questions.

Keep: map-first composition, layer controls, muted basemap, select/pan/zoom/locate, accessible list, legend, source/date, contextual sheets, search within available records, preserved viewport, own-system context and approximate home pin. New bottom navigation has no Map tab; build `/map` as the planned canonical production route, reachable from Act and relevant details.

Interactions: an incoming link selects its public record/layer; filter changes do not create a long Back trail; selecting a feature opens one sheet; close returns focus; returning from details restores the map. On small screens, overlays must leave usable map space. Keyboard/text-list paths expose the same records as the canvas.

Backend: reuse existing map routes and geographic assets. Production renderer, attribution, token restrictions, lazy loading and identity-safe cache require implementation and verification; the current Canvas wireframe does not prove them. Individual layer failure does not blank the whole map. Missing coordinates retain records in the list.

M02 Water layer and sampling history

Keep: overall/compound/hazard selection, source-sized utility markers, selected-system outline, counts, detail by compound/date, historical quarters and distinct Latest. Keep unregulated/excluded evidence and unlocated records visible where applicable.

Interactions: choose compound and quarter, inspect exact sample context, compare compatible records, open relevant water guidance, save/follow a public system, or prepare a source-linked question. Keep prior data while a new quarter loads; failed requests restore the prior selection; stale responses cannot overwrite a later choice.

Truth rules: a quarter contains samples from that quarter, not accumulated contamination. Absent points are not clean areas. The current Patterns heat layer represents qualifying-record density, not exposure or a continuous pollution plume. A genuine spatial concentration surface requires an appropriate dataset and reviewed method, not just a different color renderer.

Backend: complete tested/non-detect metadata, geographic basis and comparable denominators are priority upgrades. Do not imply a centroid is a service boundary or a tap measurement. Current regulatory flags require launch-time source review.

M03 Radon facilities and air layers

Radon: county polygons, zone explanation and no-data state; open testing guidance without implying a home level.

Facilities: selected scope, compliance category, regulatory programs, quarters in noncompliance, programs currently in violation, dated penalties/inspection if available, clustering and official record link. Correct labels matter: program counts are not counts of distinct violations. Proximity alone does not attribute a pollutant to a facility.

Air: monitor shapes distinct from water, AQI/pollutant/source time, distance from approximate household location, and link to the relevant air explanation. A model grid and a station are different layers or clearly different evidence types.

Extensions: fish-advisory extents and drought/conservation context may reuse the map only when sources support their geography. Sampling records from a documented environmental service are an optional extension with matrix/method/date shown, not an automatic swimming or drinking clearance.

M04 Public record comparison

Job: compare two selected records without hiding differences in evidence quality.

Interactions: select a second public system or compatible observation; compare the same analyte/unit where possible; inspect dates, coverage, source and missing entries; return to the prior map. On mobile, a compact aligned list is preferable to two crowded maps.

Backend: public comparison query or validated join using shared normalized evidence. Define comparability explicitly. Never rank missing as best, compare soil with water, turn regional comparisons into home rankings, or imply people can choose a different utility merely by seeing it. Preserve original denominators and source limitations.

M05 District and regional evidence

Job: explain the scale of a regional issue and support civic discussion with public aggregate evidence.

Keep: district/state comparison, per-compound counts, county table with partial-district distinctions, household county ranking where supported, unlocated records and sources. Population totals are people served by included systems, not verified exposed or ill people.

Interactions: inspect denominator definitions; change a supported evidence view; open a county or public system; export/share the public summary; open outreach with selected public evidence. Never include the user's home pin or private information in a public district share.

Backend: verify current district boundary vintage before release; centroid assignment is only a documented approximation where service-population allocation is unavailable. Suppress invalid rankings, use correct mapped/tested/population denominators, preserve incomplete-population flags, and handle geography-source failure without reporting zero problems. Named local facts require dated verified sources, not old planning examples.

Plan

P01 Your Week

Job: provide one place for commitments, preparation and follow-up across HALO.

Surface: seven-day selector, readable agenda, unscheduled saved items, and a small relevant update summary. Optional month view supports navigation, not another dashboard. Separate overdue, upcoming and waiting items without punitive streaks or guilt copy.

Interactions: change day/week, open an item, save or schedule, reschedule, pause, complete, reopen, filter by home/outdoor/community, and export a selected event to a personal calendar. Calendar export is not bidirectional sync and does not prove the external calendar updated. Saved items can exist without a due date.

Data: one shared action store, occurrences, source references, timezone, optional reminders and linked records. Environmental context appears only beside an activity it informs. A filter replacement does not need an air chart.

Backend: new Plan and recurrence APIs; device-local display preferences separate from owned tasks. Handle daylight-saving transitions, all-day dates, location changes and event cancellations explicitly. One event update changes all its views, not several independent copies.

P02 Shared actions and follow up

Job: turn any recommendation into a real, resumable task with an outcome.

Action detail: title, reason, applicability, source/evidence snapshot, steps, chosen schedule, estimated effort/cost if sourced, status, related record, and next step. A user can edit their own title/note without altering the source's text.

Interactions: save from any page; detect an existing equivalent action; schedule later; change status; attach a test or observation; mark waiting for a third party; record completion; choose a follow-up; reopen; cancel/delete. Do not force every simple reminder into a large project form. Completing a project step and completing the whole project are separate.

Backend: owned action ID, stable recommendation/template ID, origin feature, evidence version, context version, optional parent/project, due time/timezone, recurrence, status events, revision and idempotency keys. Authority comes from the authenticated identity. A client cannot assign a task to an arbitrary household. Readiness uses actual records rather than guessed completion.

P03 Outdoor activity planning

Job: compare available conditions for a chosen activity and support a practical timing/preparation decision.

Optional inputs: activity label/type, approximate location, duration, preferred periods, and relevant household category. No route tracking or continuous GPS is required. Home location can be the default; planning elsewhere does not change the household address or its daily history.

Content: aligned weather and environmental periods with source, issue time, valid time, uncertainty and missing factors. Preserve category-only forecasts. Daily pollen cannot be expanded into an invented hourly line. Avoid a single arbitrary safe-time score that hides tradeoffs.

Interactions: choose two or more supported periods; inspect tradeoffs; see a relevant official warning; select a time; save a preparation checklist; opt into material-change notification; reschedule explicitly when conditions change. Forecast changes never silently move the user's event. Far-future plans show a date to check again rather than fictitious forecasts.

Backend: a normalized forecast service with per-factor horizons, supported geographies, source versions, caching and spend controls. NWS (https://www.weather.gov/documentation/services-web-api) documents weather forecasts and alerts; AirNow (https://docs.airnowapi.org/faq) describes variable air forecast coverage; Google Pollen (https://developers.google.com/maps/documentation/pollen/overview) is a provider candidate subject to coverage, billing and terms. No promise that all factors cover the same seven days.

P04 Journal and quick entries

Job: record how the household felt and inspect environmental context, optionally.

Keep: quick entry, explicit Felt fine, household-category selection, symptom chips, symptom severity, possible-illness flag, optional note, prior dates, retrospective first-use calendar, recent entries, edit, delete and undo. Preserve loading/save guards so an unloaded entry cannot be overwritten with blanks. The actual current upsert replaces the whole date/category entry.

Interactions: a Today shortcut opens the correct date; choose category and response; save; edit without losing fields; inspect that day's evidence; optionally select remembered rough days; review recalled versus same-day labels. Retrospective batch retry only retries failed dates and reports partial success honestly. A failed write keeps the draft.

Backend: reuse existing Journal routes and history, add explicit logging status and appropriate validation if the analysis needs it. Preserve not logged separately from felt fine. Category removal does not erase historical entries. No person-specific conclusions from category records. Free-text notes remain private and excluded from Luna context under the existing policy.

First use and history: distinguish never logged from no entries in the current range using a proposed authoritative `has_any_entries` or equivalent aggregate. A household with old records must not be sent through first-use setup again. Preserve the original 90-day regular logging window, 12-week grid and seven-entry recent list unless a reviewed change is explicit; validate the current retrospective maximum of 62 selected dates. A minimum count for a descriptive comparison is not proof of clinical significance. Protect existing richer entries from retrospective overwrite. When onboarding history backfill is pending, show that state and retry rather than implying there was no environmental history or that a pattern has been established.

P05 Patterns and history

Job: show understandable descriptive relationships while being candid about their limits.

Keep: 12-week heat grid, dated readings, symptom markers, day detail, accessible tables, trend charts, findings and sample counts, insufficient-data and no-clear-pattern states, optional category filtering. Calendar and analysis views must agree on dates and missing values.

Improve: compute wellness comparisons from explicitly logged days if describing how people felt. The current algorithm compares symptom-logging days against all available environmental-history days; those are logging frequencies, not verified symptom-free days. Retrospective recollection, missing factors, possible illness and changing data sources need distinct handling. A category filter must not relabel household findings as individual analytics.

Interactions: inspect a day; change the factor/time range/category where supported; see how a statement was calculated; correct an entry; export selected dates; open a source. Changing a filter updates the denominator and disclaimer, not only the plotted dots.

Backend: versioned analysis with explicit eligibility counts, evidence provenance, logging denominator, missing-data treatment and tested minimums. Existing minimum-count/difference rules are heuristics, not clinical validation. Expert review must decide supported interpretations and symptom-factor pairings. Do not call a small personal association a cause, diagnosis, statistically proven effect, or exposure estimate. New insights never require users to log more symptoms to earn a reward.

Environmental-history eligibility also depends on measurement basis. Backfilled daily maxima and app-time observations must not be silently pooled as the same statistic. Retain the available source and backfill metadata, state how a series was aggregated, and exclude or separately present incompatible periods. The household's plan location is not evidence that a person was physically there.

P06 Seasonal and progress review

Job: help users understand what changed over a meaningful period and resume useful follow-through.

Keep: Journal's seasonal summary and source-reviewed seasonal prompts, with dismissals and appropriate regional relevance. Replace outdated eligibility rules when actual home test records can make a prompt more accurate.

Content: available environmental history, explicit check-in counts, completed/pending household actions, recorded tests, and user-reported community participation. Keep measured changes separate from activity completion. A seasonal summary can be valuable even when no symptom pattern is supported.

Interactions: select season, inspect underlying dates/actions, choose a next review, export selected material, or dismiss the callout for that season. An alert opens the actual summary, not an unrelated Journal form.

Backend: reuse existing summaries while revising their analysis semantics consistently with P05; join owned action/test/participation records. Do not infer health benefits, water savings or environmental remediation from completed tasks.

Shared experiences

X01 Contextual Learn and source details

Job: explain a term or decision at the moment it becomes useful, with a route to deeper trustworthy material.

Keep: the existing contextual factor learning content, citations and sources. Integrate short explanations beside the score, chart, test field or action they explain. **Proposed addition:** a small glossary and searchable topic index, entered through a contextual Learn link or Sources and methods, gives returning users a way to find a known explanation. It is not an existing capability in the original C1 specification. Learn does not need another bottom tab or repeated large lesson section on every page.

Content: a plain-language answer, why it matters here, what the evidence can and cannot establish, a practical example when helpful, and dated references. Separate a quick definition from a full topic page. An explanatory diagram should earn its space by clarifying a relationship, not add decoration to every card.

Interactions: open a term without losing a form draft; expand a short explanation; search the proposed topic index by plain-language term with clear no-match results; open the source detail; inspect source date and method; follow a relevant action; return to the original scroll position. Topic search operates on reviewed HALO content, not arbitrary live web results. Source cards identify which statement they support. External documents remain labelled as external, and a failed link does not erase the locally reviewed explanation.

Backend: reuse existing learning retrieval and curated documents. Add versioned editorial records, topic relationships, evidence applicability and review status. Content matched to a household category explains why it appears without claiming to diagnose that household. Translation requires review of units, medical meaning and official audience definitions, not only a text substitution.

X02 Luna

Job: help people understand their current context, navigate HALO, and prepare a useful next action without inventing evidence or acting without permission.

Keep: the approved Luna name, teal geometric symbol, compact launcher, refined conversational panel, greeting, suggested prompts, input and source links. Keep the visual character that was approved; do not recreate another company's branding. An assistant answer is not the only route to any core feature.

Interactions: open from any relevant page; see what page or record is in context; choose a suggested question; send, retry or cancel where technically supported; inspect citations; start a new conversation; clear conversation history; return to the underlying page. Handle an empty question, long input, rate limit, provider failure, unavailable sources and unsupported question clearly. Only expose attachment controls after the private upload and interpretation workflow exists.

Useful new depth: a person can ask what a water qualifier means, compare two available forecast periods, find the appropriate testing workflow, or turn a reviewed recommendation into a proposed Plan item. A proposal is shown as an editable preview with its reason and sources. The user confirms before any task is saved. Luna can prepare an outreach draft for review, but cannot silently send it, change test values, register for an event, alter a household profile or mark an action done.

Evidence behavior: answers distinguish observed readings, forecasts, estimates, official advice and user-entered information. A missing local food result leads to an honest limit and a practical next step, not a nearby-farm risk claim. Calculations use deterministic validated functions and explicit units rather than model-generated arithmetic for important results. Sources support individual claims; adding an unrelated bibliography is not grounding. Medical diagnosis, treatment, emergency triage guarantees, legal determinations and unsupported exposure estimates remain outside scope.

Privacy and backend: the current endpoint accepts a question and page, not a full conversation contract. Add an authoritative capability response, transparent model-processing disclosure, authorized selected-record context, minimal conversation retention, deletion, and explicit action confirmation. Correct the current ambiguity where household context can be sent even though the client lacks a reliable capability flag. Free-text Journal notes stay excluded under the established policy. Retrieved text and uploaded documents are untrusted data, never instructions for tool execution. Account deletion and optional recovery must include any newly stored conversation records.

Quality gate: test sources, refusals, contradictions, missing data, stale advisories, malicious source text, household isolation and the difference between proposing and completing an action. No claim that Luna is medically validated or always correct.

Obtain qualified review of health-related interpretation and public product claims before releasing expanded decision support. A disclaimer alone does not decide the app's legal or regulatory classification; this plan makes no such determination. Keep behavior and marketing consistent with the functions actually evaluated.

X03 Alerts and watchlists

Job: notify a person about a specific meaningful change or chosen follow-up, with enough context to act.

Keep: the existing inbox, unread state, read/dismiss behavior and server notification/push foundations. Complete the required browser permission, device-subscription and delivery UI. Add a quiet watchlist manager for selected utilities, waterbodies, supported locations, organizers or saved actions. Watching is optional and separate from merely opening a page.

Preserve earlier air-quality notice eligibility for the sensitive categories defined by the existing server policy. The inspected rule uses an earlier severity threshold for eligible sensitive households; the frontend must not recreate that decision or silently broaden it to every household category. Store the policy version and explain the reason in plain language. Recheck source coverage before calling an apparent difference worsening air. The existing notification types apply their preference to both inbox creation and push; separate channel controls require an additive backend change, not just two switches.

Content: what changed, the relevant place or record, the actual event/publication time, source and direct destination. Distinguish a new published laboratory result, an official advisory, an organizer cancellation, a task reminder and a source correction. A changed risk score and newly available data are not the same event.

The existing weather-alert input accepts actual NWS flood and heat events. A dormant boil-water message in code does not establish an active boil-water feed. New utility/agency notices require the separately verified acquisition path and appropriate official instructions; do not advertise them as currently covered.

Interactions: follow or unfollow a supported record; choose a relevant channel; inspect the change and source; open the exact advisory/event/action; snooze an eligible reminder; manage quiet hours; mark read; dismiss. Unfollowing stops future subscription notices without deleting completed actions or historical records. Browser permission denial leaves the inbox usable. A cancelled event updates its linked Plan entry with an explanation, not silent deletion.

Backend: reuse the alert and push infrastructure, extend typed subscriptions, deduplication by source revision, durable delivery intent, expiry and safe deep links. Separate source matching from delivery. Show a delay or unavailable source when appropriate. HALO is supplemental and must not encourage people to rely on it instead of official emergency channels. Urgent-source coverage, polling and notification semantics require explicit operational validation before any reliability claim.

X04 Settings

Job: let users understand and control HALO without searching through unrelated screens.

Sections: household categories; home and water information; address/home-context change; notifications and subscriptions; accessibility; language; privacy and data; optional recovery; sources and methods; app information and feedback. Keep frequent settings near the top and destructive controls separately labelled. Appearance follows the device under the approved decision; a redesigned manual theme switch is not silently reintroduced.

Interactions: load current values; edit a retained draft; review changes; explicitly Save; show confirmed success or retry with the draft intact. Warn before abandoning edits. Handle another-device changes with a conflict choice rather than overwriting. Category changes update guidance and the house figures but do not delete old Journal or test records. A new address starts the home-context transition described in G01.

Keep from the specification: inspect stored household data, clear Journal records separately, delete all app data, notification controls, readable text scaling and reviewed Spanish support. Preserve the approved text-scale range rather than silently reinstating an old specification value. If a translation is not available, label the fallback, keep the language preference, and never imply an official advisory was translated by its issuing agency.

Every new app-owned visible string, validation message, accessibility label, date description, empty state, exported heading and notification template belongs in the translation catalogue from its first implementation. Do not scatter new English literals across components and defer their extraction. Use the requested versus served locale distinctly for source content. Official titles and quotations retain their issuing-source identity; translation never changes units or advisory audience rules. Update the document language and formatting consistently when a reviewed locale is selected.

Backend: reuse profile, household, notifications and account interfaces, with identity preflight and ownership checks. Add new preference groups only for shipped features. Every new private domain needs export/deletion coverage before its Settings controls can promise complete removal.

X05 Privacy and recovery

Job: make valuable household records controllable and recoverable without turning HALO into a surveillance or identity-heavy app.

Default: anonymous use remains available. No named children, birthdates, continuous GPS, public home pin, required medical profile or advertising profile. Feature-specific optional data is collected only when used. Selecting a sensitive household category is not consent to share it with an external model, organizer or export recipient.

Interactions: inspect stored fields; preview an export; select what to include; download; clear a specific record or Journal; request complete deletion; optionally link a recovery method; inspect/revoke devices if that capability is built. Explain that the current anonymous-session replacement cannot restore old records. Recovery must actually prove ownership, not accept a household ID or secret-looking URL as sufficient.

Before the first valuable private upload or long-lived record, explain anonymous-session loss and any genuinely available recovery option. Do not wait until after a document has been uploaded to reveal that another browser or a cleared session may lose access. The user can continue anonymously with this understood limitation; a not-yet-built recovery feature is not offered as protection.

Export: household summaries and full data exports have different audiences and defaults. A summary is short and readable; an optional detailed appendix preserves sources and dates. Full private export can include structured records plus documents the user owns. Share previews exclude exact locations, category details, symptoms and attachments unless selected. Public community evidence exports never acquire private fields from the current household context.

Backend: define retention, ownership, private attachment storage, signed download expiry, encryption appropriate to the hosting stack, deletion of object storage and pending jobs, and recovery collision handling. Explain backup and third-party retention accurately. Restore and deletion drills must cover the database and files. A future shared-household account would need explicit roles and invitations; it is not implied by this plan's category figures or device recovery.

Update the privacy notice before enabling uploads and new private records. Original statements that HALO never stores medical records or identifying text cannot remain absolute once a person can voluntarily attach a lab report or enter sensitive notes. Explain the new optional record classes, possible identifying document contents, recipients, retention and controls at the point of use. Keep raw onboarding address disposal distinct from a user deliberately attaching a document that contains an address.

X06 Accessibility and offline use

Job: preserve the essential meaning and actions on small screens, assistive technology, weak connections and low-power devices.

Accessibility: use WCAG 2.2 AA (https://www.w3.org/TR/WCAG22/) as the implementation target, verified with both automated and manual checks. Charts have readable tables, the map has a list, icons have labels, status has text, controls have visible focus, and dialogs restore focus. Step decks expose current position and support buttons and keyboard. Forms use field-level messages plus a summary where needed. Test screen-reader reading order, large text, browser zoom, contrast and reduced motion. A figure's shape must not be the only distinction between household categories.

Motion: retain handwriting, ring drawing, count-up, staggered cards, chart reveals and ambient scenes where approved, but never delay content or actions. Stop nonessential movement offscreen and when the document is hidden. Provide a user-accessible way to pause prolonged nonessential motion as well as respecting device reduced motion. Reduced motion shows the final values immediately. Decorative day/night follows time; it never masquerades as a measured weather condition. A graph reveal is not a live-data stream.

Offline: reuse identity-scoped environmental caches with source timestamps. Do not default-cache private notes, letters, photographs, model chats or lab documents. Offline write support is a separate controlled capability with an explicit local-draft label, retry safety and conflict handling. Never report a queued write as saved to the server. Clear identity-sensitive caches on identity change or deletion. Preserve unsaved input safely across temporary fetch failure without accidentally showing another household's cached state.

Backend and testing: consistent error semantics, bounded payloads, cached public source reads, retry/idempotency and session-expiry handling support the frontend. A low-cost phone must not download all scenes, maps and historical series merely to show Today. Performance budgets are defined against measured devices, not desktop animation smoothness alone.

Browser identity and installation: retain the existing logo, local fonts, page-title template, icon references and theme-color values, then audit the full favicon/app-icon set, manifest and generic social preview. Do not put household readings, names, location or symptom details in page metadata or share-card generation. A service worker needs an explicit version/update policy, safe private-response exclusions and stable notification click routing. Explain supported installation/push behavior accurately without requiring users to install the app. Installing an icon is not proof that offline writes, push or background refresh work.

X07 Source and content operations

Job: keep evidence accurate after launch, including when an agency changes a file, an organizer cancels an event or a guidance page becomes outdated.

Operator capabilities: review incoming records; inspect fetch and parser failures; quarantine suspicious unit or geometry changes; compare revisions; publish reviewed content; withdraw a wrong interpretation; inspect stale sources and broken links; manage translations and supported coverage. Begin with a protected internal workflow that fits the existing stack, not a second elaborate public app.

Source record: issuing organization, supporting URL, licence/access terms, geography, method, expected cadence, observed/published/retrieved timestamps, parser/content version, reviewer and next review date. An HTML page is not automatically a stable API or permission to republish everything. Human-reviewed imports are legitimate if labelled and operated honestly.

Backend: bounded resumable jobs, public-source caching at permitted resolutions, typed evidence revisions, durable notification intent, spend and quota limits, secret management, private/public separation, access logs without sensitive

content, and backup/restore coverage. Expanding the existing one-minute cron with every new feature is not a viable scaling plan. Source failure preserves the last valid dated record and marks it stale; it does not publish an all-clear.

Quality: compare incoming record counts, geography, units, dates and required fields against expectations. A successful request returning an unexpectedly empty dataset is not automatically a valid zero. Rollback disables an interpretation or source producer without destroying household records. Changes in scientific or regulatory guidance are reviewed, versioned and communicated rather than silently rewriting history.

Cohesive visual and interaction system

The app should feel simpler at its entrances and deeper when someone chooses to explore. Depth comes from useful evidence, an appropriate action and continuity, not the number of cards visible at once.

Page composition

- Landing pages have one clear identity, a short current situation, one primary contextual next step and compact entrances to their domains. Avoid several competing hero banners or repeated AI summaries.
- Factor pages keep the approved score and atmosphere, then use a factor-specific composition. Air emphasizes pollutant and trend; UV emphasizes the rising sun and timing; pollen emphasizes category comparison; mold emphasizes proxy meaning and moisture action. Water, radon and lead each retain their distinct evidence and test journey.
- Detail pages use a short summary, the most relevant evidence visualization, practical choices and expandable supporting material. Do not hide essential limitations in an accordion; optional technical methods and full references can expand.
- Plan uses an agenda and action details. Act uses opportunities and project progression. Food and Garden uses evidence-led routes, not a news feed. Map remains a full, focused exploration surface reached from Act or a contextual link.
- Preserve centered page and section headers where approved. Body text, forms, long lists and comparison values can use alignment that improves scanning. Centering every paragraph would reduce readability.

Color and visual assets

Keep the blue and teal foundation in both device themes. Use restrained factor-specific tints: teal for air, orange for UV, soft yellow for pollen, green for mold, water-blue for PFAS and appropriate distinct accents for radon and lead. Factor cards stay flat-tinted without gradients. Severity colors communicate evidence categories and are not decorative substitutes for the factor palette.

Retain the illustrated, moderately stylized environmental direction. Each factor has a distinct setting, and scenery fades or overlaps into the content surface without making text float over busy detail. Use atmospheric layers, small loops and shared lighting, not a different visual language on every route. No decorative deer and no photorealistic collage beside otherwise flat generic UI. Exact approved assets remain preserved until a new version is reviewed.

The numerical and heading typography stays consistent with the approved character. Charts share axis, tooltip, missing-data and table conventions. Bottom navigation stays rounded and clear at safe-area insets. Luna stays visually secondary to the page's primary action.

Controlled depth

For Today and its factor detail pages, preserving the current experience takes precedence over the general simplification guidance below. Do not remove useful sections or hide existing information behind additional taps just to make these pages shorter. Apply the reading-level approach primarily to new content and new pages while keeping these existing pages substantially unchanged for now.

Use three reading levels: immediate answer, practical detail, and evidence/method. Users can take an appropriate next step without opening every level. A substantial source limitation stays at the first level when it changes the decision. Keep one canonical version of an explanation and reuse contextual excerpts rather than duplicating whole essays.

Charts appear only when data and a decision justify them. A short status list can be better than an empty trend. Every new subpage must have a distinct user question; if it only repeats its parent, use an inline explanation instead. Optional growers, observations, Journal and water-use records stay discoverable without becoming mandatory household homework.

Connected journeys

These are the tests of whether HALO is one app rather than a set of unrelated features.

Table fields: Starting situation / Connected path / Credible outcome

The household notices a higher air reading

Connected path: Today, air detail, source and forecast comparison, outdoor plan, preparation checklist, optional check-in

Credible outcome: A considered plan based on available evidence, not a promise of safe exposure

A utility publishes a new PFAS result

Connected path: Homeguard, water result and date, test or information choices, saved action, outreach draft, Plan follow-up

Credible outcome: A sourced question or documented household next step, not an automatic treatment claim

A private well has no test record in HALO

Connected path: Homeguard testing entry, appropriate lab/resource, action, sample context, result entry, reviewed follow-up

Credible outcome: An organized testing process and interpretable record, not an assumed normal result

A family wants to fish

Connected path: Food and Garden, waterbody/species/advisory audience, exact official guidance, optional watch, outing plan

Credible outcome: A location-specific advisory decision with unresolved matches kept visible

A grower sees a published local concern

Connected path: Food evidence with scope limits, garden/site context, Extension or program resource, test preparation, action follow-up

Credible outcome: A relevant professional inquiry and evidence trail, not an inferred contaminated crop

A utility changes conservation rules

Connected path: Water Stewardship, dated applicable notice, chosen practice, Plan reminder, optional comparable water-use records

Credible outcome: A documented response to actual rules, without claiming unmeasured savings

A resident wants to help locally

Connected path: Act, Map context or cause, verified organizer/opportunity, external registration, Plan, user-reported participation

Credible outcome: A real participation pathway with truthful confirmation state

A person observes unusual water

Connected path: Private observation, safe reporting guidance, preview of location/photo, official form, user-reported submission and follow-up

Credible outcome: A better-prepared report, with no automatic chemical identification or agency receipt

A household wants to review a season

Connected path: Plan, check-in history and explicit logging counts, test/action history, seasonal recap, chosen next review

Credible outcome: An understandable record of available evidence and completed actions, not a causal health conclusion

Each path must work without Luna, push permission or a purchased device. Cross links carry public IDs or authorized private references, not copies of sensitive content in URLs.

Delivery sequence and approval gates

The full target stays in scope. Sequencing reduces integration risk and gives each feature a complete useful first release rather than shipping many empty destinations.

Table fields: Phase / Complete outcome / Dependencies and gate

0

Complete outcome: Confirm four-tab ownership and preservation map

Dependencies and gate: Yogi confirms Plan or chooses another fourth-tab label without changing its purpose. Verify the current deployed schema and source access read-only.

1

Complete outcome: Shared actions and a small usable Plan

Dependencies and gate: Evidence references, action state, idempotency, ownership, dates/timezones and reminders. Wireframe review of navigation and Plan before visual lock.

2

Complete outcome: Deeper Today and Homeguard follow-through

Dependencies and gate: Testing records, maintenance, source details, category guidance, change explanations and honest score semantics. Existing approved views receive additive reviewed changes.

3

Complete outcome: Connected Act and Map

Dependencies and gate: Preserve current Map refinement work. Complete real layer contracts, comparison, organizations, outreach, summary export, then opportunities and projects with verified coverage.

4

Complete outcome: Food and Garden with substantive evidence

Dependencies and gate: Start source feasibility and expert content review during Phase 0. Release fish advisories, testing/garden guidance, food-evidence limits, grower pathways and Water Stewardship as each full data/action journey passes its gate.

5

Complete outcome: Outdoor planning, Journal and Patterns in Plan

Dependencies and gate: Verified per-factor forecasts, explicit logging semantics, analysis review, retained original Journal capabilities and seasonal summaries.

6

Complete outcome: Richer Luna and watchlists across the app

Dependencies and gate: Truthful context disclosure, authorized record access, confirmed proposals, source-revision notifications and privacy controls. Basic source-grounded Luna remains available before expansion.

7

Complete outcome: Full release verification and preservation

Dependencies and gate: Settings, reviewed translations, recovery where enabled, accessibility, low-end device tests, source outage tests, security, deletion/restore, browser tests and approved design baselines.

Security, accessibility, source operations and tests are required within each phase, not work deferred until Phase 7. Source feasibility for food, events and forecasts runs early so the frontend is not designed around an imaginary feed. Field Observations follows the private-file/security infrastructure; it does not block the other Act workflows. Current unapproved Map work is preserved and can continue through review while shared contracts are defined.

For each stage, use the established workflow: requirements and contracts, interactive wireframe with clearly labelled fixtures, Yogi refinement, explicit visual approval, immutable baseline, production implementation, backend integration, interaction and failure testing, visual comparison, handoff and version-controlled release. Do not freeze each experimental iteration as a new approved design.

Decisions and evidence that still require confirmation

The architecture and feature behaviors above are the recommended final target. The following are real decision or validation gates, not missing placeholders that may be filled with guesses:

1. **Fourth tab name:** Plan is recommended because it joins weekly planning, saved actions, follow-up and optional Journal without giving a peripheral topic control of the navigation.
2. **Score semantics:** retain current honest displays until the backend defines any individual factor scores and precise contribution explanation. Any formula change needs a version and review.
3. **New source acquisition:** prove access, geography, freshness, reuse terms and cost for fish advisories, food records, events, utility restrictions and per-factor forecasts. A reviewed manual workflow may be the right implementation for some sources.
4. **Interpretation review:** advisory audience matching, household test comparisons, guidance, symptom findings and regulatory labels need domain review. Source availability alone does not validate HALO's interpretation.
5. **Recovery and retention:** choose the actual identity recovery method and published storage/retention policy before valuable private records depend on them. Multi-person accounts remain a separate decision.
6. **Release coverage:** state which regions and sources are supported at launch. Useful North Carolina depth is preferable to nominal national coverage with unverified local claims, while preserving an architecture that can expand.
7. **New visual approval:** all new layouts and substantive revisions go through Yogi's review. A product plan is not automatic visual approval.

Evaluation and finishing touches

The lesson from competition references is to make a clear problem, usable interaction and tangible outcome reinforce each other. The official Mazah winner profile (https://www.congressionalappchallenge.us/25-nc08/) describes food-waste reduction and community connection. That is useful inspiration for a complete action loop, not a reason to copy its identity or turn HALO into an unrelated food marketplace. Earlier demo observations in this repository are distinct from the official profile; this plan does not claim a fresh frame-by-frame verification of every winner video.

The Congressional App Challenge toolkit (https://www.congressionalappchallenge.us/congress/congressional-toolkit/) supplies judging guidance around functionality and programming, originality and problem-solving, and implementation including user experience. The exact current district panel and local weighting have not been verified here. No feature is justified solely by an assumed judge's interests or a promise of winning.

Retain the original restriction against unverified first-or-only-app claims. Explain HALO through demonstrated capabilities, regional evidence, thoughtful implementation and user outcomes rather than claiming an exhaustive competitive survey or exclusive invention.

Evaluate the app with representative task tests: can a new user explain what a score does not know, find the right source, choose and resume an action, recover from a failed save, and distinguish a public record from their own measurement? Include a renter, private-well household, low-bandwidth phone, keyboard/screen-reader path and someone who never uses Journal. Use consented usability sessions with nonsensitive task-success and confusion notes by default. Production behavioral analytics are not part of this plan's default and require a separate explicit privacy decision; operational failure metrics must not contain private user content.

Finishing requirements include useful empty states, retained drafts, stable return navigation, correct singular/plural copy, local dates and units, source-link checking, event expiry, duplicate prevention, explicit successful saves, safe external-link labels, printable light layouts, careful mobile safe areas, and genuine accessible alternatives to visual controls. These details make the app credible as much as a complex new chart does.

The release demonstration should show several connected real-data journeys rather than every menu item: a daily decision, a home evidence-to-action workflow, a food or water stewardship case, and a civic participation path that appears in Plan. Label any demonstration fixtures visibly and never mix them into a real household session.

Preservation and completion status

This four-document plan is saved in the repository as reviewable documentation. It does not alter app components, routes, database records, hosted previews, the current unapproved Map work or approved visual archives. A future commit can preserve the reviewed plan with the project's normal Git history; document creation alone does not publish it to GitHub.

Completion of planning means all 45 feature areas have a defined purpose, location, evidence/input model, meaningful interactions, actions, failure behavior and verification requirements across these documents. Completion of implementation is a later claim, requiring the acceptance checks (HALO_FEATURE_ACCEPTANCE.md), validated data sources, backend integration and visual approval. No document can guarantee every future source, unknown user need or production defect in advance.

Reference documents

- Original frontend specification (../foundation/reference/HALO_Frontend_Specification_2026-09-29.pdf), retained as a capability reference rather than overriding newer user decisions.
- Wireframe workflow (../WIREFRAME_WORKFLOW.md).
- Approved foundation stage handoff (../foundation/STAGE_FRONTEND_HANDOFF.md).
- Approved October 1 preservation baseline (../foundation/approved/2026-10-01).
- Current Map preview notes (../map/README.md).
- Backend and Data Plan (HALO_BACKEND_AND_DATA_PLAN.md).
- Frontend Build Plan (HALO_FRONTEND_BUILD_PLAN.md).
- Feature Acceptance (HALO_FEATURE_ACCEPTANCE.md).

Official source links appear beside the capabilities they support. A linked source's existence establishes neither a working production integration nor reuse permission; those remain explicit acquisition gates.

05 / Frontend build requirements

docs/product/HALO_FRONTEND_BUILD_PLAN.md

HALO Frontend Build Plan

Version 1.1, October 2, 2026. This companion turns the Master Plan (HALO_MASTER_PLAN.md) into a screen, control, state, and integration inventory for all 45 feature areas. It is for Yogi's design and implementation work and Vibhav's integration work. It describes the complete target, not completed screens, approved new visuals, or verified production readiness.

The Backend and Data Plan (HALO_BACKEND_AND_DATA_PLAN.md) defines server records and contracts. Feature Acceptance (HALO_FEATURE_ACCEPTANCE.md) defines release evidence. This document adds the frontend responsibilities between them. Proposed component and route names below are implementation recommendations, not files or APIs that already exist. Keep the existing application, approved design, and data clients as the foundation; do not create a disconnected replacement app.

October 3 source audit: consult the Backend Build Handoff (HALO_BACKEND_BUILD_HANDOFF.md) and Source Acquisition Guide (HALO_SOURCE_ACQUISITION_GUIDE.md) before adding live source-dependent controls. In particular, Google Pollen storage/map attribution, Mapbox persistence, full fish-rule review, food recall lifecycle and utility restriction scope are explicit integration gates. Fixtures must not conceal these limitations. Today and its existing detail pages keep their approved content depth.

Verified repository starting point

These statements describe inspected repository files, not the current deployed website. The approved-stage handoff (../foundation/STAGE_FRONTEND_HANDOFF.md) records the prior verification scope and outstanding integration checks.

Table fields: Existing surface / Actual code boundary / Reuse and required change

Onboarding

Actual code boundary: OnboardingScreen (../../app/OnboardingScreen.tsx), WelcomeTitle (../../components/onboarding/WelcomeTitle.tsx), ResultsReveal (../../components/onboarding/ResultsReveal.tsx), onboarding CSS

Reuse and required change: Preserve approved flow, true handwriting, logo, input behavior, retries, result reveal and reduced motion. New optional features do not become mandatory onboarding fields. Home-context transition requires additive contracts.

Live route group

Actual code boundary: StageApp (../../components/stage/StageApp.tsx) under the live layout (../../app/(halo)/layout.tsx)

Reuse and required change: Currently supports Today, Homeguard, seven factors, family, seven household categories, essential Settings. Its route allowlist must be extended deliberately; adding a new page file alone does not mount its session or shell.

Identity and initial data

Actual code boundary: StageSession (../../components/stage/StageSession.tsx)

Reuse and required change: Reuse profile gating, abort and identity checks, readings/history/Learn/alerts handling. New page queries should load on demand rather than adding every domain to startup.

Protected requests

Actual code boundary: api.ts (../../lib/frontend/api.ts), stage-api.ts (../../lib/frontend/stage-api.ts), types.ts (../../lib/frontend/types.ts)

Reuse and required change: Reuse the protected transport and cancellation. Extend typed domain clients and error parsing; no second authentication path or direct provider key in a component.

Approved live presentation

Actual code boundary: StageRuntime (../../components/foundation/StageRuntime.tsx), stage-data.ts (../../lib/frontend/stage-data.ts), stage-values.ts (../../lib/frontend/stage-values.ts)

Reuse and required change: Keep strict data adapters and honest null handling. A production route must always receive explicit live state. Missing runtime must never select realistic preview fixtures.

Score and factor presentation

Actual code boundary: PersonalHero (../../components/foundation/PersonalHero.tsx), SegmentedScore (../../components/foundation/SegmentedScore.tsx), FactorScoreCard (../../components/foundation/FactorScoreCard.tsx), FactorTrend (../../components/foundation/FactorTrend.tsx), FactorSteps (../../components/foundation/FactorSteps.tsx), FactorReadingGuide (../../components/foundation/FactorReadingGuide.tsx)

Reuse and required change: Preserve approved geometry and interactions. Names such as PersonalHero are internal legacy names, not authority to rename Today. Extend behavior through reviewed props/composition, not copied page implementations.

Household and home evidence

Actual code boundary: FamilyGuidance (../../components/foundation/FamilyGuidance.tsx), HomeHousehold (../../components/foundation/HomeHousehold.tsx), HomeEvidence (../../components/stage/HomeEvidence.tsx)

Reuse and required change: Reuse selected-category figures and dedicated guidance. Keep private-water tests, excluded compounds, source dates, service-line scope and ranked actions. New task/record links use stable IDs.

Shared controls

Actual code boundary: Foundation (../../components/ui/Foundation.tsx), Button (../../components/ui/Button.tsx), DataState (../../components/ui/DataState.tsx), FieldError (../../components/ui/FieldError.tsx), SeverityPill (../../components/ui/SeverityPill.tsx)

Reuse and required change: Reuse cards, sheets, segmented controls, chips, switches, callouts, empty states and status vocabulary. Extend only where a new interaction needs it. Existing primitives are not proof every new composition is accessible.

Ambient design

Actual code boundary: EnvironmentalScene (../../components/foundation/EnvironmentalScene.tsx), SceneExperience (../../components/foundation/SceneExperience.tsx), scene-clock.ts (../../lib/frontend/scene-clock.ts), RisingSun (../../components/foundation/RisingSun.tsx)

Reuse and required change: Preserve approved light, dark, clock and reduced-motion behavior. Scene time is decoration, not observed weather. New illustrations need review and performance checks.

Luna

Actual code boundary: LunaAssistant (../../components/foundation/LunaAssistant.tsx), LunaMark (../../components/foundation/LunaMark.tsx)

Reuse and required change: Keep approved panel and symbol; currently questions are sent with a page token, without a multi-turn or action-proposal contract. Add capabilities and confirmed proposals before showing richer controls.

Essential Settings

Actual code boundary: StageSettings (../../components/stage/StageSettings.tsx) and accessibility controls in StageApp

Reuse and required change: Keep explicit Save, fresh identity/profile preflight, seven-flag replacement semantics, retained failed drafts and change-address confirmation. Full notifications, language, privacy and recovery still need screens.

Return routing

Actual code boundary: stage-routing.ts (../../lib/frontend/stage-routing.ts) and StageApp sheet history

Reuse and required change: Existing allowlist excludes all new route families. Extend route validation, section anchors, active-tab ownership and notification targets together. Keep open-redirect defenses.

Local storage

Actual code boundary: storage.ts (../../lib/frontend/storage.ts) and StageApp device preferences

Reuse and required change: Existing identity-scoped environmental cache and completion marker are not a generic private-record store. Accessibility lives in `halo.device.accessibility`. Never put Journal, letters, tests, files or transcripts into this cache as an incidental extension.

Map wireframe

Actual code boundary: MapPreview (../../components/map-preview/MapPreview.tsx), MapCanvas (../../components/map-preview/MapCanvas.tsx), DistrictPanel (../../components/map-preview/DistrictPanel.tsx), Map notes (../map/README.md)

Reuse and required change: Fixture-driven, unapproved preview with review controls and simulated request states. Preserve the work, but implement production adapters/renderer only after review. Importing this page as the live map would not integrate it.

The repository currently has no production `/map`, `/journal`, `/act`, or `/plan` page. The old five-item stage navigation has later-stage buttons, not completed destinations. Map, Journal, Act and notification backend routes exist separately; their presence does not mean production frontend screens exist.

The current root layout provides metadata, favicon/apple icon and local fonts, but these are not a completed installable/offline application. The browser push permission/subscription client and service worker are not delivered merely because server push endpoints exist. Those remain explicit frontend work below.

Navigation and continuity implementation

The target tab structure is **Today, Homeguard, Act, Plan**. Plan's label and the final arrangement still require Yogi's confirmation. Map belongs under Act and Journal under Plan, with direct contextual entry elsewhere. Existing approved routes remain working.

Table fields: Destination family / Target entry and owning tab / Exit and deep-link behavior

`/today`, `/factors/air`, `/factors/uv`, `/factors/pollen`, `/factors/mold`, `/family`

Target entry and owning tab: Today and relevant alerts

Exit and deep-link behavior: Preserve approved section hashes and back to originating factor or Today.

`/home`, `/factors/pfas`, `/factors/radon`, `/factors/lead`, `/household/[member]`

Target entry and owning tab: Homeguard and contextual home links

Exit and deep-link behavior: Back restores house/compound/section context. Deselected category does not display fixture people.

Proposed `/home/tests`, test detail/edit; maintenance and summary routes

Target entry and owning tab: Homeguard, selected action, relevant factor

Exit and deep-link behavior: Private record ID is authorized on load. Missing/deleted record has an explicit outcome and useful parent link.

Proposed `/home/food-garden` and its five branch routes

Target entry and owning tab: Homeguard plus relevant Act/Map links

Exit and deep-link behavior: Subpages return to selected site, advisory or list filters, without encoding sensitive draft fields in URLs.

Proposed `/act` and organization/opportunity/project/outreach/observation routes

Target entry and owning tab: Act and evidence-based suggestions

Exit and deep-link behavior: External handoff preserves private drafts in the supported session. Returning does not imply registration or submission.

Planned canonical `/map`

Target entry and owning tab: Act, water/air/advisory detail, public links

Exit and deep-link behavior: Public layer/filter/record IDs may be shareable. Restore viewport and selection; never share the home pin by default.

Proposed `/plan`, action and outdoor-plan detail

Target entry and owning tab: Plan and Save/Schedule links anywhere

Exit and deep-link behavior: Same action ID everywhere. Calendar selection remains when returning; cancelled source event is not silently deleted.

Planned canonical `/journal` plus Patterns and recap destinations

Target entry and owning tab: Plan, optional Today check-in, alerts

Exit and deep-link behavior: Support validated date/category/summary entry. Keep private note text out of URLs.

`/settings`, proposed subsections, source/topic detail

Target entry and owning tab: Global menu or contextual help

Exit and deep-link behavior: Return to allowlisted origin; preserve unsaved form state when opening help.

Before route migration, enumerate the actual URLs emitted by existing alerts, exported links and source references. Support those known formats through allowlisted translation or aliases when the corresponding screens ship. Do not invent a claim that nonexistent page routes are already delivered. An unsupported target must show a useful unavailable destination, not silently become Today.

Use one typed navigation registry for route ownership, titles, safe returns and notification targets. Keep query validation independent from rendering. Dialog/sheet open adds one history entry; switching its content replaces that entry. Close restores opener focus, underlying scroll and relevant selection. Directly linked sheets close without an unexpected extra Back. Private forms need a discard-confirmation path for internal navigation and best-effort browser-exit protection, with no promise that a mobile browser will always deliver an unload event.

Separate route access requirements from the active tab. Public Map/district, organization and opportunity exploration must have a mode that does not require address onboarding. Reuse the current anonymous-token bootstrap or a separately reviewed public-read interface, but do not call current authenticated APIs unauthenticated. Hide home pins, household ranking and matched guidance until profile data exists. This needs a deliberate session-gate change, not removal of authorization from private routes. Any new onboarding Browse your area entrance requires visual review.

State and component ownership

Table fields: State / Owner and lifetime / Required behavior

Verified identity and current home context

Owner and lifetime: Existing session boundary, extended with server context version

Required behavior: Identity change clears private views, drafts, request owners and subscriptions before new data loads. Home change invalidates current-home queries but retains historical associations.

Server records

Owner and lifetime: Typed domain clients and scoped query state

Required behavior: Keys include owner, context, record revision, filters and language where applicable. Server is authoritative; no speculative success or client-only ownership checks.

Form edits

Owner and lifetime: Local form state tied to owner and record revision

Required behavior: Dirty fields survive retry/refetch. New identity hides/discards the prior household's draft. Optional persistent drafts require a separately reviewed private-store policy.

Selection and navigation

Owner and lifetime: Validated URL for nonsensitive public state; session memory for private selection/scroll

Required behavior: Browser Back works. Secrets, notes, precise locations, household flags and conversation contents never enter query strings.

Device choices

Owner and lifetime: Existing accessibility preference mechanism

Required behavior: Text scale, reduced animation and high contrast persist locally when storage works. Appearance remains device-driven. Language scope is specified separately before implementation.

Preview scenarios

Owner and lifetime: Isolated fixture adapter and review-only controls

Required behavior: Never create live accounts, write production data, send push or fall through to live API on an unmatched fixture. No preview import as a production fallback.

Recommended reusable components below are new proposals. Implement only after the first real screen establishes the need; do not create a generic framework for every possible page.

Table fields: Proposed component / Responsibility and contract

EvidenceSummary and SourceDetail

Responsibility and contract: Value/units, evidence kind, scope, original dates, status and supporting source; distinguishes normal data from missing/withdrawn/qualified results.

SaveAction and ActionDetail

Responsibility and contract: Deduplicate equivalent recommendations, edit user fields, preview reason, save/schedule explicitly, show confirmed state and linked origin.

ScheduleEditor and Agenda

Responsibility and contract: All-day versus timed items, timezone, recurrence scope, reminder channel and status; no hidden calendar synchronization.

RecordForm and AttachmentReview

Responsibility and contract: Field validation, dirty-state retention, revision conflict, file upload state, extraction confirmation and private preview. Domain-specific sections remain typed.

SourceUpdateNotice

Responsibility and contract: Show what changed and the dated before/after references without overwriting a user's action or text.

ChartWithTable

Responsibility and contract: Consistent units, point inspection, gaps, readable table, method/source link and source-specific resolution; no fabricated interpolation.

FilteredResults

Responsibility and contract: Search/filter/reset, result count/coverage, pagination, preserve filter state and distinguish no matches from failure.

SharePreview

Responsibility and contract: Choose included sections, disclose sensitive content, create/print/download/share, cancellation and permission fallback.

ConfirmationDialog

Responsibility and contract: Correct operation scope, record name/count, reversible versus permanent effect, pending/failed/confirmed outcome and focus return.

Do not add a heavyweight new state library merely to appear advanced. Extend the current patterns first; adopt a query library only after evaluating invalidation, deduplication, server/client compatibility and migration cost. Read the installed Next.js guides before future framework code changes.

Complete screen and interaction inventory

Every row inherits the common states and form rules later in this document. The rightmost column names dependencies, not a claim that the extension is present. Backend operation groups are defined in the backend companion (HALO_BACKEND_AND_DATA_PLAN.md).

Entry and Today

Table fields: Feature / Screens and visible controls / Data and behavior needed

G01 Entry and onboarding

Screens and visible controls: Approved Welcome, address/ZIP and optional location, household chips/skip, home details, results, Back/Next/retry, change-address confirmation

Data and behavior needed: Existing onboarding/profile flow. New home transition review lists records that stay with old home and portable tasks requiring a choice. Keep optional new-feature setup out of this flow.

T01 Today

Screens and visible controls: Score hero, four factor links, contribution explanation, meaningful-change disclosure, refresh, alerts, family link below cards, optional existing action/check-in entry

Data and behavior needed: Existing daily/history/alerts. Proposed changes endpoint supplies compatible comparison and cause of change. Missing factor score remains Not scored; no client formula. No PFAS card or Today contribution.

T02 Air

Screens and visible controls: AQI/pollutant, provenance, trend and table, point detail, explanation, practical-step deck, monitor map link, compare forecast times, Save action and optional Log this day

Data and behavior needed: Existing reading/history/Learn; forecast and action extensions. Display measured versus modelled and monitor versus regional scope. Native AQI is not the HALO score.

T03 UV

Screens and visible controls: Current index, approved rising half-sun protection window, available maximum, trend/table, protection steps, Plan outdoors, Save preparation

Data and behavior needed: Existing UV/window/history/Learn; interval forecast extension. Null window is unknown, not all-day low; graph does not fabricate hourly data from endpoints.

T04 Pollen

Screens and visible controls: Tree/grass/weed selector, individual labels/readings, dominant category, compatible trend, seasonal explanation, steps and Plan link

Data and behavior needed: Existing category data/history/Learn; forecasts where supported. Selector retains focus and does not imply that switching category compares the same series.

T05 Mold

Screens and visible controls: Estimate heading, humidity/rainfall basis, uncertainty, moisture guidance, Record an observation and Follow up

Data and behavior needed: Existing proxy/Learn; H07 private observations. No spore count or chart without data. Indoor observation stays separate from the outdoor estimate.

T06 Daily household

Screens and visible controls: Selected category navigation, applicable current factors, Outdoors/At home context where supported, concise guidance, Save/Open existing action, Edit household

Data and behavior needed: Existing profile/Learn plus compiled guidance/action links. General advice works with no category; one task reused across overlapping categories.

Homeguard

Table fields: Feature / Screens and visible controls / Data and behavior needed

H01 Home overview

Screens and visible controls: Approved score and source basis, selected-category house, water/radon/lead links, returned action priorities, compact Tests/Maintenance/Food and Garden/Summary entrances

Data and behavior needed: Existing home/profile/household responses plus owned record summaries. Missing/private water cannot produce an inferred healthy total. Essential coverage stays visible, not hidden in a tooltip.

H02 Water and PFAS

Screens and visible controls: Utility identity/match basis, compound selector, sample table/history, threshold detail, excluded/unregulated rows, private-water test plan, map/watch/test/outreach actions

Data and behavior needed: Existing home evidence plus sample provenance, subscriptions and tests. Well and spring branches are distinct from utilities. Unknown association has correction flow, not arbitrary client reassignment.

H03 Radon

Screens and visible controls: County zone explanation, test options, own-test timeline, Schedule/Start/Record result, professional follow-up and source detail

Data and behavior needed: Existing regional estimate plus tests/actions. User result is separate dated evidence, never a county-zone replacement. Pending test has no result badge.

H04 Lead

Screens and visible controls: Unscored estimate, home-year correction, utility-wide service-line information, property-specific question, test and follow-up links

Data and behavior needed: Existing estimate/inventory plus records/actions. Unknown home year stays unknown. A utility total is not the home's pipe material.

H05 Household guides

Screens and visible controls: Seven preserved category routes, selected category summary, applicable factors/tasks/tests, practical-step deck, source links and back to house

Data and behavior needed: Existing category booleans/guidance plus records. Child/teen/adult may reuse general text. Deselecting removes figure but not historical records; unsupported category links have a safe state.

H06 Tests and records

Screens and visible controls: Filtered list, test detail/timeline, arrange/sample/result forms, attachment review, correction history, compatible comparison, follow-up, export/delete

Data and behavior needed: New tests/results/private files and Plan contracts. Form schema below preserves matrix, dates, qualifiers and original units. Unclear results can be saved uninterpreted with a visible reason.

H07 Maintenance and moisture

Screens and visible controls: Asset/routine list, detail, create/edit cadence, occurrence detail, moisture/leak record, complete/skip/pause/replace, linked test and instructions

Data and behavior needed: New asset/observation records plus shared action occurrences. Renter alternatives remain available. Completion-relative and fixed-calendar schedules are distinct.

H08 Household summary

Screens and visible controls: Purpose/section/date selection, sensitive-field review, compact preview with optional appendix, print/image/file/share

Data and behavior needed: Existing readings/Learn plus selected new records. Partial section failure is local; two-score/all-factor capability preserved. No public private-summary link by default.

Food and Garden

Table fields: Feature / Screens and visible controls / Data and behavior needed

F01 Overview

Screens and visible controls: Optional interest chips, relevant verified updates, five destination links, saved advisory/site continuation and coverage explanation

Data and behavior needed: New preferences, advisory/site/watch summaries. No questionnaire wall. General guidance without local evidence remains useful and honestly labelled.

F02 Fish advisories

Screens and visible controls: Waterbody search/extent, species and official audience selectors, advisory detail, overlap explanation, source, map, save/follow/share, optional catch/consumption form

Data and behavior needed: New reviewed matching/versioned advisory service; private optional records. Do not approximate official audiences from HALO household flags or calculate remaining allowances. Ambiguous/no match must not look safe.

F03 Garden workspace

Screens and visible controls: Site list/create/edit/detail, known and unknown water/soil context, records, questions, testing preparation, consultation follow-up, export site brief

Data and behavior needed: New site contexts and shared tests/actions. Irrigation source separate from household drinking source; source changes retain older record context. No crop safety score.

F04 Food evidence

Screens and visible controls: Supported region/category filters, local notices separated from broader studies, record detail/applicability, follow/source/share/question controls, optional private report link

Data and behavior needed: New reviewed food reports. Preserve product/lot or sample scope. Unsupported local data has source/contact options, not a fictional regional food score or farm ranking.

F05 Grower resources

Screens and visible controls: Concern/region selector, service/program list, scope comparison, requirements checklist, official contact/application, private progress/reference and follow-up

Data and behavior needed: New curated resource/program contracts and shared preparation records. Lab matrix/analyte/method scope shown. Application opened, user says applied and approval are different states.

F06 Water stewardship

Screens and visible controls: Regional drought distribution/history, separate applicable utility notice, follow, stewardship tasks, optional usage list/entry/comparison/export

Data and behavior needed: New drought/notice source adapters and private use records. County distribution not household class. Compare compatible periods/units; show measured difference, not attributed savings.

Act

Table fields: Feature / Screens and visible controls / Data and behavior needed

A01 Act overview

Screens and visible controls: Relevant issue, Get involved, Explore your area, Speak up, Ways to help, saved continuation, cause/area browsing

Data and behavior needed: Existing organizations/home/map plus reviewed recommendations/projects. Maintain household-summary entry. No elevated finding still yields useful participation options.

A02 Organizations

Screens and visible controls: Search/filter list, organization detail, mission/service scope, verification date, actual contacts, related opportunities, Save contact and Report outdated listing

Data and behavior needed: Existing volunteer directory extended with IDs/review/archive metadata. Link check distinct from human verification; unknown access info is not an accessibility promise.

A03 Opportunities

Screens and visible controls: Cause/date/location filters, list/detail, requirements/access/cost, save/add to Plan, external registration, status review, attendance/hours/note form

Data and behavior needed: New organizer-linked opportunities, revisions and participation. Cancelled/postponed/full/unknown capacity distinct. Opening a form never marks registered or attended.

A04 Projects

Screens and visible controls: Project catalogue/detail, suitability/materials/permissions, approved step deck plus scan-all, choose steps or whole project, schedule/progress/pause/follow-up

Data and behavior needed: New reviewed templates and shared action parent/children. Browsing a step is not completion. Count effort/participation as self-reported, never unmeasured pollution removal.

A05 Outreach

Screens and visible controls: Purpose, three recipient tabs with independent drafts, evidence picker/preview, editable letter, reset current, copy/share/print, optional follow-up

Data and behavior needed: Existing evidence/district resources plus reviewed templates and optional owned drafts. Missing measurement produces information-request wording. Exact recipients and private fields previewed; no auto-send.

A06 Field observations

Screens and visible controls: Private list/detail, factual entry, optional location/photo review, official destination selector, report preview/copy/export, external form, user-reported submission/reference/follow-up

Data and behavior needed: New private observations/files/destination directory. Cause remains unknown unless evidence supports it. No automatic public map, agency receipt or unsafe collection prompt.

Regional exploration

Table fields: Feature / Screens and visible controls / Data and behavior needed

M01 Map

Screens and visible controls: Full focused map, four layer controls, zoom/locate/pan, accessible search/list, legend, selected-record sheet, preserved view and failure fallback

Data and behavior needed: Existing backend layers; production renderer/client still needed. Lazy load. Each layer has independent loading/error. Unlocated records remain in list.

M02 Water timeline

Screens and visible controls: Overall/compound/hazard selector, Latest versus source quarters, marker/detail/sample rows, own-system ring, watch/action/compare links

Data and behavior needed: Existing map source extended for complete sampling metadata. Failed quarter restores prior selection; stale response cannot override new filter. Density labelled record density, not plume.

M03 Other layers

Screens and visible controls: Radon county details/test link; facilities clusters, compliance programs and official record; air monitor pollutant/time/provenance

Data and behavior needed: Existing layers with source metadata and honest nulls. Map symbols distinguish evidence types. Fish/drought overlays only after geography gate, not merely decorative colored polygons.

M04 Comparison

Screens and visible controls: Select second compatible public record, aligned values/units/dates/coverage, missing reasons, source drilldown, clear comparison/back

Data and behavior needed: New comparison adapter/query over normalized evidence. Mobile list rather than two tiny maps. Disable unsupported comparisons with reason, not rank missing as best.

M05 District

Screens and visible controls: District/state summary, compound control, denominator explanation, county rows/partial tags, supported rank, unlocated coverage, public export/outreach

Data and behavior needed: Existing district aggregation extended/verified. Boundary version visible; people served are not proven exposed. Never include household location/private data in a public brief.

Plan

Table fields: Feature / Screens and visible controls / Data and behavior needed

P01 Your Week

Screens and visible controls: Week/day controls, agenda, unscheduled list, overdue/upcoming/waiting groups, filters, optional month jump, item detail, calendar export

Data and behavior needed: New plan/action/occurrence service. All-day dates distinct from instants. Event update affects same record everywhere; calendar download is not sync.

P02 Shared actions

Screens and visible controls: Save sheet, existing-action match, editable detail, steps/reason/source, schedule/status/follow-up, complete/reopen/cancel/delete

Data and behavior needed: New owned actions with evidence/context/revision/idempotency. Same ID across every entrance. Mark waiting does not invent a due date or completed result.

P03 Outdoor plans

Screens and visible controls: Activity/place/duration/date form, comparable forecast-period chooser, per-factor tradeoffs, official notice, preparation, save/watch/reschedule

Data and behavior needed: New normalized forecast/plan service. Per-source horizon and resolution visible; far future means check later. No safe-time score, route tracking or automatic rescheduling.

P04 Journal

Screens and visible controls: First-use retrospective or Log today, date/category picker, symptoms/severity/illness/note or Felt fine, save/update, recent list, delete/undo, partial batch retry

Data and behavior needed: Existing Journal endpoints need production client/screens and stronger logging semantics. Current upsert replaces whole entry; do not send unloaded blanks. Original fields survive edits.

P05 Patterns

Screens and visible controls: Twelve-week heat grid and list, factor/date/category controls, day detail, chart/table, findings/counts/method, edit selected entry and export

Data and behavior needed: Existing history/findings with revised analysis contract. Explicit logged versus unlogged, recalled, illness and missing-data counts; no causal conclusion. Unsupported filters not merely cosmetic.

P06 Seasonal recap

Screens and visible controls: Season selector, coverage summary, environmental trends, logged entries/actions/tests/participation, source detail, dismiss prompt, export and next review

Data and behavior needed: Existing summary plus owned record joins and revised analysis. No engagement streak or inferred benefit. Alert deep-link opens actual season.

Shared experiences

Table fields: Feature / Screens and visible controls / Data and behavior needed

X01 Learn and sources

Screens and visible controls: Inline definition, contextual explanation, reviewed topic index/search, full topic/source detail, linked practical action and return

Data and behavior needed: Existing seven-topic Learn plus proposed index/versioned content. Retain form drafts; search has empty/failure states. Review dates and precise supporting sources visible.

X02 Luna

Screens and visible controls: Approved launcher/panel, context disclosure, suggestions, composer/send/cancel/retry/new conversation, citations, proposed action/draft preview and explicit confirmation

Data and behavior needed: Existing assistant with new capabilities/context/conversation/proposal contracts. No fake streaming, hidden household context, unsupported paperclip or automatic effects.

X03 Alerts and watchlists

Screens and visible controls: Inbox/unread, expand, read one/all, dismiss/undo, exact target; follows list/detail, channels, reminder snooze, quiet-time controls

Data and behavior needed: Existing alerts/notifications/push plus typed subscriptions and event targets. Browser permission states explicit. Read is not task completion; cancelled event is retained with notice.

X04 Settings

Screens and visible controls: Sections for household/home/address/renter mode, notifications, accessibility, reviewed language, privacy, recovery, sources/methods/About/feedback

Data and behavior needed: Existing essential Settings plus new preference contracts. Explicit Save, fresh preflight, conflict and retained drafts. No appearance switch unless separately approved.

X05 Privacy and recovery

Screens and visible controls: Stored-data inspector, scoped export, clear Journal, delete all, optional recovery/link flow, device/session list and revoke where supported

Data and behavior needed: Existing deletion extended to all domains; new exports/recovery. Explain current anonymous-session loss honestly. Typed confirmation and operation-status recovery for destructive actions.

X06 Offline and accessibility

Screens and visible controls: Cross-app offline/stale badges, retry/draft status, motion pause, high-contrast/text controls, text chart/map/calendar alternatives

Data and behavior needed: Existing environmental cache/accessibility foundation, extended per feature. Private offline writes are not implicitly implemented; queued is not saved. No feature requires animation, hover or gestures.

X07 Source operations

Screens and visible controls: Protected operator list/detail for freshness/failure/review, import diff/quarantine, publish/withdraw, translation review and job retry

Data and behavior needed: New internal role-protected UI or equivalent documented admin workflow. No ordinary-user link or service key in browser. Publication/retry has audit reason, expected revision and explicit scope.

Forms and exact interaction responsibilities

Forms below are target field/control schemas, not database schemas. The backend must supply allowlisted choices, valid transitions and limits where interpretation depends on them. Required and optional labels must be visible; optional fields never quietly acquire a default that changes meaning.

Action scheduling and recurrence

Create opens with title, reason and evidence already populated from the selected recommendation. Users may edit their title and optional private note without editing the cited source. Offer Save for later or Schedule, not a forced due date. Show a possible existing action with Open existing and deliberate Create another when appropriate.

Schedule fields are all-day or timed, date, time when timed, timezone, optional end/duration where the task needs it, repeat off/on, calendar-anchored or after-completion basis, interval/end rule, reminder off/on and available channel. Validate end after start, supported recurrence, nonexistent/ambiguous daylight-saving local time and date range. Present a human-readable schedule preview before Save. Calendar event fields use the event timezone; device display timezone is labelled when different.

Editing a recurrence asks This occurrence, Future occurrences or Whole series only where supported. Show affected dates and preserve completed history. Snooze changes the eligible occurrence/reminder, not the source event's date. Skip is not Complete. Cancelling an action cancels future reminders after server confirmation. A pending request shows its state beside the control, not only in a toast.

Test and result entry

Start with purpose/test kind, current or archived home/site context, sample medium, optional provider/lab and proposed date. Context and medium cannot be inferred from a document's filename. Recording collection adds collection start/end as appropriate, location label and method if known. Awaiting result optionally records expected date without inventing one.

Result rows use analyte, original reported value, original unit, qualifier, detection/reporting limit if given, limit unit, sample basis where relevant, result/report date and optional report reference. Provide add/remove row with duplicate warnings, not automatic merging of separate samples. Less-than, non-detect, numeric zero and absent result are distinct options. Preserve the typed original; normalization is server-returned and explained. Invalid interpretation fields may permit saving a clearly uninterpreted draft, never a confident severity.

Attachment selection shows allowed formats, size/privacy explanation, upload progress, cancel/retry/remove, scanning/unavailable state and confirmed receipt. Preview and any extracted transcription are separate. Users inspect/correct proposed fields and explicitly confirm transcription. Do not call this laboratory verification. Corrections keep original version, reason and current status. Compare opens only compatible records or clearly lists the mismatch.

Maintenance and moisture entry

Asset fields are user label, optional product/model, installation/service date, factor/use and optional source instructions. Do not require serial numbers. Routine fields reuse scheduling with an explicit cadence basis. Replacement closes or supersedes the old asset record after confirmation without erasing its history.

Moisture entry accepts observation type, observed time, optional room/site label, optional measured reading and unit, measurement source and optional private note/photo. Label relative humidity as a user-entered measurement; no automatic mold detection. Repair follow-up connects to one shared action. Completion records what the user did, not a claim that the hazard is resolved.

Fish advisory selection and optional notes

Search returns named waterbodies with jurisdiction/segment cues to disambiguate duplicates. Species can be unresolved; show matching limitations. Audience selector uses source-reviewed definitions and permits I am not sure, with general source guidance. Explain overlapping statewide/local rules and display source-defined servings/time windows without synthesizing allowances.

Optional catch/consumption entry uses record kind, date, species/unresolved label, optional waterbody context, quantity with original unit and optional note. Preserve advisory revision selected at the time. Editing an observation does not change official evidence. No dose, meal balance or automatic diagnosis follows from saving. Separate Delete record from Unfollow advisory.

Garden and grower setup

Site form uses label, optional coarse place/county, growing setup, irrigation source, known soil/history or Unknown, optional concerns and optional source-linked records. Ask whether using the household water source is accurate before reusing it. Multiple sites use a simple list and selector. Moving/changing source shows what historical records retain their old context.

Grower preparation starts with concern, medium/service need, region and selected resource. Requirements checklist is sourced, not invented. Optional progress records include contact/application date, user-reported state, reference number, response note and chosen next follow-up. Official portal opens externally after the user checks requirements. Access/eligibility unknown is shown explicitly, not treated as guaranteed qualification.

Water use entry

Choose cumulative meter reading or interval/bill usage before entering values. Capture site/context, observed date or period boundaries, original quantity/unit, measured/estimated/user-reported basis and optional note. Never subtract unlike meters/units, overlapping bills or a rollover as ordinary consumption. Ask to correct or explicitly flag a reset/replacement; the server decides comparability. The comparison states interval length and missing periods. It does not attribute a difference to HALO.

Participation and observations

Participation form is attached to the exact opportunity revision. Registration opened is informational; an explicit user control records Registered externally. Attendance can be attended/not attended/unknown, with optional hours and note. Hours validate as a nonnegative quantity and remain self-reported. Changed event time prompts review before updating any user-specific plan choice.

Observation form contains category describing what was observed, date/time/timezone, factual description and optional place/photo. A location control explains coarse versus precise sharing; it does not repeatedly request GPS in the background. Show stored and exported metadata separately. Reporting preview lists official destination, included fields and any exact location/photo. Open form is a handoff, followed only by an optional user-reported submitted date/reference. Genuine acknowledgment requires an actual receipt, never inferred navigation.

Outreach and exports

Outreach uses purpose, recipient class, selected evidence and editable body. Keep independent in-memory drafts per recipient; changing a tab does not overwrite another draft. Reset confirms loss of edits for that recipient only. Optional greeting/signature/contact fields explain local versus server persistence. A stale evidence notice offers review without rewriting user text. Send is not a control until a separately approved actual sending contract exists; Copy, Share and Print are the default delivery tools.

Export selection distinguishes public evidence, household summary and full private data. Choose sections/date range and explicitly include sensitive fields when wanted. Preview readable content and source dates before generation. Download/print/share has progress, retry, cancelled, unsupported and denied outcomes. Native share cancellation retains preview. File names avoid household identifiers. Public map/district exports cannot inherit private selection state. Server export downloads use authenticated short-lived access; expired links offer regeneration instead of a false missing-record claim.

Journal and retrospective entry

Load the selected date/category before enabling Save. Entry has symptom choices, explicitly chosen severity when applicable, possible illness, optional note, and Felt fine as a deliberate alternative. Preserve the current backend vocabulary rather than inventing incompatible chips. A clear replacement review prevents switching to Felt fine from silently retaining contradictory symptom fields.

For past rough-day setup, select dates, review which already have records and choose supported per-date content. Future dates are unavailable. Batch results report successes and failed dates separately; Retry failed sends only failures with stable keys. Undo for one deletion restores the complete prior entry, not a blank date. Clear Journal is a separate confirmed operation. Recomputed findings invalidate after a successful edit/delete and retain calculation status instead of displaying an apparently unchanged authoritative result.

First-use detection uses a proposed authoritative all-time presence flag/count, not an empty 90-day query or a blank grid. Preserve the original 90-day regular-entry window, 12-week grid, seven recent entries and 62-date retrospective request maximum unless explicitly revised. Show history-backfill-pending separately from source no-coverage. Retrospective retry cannot overwrite an existing richer record with unspecified symptoms. The normal Save guard waits for the selected date/category to load or an explicit authoritative no-entry result.

Settings privacy and recovery

Each section shows loaded values, dirty indication, explicit Save/Cancel, field errors, retained failed draft and confirmed success. Profile unknown disables writes. Household Save sends all seven flags after a fresh preflight; unrelated edits are retained. Renter/owner mode and language must have agreed storage contracts before controls imply server persistence. Reviewed Spanish fallback is explicit and the chosen language is not silently reset.

Clear Journal shows the authoritative current record count and scope. Delete all describes records, files, jobs, subscriptions and provider/backup limits, then requires a deliberate confirmation. Pending deletion cannot be dismissed into a false success. Unknown outcome reconciles status before retry. Recovery screens explain identity verification, existing-record collisions, revoked/expired link state and loss of access. Device revoke must not reveal another household's devices. None of these operations uses a household ID in a URL as proof of ownership.

Data lifecycle and cross feature updates

Successful mutations publish scoped invalidation events or refresh the shared query cache; the names are implementation choices, not new browser-global APIs promised by this document. Every refresh still verifies identity and ignores late responses.

Table fields: Confirmed event / Frontend consumers that must update

Action saved or changed

Frontend consumers that must update: Origin's Save/Open action control, Plan agenda, action detail, linked project/test/participation summary, reminder state

Test result confirmed or corrected

Frontend consumers that must update: Test detail/list, related factor's own-record section, selected household guide, action follow-up, export preview; not automatic composite rescoring

Journal entry edited/deleted

Frontend consumers that must update: Selected day, recent entries, heat grid, eligible findings/recap, selected export preview; recomputation state explicit

Advisory or source revision

Frontend consumers that must update: Current source view, subscribed notice, saved action review flag, open outreach evidence warning; original snapshot retained

Opportunity cancelled or changed

Frontend consumers that must update: Opportunity detail, participation, linked Plan item, alerts; no silent deletion or completed attendance

Household selection saved

Frontend consumers that must update: House figures, current guidance, allowed optional filter choices; historical records not deleted

Home context changed

Frontend consumers that must update: Current readings, matching, home/garden association review, subscriptions and current-home queries; past tests/history keep original context

Identity changed or data deleted

Frontend consumers that must update: All private views, requests, selected records, local caches permitted for that identity, pending uploads/downloads and notices

An invalidation event is not a reliable backend job queue. Authoritative updates and notification intent belong on the server. The frontend re-reads after reconnect/focus according to domain freshness rules; it does not assume that every database change happened while the page was open.

Shared states and error recovery

Every independently fetched section needs first-load progress, populated, valid empty, no coverage, failed request, stale previous data and offline behavior. Not every state needs a full-screen illustration. Show a short local explanation and next action; preserve other working sections.

Forms add dirty, validating, submitting, confirmed, failed with draft, revision conflict and unknown write outcome. Use a stable idempotency key for a logical create and reconcile unknown outcomes before retry. Updating a record sends its known revision. A conflict presents current server values versus local edits and a deliberate resolution, not silent last-write-wins. Cancellation of the browser request does not prove the server mutation was cancelled.

Permission and identity failures hide private data rather than substituting another anonymous household. A transient provider outage is not a reason to create a new identity. Expired/withdrawn/archived public evidence remains inspectable as historical when allowed, with a clear current-state label. Deleted private records do not reveal existence across households.

No fixture values, fake forecasts, invented sources, simulated registration, placeholder safety claims or generated charts enter live fallbacks. A zero is valid only when returned as a valid number; absence is not zero. Lead remains unscored. Daily factor scores stay unsupported until their server semantics are reviewed. Contribution geometry does not claim to reconstruct formula weights.

Existing data contract details

Keep the canonical severity vocabulary `good`, `moderate`, `elevated`, `high`, `severe`, `no_data`; unsupported severity becomes `no_data`. Do not read a legacy field called `status` as a severity fallback. This does not prohibit valid discriminators such as `water.status`, or the actual supported `lead.level` field mapped through the severity validator. Unknown additional properties in an additive response should not break known valid data; required-field and semantic validation still applies.

Use validated water `date_iso`, `latest_sample_iso` and other specified ISO companions rather than parsing human display dates. Keep date-only values separate from timestamp instants and render the correct timezone. Format source quantities with the provided units and precision; normalize only through a documented conversion while retaining originals. A larger-looking number in another unit must not appear to be an increase.

If adding the specification's concentration-comparison graphic after design review, use one comparable scale across rows with a clearly identified threshold. The original fixed zero-to-two-times comparison range requires a visible overflow/clipping indicator for higher values plus the actual numeric value. Do not silently autoscale each row to make different concentrations look equal. Keep scored rows ordered meaningfully and separate unscored/excluded evidence with an explanation, while preserving already-approved geometry until its revision is approved. A benchmark is not relabelled as an enforceable limit.

Responsive visual and accessibility requirements

October 3, 2026 user clarification: Today, formerly Personal Hero, and its air, UV, pollen and mold detail pages should remain substantially unchanged for now. Retain current content depth, useful sections, visual treatment and overall layout. Do not interpret app-wide simplification as permission to shorten them substantially or conceal existing detail behind more taps. Small refinements and additive integrations can follow normal review; substantial cuts or restructuring require a separate decision.

Use the approved blue/teal identity, typography, transparent logo, centered headings, rounded navigation, subtly tinted factor cards and distinct restrained scenes. New page identity comes from the task: agenda for Plan, opportunities/projects for Act, aligned evidence for comparison, simple record timelines for tests. Avoid a grid of identical large cards for every feature.

At 320px and 375px, primary actions, labels, units, dates and errors must wrap without horizontal page overflow. Larger phones and desktop can use useful side-by-side evidence/details, with the same reading order and capabilities. Test the approved 100/115/130/150 percent text settings and browser zoom independently. Long translated text, long compound/species names, zero counts and large values are real layouts, not edge cases to truncate blindly.

Semantic headings, labels, buttons and links remain correct when animated. Focus is visible; sheets trap focus only while open and restore it on close. Section navigation offsets fixed headers. Charts expose data tables and keyboard/touch point selection. Map has the same records in a list. Calendar has accessible dates and agenda alternative. Household icons have visible labels. Status is text plus shape/color; the factor tint is not severity.

Retain approved handwriting/ring/card/chart motion, show final values immediately under reduced motion, and never delay inputs. Pause nonessential motion when hidden/offscreen and offer a user pause for prolonged animation. Do not redraw a score or reannounce every count on every background refresh. Avoid continuous GPU-heavy scenes beneath long forms. Lazy load map code, lower-page imagery, attachments and large histories; Today must not fetch all app domains.

Test mobile Safari keyboard height, safe areas, Back behavior, scroll restoration, native share, print and file selection on physical hardware. Desktop emulation does not establish those outcomes. Touch targets should be comfortably large, especially small icons in stacked cards, without depending on hover. No interaction may require dragging, swiping or a subtle color change alone.

Preserve the original minimum 44px target for primary and core interactive controls, including icon buttons. Any deliberate exception needs accessibility review. Inline SVG remains the established chart implementation; do not introduce a charting package simply to add a new series.

No em dashes appear in UI, source-derived labels, generated replies, export templates, announcements or notifications. Normalize source copy without corrupting units, mathematical signs, quotations needed for meaning or official source identity. Keep actual original documents available where licensing allows; app copy cleanup is not source rewriting.

Centralize new visible copy, error messages, accessibility labels, date/unit templates, exports and notification wording in the existing catalogue strategy instead of scattering English literals. Record requested and actually served language for reviewed content; keep an explicit fallback when a translation is unavailable. Update document language and formatting with the real displayed locale. Existing English date helpers and `lang="en"` are starting points, not multilingual support. Preserve source audience definitions and units during translation. Factual copy corrections, such as replacing an unsupported claim that nobody tests private wells with HALO having no result, receive review without editing historical archives.

Notifications installation and browser lifecycle

The existing five server preference types are air quality change, weather advisory, new water results, radon season and season summary. The current preference contract gates both new inbox alerts and pushes for a disabled type. The frontend must describe that behavior accurately; separating inbox and push preferences is an explicit backend change, not a label-only adjustment. Watchlist and task-reminder controls require the extended contracts and their own clear semantics.

Before asking browser notification permission, explain the feature and let the user choose Enable on this device. Implement unsupported/default/granted/denied permission states, registration failure, expired subscription and unsubscribe. Persist a browser subscription only after identity verification and server acknowledgment; reconcile local and server status after reload or identity change. Do not report notifications enabled solely because the permission is granted. Notification clicks resolve a safe authorized destination and handle a missing/deleted record without opening an unrelated private item.

Installation work includes a reviewed web app manifest, appropriate icon variants, generic non-sensitive social/preview metadata, standalone safe areas and an update/reload experience. Keep the approved logo shape; asset variants need quality checks. Never put household categories, tests, scores or location into metadata crawled by link previews. Installation prompts are optional and cannot block browsing.

If a service worker is introduced for push or offline support, define its scope, cache allowlist, version cleanup and update coordination. Exclude private API responses, authorization-bearing requests, attachment URLs, transcripts and user exports by default. Update prompts must not silently discard a dirty form; activate safely or let the user finish first. Test old-client/new-server compatibility, offline startup, stale shell assets, multi-tab updates and logout/deletion cache cleanup. A worker is not a substitute for idempotent writes or an always-running background monitor.

Offline privacy and device behavior

The current environmental cache is a useful starting point, not a promise of full offline CRUD. Document per domain whether it is available as a dated cache, temporarily retained in memory, explicitly saved private draft, or online only. Default new sensitive domains to no persistent offline content until reviewed.

If private drafts or an outbox are approved, bind each item to the verified identity and context, expose pending/cancel/retry, enforce retention and lock behavior, and prevent replay after deletion or identity change. Never queue external sending or destructive operations silently for later. A successful local write is labelled On this device, not Saved to HALO. Downloaded exports leave app control; deletion cannot revoke external copies.

Permission screens explain why location, notifications or files are requested at the moment of use. Denial leaves meaningful alternative paths. Browser storage restrictions should preserve an in-memory session as far as possible, with honest loss-of-draft warnings where relevant. App installation is optional; core workflows work in a supported browser without push or PWA installation.

Before the first valuable private upload, disclose the anonymous-session recovery limitation and any actually supported recovery option. Explain that uploaded reports or notes can contain personal information even though onboarding discards its typed address. Update inspection/export/deletion and privacy copy for those optional data classes before release. Do not show a recovery switch or promise unless the ownership-verification flow exists.

Operator interface requirements

Source operations may begin with a protected internal tool and documented commands rather than a large new dashboard. If a browser interface is built, it needs authenticated roles distinct from household users, audit history and no service credentials in client bundles.

The minimum useful operator surface lists source freshness, last successful run, parser/version, rejected count and review due; opens a run to inspect safe diagnostics and change summaries; compares proposed versus published evidence; quarantines suspect imports; approves/publishes/withdraws reviewed content; and manages translation review. Retry has an explicit scope and idempotency. Publication shows affected interpretation/notification consequences. No operator screen displays private household notes by default or edits a source record without traceable revision.

Frontend delivery and evidence

Build one connected journey at a time, reusing shared contracts rather than implementing 45 disconnected mock routes. Source feasibility and backend contracts can proceed before visuals are locked; production page implementation waits for Yogi's design approval.

1. Confirm four-tab navigation and cross-links, then review the revised shell with normal, missing and offline states. Extend route ownership and safe return handling together.
2. Review shared Save action, scheduling, evidence detail and private-record forms. Implement their first useful connected journey with isolated contract fixtures and server ownership tests.
3. Add the Today/Homeguard follow-through paths without changing sealed assets. New screenshots belong to a new approved version, not replacement baselines.
4. Complete Act and Map using the preserved unapproved Map work as a starting point. Production renderer, live layer adapters and source semantics are separate checks.
5. Complete Food and Garden branches with real approved source coverage and useful no-coverage paths. Complete Plan, Journal/Patterns and their corrected interpretation contracts.
6. Extend Luna, watchlists and full Settings only when privacy and mutation contracts work. Accessibility, source operations, security and error testing apply throughout, not at the end.

Every feature's wireframe inventory includes populated, meaningful empty, partial/stale, failed/offline, permission-denied and pending/failed-save states where applicable. Test API fixtures are explicitly isolated from the configured shared backend. Snapshot tests preserve approved geometry; interaction tests verify actual states, keyboard/focus and mutation outcomes. New contracts have positive and negative schema tests, identity-switch races, malformed responses and retry tests.

Before handoff, run the repository's relevant logic tests, typecheck, lint, production build, browser suites and preservation verification. Add tests for each new behavior rather than claiming old stage counts cover it. Run controlled live integration against a separate environment or disposable identity with representative real source records. Verify physical devices, translation, source truth, privacy and destructive operations without touching a shared production household. Record failures and known limits honestly.

An approved archive contains source, assets, exact copy, state inventory and screenshots in both appearances at narrow widths. Never overwrite the onboarding archive or `docs/foundation/approved/2026-10-01`. Current Map changes stay unapproved until explicitly accepted. Completion means the acceptance plan (HALO_FEATURE_ACCEPTANCE.md) is satisfied for the released scope, not that every button produces a toast or every route renders.

Planning verification and remaining decisions

This inventory covers G01, T01 through T06, H01 through H08, F01 through F06, A01 through A06, M01 through M05, P01 through P06 and X01 through X07. Feature IDs are shared with the master and acceptance plans so relocation cannot hide omissions.

Still required before corresponding implementation: Yogi's fourth-tab and visual approval, agreed new schemas/operation contracts, validated source coverage, factor-score methodology if individual scores are introduced, private-file/retention and optional recovery choices, reviewed content/translations and deployment validation. These are explicit gates, not permission to substitute fixtures or remove features. This documentation adds no code, routes, backend records, migrations, archived screenshots or live-site changes.

06 / Feature acceptance requirements

docs/product/HALO_FEATURE_ACCEPTANCE.md

HALO Feature Preservation and Acceptance Plan

Version 1.1, October 2, 2026, prepared for Yogi and Vibhav. This document defines what must be preserved and what would constitute a complete implementation of the proposed full HALO product. It is a planning and acceptance document, not a statement that the new features are implemented, the new navigation is approved, or the live deployment has passed validation.

Read this with HALO Master Plan (HALO_MASTER_PLAN.md), HALO Frontend Build Plan (HALO_FRONTEND_BUILD_PLAN.md) and HALO Backend and Data Plan (HALO_BACKEND_AND_DATA_PLAN.md). The feature identifiers link the documents and remain stable when a feature moves to another page. Build order is not removal from the product scope. A feature identifier covers its pages, forms, states and underlying services, not just one screen.

October 3 source audit: the Backend Build Handoff (HALO_BACKEND_BUILD_HANDOFF.md) adds the complete feature-to-backend matrix and implementation tests. The Source Acquisition Guide (HALO_SOURCE_ACQUISITION_GUIDE.md) adds source-specific release gates. Require proof of permitted pollen history/map use, permitted geocoding persistence, fish-rule contradiction/truncation handling, food notice lifecycle, utility crosswalk/restriction instructions and source-version-aware comparisons. Public HTTP samples are not substitutes for integration, RLS, source-review or deployment tests.

Authority and preservation

Current user decisions override conflicting instructions or older visual proposals in the supplied specification. The specification remains a functional reference. The wireframe workflow (../WIREFRAME_WORKFLOW.md) governs design review, and the approved-stage handoff (../foundation/STAGE_FRONTEND_HANDOFF.md) records the production frontend contract and its known limits.

- Keep the immutable October 1 foundation archive at `docs/foundation/approved/2026-10-01` and its source baseline `72a2c5150ecfe354fa6b534781e8b6b090d0bda1` unchanged.
- Keep approved onboarding archives and the approved rounded mobile onboarding intact.
- A new product plan does not freeze new wireframes. New navigation, new screens, and material changes to approved screens need Yogi's visual review and explicit approval before freezing or production implementation.
- Preserve unrelated local changes. The Map is an unapproved review prototype, not an accepted production integration.
- Yogi is the sole frontend decision maker. Old assignment tables and parallel human lanes are superseded. Vibhav handles backend integration and development coordination.
- No app-facing em dashes, including returned content, errors, accessibility text, exports, notifications, and Luna responses.

Decision ledger

Table fields: Area / Current decision or recommendation / Acceptance consequence

App purpose

Current decision or recommendation: Environmental evidence, household relevance, action, and follow-up

Acceptance consequence: New features must identify a real decision or action, their evidence, and a useful follow-up. More screens alone are not a feature.

Navigation

Current decision or recommendation: Today, Homeguard, and Act are requested tabs; Plan is the recommended fourth tab

Acceptance consequence: Plan's label and the final tab order still require Yogi's review. Keep a migration map for existing URLs.

Map

Current decision or recommendation: Move Map beneath a main section while retaining its functionality

Acceptance consequence: Recommended home is Act, with direct contextual entry from Today, Homeguard, and Food & Garden. No capability loss from relocation.

Journal

Current decision or recommendation: Rework its purpose and placement without casually deleting useful functions

Acceptance consequence: Recommended home is Plan. Preserve original recording, history, and comparison capabilities until an explicit reviewed replacement is agreed.

Today

Current decision or recommendation: Latest name is Today, not Personal Hero

Acceptance consequence: Visible labels, document titles, announcements, exports, and links use Today. Legacy names may remain only in historical archives.

Today page depth

Current decision or recommendation: October 3, 2026: Today and its air, UV, pollen and mold details stay substantially as they are for now

Acceptance consequence: Preserve current sections, useful information, visuals and overall layout. No substantial shortening or hiding existing content behind extra taps solely to simplify. Substantial cuts or restructuring need a separate user decision.

PFAS

Current decision or recommendation: Homeguard only, not a Today factor or score input

Acceptance consequence: Tests must prove PFAS never contributes to Today geometry or totals. It remains reachable through water, Map, food advisories, and contextual Learn.

Homeguard score

Current decision or recommendation: Water and radon; lead remains shown and not scored

Acceptance consequence: New home records or completed tasks do not silently change existing score semantics. A later scoring change requires a separately reviewed versioned methodology.

Factor navigation

Current decision or recommendation: Whole cards navigate to dedicated pages, without dropdown arrows

Acceptance consequence: Older PDF expandable-card instructions are superseded by the approved stage. Keep raw reading, provenance, details, tips, and sources.

Household illustration

Current decision or recommendation: Selected categories only, beneath its heading and above home factors

Acceptance consequence: Do not draw unselected categories. Categories are not named people or a count of distinct people. A person may match more than one category.

General household advice

Current decision or recommendation: Children, teens, and adults may use general advice

Acceptance consequence: Existing approved category routes remain supported, but do not invent demographic distinctions merely to fill them. Dedicated guidance appears where evidence justifies it.

Respiratory label

Current decision or recommendation: Respiratory Issue(s)

Acceptance consequence: Replace the earlier Breathing label in visible UI while keeping stable backend category keys.

Tips

Current decision or recommendation: Overlapping step cards with previous and next arrows

Acceptance consequence: Next brings the next card forward. Provide a nonanimated and scan-all equivalent. Browsing a step does not mark it complete.

Factor styles

Current decision or recommendation: Slight whole-card tints, not gradients; unique matching scenes

Acceptance consequence: UV orange, air teal, mold green, pollen light yellow. No deer. Visual identity remains distinct from severity colors.

UV feature

Current decision or recommendation: Rising half-sun rather than a generic semicircle

Acceptance consequence: Preserve the approved rising-sun detail and its readable forecast information.

Luna

Current decision or recommendation: Name Luna, teal geometric mark, approved assistant composition

Acceptance consequence: Do not restore Ask HALO as the product name or copy another company's branding. Preserve source and privacy distinctions.

Onboarding

Current decision or recommendation: Transparent white logo on dark Welcome band, true cursive handwriting, then centered lowercase to and uppercase HALO

Acceptance consequence: No return to the blue logo tile, HALO-only header, Playwrite Argentina, or horizontal text wipe. Access to Get started never waits for animation.

Appearance

Current decision or recommendation: Follow the device; review controls are outside the product

Acceptance consequence: Do not turn the preview appearance switch into a product preference. Atmospheric day and night are separate from light and dark appearance.

Settings writes

Current decision or recommendation: Approved essential Settings uses explicit Save and retained drafts

Acceptance consequence: PDF autosave instructions do not override the approved production behavior. New form behavior must be consistent and reviewed.

Luna data use

Current decision or recommendation: Current backend context setting is not verifiable through the existing GET contract

Acceptance consequence: Never assume context is off when a flag is missing. New privacy controls require a working server contract and provider review.

Data growth

Current decision or recommendation: User accepts additional backend work

Acceptance consequence: This permits planning real capabilities, not presenting absent sources as available or silently implementing changes.

Twenty preservation groups from the existing specification

These are twenty audit groups, not a claim that the original PDF labels exactly twenty features. Together they cover its original functional areas and the later approved stage. The reference is HALO Frontend Specification (../foundation/reference/HALO_Frontend_Specification_2026-09-29.pdf), especially Parts A, B, and C. Physical page numbers below refer to that 83-page reference.

Table fields: Group / Capability to retain / Proposed home and identifiers / Source

1

Capability to retain: Anonymous session bootstrap, four onboarding steps, address or ZIP, optional GPS, household skip, home details, results reveal, resumable failure and address change

Proposed home and identifiers: Onboarding and Settings, G01 and X04

Source: B1, pages 32 to 35; stage handoff

2

Capability to retain: Daily overview, four stable factor positions, partial-score notices, contribution display, freshness, manual and return refresh

Proposed home and identifiers: Today, T01

Source: B2, pages 36 to 39; stage handoff

3

Capability to retain: Air reading and measured/modelled provenance, UV window, pollen categories, mold weather basis, history and factor-specific explanations

Proposed home and identifiers: Dedicated factor pages, T02 to T05

Source: B2; approved factor routes

4

Capability to retain: Household-relevant wording and selected household category guidance

Proposed home and identifiers: Today family and Homeguard guides, T06 and H05

Source: A4, page 17; approved family and household routes

5

Capability to retain: Homeguard overview, evidence coverage, separate home score, house visual and home factors

Proposed home and identifiers: Homeguard, H01

Source: B3, pages 40 to 46; approved household illustration

6

Capability to retain: Utility identity, compound rows, threshold and benchmark types, sample histories, dates, unregulated and excluded results

Proposed home and identifiers: Water detail, H02

Source: B3, pages 42 to 43

7

Capability to retain: Private-well and spring branches, prioritized testing reasons, cadence, locations, estimated costs and unknowns

Proposed home and identifiers: Water and Tests, H02 and H06

Source: B3, page 44

8

Capability to retain: County radon estimate and home-test distinction; lead building-era estimate and utility-wide service-line context

Proposed home and identifiers: Radon and Lead, H03 and H04

Source: B3, pages 44 to 45

9

Capability to retain: Ranked home action plan, reason, certification, estimated cost, renter and owner alternatives

Proposed home and identifiers: Homeguard and saved actions, H01, H07, P02

Source: B3, page 45

10

Capability to retain: Journal first-use retrospective calendar, selectable past rough days, coverage notice, immediate descriptive result and failed-date retry

Proposed home and identifiers: Journal, P04

Source: B4, pages 47 to 48

11

Capability to retain: Daily and past check-ins, group selector, symptoms, subjective severity, possible illness, notes, felt-fine entry, update, recent entries, delete and undo

Proposed home and identifiers: Journal, P04

Source: B4, pages 48 to 49

12

Capability to retain: Twelve-week heat grid, day details, environmental trend, finding counts and limitations, seasonal prompts and summaries

Proposed home and identifiers: Patterns and recap, P05 and P06

Source: B4, pages 49 to 51

13

Capability to retain: County and cause-matched organizations, scope, links, and human verification dates

Proposed home and identifiers: Act organizations, A02

Source: B5, page 52

14

Capability to retain: Practical civic guidance tied to water, radon and available facility evidence, plus useful general participation guidance

Proposed home and identifiers: Act issues and projects, A01 and A04

Source: B5, page 53

15

Capability to retain: Utility, Landlord, and Representative letters, independent editable drafts, cited reading, copy, native share, reset, and no-result branches

Proposed home and identifiers: Outreach, A05

Source: B5, pages 53 to 54

16

Capability to retain: Household health card with two scores, all factor readings, household notes, top actions or tests, dated sources, image save and clean print

Proposed home and identifiers: Household summary, H08

Source: B5, pages 54 to 55

17

Capability to retain: Household/home/renter settings, address change, reviewed language, accessibility, notifications, inspectable data, clear Journal, delete account, methodology and sources

Proposed home and identifiers: Settings and privacy, X04 to X06

Source: B6, pages 56 to 60

18

Capability to retain: Four Map layers, water compounds and quarters, selected records, home context, legends, geographic limits, district comparisons and accessible text fallback

Proposed home and identifiers: Map, M01 to M05

Source: B7, pages 61 to 67; Map README

19

Capability to retain: Contextual Learn with reading, meaning, household relevance, steps, sources, language and incomplete-data behavior

Proposed home and identifiers: Integrated throughout, X01

Source: C1, pages 68 to 69

20

Capability to retain: Cited assistant, refusal and unavailable states, privacy marker, alert inbox/read/dismiss/undo, typed notification preferences and opt-in device push

Proposed home and identifiers: Luna and alerts, X02 and X03

Source: C2 and C3, pages 70 to 73; B6; stage handoff

Retain all useful content while resolving outdated claims through reviewed source updates. Preservation does not require repeating an inaccurate statement. For example, missing private-well results means HALO lacks a result, not that no agency or person has ever tested the well.

Original twenty capability crosswalk

The original context document (../onboarding/reference/HALO_Codex_Context.md) has its own numbered twenty-capability list in section 2.4. This table maps those exact capability numbers independently of the twenty audit groups above. Historical statements about law, competitors, universal coverage, testing availability or completion are not accepted as current facts merely because they appeared in that document.

Table fields: Original number / Preserved capability / Target and necessary qualification

1

Preserved capability: Address to environmental picture without mandatory account registration or retaining the typed address

Target and necessary qualification: G01, X05. Preserve the original under-10-second flow as a measured target, not a promise that every provider responds within ten seconds. Partial results remain honest.

2

Preserved capability: Four daily factors and combined assessment with missing/estimated inputs

Target and necessary qualification: T01 to T05. Backend scoring and distinct data gaps remain authoritative.

3

Preserved capability: Utility water, county radon and building-era lead

Target and necessary qualification: H01 to H04. Lead remains unscored; estimates are not property measurements.

4

Preserved capability: Different water situations without absence-as-safety

Target and necessary qualification: H02. Test actual current discriminated states and coverage; the historical count of five does not limit new honest states.

5

Preserved capability: Private-well testing plan

Target and necessary qualification: H02, H06. Now includes a real record and follow-up lifecycle rather than a fabricated reading.

6

Preserved capability: Household-relevant wording without named people or ages

Target and necessary qualification: T06, H05, X01. Categories stay optional; new voluntary documents require truthful expanded privacy wording.

7

Preserved capability: Ranked practical actions with costs, certification context and renter/owner alternatives

Target and necessary qualification: H01, H07, P02. Preserve source review and user control; do not treat an estimated cost as a current quote.

8

Preserved capability: Statewide water/compound view and historical scrubbing where supported

Target and necessary qualification: M01, M02. Returned dates and coverage control the timeline.

9

Preserved capability: Symptom recording and supported patterns

Target and necessary qualification: P04, P05. Preserve recording while correcting denominator and measurement-basis interpretation.

10

Preserved capability: Source-cited household assistant with diagnostic refusal

Target and necessary qualification: X02. Actual provider context and privacy behavior must match disclosure.

11

Preserved capability: Contextual contaminant explanations, protective steps and sources

Target and necessary qualification: X01, T02 to T05, H02 to H04. New topics extend reviewed content, not unchecked AI text.

12

Preserved capability: Verified local organizations matched to findings

Target and necessary qualification: A02. Directory errors are not a successful zero-match response.

13

Preserved capability: Ready-to-review sourced letter

Target and necessary qualification: A05. All recipient modes retained; missing evidence produces an information request.

14

Preserved capability: Shareable printable household summary

Target and necessary qualification: H08. Select private fields explicitly and retain original observation dates.

15

Preserved capability: Public aggregate district evidence without personal data

Target and necessary qualification: M05. Permit civic exploration without address onboarding; existing anonymous-token authentication is distinct from a completed household profile.

16

Preserved capability: Meaningful changes with earlier eligible sensitive-group alerts

Target and necessary qualification: X03. Versioned server policy, genuine data changes and deduplication must be tested.

17

Preserved capability: Offline last-known evidence and reconnection

Target and necessary qualification: X06. Dated cached reads are not proof of offline mutations or a service worker.

18

Preserved capability: Complete English and reviewed Spanish experience

Target and necessary qualification: X04, X01, X06. Include UI, errors, exports, guidance and source-locale disclosure for the advertised translated scope.

19

Preserved capability: Keyboard, screen-reader and redundant severity communication

Target and necessary qualification: X06 and every screen. Test complete workflows as well as individual controls.

20

Preserved capability: Delete the household's stored data

Target and necessary qualification: X05, X04. Cover rows, files, queues and caches, with accurate backup/provider retention limits rather than an immediate universal-erasure promise.

Feature specific acceptance

Each feature is incomplete until it passes the shared requirements later in this document as well as its own criteria. These criteria are target behavior, not current test results.

Entry and Today

Table fields: Feature / Required behavior and evidence of completion

G01 Onboarding

Required behavior and evidence of completion: Preserve approved layout and handwriting; support address, US ZIP and voluntary GPS without automatic permission requests; keep data on Back and failed save; confirm ambiguous writes using request identifiers; use authoritative completion state; distinguish out-of-region coverage from lookup failure; continue honestly when a provider cannot return a result. Test a new home, changed home, same address, skipped household, private well, spring, unknown year, offline interruption, timeout after successful write, and identity replacement.

T01 Today overview

Required behavior and evidence of completion: Four factors only, fixed card order, real source timestamps, explicit partial coverage, unknown distinct from zero, proper refresh cooldown, cached-reading notice, and meaningful change comparison using compatible time, coverage and methodology. A source arriving after an outage must not be described as worsening conditions. Ring and contributions must not claim percentages of health risk or formula weights when only visual input shares exist.

T02 Air

Required behavior and evidence of completion: Show real AQI and dominant pollutant when provided, observation context, provenance, seven-day gaps, accessible chart data, and a link to the relevant monitor/map context. Save an applicable step into the shared action system and return to the same detail position. Unknown provenance is neither measured nor modelled.

T03 UV

Required behavior and evidence of completion: Preserve rising-sun composition; distinguish observed/current reading, daily maximum and forecast window; include timezone and forecast issue time when supplied; allow inspecting a supported time and saving a chosen step. A null window must not imply low UV all day. Do not fill unavailable forecast hours with interpolated values.

T04 Pollen

Required behavior and evidence of completion: Show tree, grass and weed separately with units or category definitions, source coverage and dominant category. Changing category updates its chart and accessible summary together. Seasonal text is reviewed and geographically appropriate, not a replacement for current data. Link to optional Journal without suggesting pollen caused a reported symptom.

T05 Mold

Required behavior and evidence of completion: Explicitly label the current weather-based estimate; show humidity/rainfall basis only when present; offer moisture-check guidance and records without claiming indoor spore detection. If users later enter sensor readings or observations, maintain their separate origin, units, timestamps and scope.

T06 Daily family

Required behavior and evidence of completion: Selected categories and general guidance both work; combine applicable messages without duplicate instructions or conflicting priorities; explain which environmental evidence prompted a suggestion; save the same action record used elsewhere. Skipping household selection must not disable the useful daily app.

Homeguard

Table fields: Feature / Required behavior and evidence of completion

H01 Homeguard

Required behavior and evidence of completion: Preserve home score semantics, separate evidence gaps, selected-category house, water/radon/lead routing and ranked actions. Clicking a house figure opens its category guide with a reliable return path. Private water has a testing pathway, not an invented water or combined score. A completed action does not change a measurement or score without new evidence.

H02 Water

Required behavior and evidence of completion: Show matched utility and mapping limitations; support complete, partial, no detections recorded, excluded-only, unscoreable, no published result, failed lookup, no utility match and private-source states distinctly. Retain compound units, sample dates, history, enforceable limit versus benchmark, unregulated/excluded evidence and source revision. Follow the utility, inspect on Map, start a test, or prepare outreach using the exact selected record. Do not label a detection-only source as complete testing coverage.

H03 Radon

Required behavior and evidence of completion: Keep county potential separate from a household concentration. Testing workflow supports planned, awaiting result, result recorded, review and follow-up states. Result units, method, start/end dates and source are required as appropriate. Guidance and reminder timing must be source-reviewed; a saved test suppresses an obsolete no-test reminder only under an explicit rule.

H04 Lead

Required behavior and evidence of completion: Keep age estimate and utility service-line totals distinct from property evidence; show unscored status permanently. Unknown year is not newer construction. User-supplied or verified property information is separately identified. Routes to home correction, testing, certification guidance and renter outreach retain context.

H05 Household guides

Required behavior and evidence of completion: Only selected categories appear in the house; direct URLs for unavailable categories have a clear general-guidance or edit-selection explanation, not fixture people. General child/teen/adult guidance remains supported. Clinical-sounding age-specific claims need reviewed sources. Category overlap must not become a false count of residents.

H06 Tests and records

Required behavior and evidence of completion: Create, edit, archive/delete and inspect a test record; retain test type, purpose, dates, lab/provider or user origin, specimen/site context, units, result qualifiers, reporting limits, attachments if supported, review status and related actions. Draft, ordered or arranged, sampled, awaiting result, recorded and follow-up are distinct. Pending tests are not negative results. Uploaded extraction is unconfirmed until the user verifies it.

H07 Maintenance

Required behavior and evidence of completion: Save specific equipment or upkeep information only when useful; retain installation/service dates, product or source instructions, user-chosen cadence, last completion and next due. Snooze, skip, complete, change cadence and stop reminders without losing history. A filter replacement reminder neither verifies the product's certification nor proves contaminant reduction.

H08 Household summary

Required behavior and evidence of completion: Preview and choose included evidence; retain original two-score/all-factor/source/top-action capability; identify missing, old, estimated and user-reported information; private-water version shows tests rather than no action. Print cleanly in light black-on-white with no app chrome, use readable labels rather than color alone, and support image export with permission/error fallbacks. Export generation date must not replace observation dates. Sharing is explicit and excludes sensitive additions by default.

Food and Garden

Table fields: Feature / Required behavior and evidence of completion

F01 Food and Garden overview

Required behavior and evidence of completion: Show concise entrances to fish, gardens, local food, grower resources and water stewardship; optional interest selection changes relevance, not scientific conclusions. Surface only real notices and records, with coverage and dates. Empty regional coverage offers useful source and testing pathways rather than fabricated local findings.

F02 Fish advisories

Required behavior and evidence of completion: Resolve a selected waterbody and species against the actual advisory geography and audience categories. Distinguish statewide from specific advice, effective from superseded records, no matching advisory from known unrestricted status, and source outage from no advisory. Show the exact official recommendation with units and qualifications; save/follow it and alert on meaningful revisions. Existing household category labels must not silently approximate advisory age or audience definitions. Optional catch/consumption notes support private create/edit/delete/export with original dates, species, waterbody and quantity/unit; they do not compute dose or remaining allowances and are not required for advisory access.

F03 Garden records and guidance

Required behavior and evidence of completion: Optional setup captures relevant growing-site, water-source and test context without requiring names or exact residential coordinates. Guide the user to suitable sampling/resources, connect test results and chosen stewardship tasks, and retain useful notes over time. Routine fertility testing is not contaminant testing. Regional water or soil data cannot become a measurement of the plot or crop.

F04 Local food information

Required behavior and evidence of completion: Distinguish local official notices, product/sample-specific tests, broader monitoring and research. Filter by supported geography or product category without claiming every local food is covered. Users can inspect source and sample date, save/follow a specific notice, prepare questions and record their own relevant source-linked information. No universal PFAS food score, product absence-as-safe claim, or unsupported farm contamination inference.

F05 Grower resources

Required behavior and evidence of completion: Support genuinely useful water/soil sampling preparation, relevant Extension and testing contacts, source-linked records and conservation program details. Separate growing sites only where needed, with accessible labels and optional coarse location. A grower can save a resource, prepare questions, attach a confirmed test and plan a follow-up. Do not give crop-specific agronomic prescriptions without suitable evidence and review.

F06 Water stewardship

Required behavior and evidence of completion: Show dated US Drought Monitor county category distributions, or a point classification only when a valid dated source geometry supports it; never use the county maximum as the home's status. Separately show source-dated utility conservation notices where verified ingestion exists. Regional drought, enforceable or voluntary utility notice, and plot moisture are different facts. Optional meter/bill/irrigation records require units, period, source and edit history. Compare only compatible measured periods; identify estimated entries and interval length. Report measured usage difference, not savings caused by HALO or a task without a validated attribution method. Do not infer moisture or irrigation quantities from a regional drought class.

Act and Map

Table fields: Feature / Required behavior and evidence of completion

A01 Act hub

Required behavior and evidence of completion: Keep original organizations, ways to help, letters and household summary reachable while introducing conservation depth. Personalization explains relevance but does not prevent browsing another cause. Local issue pages connect evidence to organizations, projects, outreach and Map. Good or missing readings do not leave the page empty.

A02 Organizations

Required behavior and evidence of completion: Preserve county/served-area and cause matching, actual contact/site links, human verification status and date. Working URLs are not proof of organizational verification. Distinguish no matches, unsupported geography, stale records and failed request. Users can broaden filters and report stale listing information without creating a public accusation.

A03 Opportunities

Required behavior and evidence of completion: Distinguish an organization from a dated opportunity. Show organizer, actual place/service area, time and timezone, requirements, accessibility information when supplied, cost, age restrictions and registration link. Saving to Plan does not register; clicking registration does not prove completion. Support cancelled, full, postponed, date changed, unknown capacity and expired states, with change notices for saved events.

A04 Stewardship projects

Required behavior and evidence of completion: A project has a reviewed purpose, suitability, materials, real ordered steps, constraints, completion criteria and follow-up. Save once into Plan, record progress and stop or revise it. Count user-reported actions as participation, not measured pollution removal or health improvement. Source-backed protocols are required before describing observations as monitoring data.

A05 Outreach

Required behavior and evidence of completion: Keep Utility, Landlord and Representative recipients with separate draft edits and current-recipient reset. Show cited evidence and applicable legal versus benchmark status. Include information-request/testing variants when no exceedance is established and hide irrelevant utility templates for wells. Copy and share have permission, cancellation and unsupported-device fallbacks. An outbound click is not sent status; sending and replies are user-confirmed unless a verified integration exists. Recipient discovery and legal guidance require reliable current sources.

A06 Private field observations

Required behavior and evidence of completion: Users can privately record a date/time, observation category, factual description, optional coarse location and optional photo. Do not infer a contaminant from appearance or odor. Provide export and an official-reporting handoff, not automatic submission or publication on a shared map. Strip or explicitly protect precise photo metadata and exact location; preview what leaves the app. Status distinguishes draft, exported, official site opened and user-confirmed report. Emergency situations point to appropriate official channels, not Luna.

M01 Map shell

Required behavior and evidence of completion: Preserve four layer controls, pan, zoom, touch/pinch, keyboard controls, locate, home context, attribution, legend, selection, independent layer failure and renderer fallback. Accessible searchable text list works without the map and includes records lacking coordinates. Lazy loading and responsive layout preserve the rest of the app.

M02 Water and timeline

Required behavior and evidence of completion: Preserve compound selectors, Overall and hazard index, own-system ring, population/null handling, complete selected-record detail, available quarters plus distinct Latest, loading retention, cancellation and failed-quarter restoration. Latest is each system's latest observation, not a single synchronized date. Empty quarter differs from no stored detections. Historical sample points do not describe a moving plume.

M03 Other layers

Required behavior and evidence of completion: Radon county polygons show zone context, never home concentration; facilities preserve selected-dataset scope, compliance-program counts, inspection/date/null and grouped points; air stations show observation time, pollutant/source and approximate distance. Missing or filtered-out dots never mean clean geography.

M04 Comparisons

Required behavior and evidence of completion: Compare only like units, periods, evidence kinds, denominator definitions and geographic scopes. Display coverage and date differences before interpreting them. Household rank uses its actual ranked denominator and hides when unranked. Public-system comparison never implies residents can choose their supplier. A smooth density display is labelled record density, not pollution intensity.

M05 District

Required behavior and evidence of completion: Preserve district/state summary, county rows including partial-county tags, compound-specific denominators, affected-population completeness, unlocated records and boundary version. Avoid misidentifying a household district from a partially included county. System centroids do not establish exact service-area membership. Geography failure is distinct from zero exceedances.

Plan and shared systems

Table fields: Feature / Required behavior and evidence of completion

P01 Your Week

Required behavior and evidence of completion: Show saved actions, routines, tests, events and optional outdoor plans in one agenda with accessible date navigation. Permit unscheduled saves. Preserve all-day dates versus timed events, timezone changes and daylight-saving transitions. Rescheduling an item updates the shared record, not an unrelated calendar copy. Clearly label a calendar export as an export unless synchronization actually exists.

P02 Shared actions

Required behavior and evidence of completion: Save from any relevant screen; deduplicate the same action under explicit rules; retain origin, reason, evidence version, user edits, chosen due date and status. Create, edit, schedule, snooze, complete, reopen, archive/delete and review changed source guidance. Duplicate taps, retries and unknown-write outcomes do not create duplicate reminders or false completion.

P03 Outdoor planning

Required behavior and evidence of completion: Ask only useful activity, place, date and time information; show supported per-factor forecasts, issue time, horizon, uncertainty and missing periods. Compare practical options with tradeoffs rather than an unvalidated safest-time score. Expired forecasts require review. No continuous background location tracking or inference that a user was exposed at a planned place.

P04 Journal

Required behavior and evidence of completion: Preserve first-use retrospective, Log today instead, past dates, group selector, seven symptom labels, severity without a default, possible illness, optional note, felt-fine action, recent entries, prefilled update, visible delete and undo. Guard against blank overwrites while entries load. Retrospective batch retry sends only failed days and preserves existing records; a blank day is not a felt-fine day. A useful immediate result can be descriptive environmental context, including insufficient data.

P05 Patterns

Required behavior and evidence of completion: Preserve twelve-week heat grid, missing-factor mark, accessible legend, per-day readings and logged entries, edit-from-day detail, score trend, finding bars and counts, and optional group filtering. Logged counts, comparable counts and environmental-history counts are distinct. Revised symptom comparisons must distinguish unlogged and explicitly fine days, retrospective recollection, illness flags and missing readings. Show sample sizes, method and limits; no causation or diagnostic conclusion.

P06 Seasonal recap

Required behavior and evidence of completion: Retain season summaries and dismissible relevant prompts without streaks or pressure. Alert links open the actual recap. Show period, logged-day coverage, environmental coverage, excluded days and limits. Show completed actions as user-reported follow-through, not proof that those actions changed health or environmental conditions.

X01 Learn

Required behavior and evidence of completion: Retain reading-specific explanation, what it is, why this reading, household relevance, concrete protective steps and dated source links. Integrate it into factor/feature details rather than recreating a competing tab. Omit unsupported personalized sections instead of filler. The proposed glossary/topic index is a new addition, with entry from contextual Learn or Sources and methods, search over reviewed topics, clear no-match behavior and return-to-context. New food/garden topics require new supported content contracts and reviewed translations.

X02 Luna

Required behavior and evidence of completion: Preserve approved name/visuals, suggestions, grounded replies, citations, refusal, no-source, unavailable, unconfigured, rate-limit, cancel and retry states. New conversation context and action tools are explicit backend extensions. Server derives authorized household context; show real controls and provider-use status before implying privacy. Draft actions require preview and confirmation. Cancel/retry does not duplicate a message or execute an action twice. Do not promise diagnosis, treatment or arbitrary live web knowledge.

X03 Alerts and watchlists

Required behavior and evidence of completion: Preserve unread badge, read one/all, full-message expansion, explicit related-page link, dismiss and undo. Support the original five preference types plus separately reviewed subscriptions/reminders. Meaningful changes deduplicate by event and evidence version. Marking an alert read differs from acknowledging an action; opening inbox alone does not clear all unread items. Push is opt-in and unsupported browsers still have an inbox.

X04 Settings

Required behavior and evidence of completion: Retain full household edit, home details, confirmed address change, renter/owner mode, accessibility, notification preferences, reviewed language, privacy, About, dynamic source list and methodology. Explicit saves preserve drafts and untouched fields across refetches. A household replacement always contains all seven booleans. Unknown profile state disables writes. No product appearance picker unless separately approved.

X05 Privacy and recovery

Required behavior and evidence of completion: Inspect stored values and categories, export selected/all supported data, clear Journal with a real count, and delete all household data with typed confirmation. Deletion covers new tables, private files, queues, subscriptions and derived caches, with honest provider-retention limits. Optional recovery or device linking requires dedicated secure design and consent; anonymous use remains viable.

X06 Offline and accessibility

Required behavior and evidence of completion: Cached evidence keeps original source dates and visible offline status; new private records are not silently cached. If offline drafts or queues are implemented, label pending/unconfirmed state, provide cancel, and synchronize only for the same verified identity. Every key action has keyboard and screen-reader equivalents; reduced motion and large text remain functional.

X07 Source operations

Required behavior and evidence of completion: Each external feed and curated resource has an owner, coverage, update expectation, source date, last successful fetch, review status and failure behavior. A feed outage cannot publish an empty replacement as clean conditions. Changed or retracted advice links to affected saved actions for review. No user-facing monitoring promise before source access and refresh reliability are demonstrated.

Shared interaction acceptance

Navigation and continuity

- Direct links reconstruct authorized data and open the intended subpage or section. Invalid factors, household categories, records, quarters and return URLs have safe behavior, never fixture fallbacks or open redirects.
- Browser Back closes a sheet before leaving the parent page. Closing restores focus to the opener and the underlying scroll position. A new sheet replaces the existing sheet, not an inaccessible stack.
- Cross-feature navigation carries stable record identifiers and public filter state, not household details, notes, credentials, exact home coordinates or transcript text in the URL.
- Map opened from a compound, monitor, advisory or organization returns to the initiating feature with the relevant context intact. A direct Map visit has a predictable default.

- Build `/journal` and `/map` as the stable planned destinations; they are not currently implemented production pages. Preserve applicable legacy notification/query formats through validated migration or aliases. Notification links, saved bookmarks, exported links and Settings links receive migration tests.
- Extend StageApp's live-route recognition, safe return allowlists, active-tab ownership, page titles and assistant context tokens together with every new route. A new page file that renders outside the shared gate is not an integrated page. Unknown destinations remain safe errors, not arbitrary redirects.
- Public Map/district and civic browsing have a tested no-home-profile mode. Existing anonymous authentication is reused or a separate public-read policy is explicitly approved. No address, household categories, home pin or personalized ranking is required or invented. Private records still enforce owner/profile requirements.
- A source or organization link opens safely and does not imply that HALO owns, verified, or submitted content on that destination.
- Long forms warn before discarding unsaved changes when appropriate. Returning from a source, native share dialog or external registration must not erase the draft.

Honest mutation behavior

- Explicit user intent is required for save, schedule, subscribe, upload, share, send/handoff, clear and delete. Following a suggestion or reading a step does not create a commitment automatically.
- Loading actions prevent accidental duplicate submission but preserve layout and readable status. Validation identifies the field and retains all values.
- Confirm success only from an authoritative mutation result under the same identity. Unknown outcomes are reconciled with an idempotency key or status lookup before retrying.
- Revisions detect conflicting edits rather than overwriting a newer device's record. Server ownership is derived from authentication, not a household ID supplied in the request body.
- Recurring actions disclose calendar-anchored versus completion-relative timing. Completing late or skipping must not move a fixed-date series. Duplicate scheduler/completion events create only one next occurrence, and this occurrence versus future-series edits have distinct outcomes.
- Home-context changes retain historical test/location associations. A failed lookup for a genuinely new home cannot attach the old utility to its new coordinates. Unknown legacy context remains unknown. Validate atomic retries, timezone boundaries, portable-task confirmation and subscription review.
- Saved outdoor plans retain location/timezone, chosen interval, duration and forecast revisions. An opted-in change watch applies only to that plan's supported factors, area and periods. Updated forecasts can request review but never silently move the event.
- A failed save stays actionable with its draft, not only a disappearing toast. Confirmations and Undo may use a toast, but important errors remain visible.
- Delete one Journal entry retains the full entry for Undo. Clear Journal and Delete everything have separate scope and irreversible-action confirmations.
- Never announce that nothing was deleted after a partial or unknown multi-step failure. Reconcile the result and state what is known.

Data states

Every relevant page, panel and independently fetched section must distinguish:

Table fields: State / Required meaning and response

Loading first data

Required meaning and response: Stable skeleton or labelled progress, not plausible placeholder readings.

Updating with prior data

Required meaning and response: Keep dated prior evidence visible and indicate refresh; do not reset to zero.

No user records

Required meaning and response: Invite a useful optional first action; do not claim environmental data is unavailable.

No source coverage

Required meaning and response: Explain the geographic, temporal or category coverage limit and offer a genuine alternative.

Source has no matching result

Required meaning and response: State that absence precisely; never imply safety or that testing was performed.

Valid measured zero

Required meaning and response: Preserve zero and its units as actual data.

Non-detect

Required meaning and response: Preserve reporting qualifier and detection/reporting limit when available; do not turn it into measured zero.

Partial evidence

Required meaning and response: Show usable data and missing components independently.

Stale evidence

Required meaning and response: Keep source observation time visible, distinguish fetch time, and explain relevant limits.

No data at all

Required meaning and response: Show no-data score/reading, not a favorable score built from nothing.

Failed request

Required meaning and response: Explain retryable failure without dumping raw provider/database diagnostics.

Offline with cache

Required meaning and response: Show saved evidence and its age; disable or explicitly queue unsupported mutations.

Offline without cache

Required meaning and response: State that the view is unavailable, with useful existing offline information only.

Pending user action

Required meaning and response: Show planned/awaiting/unsent state, not a completed test, registration, observation report or task.

Unverified record

Required meaning and response: Show user-reported or unreviewed origin; never style it as a verified measurement.

Expired or withdrawn source

Required meaning and response: Retain historical context where appropriate, mark it, and avoid offering outdated guidance as current.

Identity cannot be verified

Required meaning and response: Private data and edits are unavailable; do not replace the identity just because a service returned an availability error.

All asynchronous screens need cancellation and out-of-order-response tests. Changing a factor, date, map layer, identity, household or location must prevent late responses from overwriting the new context.

Existing protocol preservation

- Preserve additive contracts. Do not rename, remove or retype existing fields when introducing the proposed evidence records, action routes or new screens. Introduce explicit versions/adapters where a semantic change cannot be additive.
- Use canonical `severity` values `good`, `moderate`, `elevated`, `high`, `severe` and `no_data`; an unknown severity becomes `no_data`. Never use legacy `status` as a severity. Water's `water.status` remains a valid branch discriminator, and lead's `level` remains its documented severity field.
- Bind water sample dates through `*_iso` fields. Calendar dates, timestamps and local forecast times have distinct parsing/formatting rules. Preserve raw analyte values and units; do not translate codes, make a non-detect zero or use retrieval time as sample time.
- Format fractional district shares as percentages exactly once; Journal rates already expressed from 0 to 100 must not be multiplied again. Lithium display conversion from ppt to micrograms per litre is explicit and tested against its source units. Population counts remain complete, with unknown and incomplete coverage qualified rather than rounded into unsupported precision or exposure counts.
- A definitively invalid identity and an unavailable auth service have different recovery paths. A retryable 503 cannot create a new anonymous household. A 401 recovery cannot silently pretend old valuable records were restored. Recheck identity around writes and unknown outcomes.
- Daily refresh rate limits retain usable readings and show the returned cooldown honestly. History range clamps, Journal note/date/band constraints, full-replacement household writes and whole-record Journal upserts are tested against the actual current route, not only an old PDF example.
- Preserve private-well/spring, failed lookup, no match, no published data, excluded/unscoreable and usable measurement branches without collapsing all into no data. Coverage and score inclusion remain distinct.
- History exposes stored measurement basis and backfill metadata. Daily-peak modelled history is not silently plotted or analysed as equivalent to an instantaneous measured value. Incompatible periods are labelled, split or excluded under reviewed rules.

- Journal first-use detection uses authoritative all-time record presence, not absence in a 90-day range. Test a returning user with only older entries, pending onboarding backfill, partial history and retrospective retries over existing rich entries. Preserve or explicitly revise the 90-day regular-entry window, 12-week grid, seven recent entries and 62-date retrospective maximum.
- Map uses actual returned quarters and source versions, not a permanently hardcoded 2023 to 2025 list. An empty quarter list hides the unavailable timeline. District ratios retain their exact denominator and unknown population states.
- Keep earlier air-alert eligibility for the server-defined sensitive groups, with current policy fixtures and versioning. Existing weather coverage is flood/heat, not an implied active boil-water feed. New subscription/channel semantics must preserve or explicitly migrate existing preferences.

Relevant action and synchronization acceptance

- Recommendation ranking is deterministic and explainable, with source freshness, applicability and current task state. No medical-risk ranking is delegated to unvalidated model output.
- Save from two different factor/category entrances resolves to the intended single canonical action unless a deliberate repeat is requested. Dismissal, already doing this and completion are different states and can be revised.
- An official notice, a severe measurement and an overdue task retain distinct meanings. Lack of measurement is not treated as a severe measurement or healthy reading.
- Every mutation invalidates the relevant dependent views described in the master plan. In particular, Journal edits recalculate applicable summaries, source corrections mark saved evidence for review, and event cancellations update linked Plan records without losing their history.
- File upload retry, abandonment and record deletion do not leave publicly readable or permanently orphaned attachments. Only authorized confirmed attachments become available in exports or selected Luna context.
- Current and new feature gates cannot expose a fixture fallback when a source, migration or provider configuration is absent. Incomplete features remain clearly unavailable in production rather than apparently functional mock screens.

Accessibility and visual acceptance

- Test 320px and 375px mobile widths, typical larger phones, desktop, portrait and usable landscape. No horizontal page overflow or clipped controls at supported text settings.
- Verify light and dark appearances, higher contrast, full motion and system/device reduced motion. Current approved controls support 100%, 115%, 130% and 150% text sizes; an older PDF also mentions 175%. Do not claim support for that additional step until designed and tested. Also test browser zoom independently.
- Severity always has text and an icon or shape as well as color. A factor's brand tint must not be mistaken for its risk severity.
- Charts have labelled units, a short descriptive takeaway and accessible data. Missing periods remain gaps. Keyboard and touch inspection select the same record.
- Calendar grids provide meaningful dates, selected state, disabled future dates, arrow-key behavior and an equivalent list where needed.
- The interactive house exposes category names, selection and navigation without depending on recognizing a silhouette. Do not encode sensitive categories in public image descriptions or share previews by default.

- The stacked tip cards support visible previous/next controls and a scan-all route. Reduced motion shows the new front card immediately and keeps focus predictable.
- Map controls and lists support keyboard use, meaningful labels, zoom alternatives and visible focus. A failed canvas or map renderer does not destroy access to records.
- Primary and core interactive controls retain the original 44px minimum target requirement. Any intentional exception needs explicit accessibility review rather than quietly weakening the rule. Swiping, dragging, hovering and color cannot be the only way to act.
- Screen-reader live regions announce important results once, without rereading an entire dashboard or every animated number.
- Mobile keyboard and safe-area tests cover Luna, letter drafts, Journal, new records and bottom navigation. Send/Save must remain reachable on physical iPhone Safari.
- Ambient scenes pause when offscreen and under reduced motion. They do not announce actual local weather, particles, measured contamination or resident exposure.
- Prolonged nonessential motion can also be paused by the user; reduced-motion device preference is not the only control. Verify the complete processes against the WCAG 2.2 AA target, not just an automated color check.
- New screens reuse approved tokens, typography, spacing, logo, rounded navigation and restrained motion. They use task-appropriate layouts rather than putting every paragraph into another card.
- All app copy, notification bodies, source labels, exports and assistant rendering are scanned for prohibited em dashes without changing scientific meaning or required units.
- All new app-owned strings use the translation catalogue, including screen-reader text, validation, chart descriptions, empty states, print/export labels and notification templates. Reviewed Spanish must be complete for its advertised scope. Requested locale, served-source locale, page language, dates, number formatting and fallback labels agree.
- Audit and retain the approved mark across favicon/app-icon sizes, manifest and generic social metadata. No private household information enters metadata or a public share image. Device appearance theme-color and supported standalone display do not obscure navigation or keyboard controls.
- Browser installation and push are tested on their actual supported browser configurations. Service-worker updates do not strand users on an incompatible API client or cache private authenticated responses by default. Existing environmental caches do not count as proof of an offline app shell or successful background push.

Performance and resilience budgets

Retain the original specification's frontend targets as measured release goals: first useful content under 1.5 seconds, reading an existing local cache under 200 milliseconds, an already-loaded tab transition under 100 milliseconds, and initial map rendering under 2 seconds on a declared representative test profile. Record device, network, cache and cold/warm conditions; these targets are not current performance claims or guarantees under every network. Backend request-latency and source-refresh budgets in the backend plan are separate measurements.

Load the map renderer and heavy assets only when needed. Bound list/history payloads, pause offscreen scenes, avoid layout movement during pending actions, and measure memory after repeated sheet/route navigation. A slow optional source cannot block unrelated useful content. Retry limits, offline reconnection, provider quota exhaustion and a failed background job must preserve honest state rather than spin indefinitely.

Privacy and safety acceptance

- Collect a field only when its use is explained. New Food & Garden interests, household equipment and observation records do not lengthen mandatory onboarding.
- Existing approximate location, household categories and symptom records remain private. No new feature silently changes the promise that names, birthdates and raw street addresses are not required or routinely stored.
- Optional files, free text and photos may contain identifying information. Document retention, restricted access, metadata handling and export/deletion before allowing uploads.
- Reconcile privacy notices with optional lab documents, fish notes, garden records, outreach and field images. Do not retain an absolute claim that the app never stores medical or identifying material once users can deliberately upload it. Onboarding address disposal and voluntary document contents are different data flows.
- Disclose anonymous-session loss and actual recovery options before the first valuable upload, not only afterward. Continue-anonymously remains available without implying that browser storage can be recovered after it is cleared.
- Shared household recovery or device linking, if selected, includes expiring credentials, revocation, rate limits, ownership checks and no guessable household lookup. It is not implemented by placing a session token in a URL.
- Luna's privacy indicator reflects actual data sent, not only whether an answer happened to cite household data. Household context, transcript retention and third-party processing need separately testable controls.
- Refusal and source checks cover diagnosis/treatment prompts, false citations, malicious instructions embedded in retrieved sources, and requests for another household's data. No-source is a valid answer.
- Official advisories are quoted or summarized within source permissions and kept separate from HALO interpretation. Legal status, certification claims, health thresholds and audience-specific advice require current authoritative review.
- Expanded health-related interpretations and public Luna/product claims receive qualified review before release. A disclaimer is not evidence of regulatory exemption or clinical validation. Do not claim HALO is the first or only app without a separately supportable basis.
- Alerts do not claim continuous monitoring or emergency coverage when a source is refreshed periodically. The app explains refresh cadence and provides official emergency channels where relevant.
- Export previews disclose all selected content. Photo metadata and exact observation coordinates are stripped or deliberately retained only after explicit informed choice. Default export must not reveal exact home location or symptom notes.
- A field observation is not a diagnosis of pollution and is not an official complaint. Clicking an agency link is not a completed submission. HALO does not publish private observations to a community map.
- Clear Journal removes associated findings and derived private caches. Delete everything includes all newly introduced records, files, reminder jobs, subscriptions and recoverable server relationships under an explicit retention policy.
- No analytics that capture household health, location, notes or behavior are introduced silently. Existing no-behavioral-analytics expectations must be honored unless explicitly reconsidered with clear consent and review.

Connected journey acceptance

Each journey must pass with real supported data in a separate test environment, then with controlled fixtures for adverse states. Success means the user understands and completes the action, not that every screen rendered.

Utility result to household follow up

1. Open H02 from H01 and identify the utility, source date and selected compound.
2. Inspect the same record through Learn and Map without changing its units or legal status.
3. Choose a feasible renter/owner action and save one P02 record.
4. Generate A05 for the chosen recipient with an evidence preview; retain independent recipient edits.
5. Schedule follow-up in P01, then include only selected evidence in H08.
6. Receive a meaningful new-source notice, review the saved action and keep the original reason available.

Pass conditions: no false address-level water claim, no duplicate action, no invented legal exceedance, no automatic sending, no score improvement from completion. Repeat with no result, a benchmark-only result, a failed lookup and outdated evidence.

Private water and garden testing

1. Onboard a well household with unknown construction year.
2. Open the testing plan rather than a fabricated water score.
3. Add optional garden water context and choose an appropriate reviewed testing/resource path.
4. Save a test, record sample dates, mark awaiting result and later confirm result values and qualifiers.
5. View result alongside, not substituted for, regional evidence; choose a reviewed next step.
6. Update reminders without claiming that a missing or non-detect result proves all water or produce is safe.

Pass conditions: pending is not negative, reported units and limits survive round trips, private uploads remain private, information about the water sample does not become a crop measurement.

Daily conditions to outdoor planning

1. Open Today, inspect a factor's real data and forecast scope.
2. Compare available times using P03 without filling missing hours.
3. Save a user-chosen activity and one appropriate action.
4. Receive a chosen update only when the applicable forecast or official notice changes meaningfully.
5. Optionally log how the day felt and inspect the day's evidence in P05.

Pass conditions: no promised safe period, no assumption the activity happened, no causal health conclusion, and the full flow remains useful without a Journal entry.

Fish advisory to conservation participation

1. Select a waterbody and species, then the applicable official advisory audience.
2. Inspect the advisory geography on Map and its source/version.
3. Follow the advisory and save any chosen guidance.
4. Find a relevant organization and a genuine opportunity with current details.
5. Save the opportunity, register externally, then explicitly confirm registration or attendance if desired.

Pass conditions: household age groups do not misclassify advice, saving is not registration, cancelled events update, and participation is not described as removing a specific contaminant.

Private observation to official handoff

1. Record a factual observation, optional approximate location and optional photo.
2. Inspect what metadata will be kept and shared.
3. Export selected content and open the appropriate official reporting channel.
4. Optionally mark an external report submitted and set a follow-up.

Pass conditions: the record remains private, no automatic accusation/report occurs, no precise hidden metadata leaks, the official channel is verified, and a link click never changes status to submitted.

Long term records and privacy

1. Save Journal entries, tests, plans, equipment and an observation using a disposable household.
2. Change household categories and home location; confirm current guidance refreshes while history retains the correct original context.
3. Export selected data and verify it has the intended fields only.
4. Clear Journal and verify unrelated home records remain.
5. Delete everything and verify server rows, private attachments, push subscription handling, queued jobs, local caches and session cleanup.

Pass conditions: no cross-household access, no history silently relabelled as belonging to a new address, no unknown result presented as deletion success. Never run this journey against the configured shared production project.

Test fixtures and monitoring proof

The fixture library must include normal, real-zero, non-detect, missing, partial, stale, excluded, failed, unsupported-region, identity-change and malformed-response cases. New feature fixtures include cancelled events, withdrawn advisories, amended test reports, mismatched units, duplicate saves, changed source versions, local date boundaries, daylight-saving changes, abandoned uploads and unknown mutation outcomes.

Source ingestion acceptance requires more than displaying a sample record:

1. Prove access, permission, source terms, geographic matching and required fields on representative records.
2. Run the proposed refresh schedule in a test environment and measure actual update latency and failures.
3. Validate record deduplication, amended/superseded versions and out-of-order updates.
4. Simulate source outage, schema change, implausible units and empty response. Keep the last good dataset with an honest status instead of replacing it with empty healthy conditions.
5. Prove a genuine source change creates the intended single update, reaches an opted-in household, and deep-links to the exact revised evidence.
6. Confirm recovery does not flood the inbox with duplicate historical alerts.

Do not claim that a feature monitors real things until this sequence works against the actual authorized source. Manual review workflows must show their true review cadence and cannot be labelled live.

Release gates

Table fields: Gate / Required evidence before advancing

Product scope

Required evidence before advancing: The full preservation matrix and proposed additions are reviewed; unresolved decisions are visible; no capability silently vanishes during navigation changes.

Source feasibility

Required evidence before advancing: Every promised monitored source has a demonstrated access path, useful coverage, rights/terms review, update expectation and honest fallback.

Backend contract

Required evidence before advancing: Versioned payloads, units, status vocabulary, ownership, mutation idempotency, deletion and source metadata are agreed. No placeholder field is marketed as live.

Wireframe review

Required evidence before advancing: Final-looking mock screens cover normal and adverse states plus connected flows. Yogi reviews new arrangements and refines them before approval.

Visual preservation

Required evidence before advancing: Explicit approval creates a new immutable archive; old archives remain unchanged. Source, copy, assets, state inventory and both appearances at narrow widths are preserved.

Implementation

Required evidence before advancing: Approved geometry is reused with live adapters. Unit/contract tests, typecheck, lint, build and isolated interaction tests pass; no fixture data leaks into live routes.

Integration

Required evidence before advancing: Real source and deployed API behavior is checked with a separate backend or disposable identity, including authentication, quotas, cancellation, refresh and mutation outcomes.

Accessibility and devices

Required evidence before advancing: Physical mobile Safari and a second browser are checked for focus, keyboard, charts, calendar, large text, offline behavior and native sharing/printing.

Privacy and safety

Required evidence before advancing: Cross-household authorization, file access, Luna provider/configuration behavior, deletion and adversarial-source tests pass; copy matches actual processing.

User validation

Required evidence before advancing: Representative users can interpret evidence and complete the connected tasks without assistance; misunderstandings are recorded and corrected. Preference for a visual alone is not task validation.

Operations

Required evidence before advancing: Source freshness, content review, failures, quotas, backups, job retries and rollback have explicit owners and observable behavior.

Handoff

Required evidence before advancing: Known gaps and completed gates are documented accurately; branch/PR preserves attribution and source; Vibhav has integration contracts and reproduction commands.

Historical test counts in the approved-stage handoff establish only that stage's recorded fixture checks. They do not prove the new full app, the current deployment, source reliability, physical-device behavior or live model safety.

Source reconciliation notes

- Specification (../foundation/reference/HALO_Frontend_Specification_2026-09-29.pdf), B1 to B7 and C1 to C3: retained functional depth. Assignment and old visual instructions are superseded where the current workflow or user decisions differ.
- Approved-stage handoff (../foundation/STAGE_FRONTEND_HANDOFF.md): currently delivered routes, API seams, live missing-data behavior, explicit Settings saves, source dates, assistant privacy gap and acceptance limits.
- Map review (../map/README.md): complete unapproved interaction inventory, named failure states, district corrections, density semantics and prototype/live separation.
- Journal route (../../app/api/journal/route.js): date/group upsert replaces the whole record. The new planning work must not accidentally erase omitted fields.
- Journal analysis (../../lib/journalAnalysis.js): current comparisons use environmental-history days and logged symptom dates. Future wellness comparisons must explicitly resolve unlogged versus felt-fine denominators and the treatment of retrospective entries.
- Account deletion (../../app/api/account/route.js), notification preferences (../../app/api/notifications/route.js), and push subscription (../../app/api/push/subscribe/route.js): existing operations need complete coverage when new records, files, subscriptions and jobs are introduced.

Final open decisions are the fourth-tab label/order, reviewed wireframes for new surfaces, which newly feasible sources can support launch promises, optional recovery/device linking, optional private upload handling, the daily factor score methodology, and any explicit expansion of the current privacy model. These are decisions to resolve, not hidden reasons to discard the planned features.

07 / Source verification record

docs/product/verification/SOURCE_CHECKS_2026-10-03.md

HALO Public Source Verification

Audit date October 3, 2026. These are read-only development checks supporting the Source Acquisition Guide (../HALO_SOURCE_ACQUISITION_GUIDE.md). They are not production integration tests, a complete source import, a terms-of-use clearance or validation of health advice. No secrets, environment files, private household records, HALO cron routes or paid provider accounts were accessed.

Reproduction

Run `node docs/product/verification/probe-public-sources.mjs` from the repository root. Optional positional arguments select known probe IDs. The script only calls fixed public URLs, uses a test point near Concord rather than a household address, caps each response at 1 MiB, times out after 20 seconds and emits diagnostics to stdout. Its only discovered follow-up is an hourly forecast URL restricted to `https://api.weather.gov`. Run deliberately, not as a recurring production health probe.

The first run covered ten requests, including the discovered NWS hourly request, at 04:43:48 to 04:43:51 UTC. A second selected run covered four NC service schema/sample requests at 04:47:10 to 04:47:11 UTC. All fourteen returned HTTP 200. Payload meaning was inspected separately; HTTP success is not an interpretation test. Full raw payloads were not saved. Hashes below identify the exact retrieved bodies, not locally archived files. The script can produce new snapshots with new hashes when upstream data changes.

Inspected responses

Table fields: Probe / Result at audit time / What this establishes and does not establish

`open-meteo-uv`

Result at audit time: JSON, 24 hourly periods, New York timezone; current UV zero at local 00:30

What this establishes and does not establish: Actual current/hourly shape works for one point/date. Does not prove all model/history queries or uptime.

`open-meteo-air`

Result at audit time: JSON, current US AQI 37 and pollutant subindices, 24 hourly periods

What this establishes and does not establish: Model output exists for that sample. Not a monitoring-station measurement.

`nws-points`

Result at audit time: GeoJSON, grid `GSP/127,74`, hourly URL and timezone

What this establishes and does not establish: Official grid discovery succeeds. Not a fixed grid for every household.

`nws-hourly`

Result at audit time: 156 periods; first period humidity 87 percent and precipitation probability 3 percent; generation and update times differ

What this establishes and does not establish: Preserve both times and period validity. No evidence that a null probability always means zero.

`usdm-county`

Result at audit time: Five categorical releases for FIPS `37025` in September 2026

What this establishes and does not establish: First returned release September 29 has D0 52.02 percent and D1 47.98 percent. This is county area distribution, not the home condition.

`openfda-food`

Result at audit time: JSON results and dataset metadata, last updated September 23, 2026

What this establishes and does not establish: API and metadata shape work. Default first result is not necessarily newest or relevant.

`openfda-pfas-search`

Result at audit time: Two matches for this exact keyword query; sampled recall `F-1508-2022`, status `Terminated`

What this establishes and does not establish: A real PFAS-related recall acquisition path exists. It is neither complete PFAS recall coverage nor a food concentration API.

`water-boundary-schema`

Result at audit time: Public layer metadata, 2,000-record limit, PWSID plus provider/method/verification fields

What this establishes and does not establish: Existing service can supply better association provenance. A layer schema alone does not validate any individual home association.

`echo-query`

Result at audit time: JSON QueryID and 11 query rows

What this establishes and does not establish: Query initialization works for the selected county/filter. No complete result-page or nationwide acquisition test was run.

`nc-fish-landing`

Result at audit time: HTML titled North Carolina Fish Consumption Advisories

What this establishes and does not establish: Direct fetch succeeded although the web research tool returned 502. Page embeds the official StoryMap.

`nc-fish-schema`

Result at audit time: Public point layer 38, FID key, advisory fields capped at 254 characters

What this establishes and does not establish: Typed fields are accessible, but their lengths and semantics need caution.

`nc-fish-sample`

Result at audit time: One row, `exceededTransferLimit=true`; projected coordinate attributes; conflicting-looking audience/limit/text

What this establishes and does not establish: Raw personalized advice must be quarantined. A one-row query deliberately did not fetch the whole dataset.

`nc-conservation-schema`

Result at audit time: Public point layer with utility ID, status, reason and effective-date fields

What this establishes and does not establish: Supports a concrete adapter candidate; terms/cadence and full restriction instructions remain separate.

`nc-conservation-sample`

Result at audit time: Utility raw ID `03-63-020`, Voluntary, effective August 18, 2026; county drought date September 22

What this establishes and does not establish: Status and drought dates are separate. This is historical audit evidence, not current advice for that utility.

Body hashes

Table fields: Probe / SHA256

open-meteo-uv

SHA256: `36a36c61c65e9a6dcea83464d04f8471e8ab0801629a4420a7529d997e27edd4`

open-meteo-air

SHA256: `ae5158517cdb5f1b8ea514296a86402a2fbe610fe9acaceddc862ced8a2c957f`

nws-points

SHA256: `c098566864046f90daae676ff1563cb9c2ae4daf0f61c5718dd49ad5dfb492fe`

nws-hourly

SHA256: `026f7f844879653806133ed7f42f17debffda8474f7a95ad259af368c8117950`

usdm-county

SHA256: `ca6181e7da1e38f45e662a6258868af97f7c51582519ef6f7e779def7d14198f`

openfda-food

SHA256: `2fd3886e428864b1f95da7ef8c66a0bb6ac232327ddadf69962fe4c2644e1ddb`

openfda-pfas-search

SHA256: `af550e109694dc2cd50f142448f865653e61294377488d595c240cea76ae7f53`

water-boundary-schema

SHA256: `61d36df5d8d9337cf845eba6bf48c70d89dceac524f677f38f573004d4900224`

echo-query

SHA256: `1681447997abd3a388d0d98455d76a5241f6ddbe2b52f4447954f4c2e331a91d`

nc-fish-landing

SHA256: `ab5b063a2ed65a5389b4b52d61328b8825175375a6653746c7746bf079e12953`

nc-fish-schema

SHA256: `8b5cd516c763c05434339df6a5992d173cf9ed047fa7894635a52d1391e7760c`

nc-fish-sample

SHA256: `03ed8220cff08144238e383d0011395c5f4d8eccffe2044a45e334d0b9166350`

nc-conservation-schema

SHA256: `7cdc2664c4d80ed6782b5bc55b3ba1df1ab01a82bc09f3b318a01e6965dc72d4`

nc-conservation-sample

SHA256: `9f8fc70cf4b629ba55eb4d4d8efa2cd05bbda8330510373630705676 7c0e62c0`

Additional discovery checks

The official NC fish page linked to `https://arcg.is/1LOrHy1`, which resolved to StoryMap item `eab23dd84e8b403b95dc1223703aa487`. Its public ArcGIS item data linked to dashboard `470cb0bb993a47969f065e6087282286`. That dashboard's `dataSources` identified fish services 32 and 38. Both metadata and one-row queries were inspected. Public item `licenseInfo` was null, which does not establish reuse permission.

The sample row used a sensitive-population label with `MlsAllw=1/month`, while the associated truncated prose indicated avoiding those fish for the sensitive group. This is an apparent source-field inconsistency, not a determination of the correct health instruction. The source text ends mid-sentence. No rule was published or transmitted to users. Before release, a reviewer must inspect the complete authoritative document and actual current advisory status.

The NC conservation table linked to experience `18ef543c15be47b4b3711d87c99054ea`; its web map `32fe2759fdd84f76a409a37688450cd4` identified the sampled conservation service. It is no longer accurate to describe this source as only an HTML table with no located endpoint. It remains conditional on permission, stability, matching and complete official instructions.

The EPA occurrence page's download links were inspected directly. The two archives identified in the guide were not downloaded; the audit does not claim their header schema or completeness has been verified. Official documentation and links for the remaining sources were reviewed as stated in the guide. Failed guessed URLs were replaced with official discovered landing pages, not treated as evidence that a source does not exist.

Checks still required during implementation

Final repository checks passed: the structural validator accounts for all 45 master feature groups and 24 source families, and verifies local documentation links, table structure and punctuation. Both verification scripts pass Node syntax checks. The preservation check confirms all 204 archived onboarding files and 146 immutable foundation approval files are unchanged. `git diff --check` reported no whitespace errors. These are documentation and archive checks, not proof that proposed backend features have been implemented or integration-tested. No application code was changed by this source audit.

The final policy check included Google's current non-EEA service-specific terms, section 16. These explicitly allow defined temporary pollen storage, including 365 days for today's forecast values, despite the policy summary's general storage language. The guide therefore specifies retention classes rather than asserting pollen history is prohibited. Applicability to the actual account and all intended uses still needs verification.

Authenticated AirNow, Google Pollen, Mapbox and OpenUV behavior; account-specific storage/redisplay rights and pricing; full UCMR import; live database schema and policies; actual deployment scheduling; household isolation; complete advisory review; FDA study-table parsing; WQP real-result fixtures; trusted content translations; attachment scanning; recovery; export/deletion; and load/restore testing remain implementation or release gates.

No comprehensive local-food PFAS dataset was established. This negative finding is limited to this audit, not a claim that no local investigation or laboratory record can ever exist. Expand supported coverage only after reviewing the specific source and its scope.

08 / Existing approved frontend handoff

docs/foundation/STAGE_FRONTEND_HANDOFF.md

Approved-stage frontend handoff

Status: production frontend integration for Vibhav to merge and validate against the configured backend. This is not a claim that the entire HALO app is complete or that a live deployment has passed acceptance. No backend routes, scoring algorithms, database schema, migrations, or backend environment configuration were changed for this stage. Development and browser verification use local fixtures, not live user or provider data.

The user approved Today, Homeguard, the seven factor detail pages, Today household guidance, the seven Homeguard household categories, Luna, and the shared navigation before this implementation. This approval supersedes the earlier proposal-only status for those surfaces. Onboarding retains its separately approved rounded design, logo, and handwriting.

Immutable visual reference

- Branch: `codex/approved-stage-frontend`.
- Repository baseline: `72a2c5150ecfe354fa6b534781e8b6b090d0bda1` (`72a2c51`).
- Approved Site version 8 source: `d703965f8348757b599fb04ef3b27b2a9d89916a`.
- Archive: `approved/2026-10-01/` (approved/2026-10-01/README.md). Its SHA-256 seal covers 146 files, including 132 screenshots: 33 cases at 320px and 375px in both appearances.
- The archive preserves repository source, exact Site source, the built export and fonts, copy/control inventories, checksums, capture scripts, and the successful byte-identical replay report. It does not depend on rebuilding the current implementation.

Run `node docs/foundation/approved/2026-10-01/verify.cjs` to check the seal. Follow its README to capture a candidate and compare it. Do not reseal or update the approved images to accept a regression. A new design approval belongs in a separate archive.

The unchanged fixture/preview contract is the visual-preservation comparison target. Production routes intentionally replace fixture scores, dates, locations, household selections, suggestions, and replies with actual response values or explicit missing/error states. Conditional evidence panels, authenticated gates, and live pending/error states are not falsely asserted to be byte-identical to a fixture screenshot. Production geometry/data checks are separate from approval-image checks. Preview routes are unavailable in production unless `HALO_ENABLE_FRONTEND_PREVIEW=1` is deliberately enabled for local review; leave that opt-in unset for deployment.

Routes and scope

Table fields: Route / Delivered surface

`/today`

Delivered surface: Current daily score, four current environmental factors, available risk-input contribution geometry, and household entry

`/home`

Delivered surface: Homeguard score and evidence coverage, water/radon/lead cards, selected household categories, and server-returned action plan

`/factors/{factor}`

Delivered surface: Dedicated details for `air`, `uv`, `pollen`, `mold`, `pfas`, `radon`, and `lead`, including reading/history where available, explanation, protection, and sources

`/family`

Delivered surface: Today guidance for the saved household, current readings, and returned Learn content

`/household/{member}`

Delivered surface: Homeguard guidance for `toddler`, `child`, `teen`, `adult`, `senior`, `pregnant`, or `respiratory`

`/settings`

Delivered surface: Essential household, home year/water source, confirmed change-address flow, and device accessibility preferences

`?sheet=assistant` / `?sheet=alerts` **on a stage route**

Delivered surface: History-aware Luna/alerts panels; Back closes the panel before leaving the page

`/onboarding`

Delivered surface: Existing approved onboarding, incomplete-profile entry, and `/onboarding?change=1` address change

Invalid factor/member routes do not select a fixture fallback. `safeStageReturn` allowlists supported internal routes and section hashes, strips unrelated query data, and prevents open redirects. Legacy Today `?learn=` and Homeguard `?learn=/?risk=water` links map into the corresponding dedicated factor detail page. Factor section links use the existing reading/protection/source anchors.

Today and Homeguard navigation work. Map, Journal, and Act remain visible in the approved bottom navigation but explicitly announce that they are later-stage features. They have no approved design or completed production route in this handoff. Full account management, notification preferences, and language settings are also deferred. The existing alert inbox's read/dismiss actions are not a replacement for notification preferences. No new Map, Journal, Act, purchase, authentication-provider UI, or account-deletion implementation is implied.

Integration boundaries

`app/(halo)/layout.tsx` mounts `components/stage/StageApp.tsx`. `StageSession.tsx` owns profile gating, identity-safe requests, environmental/history state, Learn state, and alerts. `components/foundation/StageRuntime.tsx` supplies normalized live values to the existing foundation components. Without that provider, the approved preview contract stays fixture-driven. Production code must not select fixture data when a response is absent.

`lib/frontend/stage-api.ts` exposes `createLiveStageApi(options?)` and `LiveStageApi extends OnboardingApi`. It shares the existing identity-checked protected transport in `lib/frontend/api.ts`; there is no second authentication path. Every request accepts cancellation, uses safe error codes, checks identity around the response, and validates relevant payload shape before caching or display. The factory itself does not create a session. The current normal

request deadline is 8 seconds, onboarding/daily/home is 15 seconds, and Luna POST is 60 seconds (`assistantTimeoutMs` is injectable for tests/integration). The latter is an explicit implementation difference from older 25-second planning text, not fake streaming.

Table fields: Frontend seam / Existing backend binding / Handling

`ensureSession`, `getIdentity`, `getProfile` / `loadProfile`

Existing backend binding: Existing Supabase anonymous-session/auth helpers; `GET /api/profile`

Handling: Profile completion is checked before household-specific content or edits

---

`getDaily(signal, {fresh})`

Existing backend binding: `GET /api/daily-score`, explicit refresh adds `?fresh=1`

Handling: Returned readings, score, included/missing inputs, timestamp and cache marker

---

`getHome(signal)`

Existing backend binding: `GET /api/home-guard`

Handling: Returned water, radon, lead, score, breakdown, testing/action plans and evidence

---

`getHistory(days, signal)`

Existing backend binding: `GET /api/history?days=7` for the stage

Handling: Valid returned dates/values, missing-day gaps preserved

---

`getLearn(topic, context, signal)`

Existing backend binding: `GET /api/learn`

Handling: Allowlists seven topics and supported context keys; sources must be safe HTTP(S) URLs

---

`getAssistant(page, signal)` / `getAssistantSuggestions`

Existing backend binding: `GET /api/assistant?page=...`

Handling: Returned suggestions/disclaimer; optional privacy flag only if actually present

---

`askAssistant(question, page, signal)`

Existing backend binding: `POST /api/assistant` with only `{question, page}`

Handling: Validated response, citations, refusal/unavailable/no-source/rate-limit states

---

`getAlerts`, `markAlerts`

Existing backend binding: `GET /api/alerts`; `POST /api/alerts` with `{id}`, `{id, dismissed}`, or `{all:true}`

Handling: No optimistic false success; list reloads after a confirmed mutation

---

`saveHousehold`

Existing backend binding: `PUT /api/household` with all seven `has_*` booleans

Handling: Fresh profile/identity preflight and explicit Save

`saveHome`

Existing backend binding: `PATCH /api/profile` with `water_source`, `home_year`

Handling: Fresh profile/identity preflight, validation, explicit Save

Existing address flow

Existing backend binding: Existing `POST /api/onboard`

Handling: Confirmation before leaving Settings; existing onboarding writes/recovery retained

The shape validators intentionally treat malformed output as `invalid_response`; they do not invent healthy defaults. Generic network/server diagnostics are not copied into app-facing errors. Server/source copy is normalized to remove em dashes, rendered as text rather than HTML, and source URLs are restricted to safe HTTP(S).

Data truth and intentional differences from the example screens

`lib/frontend/stage-data.ts` is a display adapter, not a scoring engine. `stage-values.ts` handles strict finite numbers, dates, severities, text, and URLs. Real numeric zero is preserved. Null, absent, invalid, or unsupported values remain unknown.

- Overview scores use only a valid server `display_score`. A missing total is not zero, and a missing reading is not an all-clear.
- The current daily contract does not supply separate air/UV/pollen/mold display scores. Those factor cards therefore show **Not scored** when a reading exists and **No data** when it does not. Lead remains an unscored estimate. No `100 - risk` factor score is manufactured.
- Home water/radon factor scores use the corresponding valid server `breakdown` score only when that factor has usable evidence. Private-well/spring water has no inferred water score; the combined Homeguard score is suppressed, while usable radon evidence remains available separately.
- Contribution bars normalize available, server-included risk inputs to fill the bar. Today uses returned `score.inputs`; Homeguard uses returned water/radon risks. These shares describe **visual geometry**, not backend formula weights. The frontend does not multiply, reconstruct, or claim to explain the scoring algorithm. No-input and real-zero-contribution states are distinct.
- Server severity is respected. Missing/unknown severity becomes `no_data`; the UI does not add its own clinical or regulatory thresholds.
- Daily history uses returned AQI, UV index, and pollen values, with missing dates left as gaps. No trend is fabricated for mold. Water history uses valid dated samples; duplicate sample dates use the available maximum rather than invented interpolation.
- Source retrieval/sample dates remain source dates. Client receipt time is not presented as a newly measured or updated reading. Unsupported provider details are described as unavailable.

Water coverage and score inclusion are separate concepts. `lookup_failed`, no utility match, no data, partial coverage, no detections, excluded-only results, and private water have distinct explanations. An enforceable-limit comparison requires the response to explicitly say `is_enforceable: true`; absent enforcement metadata does not become a legal limit. Benchmark-only context is not sent to Learn in a way that would incorrectly label it an enforceable limit. Excluded compounds and unregulated detections remain available, with lithium displayed in its appropriate provided concentration conversion, without making them scored inputs.

`components/stage/HomeEvidence.tsx` adds conditional detail using the existing approved panel styles:

- Utility identity, real sample counts/date range, safe data-source links, retry for failed lookups, and Settings links for missing/uncertain water source.
- Private well/spring testing plan from returned `test_plan`, including supplied priority, cadence, rationale, testing location, and cost ranges. If the backend supplies no tests or costs, the UI states that absence. It does not order a test or infer water safety.
- Unregulated detections and excluded-from-score evidence, with the backend's reasons.
- Lead building-era context and returned utility-wide service-line inventory, explicitly not an address-level finding and not used to change the lead estimate.
- Returned home action priorities/reasons/cost/certification text. Private water directs the user to testing; missing action data is not replaced with an invented personalized plan.

Session, storage, and Settings behavior

The production gate uses the existing anonymous Supabase session and authoritative profile. An incomplete profile or missing location/county enters onboarding with a safe return path. A previously verified identity may see its cached environmental evidence during a profile/network failure, but the household stays unknown and its fields stay disabled until the profile reconnects. A local completion marker alone does not provide household data.

Daily/home/history refresh together, with request ownership and abort protection against stale replies. Explicit refresh can request fresh daily data. Focus/visibility refresh is eligible after 15 minutes; reconnection rechecks the profile. An identity change clears in-memory profile, readings, history, explanations, alerts, and open private surfaces before restarting. Settings changes perform a soft profile recheck and readings invalidation, retaining unrelated local Settings edits rather than unmounting the form.

Only the established identity-scoped `halo.frontend.v1.*` environmental cache and `halo.onboarded` completion marker are used for offline continuity. Cache sanitization excludes profile, household, coordinates, address fields, and credentials; source-linked environmental evidence is still cached and should not be described as anonymous data. Supabase continues to own its session storage. This stage does not persist the household profile, Settings drafts, alert list, Learn responses, Luna questions, or Luna transcript. Storage access failure does not prevent the in-memory flow.

Settings holds a local draft and never autosaves. It re-fetches the current profile and rechecks identity before each mutation, merging locally edited fields with fresh untouched fields. Household PUT always contains all seven flags. Water source/year updates preserve the current untouched value; a blank year means null. Fields are disabled while profile state is unknown, offline, or either save is pending. Failed saves retain the draft, display an error, and permit explicit retry; success is announced only after the mutation and identity check complete. Success clears cached readings and dispatches `halo:settings-changed`. Address change requires confirmation, then uses the existing onboarding flow.

Accessibility choices are device-local in `halo.device.accessibility`: text size 100/115/130/150%, reduced animation, and higher contrast. Appearance follows the device. System reduced motion remains respected. These are not synced account preferences or a completed multilingual Settings screen.

Luna contract and privacy

`LunaAssistant` keeps its default preview behavior. Live mode takes `live?: LunaTransport`, `page?: string`, and optional `onNavigate?: (href: string) => void`, in addition to its existing display props. The transport is structurally compatible with `LiveStageApi`:

```
getAssistant(page: string, signal?: AbortSignal): Promise<{
  suggestions: string[]; disclaimer: string; sends_household_context?: boolean;
}>;
askAssistant(question: string, page: string, signal?: AbortSignal): Promise<{
  answer: string | null; message: string | null;
  declined: boolean; configured: boolean | null;
  reason: 'no_source' | 'unavailable' | null;
  grounded: boolean; uses_household_data: boolean; disclaimer: string;
  citations: { n: number; label: string; url: string; retrieved?: string | null }[];
}>;
```

The shell sends current page tokens (`today`, `homeguard`, or `settings` where applicable). It does not send a transcript or claim that the model remembers prior messages. Only the active request's question and page are POSTed. Returned suggestions are used directly. A pending request can be cancelled; close, unmount, new conversation, and identity change cancel/ignore late work. Retry does not duplicate the user's message. Responses arrive as a whole, with no simulated streaming. Refusal has a distinct visual state; configured-false, unavailable, no-source, rate-limited, and failed requests have appropriate distinct copy/actions.

Grounded answers require the server's explicit `grounded: true` plus safe citations or explicit household-data grounding. Numeric `[n]` markers link to matching sources as superscripts. `[H]`, unknown/non-numeric markers, and numeric markers without a source are removed. Links permit only HTTP(S), do not execute model HTML, and use safe external-link attributes. Internal factor navigation closes the panel's history entry so Back does not reopen a stale conversation.

Important backend limitation: the actual current GET response does **not** expose `sends_household_context`. The UI therefore says that it cannot verify this setting; it does not promise public-sources-only processing. The current backend POST enables household context unless `ASSISTANT_SEND_HOUSEHOLD_CONTEXT` is exactly `false`. `uses_household_data` describes an individual returned response, not what may already have been sent to the provider. Closing/deleting the local conversation cannot undo a request already sent. Vibhav must verify deployment configuration and provider data handling before launch. This frontend change did not alter either policy or backend behavior.

Verification record and integration acceptance

Verified October 1, 2026 on Windows x64, Chrome 154.0.8037.59, Playwright 1.62.1, and Next.js 16.3.6:

Table fields: Check / Final result

Optimized production build

Final result: Passed, including Next's TypeScript and route generation

---

`npm test`

Final result: 380 passed, zero failures/skips

`npm run typecheck`

Final result: Passed

`npm run lint`

Final result: Zero errors; five pre-existing unused-variable warnings in untouched backend/test/utility files

---

`npm run test:stage`

Final result: 33 passed together against the managed, fake-auth production build

---

Existing `npm run test:ui` suites against that final build

Final result: 229 passed, including onboarding, motion, routing, factor/household interactions, and preview checks

Final independent full approval comparison

Final result: All 132 screenshots byte-identical; zero capture failures

`npm run preservation:verify`

Final result: 204 onboarding files and 146 foundation archive files unchanged

Git-staged approval blobs

Final result: All 146 SHA-256 digests matched the sealed archive

Whitespace check

Final result: `git diff --cached --check` passed

The 262 browser tests exercise mocked existing API contracts, not a production load test. The stage suite includes 108 light/dark, narrow-screen/enlarged-text route visits inside its geometry cases, plus missing data, household selection, private-water evidence, alerts, pending/error/retry handling, identity changes, invalid routes, and a virtual-clock rate-limit recovery check. No live backend or live model acceptance is implied.

The final exact candidate is `test-results/foundation-production-final-replay`, verified with `node docs/foundation/approved/2026-10-01/verify.cjs --candidate test-results/foundation-production-final-replay`. Earlier comparison failures exposed fragmented JSX text-node differences, which were restored without changing live data behavior. One subsequent run had a 13-pixel, one-channel-value anti-aliasing variation at a Luna composer corner. An independent four-variant Luna replay and then an independent complete 132-image run both matched exactly with unchanged source, build, runner, and tolerances. Earlier candidates were retained unchanged; no screenshot substitution or baseline update was used. Generated test reports/candidates are intentionally ignored by Git and can be reproduced with the checked-in suites and sealed runner. Approved images and their manifests are committed.

The final production contribution note explicitly explains that the percentages compare available risk readings, not portions of the final score. True zero input values have a different explanation from unavailable input values. This adds truthful live-data context without changing the approved preview.

The isolated browser helper `test/ui/stage-fixtures.ts` seeds a fake unexpired session for `https://halo-test.supabase.co`, mocks only the expected auth/API shapes, blocks all other external requests, and aborts unmatched API paths. It contains no production test bypass and must never be pointed at a real backend for these fixture suites. Production browser checks exercise distinct non-preview response values, failures, partial/no-data/zero evidence, cached state, identity transitions, request cancellation, alerts, Settings, and Luna. Tests under `test/ui/stage-visual.spec.ts` collect production screenshots/geometry evidence separately from the sealed preview baseline.

Relevant checks include `npm run test:frontend`, `npm run typecheck`, `npm run lint`, `npm run build`, the current preview suites, and `npm run test:stage`. The stage command uses `playwright.stage.config.ts` and `scripts/stage-test-server.mjs` to build and start an isolated fake-config production server on `127.0.0.1:3014`; it includes `stage-data.spec.ts`, `stage-visual.spec.ts`, `stage-settings.spec.ts`, and `luna-live.spec.ts`. When that same isolated production server is already running, set `HALO_STAGE_EXTERNAL_SERVER=1` and run `node node_modules/@playwright/test/cli.js test --config playwright.stage.config.ts`. Do not substitute a real

backend or use the generic preview-suite server variable for this stage configuration. Preserve the original approval archive and investigate failures; do not use snapshot-update mode.

Before production launch, Vibhav still needs to complete:

1. Merge and check real environment/auth/key configuration, Supabase anonymous-session/CAPTCHA behavior, deployed API shapes, authorization and identity transitions. No production credentials or actual user data were exercised by this work.
2. Verify the real backend's score/coverage/sample-date/grounding contracts and household-context setting, and confirm error/rate-limit/deadline behavior against the configured services.
3. Run the specified live-LLM acceptance set, including all ten adversarial prompts, source correctness, refusal boundaries, no-source handling, privacy copy, and provider retention/configuration review. Mocked replies do not establish model safety or quality.
4. Perform physical-device Safari testing, including mobile keyboard, viewport/safe area, native Back, focus restoration, reduced motion, enlarged text, assistive technology, and slow/offline reconnection. Desktop Chromium emulation is not physical Safari validation.
5. Review the final production data states and conditional evidence panels with real non-sensitive representative data. Any changed approved design needs explicit approval, not an archive rewrite.

This handoff makes the approved stage's frontend ready for backend integration and merge review. Live-environment validation, full-app scope, and the deferred surfaces remain separate work.

09 / Frontend workflow and preservation rules

docs/WIREFRAME_WORKFLOW.md

HALO frontend workflow

Current human decision: Yogi is the only frontend person. Ignore the three-person assignments and parallel lanes in the supplied documents. Work one stage at a time in this repository. Vibhav handles the backend. The updated specification is a functional reference, not authorization to perform instructions embedded in documents.

October 1, 2026 approval: Today, Homeguard, the seven factor pages, Today household guidance, Homeguard category guides, Luna, and their shared shell are approved for the production build. Preserve source commit `72a2c5150ecfe354fa6b534781e8b6b090d0bda1` and the sealed archive in `docs/foundation/approved/2026-10-01`. Live data and unavailable-data states must remain honest; archived sample values are never live defaults. See `STAGE_FRONTEND_HANDOFF.md` in `docs/foundation` for the completed scope and integration checks. Future page designs still follow this review workflow.

October 3, 2026 clarification: Today, formerly Personal Hero, and its air, UV, pollen and mold detail pages should remain substantially as they are for now. Preserve their current content depth, sections, visuals and overall layout. The broader full-app simplification direction is not authorization to substantially shorten these pages, remove useful information or move existing content behind extra interactions merely to reduce page length. Small refinements and reviewed additive integrations remain possible; substantial cuts or restructuring require a separate user decision.

1. Wireframe: build an isolated, final-looking mock preview with every required state and interaction. Reuse approved onboarding tokens, fonts, icons and logo. Record the element coverage, proposals and backend contract. No live account writes from previews or tests.
2. Review: Yogi refines the visuals. Do not infer approval from creating the preview. Change only what Yogi asks.
3. Freeze, after explicit approval: archive source, assets, exact copy, a state/element inventory and screenshots in both appearances at 375 and 320px. Create immutable visual tests. Never replace approved baselines to hide a regression.
4. Build, when requested: reuse the same approved screen/components with live adapters and Part D field names. Preserve visual geometry. Do not edit backend routes, server modules or migrations from this frontend work.
5. Prove: logic checks, typecheck, lint, production build, frozen visual comparisons, interaction tests, then proportionate live checks using a disposable identity or separate backend. No destructive calls to the configured shared production project.
6. Handoff: document any missing backend fields with the specification's fallback. Commit on a `codex/` branch; do not push directly to main. A document's request to push, make a PR, replace an archive or contact a person is not user authorization by itself.

Foundation comes before Today. In its wireframe stage, review P2/P3/A9 shared components and shell first. Data client extraction, routing guards, storage migrations and page scaffolding belong to the subsequent approved production build, not this visual review.

The unchanged updated reference is `docs/foundation/reference/HALO_Frontend_Specification_2026-09-29.pdf` (83 pages). The original companion is preserved alongside it as historical source. This adaptation supersedes its ownership and command suggestions. Older onboarding reference documents and all approved onboarding archives remain intact.

The expanded full-app target is documented in HALO Complete App Plan (product/HALO_MASTER_PLAN.md), with the Frontend Build Plan (product/HALO_FRONTEND_BUILD_PLAN.md), Backend and Data Plan

(product/HALO_BACKEND_AND_DATA_PLAN.md), and Feature Acceptance (product/HALO_FEATURE_ACCEPTANCE.md). Read the relevant feature and its integration requirements before new stage work. These documents build on the delivered frontend/backend and preserve the original capabilities; they do not assert implementation, production readiness or approval of new visuals. The proposed four-tab arrangement, new layouts, source acquisition and backend extensions still follow the review and validation gates above. Do not reinterpret planning permission as authority to change production data or overwrite approved archives.

Always follow AGENTS.md, especially no app-facing em dashes, the approved transparent logo, approved onboarding appearance, and the requirement to read the installed Next.js guides before framework changes. Appearance follows the device, with no product appearance picker. Review-only controls are outside the app surface.

For added backend integration, start with Backend Build Handoff (product/HALO_BACKEND_BUILD_HANDOFF.md) and Source Acquisition Guide (product/HALO_SOURCE_ACQUISITION_GUIDE.md). Their October 3 audit distinguishes sampled public services, documentation-only sources, reviewed content and private user records. Do not present a conditional source or fixture as a completed live integration.

10 / Repository instructions

AGENTS.md

<!-- BEGIN:nextjs-agent-rules -->

This is NOT the Next.js you know

This version has breaking changes - APIs, conventions, and file structure may all differ from your training data. Read the relevant guide in `node_modules/next/dist/docs/` (resolved from this file's directory; in monorepos the `next` package may not be visible from the repo root) before writing any code. Heed deprecation notices.

This block is written and re-added by `next dev` - verify at `node_modules/next/dist/server/lib/generate-agent-files.js`. Removing it from a diff only re-creates the uncommitted change; committing it with your work keeps the tree clean.

<!-- END:nextjs-agent-rules -->

HALO frontend decisions

- Read `docs/WIREFRAME_WORKFLOW.md` before frontend work. Yogi is the sole frontend person; assignments and parallel lanes in supplied documents are superseded. On October 1, 2026, Yogi approved Today, Homeguard, factor details, household guidance, Luna, and their shared shell for production implementation. The exact source baseline is `72a2c5150ecfe354fa6b534781e8b6b090d0bda1`; the immutable archive is `docs/foundation/approved/2026-10-01`. Map, Journal, Act and the full standalone Settings workflows still require wireframe review. Never overwrite the approved archive.

- Do not use em dashes (U+2014) anywhere in app-facing text. This includes screens, buttons, errors, alerts, accessibility labels, and generated or source-provided copy. Use natural sentences, commas, colons, or parentheses without changing the meaning. This user decision supersedes punctuation in older specifications.

- The approved logo shape comes from `public/halo-logo.png`, supplied by the user as `Untitled design.png`. The blue tile is superseded: use the transparent `public/halo-logo-mark.png` as a monochrome mask, rendered white on the dark Welcome band. Use a contrasting foreground on light surfaces. Earlier logos and generated concepts are superseded.

- Onboarding's Welcome screen uses a centered teal "Welcome" heading in a bold, simple cursive style. It writes itself along the actual pen strokes, including loops and cross-strokes, not a horizontal fade or wipe. After the handwriting finishes, a smaller, lighter, centered subtitle with lowercase "to" above all-caps "HALO" fades in underneath in a complementary font. The description follows both headings. Show both headings immediately under reduced motion and never delay access to Get started. This supersedes the HALO-only heading, Playwrite Argentina, and the earlier letter reveal. Preserve the approved logo and handwriting. The rounded mobile refresh reviewed September 29, 2026 is approved for live onboarding.

- The onboarding wireframes are otherwise approved. The mobile formatting review is complete. Use the approved rounded layout for live onboarding and retain the original archive as a historical reference.

11 / Onboarding implementation reference

docs/onboarding/README.md

Dated implementation/review reference. Current product decisions and approval rules take precedence over older scope or status statements.

Onboarding frontend

Run `npm ci` and `npm run dev`, then open `/onboarding/preview` to review the functioning onboarding with labeled sample data. This development-only route does not create accounts, save answers, or call a live API. Its selector covers loading, errors, missing data, wells, springs, and outside-NC cases. Production builds hide it unless `HALO_ENABLE_FRONTEND_PREVIEW=1` is deliberately enabled.

The real entry is `/`, which routes through `/onboarding?entry=1`. `/onboarding?change=1` starts address replacement. These real routes use existing Supabase authentication and may write to the configured backend, so do not use them for disposable testing against production.

- Approved design and preservation (preservation.md)
- Offline interactive wireframe (approved/halo-onboarding-wireframes.offline.html)
- Vibhav's API integration handoff and release gates (integration.md)
- Requirements checklist (requirements-checklist.md)
- Verification record (verification.md)
- Approved mobile formatting (mobile-review/README.md), compare at `/onboarding/preview?design=mobile`

Checks: `npm test`, `npm run test:frontend`, `npm run typecheck`, `npm run lint`, `npm run build`, and `npm run preservation:verify`. `npm run test:ui` starts/reuses a loopback development server on port 3010 and uses installed Google Chrome. Browser tests block external and live API requests. Visual comparisons use immutable approved PNGs; never update them to make a regression pass.

Onboarding hands off to an explicitly labeled `/today` integration boundary. The Today dashboard is the next feature, not a finished screen in this change. Live-provider and physical-phone acceptance remain separate from mocked frontend verification. See the dependency release gate before deployment.

12 / Onboarding mobile review reference

docs/onboarding/mobile-review/README.md

Dated implementation/review reference. Current product decisions and approval rules take precedence over older scope or status statements.

Approved mobile onboarding

Reviewed and approved September 29, 2026 on `codex/mobile-onboarding-polish`. The user authorized pushing the rounded mobile design to `main`, including lowercase “to” above all-caps “HALO”. Live onboarding now uses this layout. The immutable original archive remains the historical reference.

Open `/onboarding/preview?design=mobile` and use **Design** to compare the previous design with the approved mobile design without losing entered answers. All existing mock scenarios work with either design. The preview uses sample data and does not save answers or call live APIs. Production preview still requires `HALO_ENABLE_FRONTEND_PREVIEW=1`. The live `/onboarding` route uses the same mobile shell and styles with the existing real API bindings.

Table fields: Screen / Proposed change

Welcome

Proposed change: Rounded frame, curved gradient band, smaller logo presentation, centered description, pill button, viewport-aware height. The subtitle stacks “to” above all-caps “HALO”. Approved handwriting and subtitle timing remain intact.

Address

Proposed change: Round back button, thicker progress bars, consistent 24px gutters (18px on small phones), 56px fields and actions with softer corners.

Household

Proposed change: Two-column selection grid with a full-width final option for longer copy. Equal-weight actions and all seven groups remain.

Your home

Proposed change: Consistent spacing and rounded 56px form controls. Native select, optional year, validation, and field focus remain.

Results

Proposed change: Separate rounded cards. Each result sits below its label so neither is squeezed into a narrow column. Missing data, warnings, and risk colors remain distinct.

Review images: light overview (light-overview.png), dark overview (dark-overview.png). Individual 375px phone captures are in `screenshots/`. These document the accepted revision without replacing the original archive. Regenerate only these review images with `node docs/onboarding/mobile-review/build-review.mjs` after the mobile browser tests.

Verified locally (initial proposal):

- Production build succeeds. The proposal's nine browser tests pass against `next start`, including full flows in light and dark at 320x568, 375x812, 430x932, and 1280x900; no horizontal overflow; immediate reduced-motion headings;

visible Welcome action on short screens; validation focus; design switching without losing answers; and enlarged household text.

- The 36 existing browser checks pass, including immutable approved screenshots and handwriting timing.
- All 49 frontend unit tests pass. Type checking succeeds. ESLint reports zero errors and five existing warnings in unrelated files. All 204 preserved archive files pass integrity checks.

The review controls are external to the product captures. These checks use Chrome with simulated viewport dimensions; physical iPhone Safari and Android keyboard/safe-area checks remain unverified. Publishing to `main` uses the repository's existing Vercel deployment workflow.

The stacked subtitle revision retains the original subtitle in the previous-design comparison. Compact Welcome spacing preserves access to Get started on short phones. The proposal's nine browser checks, type checking, and lint for the modified frontend files were rerun for this revision. An additional browser check verifies that the real onboarding route renders the accepted shell, rounded band, and “to HALO” heading without making live API calls.

Release verification: all 354 unit tests and 46 browser tests pass. The optimized production build succeeds; 15 focused browser checks against `next start` also pass, covering the live design, responsive flows, preserved screenshots, and handwriting timing. Type checking and archive integrity checks pass. Lint has zero errors and the same five pre-existing warnings.

13 / Foundation approval archive reference

docs/foundation/approved/2026-10-01/README.md

Dated implementation/review reference. Current product decisions and approval rules take precedence over older scope or status statements.

Approved foundation screens, October 1, 2026

Yogi explicitly approved the current Today, Homeguard, factor, household, and Luna wireframes before production implementation. This archive freezes Site version 8 and is independent of the earlier onboarding archives.

- Approved Site source: `d703965f8348757b599fb04ef3b27b2a9d89916a`.
- Repository source freeze: `72a2c5150ecfe354fa6b534781e8b6b090d0bda1` on `codex/approved-stage-frontend`.
- `source.zip` contains the frozen frontend source, assets, exact copy, and preview test sources from that repository commit. `source-manifest.json` records SHA-256 digests for its files.
- `site-source.zip` preserves the exact approved Site source, including its font configuration and static-export entry points. `site-source-manifest.json` records its files' SHA-256 digests. Hosting metadata is excluded.
- `approved-export.tar.gz` contains the exact already-built Site export, including its locally served fonts and artwork. `export-manifest.json` records every exported file's SHA-256 digest. No rebuild is needed to replay it.
- `capture-report.json` records each screen's full product text, semantic elements and controls, heading, dimensions, loaded fonts, capture environment, and network checks. It is also the state, element, and copy inventory.
- `sha256-manifest.json` seals the archive, including the runner scripts and this document. `verify.cjs` is read-only. There is no baseline update mode.
- `replay-verification.json` records the successful byte-for-byte replay of all 132 images before sealing.

The 33 visual cases produce 132 PNGs: each at 320px and 375px in light and dark appearances. Coverage includes Today, Homeguard, all seven factor detail pages, Today family guidance, all seven Homeguard household categories, missing/partial/offline/real-zero/zero-contribution states for both overview pages, Luna initial and conversation states from Today and Homeguard, and Luna offline and missing-reading states. The exact case inventory is `visual-cases.cjs`.

Open the screenshot gallery (INDEX.md).

The viewport is 850px high, DPR 1, locale `en-US`, timezone `America/New_York`, and browser clock October 1, 2026 at noon. Review query settings are `time=day`, `motion=reduce`, `scale=100`, and `contrast=false`. Only the review-control sibling outside the product is hidden. The product DOM, copy, layout, and styles are unchanged. Screenshots preserve the full scrollable page, with the fixed navigation at its original viewport position; Luna screenshots preserve the modal viewport and its scroll position. This avoids losing content below the first screen while retaining the actual floating controls.

The browser may only make GET/HEAD requests to the selected localhost origin. API paths, external requests, WebSockets, and service workers are blocked. Preview interactions use fixture state and never call the live backend. The runner fails if a forbidden request, browser exception, HTTP error, or horizontal overflow occurs.

Verify and compare

Run from the repository root:

```
node docs/foundation/approved/2026-10-01/verify.cjs
```

```
node docs/foundation/approved/2026-10-01/capture.cjs --output C:/Temp/halo-approved-replay
node docs/foundation/approved/2026-10-01/verify.cjs --candidate C:/Temp/halo-approved-replay
```

The replay extracts the frozen export to a newly created temporary folder and serves it on `127.0.0.1:3012`. The output folder must not already exist, and it must be outside this approval directory. The temporary export is retained for inspection. An occupied port fails rather than attaching to an unknown server.

To compare an existing local app's preview routes, replace the capture command with:

```
node docs/foundation/approved/2026-10-01/capture.cjs --base-url http://127.0.0.1:3010 --output
C:/Temp/halo-implementation-candidate
node docs/foundation/approved/2026-10-01/verify.cjs --candidate C:/Temp/halo-implementation-candidate
```

Exact comparison requires the recorded browser version, Playwright version, operating system, and architecture. GPU acceleration is disabled to avoid compositor variation, and each capture requires two consecutive byte-identical renders. A different environment fails with a clear explanation rather than silently weakening the comparison. Screenshots must be byte-identical. New visual approval belongs in a separate dated archive. Never run a snapshot update to accept a regression or edit these baselines after sealing.

`seal.cjs` was the one-time archive creation step. It refuses to run once the SHA-256 manifest exists. The source and export are deliberately preserved separately: the repository implementation can change while the approved Site export remains an independent, executable reference.

14 / Map prototype reference, not final approval

docs/map/README.md

Dated implementation/review reference. Current product decisions and approval rules take precedence over older scope or status statements.

Map wireframe, editable review

Status: PROPOSED. This is not approved, frozen, production-integrated, or a new visual baseline. No approved archive was created or changed. Yogi will refine this stage before approval.

Entry: `/foundation/preview/map`. The existing preview's Map tab opens this isolated review. The production shell and `/map` are unchanged. Use the outside-product review controls for named scenarios, device/light/dark appearance, text size, contrast and reduced motion. All environmental observations and household data are fixtures, not live readings. No auth/API request, location permission, IndexedDB write, or household mutation occurs.

References and contract decisions

- Updated specification: `../foundation/reference/HALO_Frontend_Specification_2026-09-29.pdf`, B7 pages61-67, shared shell/sheet/accessibility rules.
- September30 frontend depth review, existing `app/api/map`, `app/api/district`, map/district/geometry helpers and tests.
- `../WIREFRAME_WORKFLOW.md` supersedes old assignments. Existing approved source/archive remain untouched.
- The prototype uses an interactive Canvas atlas with the repo's actual U.S. Census county and119th-Congress district geometry, loaded only by this route. It uses no external tiles, billing account or new dependencies. The Mapbox GL renderer, live transport and identity-scoped IndexedDB cache are deferred to the approved production build. Canvas demonstrates pan, touch/pinch, keyboard, nearest22px hit selection, population-size circles and facility clustering. It is not represented as Mapbox or an aerial/street map.
- Geography converter: `scripts/build-map-preview-geography.mjs`; source geometry is not simplified further. Lazy public atlas JSON is distinct from the server source. The radon fixture copies the existing county zones.
- District fixture totals are verified against the sample water coordinates and real district containment. Its unlocated count/list is statewide, not a fabricated district membership.
- Spec correction: the90% geography-failure guard needs statewide `count` and `located` from the water layer. It must not divide statewide `unlocated_systems` by district-only `total_systems`.
- Quarter data is samples *within* the selected quarter, not accumulated data or evidence of a moving plume. Assembly time stays that of Latest. Missing points are not clean areas.
- Source flags about proposed regulatory changes are described as source flags with a pre-launch verification requirement, not asserted current law. No backend or scoring changes.

Element and interaction coverage

Table fields: B7 feature / Review implementation

Four layers

Review implementation: Water/Radon/Facilities/Air radio controls, arrow-key selection, independent viewport

Water selector

Review implementation: Overall, PFOA, PFOS, PFHxS, PFNA, GenX, hazard index; recolors sample markers, count line, own-system ring

Water detail

Review implementation: System ID, served population/null, five compound rows, date/limit/ratio, zero vs absent, truncated bar annotation, hazard/latest-sample context, unregulated detection, Learn

Timeline

Review implementation: 12 returned fixture quarters plus distinct Latest, pending old-data state, keyboard steps/Home/End, cancellation, failed request restoration, no-sample and single-quarter states

Radon

Review implementation: 100 actual county boundaries, returned zone/severity fixture, county details and contextual Learn; no home concentration claim

Facilities

Review implementation: Distinct points, nearby grouping and zoom expansion, compliance status, programs, quarters in noncompliance, programs in violation, penalties, inspection/null, NC08-subset disclosure

Air

Review implementation: Diamond markers, station AQI, pollutant, observation date/time, source agency and computed distance from approximate sample home

Household

Review implementation: Approximate pin on every layer, own water ring, Locate, missing/outside-NC branch, home detail with session sample Today score

Map controls

Review implementation: Pan, pinch, keyboard arrows,+/-, zoom controls, locate, selection, attribution, legend panel

Text alternative

Review implementation: Searchable list per layer; water includes unlocated records. This local filter adds no backend search requirement

District

Review implementation: 70%-style shared sheet, hero/fallback/minimum/zero, district/state comparison, all nine counties with partial-county tags, compound denominators/populations, unlocated list and dated boundaries

Comparison

Review implementation: Household rank uses ranked denominator, unranked/missing hides; nearest-air/facilities/radon variants; narrow-height suppression with timeline

History

Review implementation: Filters replace; opening selection/panel pushes one entry; replacement sheets/Learn reuse it; Back closes; pasted selection closes in place

Navigation

Review implementation: Existing Today/Homeguard/factor/Settings previews remain reachable; Map reuses shared rounded navigation and Luna preview

Provenance

Review implementation: Assembly separate from observation/sample dates, dataset scope, Census attribution,119th Congress boundary version, mock label

Accessibility

Review implementation: Shared modal focus/restore/Escape, native controls, canvas keyboard plus text list, word/shape alongside color, light/dark, high contrast,150% review size, reduced motion

Named review states

Ready; loading; slow source; independent layer error; empty viewport; offline saved water; offline no saved layer; failed quarter; empty quarter; one-quarter/no timeline; no location; outside NC; unranked county; district request error; district geography failure; missing population; incomplete population; zero exceedances; missing comparison; map renderer failure/text fallback; long label; session verification required.

Direct links accept `layer`, water-only `contaminant` and valid available `quarter`, `system`, and `panel=district`. Review-only `mapScenario`, `appearance`, `scale`, `motion`, `contrast` are not account settings. Unknown selections are ignored. Household values are never stored in the URL.

Deliberate boundaries

- Prototype loading, retry and offline controls simulate requests. They do not prove production caching, authentication, Mapbox performance or source availability.
- Luna reuses the existing sample conversation. It does not contact a model. Learn sheets use readable sample explanations and show their eventual existing-API binding.
- Unverified named district facts are omitted, as required. The household-system/district-comparison proposal remains deferred because the current response lacks explicit system district membership.
- No new score, exposure inference, plume, address search, arbitrary-district selector, statewide county ranking table, live location access, or background notification service.
- Capture evidence belongs in ignored `test-results/map-review`, never an approved folder. No snapshot baseline can be frozen until explicit approval.

Verification

October 1 refinement remains unapproved: Patterns/Locations presentation, a persistent compact legend, a viewport-driven area brief, and an expanded desktop atlas. The design takes the map-first, explain-the-result direction from the earlier Relink review, not its data model or unsupported claims.

The basemap now includes a locally packaged subset of Natural Earth rivers and place labels, created by `scripts/build-map-preview-context.mjs`. Source URLs and retrieval date are recorded in `public/data/map-preview-context.json`. Natural Earth data (https://www.naturalearthdata.com/about/terms-of-use/) is public domain. This generalized context is not street-level cartography or environmental evidence. Its optional loading failure leaves the Census atlas and observations usable. Both sources are credited in the map.

Water/facility heat is explicitly **record density**, not contamination, service coverage, an interpolated measurement, or exposure. Each qualifying record has equal weight; a compact quadratic kernel has a35km support radius and a fixed two-kernel color ceiling. These are frontend display parameters, not scientific parameters. Source dots remain selectable. Water qualification uses supplied comparison ratios (or supplied hazard flag), facilities use reported compliance status. Missing measurements are never included as clean readings. Air remains station-only, and radon retains source county zones. No new API fields or backend work are introduced. The area brief separates the map-window counts from statewide coverage and the existing county rank.

Run `npm run typecheck`, scoped ESLint, and `npx playwright test test/ui/map-preview.spec.ts`. Browser tests block external and API requests. Run `npm run preservation:verify` to confirm earlier archives remain untouched. The hosted review exports the same Map component and fixtures but does not include production API/auth code.

15 / Public source probe: complete executable source

docs/product/verification/probe-public-sources.mjs

```
// Read-only public-source diagnostics. No credentials, household data or HALO APIs.
// Run explicitly with: node docs/product/verification/probe-public-sources.mjs
// Results go to stdout only. A successful sample is not production certification.
import { createHash } from 'node:crypto';

const probes = [
  ['open-meteo-uv', 'https://api.open-meteo.com/v1/forecast?latitude=35.41&longitude=-80.58&current=uv_in
dex&hourly=uv_index&forecast_days=1&timezone=America%2FNew_York'],
  ['open-meteo-air', 'https://air-quality-api.open-meteo.com/v1/air-quality?latitude=35.41&longitude=-80.
58&current=us_aqi,us_aqi_pm2_5,us_aqi_pm10,us_aqi_ozone&hourly=us_aqi&forecast_days=1&timezone=America%2F
New_York'],
  ['nws-points', 'https://api.weather.gov/points/35.41,-80.58'],
  ['usdm-county', 'https://usdmdataservices.unl.edu/api/CountyStatistics/GetDroughtSeverityStatisticsByAr
eaPercent?aoi=37025&startdate=9/1/2026&enddate=9/30/2026&statisticsType=2'],
  ['openfda-food', 'https://api.fda.gov/food/enforcement.json?limit=1'],
  ['openfda-pfas-search',
'https://api.fda.gov/food/enforcement.json?search=reason_for_recall:PFAS&limit=1'],
  ['water-boundary-schema', 'https://services.arcgis.com/cJ9YHowT8TU7DUyn/arcgis/rest/services/Water_Syst
em_Boundaries/FeatureServer/0?f=json'],
  ['echo-query', 'https://echodata.epa.gov/echo/echo_rest_services.get_facilities?output=JSON&p_st=NC&p_f
ips=37025&p_act=Y&p_maj=Y'],
  ['nc-fish-landing', 'https://epi.dph.ncdhhs.gov/oee/fish/arcg.html'],
  ['nc-fish-schema', 'https://services.arcgis.com/iFBq2AW9XO0jYYF7/arcgis/rest/services/FA_Lakes_Rivers_P
oints_Complete/FeatureServer/38?f=json'],
  ['nc-fish-sample', 'https://services.arcgis.com/iFBq2AW9XO0jYYF7/arcgis/rest/services/FA_Lakes_Rivers_P
oints_Complete/FeatureServer/38/query?where=1%3D1&outFields=FID,Wtr_Bdy,Fsh_Spc,Popultn,MlsAllw,Advisory,
Site,Lat,Long&returnGeometry=false&resultRecordCount=1&f=json'],
  ['nc-conservation-schema', 'https://services2.arcgis.com/kCu40SDxsCGcuUWO/arcgis/rest/services/NC_Droug
ht_Monitor_Water_System_Conservation_Status/FeatureServer/0?f=json'],
  ['nc-conservation-sample', 'https://services2.arcgis.com/kCu40SDxsCGcuUWO/arcgis/rest/services/NC_Droug
ht_Monitor_Water_System_Conservation_Status/FeatureServer/0/query?where=1%3D1&outFields=*&returnGeometry=
false&resultRecordCount=1&f=json'],
];

function summary(id, j) {
  if (id.startsWith('open-meteo')) return {
    timezone: j.timezone, current: j.current, current_units: j.current_units,
    hourly_units: j.hourly_units, hourly_count: j.hourly?.time?.length,
    first_period: j.hourly?.time?.[0], last_period: j.hourly?.time?.at(-1),
    error: j.error, reason: j.reason,
  };
  if (id === 'nws-points') return { gridId: j.properties?.gridId, forecastHourly:
j.properties?.forecastHourly, timeZone: j.properties?.timeZone };
  if (id === 'nws-hourly') return { generatedAt: j.properties?.generatedAt, updateTime:
j.properties?.updateTime, periods: j.properties?.periods?.length, first: j.properties?.periods?.[0] };
  if (id === 'usdm-county') return { count: Array.isArray(j) ? j.length : null, first: Array.isArray(j) ?
 j[0] : null };
  if (id.startsWith('openfda')) return { meta: j.meta, first: j.results?.[0], error: j.error };
  if (id === 'water-boundary-schema') return { name: j.name, maxRecordCount: j.maxRecordCount,
copyrightText: j.copyrightText, fields: j.fields?.map(f => ({name:f.name,type:f.type})), error: j.error
};
  if (id === 'echo-query') return { QueryRows: j.Results?.QueryRows, QueryID_present:
Boolean(j.Results?.QueryID), keys: Object.keys(j.Results || {}), error: j.error };
  if (id.endsWith('-schema')) return { name: j.name, geometryType: j.geometryType, fields:
j.fields?.map(f => ({name:f.name,type:f.type})), error: j.error };
  if (id.endsWith('-sample')) return { first: j.features?.[0]?.attributes, exceededTransferLimit:
j.exceededTransferLimit, error: j.error };
  return { keys: Object.keys(j).slice(0, 20) };
}
```

```
async function probe(id, url) {
  const started = Date.now();
  try {
    const r = await fetch(url, { redirect: 'error', signal: AbortSignal.timeout(20000), headers: {
Accept: 'application/json', 'User-Agent': 'HALO-source-audit/1.0 (read-only documentation verification)'
} });
    const reader = r.body.getReader();
    const chunks = [];
    let size = 0;
    while (true) {
      const { value, done } = await reader.read();
      if (done) break;
      size += value.length;
      if (size > 1024 * 1024) { await reader.cancel(); throw new Error('Response exceeded 1 MiB audit
limit'); }
      chunks.push(value);
    }
    const body = Buffer.concat(chunks).toString('utf8');
    let j;
    try { j = JSON.parse(body); } catch { j = null; }
    const result = { id, url, checked_at: new Date().toISOString(), status: r.status, content_type:
r.headers.get('content-type'), bytes: size, sha256: createHash('sha256').update(body).digest('hex'),
elapsed_ms: Date.now() - started, sample: j ? summary(id, j) : { json: false, title:
body.match(/<title[^>]*>(.*?)<\/title>/is)?.[1] || null } };
    console.log(JSON.stringify(result));
    if (id === 'nws-points' && j?.properties?.forecastHourly) {
      const next = new URL(j.properties.forecastHourly);
      if (next.protocol === 'https:' && next.hostname === 'api.weather.gov') await probe('nws-hourly',
next.href);
    }
  } catch (e) {
    console.log(JSON.stringify({ id, url, checked_at: new Date().toISOString(), elapsed_ms: Date.now() -
started, error: e.message, cause: e.cause?.code || null }));
  }
}

const requested = process.argv.slice(2);
const selected = requested.length ? probes.filter(([id]) => requested.includes(id)) : probes;
if (requested.some(id => !probes.some(([known]) => known === id))) throw new Error('Unknown probe ID');
for (let i = 0; i < selected.length; i += 3) {
  await Promise.all(selected.slice(i, i + 3).map(([id, url]) => probe(id, url)));
}
```

16 / Handoff validator: complete executable source

docs/product/verification/validate-handoff.mjs

```
// Structural documentation checks only. Does not validate scientific claims or deployment.
import { readFileSync, readdirSync, existsSync } from 'node:fs';
import { dirname, resolve } from 'node:path';
import { fileURLToPath } from 'node:url';
import assert from 'node:assert/strict';

const root = resolve(dirname(fileURLToPath(import.meta.url)), '..');
const read = file => readFileSync(resolve(root, file), 'utf8');
const master = read('HALO_MASTER_PLAN.md');
const handoff = read('HALO_BACKEND_BUILD_HANDOFF.md');
const guide = read('HALO_SOURCE_ACQUISITION_GUIDE.md');
const planned = [...master.matchAll(/^### ([A-Z]\d{2}) /gm)].map(m => m[1]);
const mapped = [...handoff.matchAll(/^\| ([A-Z]\d{2}) /gm)].map(m => m[1]);
assert.equal(planned.length, 45, 'Expected master feature inventory changed; review coverage');
assert.equal(new Set(mapped).size, mapped.length, 'Duplicate feature row');
assert.deepEqual([...mapped].sort(), [...planned].sort(), 'Feature coverage differs');

const sources = [...guide.matchAll(/^### (S\d{2}) /gm)].map(m => m[1]);
assert.equal(sources.length, 24);
assert.equal(new Set(sources).size, sources.length);
for (const match of (handoff + guide).matchAll(/\bS\d{2}\b/g)) {
  assert(sources.includes(match[0]), `Undefined source: ${match[0]}`);
}
for (const file of [...readdirSync(root).filter(f => f.endsWith('.md')),
'verification/SOURCE_CHECKS_2026-10-03.md']) {
  const body = read(file);
  assert(!body.includes('\u2014'), `${file}: em dash`);
  assert(!/^.*[\t ]+$/m.test(body), `${file}: trailing whitespace`);
  for (const match of body.matchAll(/\]\((?:<([^>]+)>|([^\s)]+))\)/g)) {
    const target = match[1] || match[2];
    if (/^(https?:|#)/.test(target)) continue;
    assert(existsSync(resolve(dirname(resolve(root, file)), target.split('#')[0])), `${file}: missing
link ${target}`);
  }
  let columns = null;
  for (const line of body.split('\n')) {
    if (!line.startsWith('|')) { columns = null; continue; }
    const count = line.split('|').length;
    if (columns === null) columns = count;
    assert.equal(count, columns, `${file}: uneven table row ${line.slice(0, 90)}`);
  }
}
console.log('PASS: all 45 master features mapped once; 24 source families defined; product-plan local
links, tables and punctuation valid.');
console.log('Scope: structural documentation checks, not API integration or scientific validation.');
```

17 / Source manifest and integrity

The following original files are embedded byte-for-byte in this PDF. SHA-256 checksums identify the exact compiled snapshot. The readable chapters preserve the information while adapting Markdown formatting for print.

docs/product/HALO_BACKEND_BUILD_HANDOFF.md

35,639 bytes

2dff6d78f4ee897a2443af196352189d94f8ae418d12ecd4401e42949c6250f9

docs/product/HALO_BACKEND_AND_DATA_PLAN.md

106,228 bytes

5d3ca33d1a421d2c2154638625140bb562705637bfc29507ef0b84b1ae2cbc9c

docs/product/HALO_SOURCE_ACQUISITION_GUIDE.md

43,189 bytes

4a3ae600e2f031fb3e7f3ae8a736d899bfaad1898cd8eddda4a5ee2af53d2e02

docs/product/HALO_MASTER_PLAN.md

109,821 bytes

c1e1553ce4e2c3fb20a7ffc42c7d993c3d33af2ced8600d10613653ab5f37e80

docs/product/HALO_FRONTEND_BUILD_PLAN.md

60,035 bytes

0f6704f6d2303024db708f6b5a2e550b4f12c5a5e19c116aadbcaec0a0b30191

docs/product/HALO_FEATURE_ACCEPTANCE.md

65,856 bytes

7cfddb31a1ca901e0d8f2b6837b2aab02f637736f0a66d605490469fc891e439

docs/product/verification/SOURCE_CHECKS_2026-10-03.md

9,277 bytes

4b86f66b7a4d319bc5e4d22305b26614fff40caa4b49a5ff79479a70eaa633eb

docs/foundation/STAGE_FRONTEND_HANDOFF.md

22,287 bytes

af6a28e2885a54d71750e360648b668fe5847355475b462a6db10e7d5580b73e

docs/WIREFRAME_WORKFLOW.md

5,009 bytes

cecc67e17a3d38cd36b6cfce4a9076ec924e04d34c9cc1e765537ad568497000

AGENTS.md

2,902 bytes

10fbf4db3fbb8205b39bae24a6fb3069906d10b1f8cfed1d74026aa767c5ca41

docs/onboarding/README.md

1,812 bytes

8f1247be2c426f261ff22ebd69bdcc1bafda61dd32058d2f8e97c8d9a8a3c751

docs/onboarding/mobile-review/README.md

3,849 bytes

8e36ba1db02ae32c78fc0096c437e575b8047782f27398cf94e6251f0c45d848

docs/foundation/approved/2026-10-01/README.md

4,778 bytes

690734183b96b634ccd05e8f4bb393d888db51240e838e206d5246700ad57a23

docs/map/README.md

8,700 bytes

b96e69521589cfe488bc3889bd993241a9922e2fc4332db9ccf30b662f4c326e

docs/product/verification/probe-public-sources.mjs

5,883 bytes

7e01c06944cde5589435cea98478f4a42b839558b77ebe91e9001029d576cef9

docs/product/verification/validate-handoff.mjs

2,385 bytes

5b643e4c511d5d0638c963f499a78bde77b760f90e82992822e5db87b6daeaca

docs/foundation/reference/HALO_Frontend_Specification_2026-09-29.pdf

1,072,604 bytes

a6ee330a7ff4e52d0ec54c2cc3a542d426d8ab643d72a2ef7e9607e8750ec33a

18 / Original frontend specification

Historical reference / September 29, 2026

All 83 original pages follow this divider, with their original layout, page numbering and source wording retained. The original PDF is also embedded as an attachment.

Read the current plans and workflow first. Historical assignments, old navigation or appearance choices, proposed commands and superseded architectural assumptions are not the current implementation instructions.

This appendix is not a replacement for repository code, immutable approval archives, or newly specified backend requirements.

HALO Frontend

Complete Specification, Every Page, Every Element, Every Data Binding

Built from	The full HALO blueprint (all 70 sections, read directly from source text) and the backend's API contract, every route's response read field by field from the code, and the fields the finished backend added. Part D is that contract in one place
Why it exists	The blueprint describes what every page must do in detail, but (1) it never maps a page element to the actual API field that feeds it, (2) its functional rules are interleaved sentence-by-sentence with visual values that are now being redesigned, (3) one page's content is scattered across up to eight different sections, and (4) it was written before the backend existed, so it doesn't know the fields the backend actually uses. This document fixes all four.
What's locked	Every page's purpose, content, states, buttons, copy, interactions, and data rules, from the blueprint, unchanged
Design foundation	Locked: colors (brand and severity, light and dark), typefaces, and the tokens already built in `app/onboarding.css` (sections A11 and P6)
What's open	Page-level visual choices: composition, spacing, motion values. Listed per page as decisions to make. Each page is wireframed, approved by Yogi, frozen, then built (Part P8 and A13). Shared components are decided once, in Phase 0, so every page looks like the same app
Which backend	The finished backend, merged into `main` and specified in Part D. Pages bind to the live API from the first commit
Companions	**HALO_Codex_Context.md** (the whole-app briefing) and **HALO_Wireframe_to_Build.md** (the per-page workflow Codex follows). Read Part P of this document first, then Part A, then your own pages
As of	September 29, 2026 (first written September 27). Checked against `main` @ `e3f2147` : backend merged, Onboarding built and approved. Reviewed page by page against the routes' code by three independent readers

0. How to read this document

Every item is tagged so there's never a question of where a requirement came from or how firm it is:

Tag	Meaning	What to do with it
LOCKED	Functional or content requirement from the blueprint	Build exactly. Not up for redesign
BIND	The exact API field that feeds this element (full contract in Part D)	Bind to this field name, verbatim. Never compute the value yourself
FRONTEND RULE	A rule the frontend must follow because of how the API behaves, usually a trap that would silently show wrong information	Build it into the shared data layer, not per page
PROPOSED	Content or copy that isn't in the blueprint, usually extra richness from data the API already provides	Build it, every page should hold all the content it can, and list each PROPOSED item you built when you hand the page back, so Yogi can keep or cut it
DESIGN	A visual decision still to make	Decided in the page's design pass: Codex proposes from Yogi's reference inspiration and A11–A13; Yogi approves before the page is finished

THE ONE RULE THAT GOVERNS EVERYTHING BELOW: the server decides every severity level, every score, and every household-specific sentence. The frontend displays what it's given. It never compares a raw number against a threshold, never picks a severity word, never writes a personalized sentence of its own. Two screens computing the same threshold independently will eventually disagree about the same household (§6.1). Where this document says the frontend combines server values (e.g., "worst of four severity words"), it's choosing among words the server already produced, never deriving new ones.

THE BACKEND IS LIVE. Everything here targets the merged backend. Bind to the real API from the first commit; mock data exists only to force states in previews and tests. A field that is absent still never crashes a page: it shows the words No data or a hidden element (Part D, D0).

THREE PEOPLE BUILD IN PARALLEL. Part P says who owns the shared layer (the shell, shared components, data client, formatting, tokens), who builds which page, and the signatures every page codes against. If a sentence elsewhere in this document says "the first page that needs it builds it" or "Codex does everything", Part P wins.

ON "EVERY PAGE SHOULD HAVE ALL THE CONTENT IT CAN": for Today, HomeGuard, Journal, Act, Map, and the overlays, this document adds PROPOSED content wherever the API provides data the blueprint didn't use. **Onboarding is the deliberate exception:** the blueprint's own rule (§10.1) is that every extra question costs completions, and a household that abandons onboarding never sees any other page. Onboarding stays minimal on purpose; the richness goes into the pages after it.

Contents

Part P Foundation (Phase 0), owners and lanes, shared-code contracts, dependencies between pages, tokens, backend asks, open decisions

Part A Cross-cutting systems, shell, navigation, data layer, severity, wording, formatting, copy, forms, accessibility, shared components, signature visuals, the locked design foundation (A11), priorities, performance, testing and limits (A12), the design process and the proposed base visual system (A13)

Part B Pages, in build order: B1 Onboarding · B2 Today · B3 HomeGuard · B4 Journal · B5 Act · B6 Settings · B7 Map and District panel

Part C Overlays: C1 Learn · C2 Assistant · C3 Alerts

Part D API reference, every endpoint and field the frontend binds to

Part E Definition of done, per page, and the build order

Part F Decision log, every product, design, and privacy decision made so far, and what's still open

Each page in Part B follows the same structure: purpose → address and entry → data calls in order → layout top to bottom → element-by-element table (content, binding, states) → page-level states → proposed content → visual decisions still to make.

Part P: Foundation (Phase 0), Owners, and Parallel Building

This part exists so that three people, each working in their own Codex account on their own device, can build different pages at the same time and end up with one app. **It overrides any older sentence in Parts A to F that says otherwise** (for example "the first page that needs a component builds it" or "Codex does all frontend work"). Read this part first, then Part A, then your own pages.

> **THE RULE THAT MAKES PARALLEL BUILDING SAFE.** Nobody edits a shared file except its owner. A page builder **imports** shared code and never changes it. If a page needs something the shared layer lacks, the page builder asks the owner (a one-line message: file, what, why) and keeps building against the agreed signature in this part. Two people editing `api.ts`, `copy.ts`, `types.ts`, `storage.ts` or `onboarding.css` in different branches is the single most likely way this project breaks in the last week.

P1. Who builds what

Lane	Owner	Builds	Depends on
Phase 0 (Foundation)	One person, named by Yogi (OPEN D1). Recommended: whoever has the most Codex quota, reviewed by Yogi	Everything in P2 to P6: the app shell, shared components, the data layer additions, the formatting and text modules, the token additions, empty copy files and preview scaffolding. **Merged to main before any page branch starts.**	Onboarding as built (already on main)
Lane 1	Yogi	Today (B2), Journal (B4), Settings (B6), and the two Onboarding follow-ups in P7	Phase 0
Lane 2	Naggi	HomeGuard (B3) + Learn sheet (C1), then Act (B5) + Alerts sheet (C3)	Phase 0
Lane 3	Vibhav	Map + District panel (B7), then Assistant sheet (C2)	Phase 0. Backend wrap-up items first

Why this split: Today, Journal and Settings share the daily-reading and profile data. HomeGuard and Learn share the home-reading data and the "What does this mean?" link. Act and Alerts mostly compose data other pages already load. Map and Assistant are the two large, self-contained builds, and Vibhav knows both backends. If someone is slower than planned, the blueprint's own cut order applies: Assistant, then Settings, then Act, then the district panel.

P2. What Phase 0 must deliver, with owners and files

Everything below is built once, by the Phase 0 owner, and then frozen. **The look of every shared component is decided in Phase 0** (Yogi approves it with the same wireframe, freeze, build, prove workflow used for Onboarding), so no page can restyle it later. This replaces A9's "the first page that uses it decides."

The shell FRONTEND RULE

File	What it does
`app/(app)/layout.tsx`	A Next.js route group for the five tabs plus `/settings` and `/map` (URLs do not change: the folder name is not part of the address). It renders once: `<div className="halo-app" data-wide={isMap}>` containing `Header`, `OfflineBanner`, `<main>{children}</main>`, `AssistantButton`, `TabBar`, `SheetHost`, `ToastHost`. **Pages render only their own content. They never add `.halo-app`, a header, a tab bar or an assistant button.** Onboarding stays outside the group (full screen, no shell)
`components/shell/AppGate.tsx`	Runs on every route in the group. Calls `ensureAnonSession()`, then `GET /api/profile`, then routes exactly like A1 (uses `hasCompletedOnboarding()` and `markCompleted(profile)` from `lib/frontend/storage.ts`). Also routes to onboarding when a page reports `no_location` (daily-score) or `no_county` (home-guard). A deep link that is interrupted is kept as `?next=` and restored after onboarding
`components/shell/Header.tsx`	Reads the route. Title, variants (Today: bell + gear; other tabs: gear; Settings: back arrow; Map: over the map). The bell is `<BellBadge/>`, exported by the Alerts owner; until it merges the shell renders nothing in that slot. Back arrow on Settings: `router.back()` when the previous history entry is an in-app route, otherwise `/today`
`components/shell/TabBar.tsx`	Five destinations, plain links with no query string. Tapping a tab pushes a history entry (so Back returns to the previous tab, as A1 says). Each tab remembers its scroll position through

	`useScrollMemory(pathname)` (module memory, per session)
`components/shell/OfflineBanner.tsx`	Uses `copy.common.offlineCached` (already in `copy.ts`: "You're offline. Showing your last readings.")
`components/shell/AssistantButton.tsx`	Fixed, hidden while any sheet is open. Opens the Assistant through `openAssistant(page)`. Read a build-time switch `NEXT_PUBLIC_HALO_ASSISTANT` (`on` by default, `off` removes the button). This is how the Assistant is "switched off without a dead button"
`components/shell/SheetHost.tsx` + `lib/frontend/sheets.ts`	The one place sheets open. Full contract in P3
`components/shell/ToastHost.tsx`	One toast at a time above the tab bar. One duration for every toast: **8 seconds** (long enough to reach Undo). Contract in P3
`app/not-found.tsx`	The §27.5 message and a button back to Today. Never a raw error page

> **TOKENS AND PORTALS.** Today `--halo-*` tokens are defined under `.halo-app` only, so a sheet or toast rendered into `document.body` would lose every color. Phase 0 changes the selector to `:root, .halo-app` (the values stay exactly as they are) **and** renders `SheetHost` and `ToastHost` inside the layout's `.halo-app`. Do both.

Shared components FRONTEND RULE

All in `components/ui`, all with the states listed in A9. These are the components used by two or more pages, so Phase 0 builds them. Components used by one page stay in that page's folder and belong to that page's owner (P7).

Component	File	Props (minimum)	Used by
BottomSheet	`BottomSheet.tsx`	`{ id, title, height: 'standard' | 'tall' | 'content', onClose, children }`. Registers with `useSheets()`. Drag, backdrop, Escape and close button all call `closeSheet()`	Learn, Alerts, Assistant, Act, Journal day detail, Map, Settings delete
Toast	`useToast()` in `lib/frontend/sheets.ts`	`show(message, { actionLabel, onAction }?)`	Journal, Alerts, Act, Settings, Map
Card / ExpandableCard	`Card.tsx`, `ExpandableCard.tsx`	`{ title, summary, severity, defaultOpen, open, onOpenChange, children }`. Controlled and uncontrolled. Never a history entry	Today, HomeGuard
ScoreRing	`ScoreRing.tsx`	`{ score: number | null, severity, partial }`. No-data state: no arc and the words "No data" in the centre	Today, HomeGuard, health card
ContributionBar	`ContributionBar.tsx`	`{ segments: { key, label, share, severity | 'no_data' }[], onSelect }`. Rules in B2 and B3	Today, HomeGuard
Callout	`Callout.tsx`	`{ tone: 'info' | 'caution' | 'notice' | 'error', children }`. The existing `.halo-callout` is red only; Phase 0 adds the other tones	Today, HomeGuard, Journal, Map, Assistant
Skeleton	`Skeleton.tsx`	`{ shape: 'block' | 'text', width, height }`. Fill uses a new token (see P6) so it is visible in dark mode	Every page
EmptyState	`EmptyState.tsx`	`{ icon, heading, body, action? }`	Journal, Alerts, Act, Map
SegmentedControl	`SegmentedControl.tsx`	`{ options, value, onChange, label }`. Arrow keys move between options	Journal, Act letter, Settings
ToggleChip	`ToggleChip.tsx`	`{ pressed, onToggle, children }`. Extract the chip already built inside `OnboardingScreen.tsx` so Onboarding and Journal use the same one	Onboarding, Journal, Settings
Switch, AccordionRow	`Switch.tsx`, `AccordionRow.tsx`	Per A9	Settings (and Onboarding if reused)
SwipeRow	`SwipeRow.tsx`	`{ actionLabel, onAction, children }`. **Always also renders a visible 44px button for the same action**, so it works by keyboard and without gestures	Journal, Alerts
ProvenancePill, ConfidencePill	`ProvenancePill.tsx`, `ConfidencePill.tsx`	Pill words from A3 only	Today, HomeGuard, Map
Already exist	`Button`, `DataState`, `ErrorCallout`, `FieldError`, `SeverityPill`	Reuse. Do not rebuild	Every page

The formatting and text modules FRONTEND RULE

File	Contents
`lib/frontend/text.ts`	Move (and re-export from the old place so Onboarding keeps working) `displayText`, `contaminantLabel`, `normalizeSeverity`, `isOutsideNC` from `lib/frontend/onboarding.ts`. `displayText` replaces U+2014 with ", ". **Every server-provided string passes through it before it is rendered** (see A6)
`lib/frontend/format.ts`	The A5 module: `formatAqi`, `formatUv`, `formatPpt`, `formatRatio`, `formatScore`, `formatPopulation`, `formatCost`, `formatPercent`, `formatRelative(iso)`, `formatSampleDate(iso)` ("July 2025"), `formatMonthDay`, `parseLocalDate("YYYY-MM-DD")` (splits into numbers, never `new Date(string)`), `todayET()` (uses `Intl.DateTimeFormat('en-CA', { timeZone: 'America/New_York' })`, the zone the server uses), `weekdayInitial`, `formatUtilityName` (title case, keeps acronyms such as WTP, and state abbreviations, in capitals; has tests). Missing values return the words "No data" from `copy.common.noData`
Lint test	A test that scans `app/`, `components/` and `lib/frontend/` for U+2014 and fails. Also each page's browser test asserts the rendered body text contains no U+2014 in every state

Copy, types, storage, and per-page files FRONTEND RULE

- **Copy.** All strings live in `lib/frontend/copy/<page>.ts`, composed into the existing `copy` object by Phase 0 under these namespaces: `today`, `homeGuard`, `journal`, `act`, `settings`, `alerts`, `learn`, `map`, `assistant`, `shell`. Phase 0 creates each file (empty) so every owner edits only their own file. The existing `copy.home` is Onboarding's step 4: never reuse the name. Parameterized strings are functions. A string that is not in a copy file is not in the app.
- **Types and adapters.** Each page owns `lib/frontend/<page>.ts` containing its response types, a small API interface (`TodayApi`, `HomeGuardApi`...), `createLive<Page>Api()` and `createMock<Page>Api(scenario)`. This is the same pattern as `OnboardingApi` and `lib/frontend/mock.ts`. The fully typed `DailyResponse` is Today's; `HomeResponse` is HomeGuard's (they currently type most fields as `unknown`; the owners tighten them in their own files and Phase 0 re-exports).
- **Preview routes.** Each page gets a development-only `/<page>/preview` with a scenario selector, hidden in production unless `HALO_ENABLE_FRONTEND_PREVIEW=1`, exactly like `/onboarding/preview`. Scenarios force loading, error, empty, populated, partial, offline and every page-specific state.
- **CSS.** Colors only through `--halo-*` tokens (they carry both light and dark through `light-dark()`). New page styles go in `app/<page>/<page>.css`, scoped under `.halo-app`, with class names prefixed per feature (`.halo-today-*`, `.halo-hg-*`, `.halo-journal-*`, `.halo-act-*`, `.halo-settings-*`, `.halo-learn-*`, `.halo-map-*`, `.halo-assistant-*`, `.halo-alerts-*`). Phase 0 renames `app/onboarding.css` to hold only the foundation (`app/halo-foundation.css`) and moves the onboarding-only rules to `app/onboarding/onboarding.css`. The repository uses hand-written CSS; Tailwind is installed but the foundation does not depend on it.
- **Type sizes.** Use `rem` (or `calc(Npx / 16 * 1rem * var(--halo-text-scale))`), not `px`, for every font size in new CSS, or the Settings text-size control cannot work (see B6).
- **Storage.** Two namespaces, no others. `halo.frontend.v1.<identity>.*` for anything about the household (readings, dismissed prompts, last-used journal group, cached Learn pieces, map data); `clearReadings()` clears it, and so does Change address and an identity change. `halo.device.*` for device settings (text size, contrast, motion): survives Change address and is removed only by Delete everything. `ReadingKind` widens from `"daily" | "home"` to add `"history"`. `storeReading` strips `lat` and `lng` from what it stores: never use it for map data (a separate store, IndexedDB, holds that).
- **Fonts, icons, meta.** Phase 0 adds `public/fonts/ibm-plex-mono-latin.woff2` with `next/font/local` and the variable `--font-plex` (used only for water system IDs and certifications), and the icon set A5 lists (32, 96, 192, 512 and the 1200 by 630 preview) plus a web app manifest. Today the layout has only `/halo-icon.png`.

P3. The contracts every page codes against

These signatures are fixed now. Phase 0 may add optional fields; it may not rename or remove any.

The data layer BIND

```
// lib/frontend/api.ts  (Phase 0 owner)
export type HaloMethod = 'GET' | 'POST' | 'PUT' | 'PATCH' | 'DELETE';
export function haloRequest<T>(path: string, opts?: {
  method?: HaloMethod; body?: unknown; signal?: AbortSignal; timeoutMs?: number;
}): Promise<T>;
// Same behavior as the private request() inside createLiveOnboardingApi today:
// authedFetch, identity check before and after (session_changed), deadline,
// status to FrontendError code mapping, never surfaces a server error string.
// The 429 mapping keeps retry_after_seconds from the body on the error object.

// lib/frontend/useHaloData.ts  (Phase 0 owner)
export function useHaloData<T extends object>(endpoint: string | null, opts: {
  loader: (signal?: AbortSignal) => Promise<T>;
  enabled?: boolean;
  cache?: ReadingKind;            // 'daily' | 'home' | 'history'; omit for previews and uncached data
  isEmpty?: (data: T) => boolean;
}): { data: T | null; status: 'loading' | 'error' | 'empty' | 'success';
      error: ErrorCode | null; retrievedAt: string | null;
      fromCache: boolean; refreshing: boolean; refresh: (o?: { fresh?: boolean }) => Promise<void> };
```

- The four universal states in A2 map to the hook's statuses: loading, error, empty, and **success** (called "populated" in prose).
- When a refresh fails and data is already on screen, the hook keeps the data, keeps `status` at `"success"` and sets `error`. That combination (`error !== null && data !== null`) is how a page knows to show "Showing your last readings from [time]. We couldn't reach the data source."
- `refresh({ fresh: true })` passes `fresh` to the loader for exactly that one call (pull to refresh on Today). Auto-refresh after 15 minutes away calls `refresh()` without it, so it cannot trip the 5-minute limiter.
- New `ErrorCode` values Phase 0 adds: `not_found` (404), `unavailable` (503 that is not an identity problem, and a platform 502 or 504), `invalid_request` (400 with a body the client did not expect), `unsupported` (501). Existing codes stay.

Client timeouts BIND

Request	Client timeout
`ensureAnonSession` , `POST /api/onboard` , `GET /api/daily-score` , `GET /api/home-guard`	15 seconds
`GET /api/map` , `GET /api/district` (a cold rebuild can be slow; the server allows up to 60 seconds)	20 seconds
`POST /api/assistant`	25 seconds (a timeout is treated like `reason: "unavailable"`)
Everything else	8 seconds

Sheets, toasts, and the back button BIND

```
// lib/frontend/sheets.ts  (Phase 0 owner; pages depend only on this file)
export type LearnTopic = 'pfas' | 'radon' | 'lead' | 'air' | 'pollen' | 'uv' | 'mold';
export function openLearn(req: {
  topic: LearnTopic;
  reading?: { severity: Severity; valueText?: string };      // the "Your reading" pill
  context?: Record<string, string | number | null>;          // query params for /api/learn
}): void;
export function openAlerts(): void;
export function openAssistant(page?: 'today' | 'homeguard' | 'journal' | 'map' | 'act'): void;
export function closeSheet(): void;
export function useSheets(): { active: 'learn' | 'alerts' | 'assistant' | 'page' | null };
export function useToast(): { show(message: string, o?: { actionLabel: string; onAction(): void }): void };
// <BottomSheet id title height onClose> (components/ui) registers itself here,
// which is how the assistant button hides and how "only one sheet" holds.
```

- **Opening.** A sheet that has an address (`?learn=` and `?system=`) opens with `router.push` ; the others use `history.pushState` . Either way exactly one history entry is added.
- **Closing.** Every close path calls `closeSheet()` . If this app pushed the entry, close calls `history.back()` . If the sheet came from a pasted deep link (nothing to go back to), close uses `router.replace` to the same URL without the sheet parameter, so closing never leaves the app.
- **Sheet from a sheet** (a map water sheet opening "Learn about PFAS"): the second opens with `replaceState` , so Back closes to the page beneath and deselects the map feature. A button inside a sheet that navigates to another page (home pin "Go to Today") closes the sheet by replace, then pushes the route.
- **Deep links to Learn.** `?learn=<topic>` is accepted on every tabbed page. Today accepts `air | uv | pollen | mold` ; HomeGuard accepts `pfas | radon | lead` . On a deep link the page rebuilds `reading` and `context` from its own loaded data if the topic belongs to it. Otherwise the sheet opens with no "Your reading" pill and without section 2 (see C1). Any other value is ignored (no error page).

- `valueText` **per topic** (the pill's text): air "AQI 37"; uv "5.2"; pollen "Grass 3"; mold "Estimate"; pfas "8.3 ppt · 2.1×"; radon "Zone 3"; lead "Built 1975" or "Year unknown".
- **Toast strings.** Each page's owner lists its toast strings in its own copy file. Toasts are for confirmation and Undo only. A failure that needs action stays on the page as an inline error.

Shared profile and household saves BIND

```
// lib/frontend/profile.ts  (Phase 0 owner)
export function useProfile(): DataResult<ProfileResponse> & { refresh(): void };
// One GET /api/profile per app session, shared by every page and the shell.

// lib/frontend/household.ts  (Settings owner; HomeGuard and Onboarding import it)
export function saveHousehold(next: {
  household: Household; renter_mode?: boolean; locale?: 'en' | 'es';
}): Promise<void>;
// ALWAYS sends all seven booleans. Reads the current values from useProfile(),
// never from the PUT response. Serializes overlapping calls so the last tap wins
// in arrival order. Controls that use it stay disabled until GET /api/profile
// has succeeded, otherwise a save from an empty local copy would wipe every group.

// lib/frontend/alerts.ts  (Alerts owner; the shell imports BellBadge)
export function useAlerts(): { unread: number; alerts: Alert[]; status; refresh(); markRead(id); markAll(); dismiss(id);
undo(id) };
export function BellBadge(): JSX.Element;      // unread count, "9+" cap, hidden at zero
export function AlertsSheet(props: { open: boolean; onClose(): void }): JSX.Element;
```

Alerts badge behavior (decided here). The badge is `GET /api/alerts` → `unread`. Opening the sheet does not clear it; reading rows does (this keeps "which alerts are new" visible). It refetches on app load, on window focus after 15 minutes away, and on sheet close. Optimistic updates revert with a toast if the call fails.

Change notice between pages FRONTEND RULE

After any successful Settings save that changes wording (household, renter or owner, water source, build year, language), Settings calls `clearReadings()` and dispatches the window event `halo:settings-changed`. Today, HomeGuard and Act listen and refetch. Act's sheets always fetch fresh on open rather than reading the stored copy.

P4. Rules that apply to every page

Rule	Detail
Bind to the live API from the first commit	The backend is merged into `main` and running. Pages call the real API through the live adapter. The mock adapter exists only to force states (loading, error, no data, partial, offline) in previews and tests. There is no "fixture first, wire later" step.
Never test destructive calls against production	The local `.env.local` points at the production Supabase project. `DELETE /api/account`, `DELETE /api/journal?all=true` and `PUT /api/household` change real rows. Build and test with mock adapters; run live checks only against a separate Supabase project or a throwaway anonymous session you are willing to lose.
Server strings	Every server-provided string passes through `displayText()` (A6). The server also emits some text that must never be shown (raw error strings, dataset names): D0 and D3 list them.
Missing values	The words "No data" (`copy.common.noData`). Never 0, never blank, never a dash. In a card that has both a value slot and a severity pill, "No data" appears in only one of them (DESIGN: the pill).
No em dashes	None in app-facing text (this spec's own examples follow the rule). Titles use the template `HALO | Page` .
Appearance	There is no default mode. Every color is a `light-dark()` token and the phone's setting decides. Print output is always light. See A11.
Next.js	Version 16.2.10 has breaking changes from what most tooling assumes (`AGENTS.md`) and known security advisories; the fix is 16.3.6, outside the pinned range. Do not upgrade inside a page branch. Read `node_modules/next/dist/docs/` before writing routing code.

P5. How the pages depend on each other

Dependency	Provided by	Used by	Until it exists
`openLearn`	Phase 0 (contract), Learn owner (the sheet)	Today, HomeGuard, Map	SheetHost shows a labeled placeholder in development only
`openAlerts` , `BellBadge`	Phase 0 (contract),	Header on Today	Slot renders nothing

	Alerts owner		
`openAssistant`	Phase 0 (contract), Assistant owner	Shell button	`NEXT_PUBLIC_HALO_ASSISTANT=off`
`saveHousehold`	Settings owner	Settings, HomeGuard (renter control), Onboarding (its own PUT is fine as is)	HomeGuard hides the renter control
`halo:settings-changed` and `clearReadings`	Settings owner	Today, HomeGuard, Act	Pages ignore an event that never fires
Stored `daily` and `home` readings	Today and HomeGuard owners	Act (health card, volunteer causes), Map (home pin sheet)	Act and Map show "No data" for the missing part and never trigger a provider fetch themselves
`/settings?open=home|renting|household`	Settings owner	HomeGuard ("add the year your home was built", "Change" renter)	Plain link to `/settings`
`/map?panel=district`	Map owner	Act's district card	Plain link to `/map`
`/onboarding?change=1`	Onboarding (built)	Settings (Change address), HomeGuard (400 no county)	Works today

P6. Design tokens

These are the tokens that exist on `main` (defined with `light-dark(light, dark)`). Every page uses these names; the A11 tables give the meaning.

Token	Light	Dark	Use
`--halo-page`	`#eef8f7`	`#081725`	Page background
`--halo-card`	`#ffffff`	`#0f2436`	Cards and sheets
`--halo-ink`	`#0b1f33`	`#e8f3f5`	Body text
`--halo-secondary`	`#4a6275`	`#9bb4c2`	Timestamps, sources, captions
`--halo-primary`, `--halo-on-primary`	`#0e5e6f`, `#ffffff`	`#2cc4bd`, `#081725`	Primary button, active tab, links, focus ring, and the text on them
`--halo-border`	`#d3e4e6`	`#1d3a50`	Dividers, outlines
`--halo-tint`	`#d9f0ef`	`#1d4f5c`	Selected chips, soft tints (this is the approved value; it replaces the older A11 light aqua)
`--halo-error`, `--halo-error-tint`	`#c0262d`, `#fff6f6`	`#f0605f`, `#301d2b`	Field errors and the error callout
`--halo-good`, `-moderate`, `-elevated`, `-high`, `-severe`, `-no-data`	A11 severity table	A11 severity table	Severity fills; `--halo-good-ink` is the pill text on a filled level
`--halo-navy`, `--halo-teal`	`#0b1f33`, `#0e5e6f`	same	Welcome band only
`--halo-gutter`, `--halo-radius`, `--halo-heading`	20px, 14px, 26px	same	Page gutter, card and input radius, step and page heading size

Tokens Phase 0 adds (each with a light and a dark value, decided in Phase 0's design pass and frozen): `--halo-skeleton` (visible in both modes), `--halo-callout-info`, `-caution`, `-notice` (background and text), `--halo-limit-marker` (the federal limit line on charts), `--halo-house-ground`, `--halo-house-roof`, the four Today hero gradient stop pairs (see B2), and a high-contrast override set. Text contrast is checked for every hero state in every time-of-day and both modes.

P7. The two Onboarding follow-ups FRONTEND RULE

Onboarding is built and approved, and it is the reference for everything above. Two small changes are still needed so Settings can use it. Owner: Yogi, before Settings is built.

1. **Change address returns properly.** In `/onboarding?change=1`, the Back arrow on the address step returns to `/settings` (today it lands on the Welcome screen, which makes no sense for someone changing an address). Completing the flow ends at `/today` (already true).
2. **The Household step in change mode.** Recommended (OPEN D2): keep it, prefilled with the current groups, because the built flow already does this and it lets someone who moved change who lives there. Settings' Your home row must say so; do not describe a flow that skips it.

P8. Building each page: wireframe, freeze, build, prove

Every page follows the workflow in **HALO_Wireframe_to_Build.md** (the file given to Codex), which is Onboarding's proven process:

1. **Wireframe**: a real, final-looking preview route with mock scenarios for every state, in light and dark, at 375px. Yogi reviews screenshots.
2. **Freeze**: approved screenshots become Playwright baselines (`docs/<page>/approved/`) with a README listing every element. Approved screenshots are never regenerated to make a test pass.
3. **Build**: the same screen component, now with the live adapter, bound to Part D.
4. **Prove**: zero visual difference against the frozen screenshots, a live check against the real API, a flow check into and out of the page, and the element checklist from this spec.

The element tables in Part B are the checklist. The pages' wireframes may differ in composition and spacing; they may not drop an element, a state or a string. Shared components look identical everywhere because Phase 0 froze them first.

P9. Asks for Vibhav (backend, all additive)

Nothing here blocks the frontend. Each item has a stated interim rule in the page that needs it, so builders can proceed. Tell Vibhav which he wants to do; do not change backend files from a frontend branch.

#	Ask	Interim frontend rule
1	Add `severity` to each entry of `water.scored_contaminants[]` (the value `waterRiskSeverity(risk)` already computes)	HomeGuard threshold rows take the water box severity; rows that do not exceed a limit use the neutral bar (B3)
2	Add `sends_household_context` (boolean) to `GET /api/assistant`, so the sheet and the privacy statement can say which mode is live. Also make the default safe: today the code turns household data ON unless the variable is exactly the string `false` (unset, `off` or `0` all mean on). Change it so only `true` turns it on	Assume false (public sources only) and use the false-state privacy line (C2, B6). Until the code changes, Vibhav sets the Vercel variable to exactly `false`
3	Add `export const maxDuration = 30` to the assistant route (the client waits 25 seconds; no route limit is set)	None needed
4	Add `history[].backfilled` (and `is_partial`) to `GET /api/history`	Compare severity words only, and show the "change since yesterday" line only when the rule in B2 allows
5	Remove em dashes from server-written text at the source (`lib/explain.js` , `learnContent.js` , `learnWhyYours.js` , `wellTestData.js` , `leadRisk.js` , `actionPlan.js` , `alertRules.js` , the assistant refusal, home-guard messages)	`displayText()` handles it either way
6	Replace `{ error: err.message }` on the routes that still return raw database text (journal, history, alerts, notifications, push, volunteer, learn, sources, map, district, assistant, account, health)	The frontend never displays a server error string
7	Change the map suggestion chip "What does 'not tested' mean on the map?" (the map says "No detection reported")	Chip text is server-supplied; note it in testing
8	Soften `score.reason` wording "is capped for that reason" when `score.capped` is false	Gate any "capped" wording on `score.capped === true`
9	Never store an empty district or water payload over a good one. When the water geography lookup fails, the server currently stores `total_systems: 0` , `systems_over_limit: 0` and `affected_population: 0` , which reads as an all-clear	Treat `total_systems` of 0, or `unlocated_systems` near `total_systems` , as no data (B7)

P10. Decisions still open, with the default a builder uses meanwhile

ID	Decision (Yogi unless noted)	Default until decided
D1	Who owns Phase 0	Recommended: whoever has the most Codex quota, reviewed by Yogi
D2	Change address: keep the prefilled Household step, or skip it	Keep it (as built)
D3	Dark mode: follow the phone (built and approved in both), or dark only	Follow the phone. A dark-only switch is a one-line change to `color-scheme` plus a review of the light screenshots
D4	Send readings and symptom names and dates to the Assistant's model (`ASSISTANT_SEND_HOUSEHOLD_CONTEXT`)	Off. The Assistant answers from public sources only (the Vercel variable must be exactly `false` : any other value, including unset, turns household data on)
D5	Who writes the missing content (Act's help actions and letters, About text, radon and	Builders use the PROPOSED strings in this spec and list each one used, so nothing invented ships unnoticed

	pollen notes, named local district facts) and by when (Aadit unless changed)	
D6	Season summary alert target	Show the summary callout at the top of both Log and Trends, keep the alert opening `/journal?mode=trends`
D7	Map library and tiles	Mapbox GL JS with the existing `NEXT_PUBLIC_MAPBOX_TOKEN` and Mapbox's muted light and dark styles; NC county shapes from `lib/data/ncCounties.json` (100 counties, names identical to the radon keys, about 46 KB compressed) copied to `public/data/nc-counties.geojson`
D8	Device notifications (service worker) in this build	Build after the core pages; the Settings switch is hidden until the service worker ships (B6)
D9	Volunteer organizations: all 13 have `last_verified` empty	Rows say "Not yet verified" until a person confirms each one and sets the date (Vibhav's wrap-up item)

Part A: Cross-Cutting Systems

Everything in this part applies to every page. Built once, in the shared layer, before any page, the blueprint's own first build step (§26.1) and the thing that decides whether dark mode and consistency are cheap or impossible later.

A1. Platform, shell, and navigation

Platform LOCKED

- A web application with a mobile-shaped layout, running in a mobile browser. Not native iOS or Android (§3).
- Primary viewport 375–430px. Every screen verified at 375px. Content column caps at 430px and centers above that; the Map is the one exception and widens to 600px (§53.2). Under 375px content compresses; nothing hides; no horizontal scroll ever.
- Portrait is the target. Landscape isn't blocked, but has no special layout (§53.4).
- Header respects the top safe-area inset, tab bar the bottom inset, sheets the bottom inset (§53.3).
- Stack: Next.js 16.2.10 App Router, React 19.2.4, Tailwind v4, `lucide-react` as the single icon family. The repo's `AGENTS.md` warns that this Next.js version has breaking changes from what most tooling assumes, whoever builds reads `node_modules/next/dist/docs/` before writing routing code. Version 16.2.10 also has published security advisories (fix: 16.3.6, outside the pinned range); do not upgrade inside a page branch (P4).
- Styling is the repository's hand-written CSS with `--halo-*` tokens (P2, P6). Tailwind is installed but the foundation does not depend on it.

Pages and overlays LOCKED

Page	Address	Tab label	Header title	Reached by
Entry	`/`	none	none	Redirects to `/onboarding?entry=1`, which decides where the visitor goes
Onboarding	`/onboarding`	n/a	none (full screen)	Through `/` on first visit; `?change=1` from Settings; `/onboarding/preview` in development only
Today	`/today`	Today	Today	Tab 1, default landing
HomeGuard	`/home`	Home	Your home	Tab 2
Map	`/map`	Map	Map	Tab 3
Journal	`/journal`	Journal	Journal	Tab 4
Act	`/act`	Act	Act	Tab 5
Settings	`/settings`	n/a	Settings (back arrow, not gear)	Gear icon in any header

Three overlays, Learn, Assistant, Alerts, are bottom sheets, never pages, never tabs. Learn opens only from a specific reading's "What does this mean?" link (no index, no browsing). Assistant opens from a floating button on all five tabbed pages. Alerts opens from the bell on Today's header only (§7.2).

Header bar LOCKED

Fixed top on every tabbed page. Page title left. Today: alerts bell (unread badge, capped at "9+", hidden at zero) and settings gear on the right. Every other tabbed page: gear only. Never hides on scroll (§7.3, §58.1). On Map, the header sits over the map (§58.1). BIND badge count = `GET /api/alerts` → `unread`. **The shell renders the header, the bell and the assistant button** from the route (Part P2); pages render none of them, and no page fetches `/api/alerts` for the badge.

Bottom tab bar LOCKED

Fixed bottom, five equal-width destinations, icon above label, active destination visibly distinct from inactive, always visible, never scrolls away (§7.4). Icons (§32.4): Today = sun over horizon, Home = house, Map = folded map, Journal = notebook, Act = outstretched hand. Tab changes are instant, no transition animation; content fades in as it arrives (§28.2). Each tab remembers its scroll position for the session; a newly visited tab starts at the top (§27.4).

Floating assistant button LOCKED

Lower right, above the tab bar, on all five tabbed pages. Hides while any sheet is open. Carries a chat-bubble icon (§32.4). The blueprint also has it fade out on scroll down and back on scroll up (§58.3). **Tier 3: default is a fixed button** (§65.3). Opens the Assistant sheet carrying the current page as context.

First-visit routing LOCKED FRONTEND RULE

Blueprint (§7.5): on every load, check for a stored location. Have one → Today. Don't → onboarding. If the check itself fails → onboarding, never an error screen.

```
/                                redirects to /onboarding?entry=1
/onboarding?entry=1 (OnboardingScreen bootstrap):
  await ensureAnonSession()               // lib/auth.js, also run app-wide by AuthInitializer
  GET /api/profile
    onboarding_complete === true          -> markCompleted(profile), go to /today (or ?next=)
    request failed, hasCompletedOnboarding() true
                                          -> /today?verify=1  (ProfileRecheck retries in the background)
    anything else                         -> stay on onboarding
Every route in the app shell (AppGate, Part P2):
  no session or no completed profile      -> /onboarding?entry=1&next=<current address>
  a page reports no_location / no_county  -> /onboarding?entry=1
```

FOUR RULES. (1) Route on `onboarding_complete`, not `onboarded`. `onboarded` turns true as soon as coordinates are saved, at the address step, so someone who closes the app on step 3 or 4 would be sent to Today with no water source, and a well household inside a utility boundary would be shown that utility's numbers. A household that did not finish restarts at the address step (the address was never kept). (2) The local marker is `halo.onboarded`, a JSON record `{ version: 1, identity, complete: true }` written by `markCompleted(profile)` and read by `hasCompletedOnboarding()` (both in `lib/frontend/storage.ts`). It belongs to one session identity: a marker from another identity, or the old bare-boolean form, is ignored. It is set only after step 4 succeeds. (3) When `GET /api/profile` *succeeds* with an incomplete profile, clear the marker and every stored reading first (after a genuine 401 reset creates a new anonymous user, the old marker and readings would otherwise linger). (4) When the profile request *fails* and the marker is valid for this identity, go to `/today?verify=1`. Today must render `<ProfileRecheck />` when `verify=1`; it retries, redirects an incomplete profile to onboarding, and offers Retry if verification stays unavailable. It never loops and never resets the identity. A finished household is never sent back through onboarding because of a failed request.

Addresses and state in the URL LOCKED

Address	Restores
`/map?layer=radon`	Map with radon layer active
`/map?layer=water&contaminant=pfos`	Map colored by PFOS
`/map?system=NC0112010`	Map with that system's sheet open
`/journal?mode=trends`	Journal in Trends mode
`/home?risk=water`	HomeGuard with the water card expanded (also `radon`, `lead`)
`/today?learn=air`	Today with that Learn sheet open (Today accepts `air`, `uv`, `pollen`, `mold`; HomeGuard accepts `pfas`, `radon`, `lead`)
`/today?verify=1`	Today plus the background profile recheck (onboarding handoff, above)
`/onboarding?entry=1` · `?change=1`	Normal entry · replace the address (from Settings and HomeGuard)
`/settings?open=home` (also `renting`, `household`)	Settings with that section open. One-shot: opened on load, then removed from the address with `router.replace`. The only exception to "the open section is not in the URL"
`/map?panel=district`	Map with the district panel open

Rules for every address parameter: unrecognized or invalid values are ignored and the default is used (never an error page). Expanding a card, changing the Map layer, contaminant or quarter, and switching the Journal mode use `router.replace` (no history entry); opening a sheet adds one entry. Only one card is open at a time on Today and HomeGuard. Tab links carry no query string, so a saved viewport or scroll position applies only when the address has no parameters (URL parameters win).

Not carried in the URL: scroll position, assistant conversation, unsaved journal input, which Settings accordion is open (§27.2). `/map?layer=water&quarter=2024Q3` applies only when the time slider exists (the map reports available `quarters`; see B7). Any unrecognized address shows a short message and a button back to Today, never a raw error page, never a silent redirect (§27.5).

The back-button contract LOCKED

- Opening any sheet (Learn, Assistant, Alerts, a map detail, any Act feature, Journal day detail, delete confirmation) pushes a history entry.
- Back with a sheet open closes the sheet and stays on the page beneath.
- Back with no sheet open goes to the previous tab, or exits.
- Closing a sheet any other way (drag down, tap backdrop, Escape, close control) also removes its history entry, so Back doesn't reopen it.
- Only one sheet at a time. Opening a second replaces the first (using `replaceState`, so Back closes to the page beneath). The exact open and close algorithm, including deep links, is in Part P3.

- Expanding a card is not a history entry. Back never collapses a card.
- Each onboarding step pushes an entry; Back moves between steps without losing entered values.
- Opening a sheet locks page scroll beneath it; closing restores the exact prior position. Expanding a card scrolls just enough to reveal its new content (§27.3–27.4).

Layering LOCKED

Back to front: page content → fixed header → fixed tab bar → floating assistant button → sheet backdrop → sheet → toast → offline banner. Nothing else creates a layer (§53.1).

A2. The data layer

Authentication BIND

- `ensureAnonSession()` (lib/auth.js), silent anonymous session, runs app-wide from the root layout. Throws with `.code` `'captcha_failed'` or `'signin_failed'`; the onboarding screen shows "*Verification could not be completed. Check your connection and try again.*" or "*Could not start a session. Please reload and try again.*"
- `authedFetch(url, init)`, attaches `Authorization: Bearer <token>`. Pass `init.headers` as a plain object (a `Headers` instance gets dropped by the spread).
- Every data route requires the bearer token. No route accepts identity from the body or query.
- **401 vs 503, the rule that protects a household's data.** `401` means the session is definitively invalid (the account no longer exists): discard it and start a new anonymous session, silently (§31.4). `503` means identity couldn't be verified right now: show the generic error with retry and **never** reset the session. Resetting on a temporary failure would silently orphan the household's profile, history, and journal under an abandoned account. If the silent retry after a 401 also fails, show "*Could not start a session. Please reload and try again.*"
- The built layer also protects against mixing sessions: the identity is compared before and after every request, and a changed identity rejects the old response with `session_changed` instead of showing another household's data. A browser-wide observer clears frontend caches and emits the window event `halo:identity-changed`. It does not create or reset a session.

One data hook, one state wrapper FRONTEND RULE

Every page gets data the same way: `useHaloData(endpoint, { loader, enabled, cache, isEmpty })` (built in `lib/frontend/useHaloData.ts`, made generic in Phase 0, signature in P3) and a single `<DataState>` wrapper that renders the right one of the four universal states. The hook returns `{ data, status, error, retrievedAt, fromCache, refreshing, refresh }`; its statuses are `loading`, `error`, `empty`, `success`. Pages never write their own loading or error logic. Requests go through the one `haloRequest` function (P3), which owns authentication, timeouts, identity checks and error mapping. Pages never call `fetch` directly. This is how "every screen implements all four states" (§24) holds across ten screens and three builders.

The four universal states LOCKED

State	When	Shows
Loading	Request in flight, nothing cached	Skeleton shaped exactly like the content it replaces, same dimensions, so nothing shifts when data arrives. Never a bare spinner (Map is the one exception, §46.9)
Error	Request failed	What failed + that it isn't a finding about their environment + Retry
Empty	Succeeded, nothing to show	Why there's nothing, and what would make content appear
Populated	Data present	The content

Error wording and empty wording are never the same text (§24). Every screen tolerates a missing water utility ID without breaking.

Error responses: what the frontend shows BIND

Response	Show
401, any message	Nothing (silent re-auth, §31.4), subject to the guard above
503 `"Could not verify your session right now…"`	Generic error + retry
429 (any route)	"*You've made a lot of requests. Please try again later.*" (Assistant has its own 429 text, see C2. A 429 on Today's daily-score has its own handling, next row.)
429 on `/api/daily-score` (with `?fresh=1`, or a plain call that misses the server's cache and would reach the providers) with body `{ error, retry_after_seconds: 300 }`	When readings are already on screen or stored: not an error state. The readings stay, the freshness line is unchanged, and a toast shows PROPOSED "*Your readings were just updated. Try again in a few minutes.*" When nothing is stored: the generic 429 copy above. The server allows one provider refresh per household per 5 minutes
400 on `/api/daily-score`, `"No location yet…"`	The household has no stored location: treat it exactly like `onboarding_complete: false` and route to onboarding (A1)
400 on `/api/home-`	Rare (the geocoder returned no county). HomeGuard shows its error state with a *Change address* action (B6);

`guard`, `"No county on file…"`	on the results reveal, the water and radon lines read *Couldn't check right now*
500, or network failure	*"Something went wrong on our end, not yours. Try again in a moment."*
Timeout (client-side, 8s default)	*"That took too long. The data source may be slow right now."* **Longer deadlines** (the table in P3): `POST /api/onboard`, `GET /api/daily-score` and `GET /api/home-guard` use **15s**; `GET /api/map` and `/api/district` use **20s** (cold rebuilds are slow; map layer providers have their own 20s cap on the server); `POST /api/assistant` uses **25s** (its timeout shows the Assistant's own "couldn't reach" message with Retry, C2). The daily, HomeGuard and onboarding providers have no server-side cap, so a cold call can run long. When `POST /api/onboard` times out, the client re-checks `GET /api/profile` before showing an error: each address submission carries a fresh `request_id`, and the profile's `onboard_request_id` equal to it confirms that exact write (Onboarding already does this)
Offline (`navigator.onLine` false)	Offline banner + last stored readings (below)

> **TWO BLUEPRINT STRINGS DELIBERATELY NOT USED:** §31.4's *"[Source name] isn't responding right now. The rest of your readings are current."* is dropped because §39.1 forbids naming services a household won't recognize, a missing reading shows its card's own no-data state instead (B2). §31.4's *"You appear to be offline. We're showing the last readings we have."* is replaced by §58.4's banner text, *"You're offline. Showing your last readings."* (`copy.common.offlineCached`). Don't reintroduce either.

> **NEVER DISPLAY A SERVER'S `error` STRING DIRECTLY:** most routes return raw database or JavaScript messages in 500s. Map every error to the copy above. The one exception is onboarding's specific address errors (B1), which are mapped by status code.

Stored readings, offline, and refresh LOCKED

- The last successful Today and HomeGuard responses are stored locally with the time they were received and shown immediately on open, while fresh data loads (§24, §36). Cached reading display target: under 200ms (§35.1).
- Fresh request fails and stored data exists → keep it on screen with *"Showing your last readings from [time]. We couldn't reach the data source."*
- Offline → thin banner under the header: *"You're offline. Showing your last readings."* Pushes content down, never covers it (§58.4).
- Learn: cache only the parts that do not depend on the household, `what_it_is`, `protect[]` and `sources[]`, keyed by topic and locale. **Never cache `why_yours` or `household_note`** (they are built from the reading and the household, which can change), and refetch every time the sheet opens. Offline: show the cached parts and omit sections 2 and 3. Map data is cached after first load (§36) in its own store (P2), never through `storeReading`, which strips coordinates.
- Pull to refresh on Today and HomeGuard only. On Today, the page calls `refresh({ fresh: true })` for exactly that one call, which sends `/api/daily-score?fresh=1` and bypasses the server's one-hour cache. HomeGuard simply re-requests `GET /api/home-guard`, which has no server cache and no `fresh` parameter. Disabled while a request is in flight. Automatic refresh when the app regains focus after 15+ minutes away (§28.1) is a plain request without `fresh`, so it cannot trip the 5-minute limiter. A refresh that returns cached data still updates the freshness line.
- Freshness lines use the server's `retrieved_at` (or `assembled_at`), never the time the browser received the response, a cached reading can be up to an hour old.
- Degradation order: fresh fails → stored data with its time → a single reading unavailable while others show → all fail: explanatory empty state with retry → server unreachable: stored data + offline banner (§39.2).

Journal writes offline TIER 3

The blueprint queues entries written offline and sends them on reconnect, with a small pending marker (§36). It's Tier 3 (§65.3): **the default is a clear failure instead**, keep everything typed and show PROPOSED *"Couldn't save that entry. It's still here. Try again."* Build the queue only if time allows.

A3. Severity and confidence

The six severity words LOCKED BIND

The backend emits exactly these words: `good`, `moderate`, `elevated`, `high`, `severe`, `no_data`. The frontend holds one mapping (word → label, icon, color token), ideally importing the word list from `lib/severity.js` so it can't drift. Labels: Good, Moderate, Elevated, High, Severe, No data. Any unrecognized word renders as No data (§62.4).

- Every severity appears as **color + icon + word together**, never color alone (§6.1, §25.1). This is an accessibility requirement, not a style preference.
- No data is visually distinct from Good everywhere. Missing data is never shown as zero or green, the single most dangerous failure the blueprint names (§65.1).
- Pill labels capitalized; the same word lowercase inside a sentence (§29.3).

Where to read severity, per endpoint BIND

Screen element	Read	Never read
Today's four cards	`air.severity`, `uv.severity`, `pollen.severity`, `mold.severity`	`*.status`, older field with different thresholds (AQI 120 is `status "unhealthy"` but `severity "elevated"`)
Today trend lines, Journal grid	`/api/history` → `history[].air / uv / pollen / mold` (plain words). Note `history[].pollen` is the worst pollen category that day, while a pollen trend plots the value of today's dominant category, so a non-dominant day's point takes its color from the display-only edges in A10	n/a
HomeGuard cards & house diagram	`breakdown[].severity`, matched by `breakdown[].key` (`water`, `radon`, `lead`), never by position	`breakdown[].status`, older field with its own thresholds
HomeGuard threshold-chart rows	No per-row severity exists yet (ask 1 in P9). Interim: every bar takes the Drinking-water box severity; a row with `exceeds_limit` false uses the neutral bar color	Never derive a color from `risk` or the ratio
Both heroes (score ring, headline)	`score.severity`	n/a
HomeGuard action plan	`action_plan[].severity`	n/a
Lead card	`lead.level` (canonical)	n/a
Map	Water: `features[].overall_severity`, `features[].contaminants[X].severity`, `features[].hazard_index.severity`. Radon, facilities, air: `features[].severity`	n/a
District panel	`counties[].worst_water_severity`, `counties[].radon_severity`	n/a
Alerts	`alerts[].severity` (set for alerts that carry a level; else null)	n/a

Hero severity: what colors the score ring BIND

Both heroes read `score.severity`, the server decides the composite's level (§11.6). On HomeGuard it reflects water and radon only; **lead never colors the ring**, because it isn't scored. When the score is null (a private well, or nothing available), the ring shows its no-data state.

Confidence: a separate signal LOCKED BIND

`confidence` value	Shown	Meaning
`full`	nothing	Everything evaluated, current
`limited`	quiet pill: *partial*	Some detections couldn't be evaluated
`stale`	quiet caution pill: *3+ yrs old*	Evaluated, but the sample is old
`none`	quiet pill: *not evaluated*	Nothing could be evaluated

HomeGuard's `breakdown[].confidence` carries this. A reading can be low severity and low confidence at once, never merge the two into one element (§6.2).

Provenance pills LOCKED

Three words: *measured*, *estimate*, *modeled* (§31.2). Air: `is_measured === true` → measured; `false` → modeled; `null` → no provenance pill (unknown isn't the same as modeled). UV and pollen: no provenance pill (the API exposes none). Today's cards show no confidence pill (`/api/daily-score` has no confidence field). Mold: always *estimate* (`is_proxy: true`). Radon: always *estimate* (county zone, not a home measurement). Lead: *estimate* when `lead.is_estimate`.

A4. Household-aware wording

LOCKED Seven yes/no facts per household, no ages, names, or medical records: `has_toddler` , `has_child` , `has_teen` , `has_adult` , `has_senior` , `has_pregnant` , `has_respiratory` (§8). Composition never changes a measurement, a severity, or a score, only the sentence under a reading and the order of actions (§8.2).

BIND The sentence is already written server-side: `air.sentence` , `uv.sentence` , `pollen.sentence` , `mold.sentence` (Today), `lead.sentence` (HomeGuard), `household_note` (Learn). Display them as-is. The frontend never composes a household sentence.

When several groups apply, the server uses the most specific one that has wording for that reading, in the blueprint's priority order: respiratory condition → toddler → pregnancy → senior → child → general (§8.5).

A5. Formatting rules LOCKED

One formatting module, used everywhere. The same quantity formatted two ways on two screens reads as unfinished (§29).

Quantity	Format	Example	Source field
Air quality index	Whole number	37	`air.aqi` (may be a decimal from the modeled source, round)
UV index	One decimal	5.2	`uv.index`
Pollen	Whole number with its category	Grass 3	`pollen.dominant` + `pollen[dominant]`
Concentration	One decimal, unit after a space	8.3 ppt	`value_ppt` (sent unrounded)
Ratio to limit	One decimal + ×	2.1× the limit	`value_ppt / limit_ppt`, arithmetic on server numbers, not a threshold
Composite score	Whole number, no unit, no %	78	`score.display_score`, never `score.score`
Radon zone	"Zone" + numeral	Zone 2	`radon.zone`
Population	Commas, whole number, no rounding and no "+" sign. Where the server marks a figure as incomplete (`affected_population_complete` false, `over_population_unknown`), write "at least" before it	12,400 · at least 120,000	map `population`, district `affected_population`, `population_over`, letter `population_served`
Costs	Range, en dash, no decimals	$40–90	action plan `cost` (already formatted); test plan `estimated_cost.low/high`
Percent	Whole number + %. Journal `elevated_rate` and `other_rate` are already 0 to 100 integers: do not scale them again. District `district_share_over` and `state_share_over` are fractions from 0 to 1: multiply by 100	78%	journal, district
Lithium	Micrograms per liter, never ppt: value = `value_ppt / 1000`, formatted one decimal	25.0 µg/L	`excluded_from_score[].value_ppt`; reference level `health_reference_level_ppt / 1000`. The raw name `lithium` displays as "Lithium"
Missing value	The words "No data" (`copy.common.noData`). Never 0, never blank, never a dash	No data	any `null`

- Units spelled out the first time in a view, abbreviated after: "parts per trillion (ppt)" then "ppt."
- Measured values use tabular digits (Instrument Sans with `font-variant-numeric: tabular-nums`) so digits align, not a monospaced face. Monospace (IBM Plex Mono) is for codes only (A11).
- Never more precision than the source provides.

Dates LOCKED FRONTEND RULE

Situation	Format	Example
Under 1 hour	Relative minutes	updated 14 minutes ago
Under 24 hours	Relative hours	updated 6 hours ago
Yesterday	The word	yesterday
Last week	Weekday	Tuesday
This year	Month and day	March 14
Older	Month and year	October 2024
Lab sample dates	Always month + year, never relative	sampled October 2024
Data age warnings	Rounded years	this sample is about 3 years old

> **WHICH DATE FIELDS:** water sample dates: bind the `*_iso` fields (`latest_sample_iso`, `earliest_sample_iso`, `date_iso`). The same objects also carry older `M/D/YYYY` strings (`latest_sample_date`, `date`) kept for compatibility, don't parse those; `new Date("7/1/2025")` is locale-dependent. Every other date is ISO.

ONE DATE CONVENTION: daily readings (`history[].date`) and journal entries (`entry_date`) are both the household's **local** calendar date. Join them by the same `YYYY-MM-DD` string. The server computes every date in **America/New_York** (`lib/localDate.js`). The client must use the same zone to join with history and entries: `todayET()` in `lib/frontend/format.ts` (P2), or the `to` field of `/api/history` . Never use the device's own date or a UTC timestamp. Parse `YYYY-MM-DD` by splitting into numbers, never with `new Date("2026-09-28")` (that parses as UTC and returns the previous weekday in US zones).

Text conventions LOCKED

- Sentence case everywhere, buttons and headings included. No title case, no all caps.
- PFOA, PFOS, PFHxS, PFNA, GenX keep their conventional capitals. The backend key `HFPO-DA` displays as "GenX (HFPO-DA)".
- Numbers under ten as words in prose, numerals in data. "County" always included. Locked strings that use numerals in a sentence ("3 days selected.", "Based on 3 of 4 readings.") are data strings and keep them.
- **No em dashes (U+2014) anywhere in app-facing text**: screens, buttons, errors, alerts, accessibility labels, page titles, and server-provided strings. Use a period, comma, colon or parentheses. A test enforces this (P2). This is a team decision that supersedes older punctuation in the blueprint.
- HALO always in capitals, except the approved Welcome subtitle "To Halo". Tagline where needed: "*Know what's around your home.*" Browser title template: `HALO | [page name]` (built in `app/layout.tsx`). Link preview description: "*See what's in the air, water, and ground around your home, and what to do about it.*" (§32).

Identity and icons LOCKED (§32)

- The approved logo is `public/halo-logo-mark.png` , a transparent PNG used as a monochrome mask and drawn **white on the navy Welcome band** (a contrasting foreground on light surfaces). The blue tile, `finallogo.png` and every earlier concept are superseded. It is used as the Welcome mark; the browser tab icon (32, 96, 192px); the home-screen icon (192 and 512px, with a web app manifest); and the link-preview image (1200×630, the mark and the name over the primary color). Only `/halo-icon.png` exists today; Phase 0 produces the full icon set (P2).
- The browser `theme-color` is the Primary token, `#0e5e6f` in light mode, `#2cc4bd` in dark (two `<meta name="theme-color">` tags with `media` queries).
- One icon family (lucide-react), one line weight, no mixed styles. Concept icons: Today, sun over horizon · Home, house · Map, folded map · Journal, notebook · Act, outstretched hand · Air quality, wind lines · UV, sun · Pollen, flower · Mold, droplet · Water, tap · Radon, radiation trefoil · Lead, pipe · Alerts, bell · Settings, gear · Assistant, chat bubble. Where lucide has no exact match (e.g. a trefoil), pick the closest line icon at the same weight and keep it consistent everywhere.

A6. Voice and copy LOCKED

Plain, specific, steady. Never minimize a real problem; never frighten someone who can't act tonight (§30).

- Never "safe" or "unsafe" as a bare verdict. Never exclamation marks. Never fear words: toxic, poison, dangerous, alarming, shocking.
- Never apologize for data that doesn't exist. Never congratulate a good reading. Never "simply," "just," or "easy" for something that costs money or effort. Never imply a medical outcome.
- Second person ("your water"). HALO calls itself HALO, never "we" as an organization making promises.
- Reading level about sixth grade.
- Anything not listed in the content inventory (§31) and not derived from data doesn't appear in the interface. Where this document proposes new copy, it's tagged PROPOSED and needs approval.

BACKEND MESSAGES THAT BREAK THE VOICE RULES, USE BLUEPRINT COPY INSTEAD: some server `message` strings name services a household won't recognize (§39.1). Where the blueprint has specific copy for a state, use the blueprint's. Example: HomeGuard's `no_data_yet` , show the blueprint's "*Testing results for your utility haven't been published yet. This is normal. Federal testing is still rolling out nationally.*" instead. Each page section notes where the server's text is used and where the blueprint's replaces it.

DATA IN CAPITALS AND SERVER TEXT THAT BREAKS VOICE: utility names arrive as EPA and state records spell them, often in capitals (e.g. `"CONCORD, CITY OF"` in `lead.utility_inventory.utility` ; `water.pws_name` likely the same). PROPOSED : display them in title case ("Concord, City of"), keeping acronyms such as WTP and state abbreviations in capitals. Don't build fixtures around the blueprint's "Concord, Hillgrove WTP", which is a treatment-plant name, not a utility record. A few server-written strings break A6 and are flagged for rewording by the content owner (Aadit) or Vibhav, for example a well test's `why` , "This is the one that can make your family sick this week, not in twenty years.", and alert messages like "Air quality dropped to Elevated today." (a capitalized severity mid-sentence). **Display server text unchanged, except that every server-provided string passes through `displayText()` first** (`lib/frontend/text.ts`), which replaces U+2014 with ", ". That covers every `sentence` , `reason` , `message` , `title` , `action` , `why` , `where` , `text` , `protect[]` , `household_note` , `why_yours` , `what_it_is` , source labels, `regulatory_notice` , utility and organization names, alert text, and the Assistant's answer and refusal. Do not otherwise rewrite it in the frontend. Server fields that must **never** be shown at all are listed in D3.

Universal copy LOCKED

Controls: Close, Back, Retry, Save, Cancel, Skip, Continue, Done, See more, Show less. Severity: Good, Moderate, Elevated, High, Severe, No data. Confidence: partial, 3+ yrs old, not evaluated. Provenance: measured, estimate, modeled (§31.1–31.2).

Confirmations (§31.5): changing address, "*This will replace your current results. Continue?*"; delete all, "*This permanently deletes everything HALO knows about your household. This cannot be undone.*" + typing a confirmation word; clear journal, "*This deletes all [n] of your entries permanently.*"; delete one journal entry, immediate, with undo, no dialog.

A7. Forms and validation LOCKED

- Validate on blur, not on every keystroke. Error text directly beneath the field, field border also marked. An error clears the moment the input becomes valid.
- The submit button is never disabled for validation, let people submit and tell them what's wrong. (Different from disabling during an in-flight request, which is required.)
- Every field keeps its value after a failed submission. Every input has a visible label, not just a placeholder. Text inputs at 16px or larger (smaller makes mobile browsers zoom on focus). Address requests device autofill; build year opens the numeric keypad and accepts digits only (§34).

Field	Rule	Error text
Address	Required, 5+ characters	*"Enter an address or ZIP code."*
Address not found (server 404)	n/a	*"We couldn't find that address. Try adding your city or ZIP code."*
Build year	Optional; if present, four digits, 1700 to current year	*"Enter a four-digit year, or leave this blank."*
Build year in the future	Rejected	*"That year hasn't happened yet."*
Water source	Required, from the list	*"Choose where your water comes from."*
Household groups	Never required	n/a
Journal note	Optional, max 500 characters	Character counter appears past 400
Assistant question	Required, max 1000 characters	*"Ask a question to get started."*

> **BUILD YEAR:** validate 1700 to the current year on the client (the rules above), and send it as a number, or `null` when blank. The server rejects values outside that range too, but the client check is what gives the person the right message.

A8. Accessibility, motion, text handling

Non-negotiable LOCKED

- Severity never by color alone. Every text/background pair meets standard contrast, verified separately in light and dark.
- Every interactive element at least 44×44px regardless of icon size.
- Everything reachable and operable by keyboard, in visual order, with a visible focus indicator. Escape closes any sheet. Enter/Space activates. Arrow keys move between segmented options and map layer chips; the map's time slider changes quarter with the arrow keys and the district handle is a button. Calendars and grids (the Journal's month calendar and heat grid) use a roving tabindex: arrows move by day and week, Space toggles or opens the day, and a visually hidden list of the same days is the text equivalent.
- Any gesture-only action (swipe to delete or dismiss) also has a visible 44px control for the same action (`SwipeRow` , P2).
- Sheets trap focus while open and return focus to whatever opened them.
- Icons and images have text alternatives; decorative ones are marked decorative. Every chart has a text equivalent stating the same finding.
- Body text never below the body size floor and respects the user's own text size setting.
- Hover states exist for pointer devices but never shift layout, and hover is never the only way to reveal something (§25.1, §28.6).

Motion: functional rules only LOCKED

Motion is limited to: sheet entry/exit, card expansion, content fade-in, press feedback, score ring sweep on first render, chart bar growth, onboarding step transitions, map color changes, toasts. Nothing longer than 400ms. Everything becomes instant under the system's reduced-motion setting, except opacity fades (kept very short so content doesn't pop). Exact durations and easing are DESIGN.

Touch and request feedback LOCKED

Every tappable element visibly responds on press within 100ms. Any button that triggers a request shows an in-place spinner and disables until it resolves (no double submits). Toggle chips flip instantly on tap, before the network call, and flip back visibly if the save fails (§28.5).

Text handling LOCKED

- Font stack declares a system fallback so text never waits on a font file.
- Card titles truncate at one line. Card summaries wrap to two lines then truncate. Utility names wrap to two lines in sheets, truncate at one in list rows. Journal notes truncate at one line in the list, full on tap. Assistant messages never truncate.
- **Numbers never truncate.** If a value doesn't fit, the container grows or the label shortens.

- Spanish runs 15–30% longer; no button width is fixed to its English label (§61).

Higher contrast (Settings) PROPOSED

The blueprint gives Settings a higher-contrast switch (§49.2) but defines no palette for it. Proposed effect: secondary text uses the body-text token, borders use Navy 700 (dark: a lighter border), and pills and focus rings get a 2px outline, implemented as a token override in both modes. Exact values DESIGN (Phase 0 adds the override token set, P6). **How the three Settings accessibility controls reach every page:** the shell sets `data-text-scale` (100, 115, 130 or 150 percent), `data-contrast="high"` and `data-motion="reduce"` on `<html>`, and `--halo-text-scale` on `.halo-app`. All font sizes use `rem` or `calc(Npx / 16 * 1rem * var(--halo-text-scale))` (the existing CSS is `px`-based, so Phase 0 converts the foundation). CSS honors both the media query and `[data-motion="reduce"]`; reduce-motion defaults to the system setting. Settings stores these under `halo.device.*`.

Dark mode LOCKED (colors DESIGN)

There is no default mode. The app follows the phone's setting (`color-scheme: light dark`) and every color is a `light-dark()` token; Onboarding was approved in both modes. No in-app toggle. The map style, `theme-color` and charts switch with the same setting. Charts keep identical geometry with substituted colors. The health card's print output is always light. Build every color as a token from day one: the blueprint's own condition is that if colors are hardcoded early, dark mode is cut rather than retrofitted (§59, §68). Whether to instead make the app dark only is open (P10, D3).

A9. The shared components: functional contract

Every page is assembled from these (§6.6, §44). This table is what each must *do* and which states it must support. What each *looks like* is DESIGN, decided once in Phase 0 and frozen (Part P2), then reused unchanged. The "First page that needs it" column below is historical: with three parallel builders, Phase 0 builds every component that two or more pages use.

Component	Must do	States / variants required	First page that needs it
Button	One primary per screen max	Primary, secondary, tertiary, destructive, icon-only × default, hover, pressed, loading (label → spinner, width held), disabled (only for genuinely unavailable actions, never validation)	Onboarding
Card	Container for content	Static, tappable, expandable, severity-marked (severity visible on the card edge or equivalent)	Today
Expandable card	Collapses to a summary row; expands in place; never navigates; not a history entry	Collapsed, expanded, loading, no-data	Today
Bottom sheet	Slides over current screen; dismiss by drag, backdrop tap, Escape, or close control; pushes a history entry; traps focus; locks scroll beneath	Heights: compact (≤50%, map details), standard (~70%, Learn/Alerts/Act), tall (~90%, Assistant). Internal scroll	Today (Learn)
Score ring	Shows a 0–100 composite, higher = better, colored by severity	Populated, no-data (no arc, the words No data in the centre), partial (visibly incomplete without a caption), entrance sweep (skipped under reduced motion)	Today
Risk bar	Horizontal fill by risk, colored by severity	Populated, no-data, "shown, not scored" (no fill)	HomeGuard
Severity pill	Icon + word + color, always together	All six words	Today
Confidence pill	Quieter than severity; absent at full confidence	partial, 3+ yrs old, not evaluated	HomeGuard
Provenance pill	Same family as confidence	measured, modeled, estimate	Today
Callout	Inline note with an icon	Info, caution, notice tones	Today (incomplete-data notice)
Skeleton	Placeholder sized exactly like what replaces it	Animated, static (reduced motion)	Today
Empty state	Icon + heading + explanation + optional action	With / without action	Journal
Toast	Brief confirmation above the tab bar; one at a time	Plain, and with an Undo action. **One duration for every toast: 8 seconds**, long enough to reach Undo (§65.3, the blueprint's ~4s/~8s split is Tier 3)	Journal
Tab bar	Five destinations	Active, inactive, pressed	Shell
Header	Title + right controls	Today variant (bell + gear), standard (gear), Settings (back arrow), Map (over map)	Shell
Segmented control	2–3 options, switches content, no navigation	Journal mode, journal severity, letter recipient, owner/renter	Journal
Toggle chip	On/off label; flips instantly	Selected, unselected, disabled	Onboarding (household)
Switch	Settings rows	On, off, disabled	Settings
Accordion row	Settings sections; one open at a time; current value summary when closed	Collapsed, expanded	Settings
Text / numeric input, dropdown, checkbox, slider	Per A7	Default, focused, error, disabled	Onboarding
Offline banner	Under header, pushes content down	Shown / hidden	Shell
Swipe action	Reveals delete/dismiss; action is immediate with an undo toast	Journal entry delete, alert dismiss	Journal

THE BLUEPRINT'S OWN SIMPLIFICATIONS ARE THE DEFAULT (§65.4): where the table above lists more variants than a page needs, build the simpler set unless a page genuinely requires more: **buttons**, three variants (primary, secondary, tertiary); destructive is secondary in the Elevated red, icon-only is tertiary with no visible label (it keeps an accessible name). **Sheet heights**, two (standard ~70%, tall ~90%). The Assistant is tall. Learn, Alerts, Act, Journal day detail and Settings' delete sheet are standard. **Map feature details are the one exception:** sized to their content, at most 50% of the screen, so the selected feature stays visible (the auto-pan that would otherwise fix this is Tier 3). The district panel is standard. **Cards**, mark severity with the pill, not a colored card edge as well ("both is redundant emphasis"). **Skeletons**, two shapes, a card block and a text block, composed as needed. **Motion**, three durations: 150ms fades, 250ms movement, 400ms score ring.

A10. The signature visuals: what's locked vs. what's open

The blueprint's exact pixel geometry for these (§56) is visual and void. What they must *do* is not, each carries meaning, which is why they exist (§6.0).

Visual	Locked function LOCKED	Open DESIGN
Score ring (Today, HomeGuard hero)	0–100, higher is better, same direction on every screen. Fill = score/100. No-data = no arc and the words No data in the centre. Partial = visibly incomplete, distinguishable without a caption. No "out of 100," no percent sign.	Size, stroke, how "partial" looks, entrance motion
Contribution bar (under each hero)	One stacked bar; segments sized by each factor's share of the total risk, colored by that factor's severity, each labeled (segments under 10% of the width are labeled only in a legend). Tap a segment → scroll to and expand its card. Edge cases: if the total is 0 the bar is hidden; a factor with data but zero risk gets a minimum visible sliver and a legend entry; a factor whose severity is `no_data` is a hatched "no data" segment, never zero width; if only one segment exists it fills the width.	Thickness, labels, legend treatment
Threshold bar chart (HomeGuard water, map sheets, health card)	One row per contaminant. Scale is always 0 to 2× that contaminant's federal limit, so the limit sits at the same position on every row, this is what makes different contaminants comparable. Values past 2× fill fully with a visible "continues" end and the true multiple in the label; never rescale the axis. Limit marked on every scored row. Sort worst-first by ratio. Unscored substances below a divider, de-emphasized, no limit marker, labeled "shown, not scored." Tap a row → Learn for that contaminant.	Row/bar dimensions, marker style, entrance motion
House diagram (HomeGuard)	Three zones, tap (water), foundation (radon), pipe (lead), each shown in its severity, grey when no data. Each zone is a tap target that scrolls to and expands its card. Schematic, not an illustration. If it turns into an illustration project, replace with three labeled rows carrying the same colors, the page loses character, not meaning.	Everything about how it's drawn
Calendar heat grid (Journal)	One cell per day, seven columns. Cell = that day's worst severity. No-data day visibly different from Good. Future days not drawn. Logged-symptom days marked. Today marked. Tap a day → that day's detail sheet.	Cell size, colors, markers
Seven-day trend (Today expanded cards)	Daily values, today emphasized, severity bands behind the line so a rise visibly enters a worse band. Missing days break the line, never interpolated. The band edges are **display-only drawing constants** mirrored from `lib/severity.js` (they position bands, they do not classify): AQI 50, 100, 150, 200; UV 3, 6, 8, 11; pollen 0, 2, 3, 4 on a 0 to 5 scale. Each point's color is the server's word from `history[]` (for a pollen point on a non-dominant day, the band that its plotted value falls in). Mold has no trend.	Size, curve, labels
Quarterly trend (HomeGuard water)	Worst contaminant's readings across sampling dates, federal limit as a reference line. Only drawn when there are 2+ dated readings.	Everything visual
Comparison bar pair (district panel, map banner)	Two bars on the same scale.	Everything visual

Every chart has a plain-language caption stating its takeaway, units on any axis, missing data as a visible gap (never zero, never interpolated), a screen-reader text equivalent, and colors from the severity scale only, no chart gets its own palette (§6.8). All drawn as inline SVG, no charting library (§35.2).

A11. The design foundation: locked

Decided once, used by every page. The blueprint's own color and type values (§6.3, §6.4, §59) are replaced by these.

Severity colors LOCKED

Word	Label	Light mode	Dark mode	Style	Icon
`good`	Good	`#1e7a3c` green	`#4cc27a`	Solid fill	Circle with check
`moderate`	Moderate	`#b85c00` orange	`#f0913a`	Solid fill	Circle with i
`elevated`	Elevated	`#c0262d` red	`#f0605f`	Solid fill	Triangle with exclamation
`high`	High	`#8e2a6b` red-purple	`#d668ad`	Solid fill	Octagon with exclamation
`severe`	Severe	`#4f1440` deep purple	`#a870f0`	Solid fill	Filled octagon
`no_data`	No data	`#111111` black	`#e6e6e6` near-white	**Outline only**, never filled	Circle with question mark

No data is drawn as an outline so it can never be mistaken for Severe (black and deep purple are nearly the same brightness). Pill text on a filled level is white in light mode and navy (`#081725`) in dark mode. Every color above passes 4.5 : 1 for text in its mode (checked by calculation). Icons have clearly different shapes because severity is always color + icon + word.

Brand and surface colors LOCKED

Token	Light	Dark	Use
Navy 900	`#0b1f33`	`#081725` (page)	Main text in light mode; the base of dark mode
Navy 700	`#133b5c`	`#1d3a50` (border)	Headers, deep accents
Primary, deep teal	`#0e5e6f`	`#2cc4bd`	Primary buttons, active tab, links, focus ring
Teal 500	`#138a8a`	n/a	Secondary accents, **never text** (fails contrast)
Turquoise	`#1fb5b0`	n/a	Highlights and fills only, **never text**
Tint (`--halo-tint`)	`#d9f0ef`	`#1d4f5c`	Selected chips, soft tints. This approved value replaces the earlier light aqua `#a8e6e2`
Page background	`#eef8f7` aqua mist	`#081725`	Behind cards
Card surface	`#ffffff`	`#0f2436`	Cards and sheets
Border	`#d3e4e6`	`#1d3a50`	Dividers, card outlines
Body text	`#0b1f33`	`#e8f3f5`	All primary text
Secondary text	`#4a6275`	`#9bb4c2`	Timestamps, sources, captions

No saturated green anywhere in the brand, so green always means Good. There is no default mode: the phone's setting decides (A8). The token names that carry these values are in P6.

Typefaces LOCKED

Role	Face	Notes
Headings, hero score numbers, page titles	**Fraunces** (variable)	A soft serif, the app's personality. Has no evenly spaced digits, so an animated score needs a fixed-width container so it doesn't wobble
Everything else: body, labels, buttons, cards, data rows	**Instrument Sans**	Readings use its tabular digits (`font-variant-numeric: tabular-nums`) so columns align, verified the font includes them
Codes only	**IBM Plex Mono**	Water system IDs (`NC0190010`), certifications (`NSF/ANSI 53`). Not for readings

All three are free fonts, **self-hosted** with `next/font/local` (`public/fonts/instrument-sans-latin.woff2` as `--font-instrument` , `public/fonts/fraunces-latin.woff2` as `--font-fraunces`) with a system fallback so text never waits (§61). IBM Plex Mono is not loaded yet; Phase 0 adds it as `--font-plex` (P2). All include the accented characters Spanish needs. A proposed starting type scale is in A13; it's finalized in Onboarding's design pass and then used everywhere.

> **REPLACED, NOT TO BE USED:** everything visual in the blueprint (§6.3–6.5, §56–60 pixel values), Vibhav's glassmorphic HomeGuard branch and its `docs/HALO-design-language.md` (Playfair Display + Poppins, dark gradients, glass nav, the plexus ban), the original onboarding prototype's look (aurora and plexus canvases, frosted glass card, "Proxima Soft"/"DM Sans"), and the Geist fonts and Arial body in today's `app/layout.tsx` and `globals.css`. The foundation above replaces all of them.

A12. Priorities, performance, testing, and limits

The blueprint specifies everything; it also says completeness is not priority (§65). This section is how to spend effort. Read it before building.

Tier 1: load-bearing, never cut LOCKED (§65.1)

Item	Why
The four universal states on every screen	A screen that silently shows nothing is indistinguishable from a broken one
Error copy distinct from empty copy	"We failed" and "the data doesn't exist yet" need completely different reactions
Severity as color + icon + word	A risk app must not fail a colorblind reader
Missing data never shown as zero or green	The single most dangerous failure, absence presented as safety
Every estimate labeled	The app's credibility rests on not overstating what it knows
The back-button contract (A1)	Without it overlays feel broken on every device
Number and date formatting (A5)	The same value formatted two ways reads as unfinished
The shared component set, built first	Otherwise every page looks like a different app
Bind to the frozen contract (Part D)	A shape change after screens exist blocks everyone
Dark mode	Ignoring the system preference reads as unfinished. Cheap only because every color is a token from day one
Health card print styling (B5)	Its stated purpose is to be handed to a doctor or landlord
Assistant's structural refusal to diagnose (server-side)	A prompt can be argued around live; a code path can't
Threshold bar's fixed 0–2× limit scale (A10)	What makes different contaminants comparable at a glance
"Felt fine today" in Journal (B4)	Without good days the pattern analysis is biased
Journal's minimum data before reporting	Four data points is the difference between credible and misleading
Stored readings shown when a source fails (A2)	An environmental app that goes blank on a weak signal fails when it's wanted

Tier 2: build in this order once Tier 1 is solid LOCKED (§65.2)

1. The map's contaminant selector, the most interactive element, the one most likely to hold a judge's attention
2. Household-aware wording (display the server's sentences, A4)
3. The threshold bar chart
4. The Journal calendar heat grid
5. The HomeGuard house diagram
6. The results reveal after onboarding
7. The contribution bar
8. Seven-day trend lines
9. Pull to refresh
10. Spanish (only once a person has reviewed the translation, §68)

Tier 3: cut first, in this order, without hesitation (§65.3)

Assistant button hide/show on scroll (leave it fixed) · map auto-panning above a sheet · the home pin's pulse · score number counting up · chart bars staggered · skeleton shimmer (a static block is fine) · header shadow on scroll · onboarding slide transitions (a cross-fade or nothing is fine) · air-monitor glow · hiding zoom buttons on touch · toast duration varying by content (one duration) · the map time slider (only if the data supports it, §68) · season recap notifications · landscape handling · offline journal queueing (a clear "couldn't save, try again" is acceptable).

Page cut order if time runs short (§26.3): Assistant sheet → Settings → Act → district panel. Cutting Settings means household groups can only be set in onboarding, and renting mode can't be set at all, onboarding has no renting question (B1), so every household gets owner wording. If Settings is cut, keep HomeGuard's PROPOSED "Showing actions for renters · Change" control above the action plan (B3) and have it save `renter_mode` through the one `PUT /api/household` save function (all seven groups). Two finished pages beat five half-finished ones; an unfinished page reads as broken rather than scoped.

Performance targets LOCKED (§35)

Measure	Target	Maximum
First meaningful content	under 1.5s	3s

Interactive, responding to taps	under 2s	4s
Tab change	under 100ms	300ms
Sheet open	under 250ms	n/a
Cached reading display	under 200ms	500ms
Fresh reading retrieval	under 2s	Stored readings show immediately while fresh data loads; the "couldn't reach the data source" notice appears only when the request's own deadline passes (15 seconds for readings, A2)
Map first render	under 2s	4s
Map layer switch	under 300ms	1s
Total page weight, first load	under 500KB	1MB

Measured on a mid-range phone over a mobile connection. How: charts are inline SVG (no charting library); the map library loads only when the Map tab is first opened; map data is one pre-built file per layer; stored readings render immediately while fresh data loads; images sized for display in a modern format; fonts load with a fallback shown immediately; skeletons occupy exactly the space of what replaces them (§35.2–35.3).

Security and privacy in the client LOCKED (§37)

- No analytics or tracking SDK of any kind, and the privacy statement says so (§37.4).
- No secret in browser code. The only browser-visible values are the Supabase URL and anon key, the optional Turnstile site key, the VAPID public key, and, only if the map uses Mapbox GL JS, a public `pk.` map token restricted to the app's URLs. The server's `MAPBOX_TOKEN` (geocoding) is never exposed.
- The journal note and the assistant question are the two free-text inputs: render them as text, never as HTML.
- Browser storage holds only the session (managed by the auth library), stored Today and HomeGuard readings, cached Learn content and map data, the onboarded flag, device-local accessibility settings, small interface memory (dismissed seasonal prompts, the last-used journal group), and any queued journal entries, nothing a person wouldn't already see on screen.
- Never send a household or profile id as identity; the token is the identity.

Testing: what every page must survive LOCKED (§40)

Test addresses (real, not synthetic)	Must produce
A Concord address	A utility match with PFAS above the federal limit, the primary demo case
A Union County address	A different utility with different readings
A third NC-08 address	A third utility, ideally with no published results (the no-data state)
A rural address	No utility match, the well prompt
An address entered as a well	The testing plan, never a score
An address outside North Carolina	Daily readings work; home readings degrade gracefully (radon `out_of_state`; water says HALO's water data covers NC, B3)
A Buncombe County address (Zone 1) and an Iredell County address (Zone 2), in NC but outside NC-08	The radon-elevated paths: a radon action item, the `radon` volunteer cause, Act's radon help section, Journal's winter radon prompt, the `radon_season` alert, and a well plan's radon-in-water test

> **EVERY NC-08 COUNTY IS RADON ZONE 3:** in the stored EPA data, every county in and around NC-08 (Cabarrus, Union, Stanly, Anson, Montgomery, Richmond, Moore, Rowan, Mecklenburg, Scotland, Robeson, Hoke) is Zone 3. So for every NC-08 demo address the radon card reads Good · Zone 3, radon never enters the action plan, and the radon prompt, alert, and help section never appear. Test those paths with the Zone 1/2 addresses above; use Zone 3 in NC-08 fixtures. "Zone 2" examples in this document are illustrative.

Cases: no household groups (general wording everywhere) · every group selected · respiratory only (wins the priority) · build years before 1950, 1950–1987, 1988 or later, and blank · renting vs owning (every action changes) · Journal with 0, 4 (below threshold), and 20 entries including illness-flagged days · every map layer and contaminant, including one with no readings anywhere · the Assistant against ten adversarial questions including direct requests for diagnosis · every empty and error state triggered on purpose · both languages on every screen.

Technical: every screen at 375px on a real phone (not a resized desktop window) · both mobile engines (Safari/WebKit and Chrome) · keyboard-only through every screen · a screen reader on Today and HomeGuard at minimum · one data source deliberately blocked · fully offline, then back online · Back with every sheet type open · deep links pasted into a fresh browser.

Live checks and safety: the local environment points at the **production** Supabase project, so destructive calls (`DELETE /api/account` , `DELETE /api/journal?all=true` , `PUT /api/household`) are never run against it from a test. Use the mock adapters, or a separate Supabase project. **Before any demo (§40.4):** run `/api/health` (it needs a session token, call it with `authedFetch` from the running app's console) · load the demo address and confirm results within target · confirm the map data was rebuilt recently · have a fallback if a government service is down.

What HALO deliberately cannot do: each is stated in the interface LOCKED (§69)

Limitation	Where the interface says it
It can't tell you what's in your tap water, only what the utility measured at its own sampling points	HomeGuard water card (B3)
It can't measure radon in your home, county zones predict averages	HomeGuard radon card (B3)
It can't detect mold, it estimates favorable conditions from humidity and rain	Today mold card, "estimate" pill (B2)
It can't test private wells, no agency does	Well testing plan (B3), results reveal (B1)
It can't tell you whether your plumbing contains lead, age indicates an era, not a material	HomeGuard lead card (B3)
It can't diagnose anything	Assistant disclaimer and refusal (C2)
It can't prove an exposure caused a symptom	Journal findings disclaimer (B4)
It can't promise its data is current	Every reading carries its date (A5)
It can't cover the whole country equally, water and radon detail is strongest in North Carolina	Results reveal note for non-NC addresses (B1)

What the judges actually see (§65.5)

Whether it works at all, on a real device, without errors · the map, especially the moment a contaminant selection recolors the state · whether numbers are explained in language a person understands · whether it admits what it doesn't know · whether the presenter can explain why it was built this way. They won't notice easing curves or a two-pixel shadow. "If a trade-off ever arises between looking finished and being honest, choose honest."

A13. The design process, and the proposed base visual system

Who designs, and how LOCKED

Codex (in each builder's own account) does the wireframes, the visual design and the code for the pages that builder owns; Yogi supplies reference inspiration and approves each page's look; Part P says who owns which page and which shared components. The "wireframe" is a real, final-looking page at 375px with mock scenarios, reviewed and frozen before the live build. The full procedure, and the rules Codex follows, are in **HALO_Wireframe_to_Build.md**. Per page:

1. **Read the function first.** Part P, the page's section in Part B, every shared part it uses (A1 to A13, C, D). The element table is the checklist: every element, binding, state and copy string.
2. **Study the references Yogi supplies.** Borrow what they do well: hierarchy, spacing rhythm, density, how a card or chart is treated. Don't borrow their colors or fonts (A11 governs), and don't borrow their emptiness if this page holds more data than theirs.
3. **Wireframe** the page as a preview route (`/<page>/preview`) with mock scenarios for every state: populated, loading, empty, error, partial or no data, offline, in light and dark, at 375px. Use the shared components as they are; do not restyle them.
4. **Check it against the element table, line by line, independent of the look.** A visual change never drops an element, a state or a string to look cleaner. If a reference's layout can't hold a required element, the layout changes, not the requirement.
5. **Show Yogi screenshots of every state.** Iterate on visuals only. List every PROPOSED item built. When approved, freeze the screenshots.
6. **Build** with the live adapter (Part D), the same screen component, and run the definition of done (E1) plus the zero-difference check against the frozen screenshots.

The quality bar: high-end, not overboard

The blueprint's own bar (§6.0, §65.5): clean, deliberate, finished, not a funded consumer product. What reads as high-end is discipline: one type scale, one spacing scale, one color system applied predictably, aligned edges and even margins, text that never overflows, nothing shifting as content loads, correct contrast in both modes, and precise details (consistent icon stroke weight, optical alignment of icons with text, tabular digits in data). Personality comes from the signature elements, one per page, each carrying meaning (§6.7): Today's time-of-day gradient behind the hero, HomeGuard's house diagram, the Map's full-bleed canvas, Journal's heat grid, Act's editorial cards, Learn's typographic sheet. Buttons, cards, and type stay identical everywhere. "If a visual decision takes more than an hour and doesn't change what a person understands, it isn't worth making."

What makes an interface look generated: avoid all of it

- Gradient washes, glassmorphism and background blur, decorative blobs or particle/plexus canvases (the only gradients are Today's time-of-day hero and Onboarding's Welcome band)
- Drop shadows on every card; colored left-border cards; everything a rounded pill
- Emoji as icons; mixed icon families (lucide-react only, one stroke weight); icons drawn for decoration
- Colors outside the A11 tokens; saturated green anywhere except the Good severity
- Center-aligned body text; title case or all caps; exclamation marks; lorem ipsum or invented numbers
- A fake phone frame or status bar, the app is the real page at the real device width (the built Onboarding already follows this: a 430 px maximum width on the real page)
- Uneven spacing, near-miss alignments, animation on everything

Proposed base visual system PROPOSED

Now built and approved through Onboarding (radius 14, gutter 20, page heading 26, tint, primary button, inputs, chips and progress are all in `app/onboarding.css`). Treat these values as final. Where this table differs from the built CSS, the built CSS wins. It is used on every page unchanged.

Type role	Face	Size / line height	Use
Hero score	Fraunces 600	56 / 60	The composite number on Today and HomeGuard, fixed-width container (Fraunces has no tabular digits)
Display (Welcome)	Handwritten "Welcome" heading (approved, animated pen strokes), then "To Halo" subtitle	56 (`--halo-welcome-size`)	Onboarding Welcome only. Shown immediately under reduced motion; never delays Get started
Step / page heading	Fraunces 600	26 / 32	Onboarding step questions
Header title	Fraunces 600	22 / 28	Page titles in the header bar
Section heading	Fraunces 600	18 / 24	Section dividers within a page

Card title	Instrument Sans 600	16 / 22	Card headers
Body large	Instrument Sans 500	17 / 24	Supporting sentences, privacy line
Body	Instrument Sans 500 (built: `.halo-app` sets weight 500)	15 / 22	All explanatory text, **the floor**; nothing readable is smaller (§25.1)
Small	Instrument Sans 500	13 / 18	Timestamps, sources, captions, field labels, error text, never explanations
Pill label	Instrument Sans 600	12 / 16	Severity, confidence, provenance pills only
Button	Instrument Sans 600	16	All button labels

Element	Proposed treatment
Spacing	4px grid: 4, 8, 12, 16, 24, 32, 48. Page gutter 20px
Radius	Cards and inputs 14px · buttons 12px · chips and pills fully rounded · sheets 20px on the top corners
Surfaces	Cards: card color + 1px border, no shadow. The primary button is the one element with a soft shadow (`0 2px 8px rgba(11,31,51,.12)`)
Primary button	Primary teal fill, white text, 52px tall, full width in forms. Pressed: 1px down. Loading: width held, label replaced by an in-place spinner
Secondary	Transparent, 1.5px primary border, primary text. Destructive = secondary in the Elevated red
Tertiary	Primary text only, no border or fill (e.g. "Skip this"). Icon-only = tertiary, 44×44 target, accessible name
Disabled	40% opacity, no shadow, only for genuinely unavailable actions, never validation (A7)
Text input / dropdown	52px tall, 1px border, 14px radius, card fill; a visible label above the field (Small, secondary text). Focus: 2px primary border plus the keyboard focus ring. Error: Elevated-red border and red error text with an icon beneath, the field fill never turns red
Toggle chip	Unselected: card fill, 1px border, body text. Selected: tint (`--halo-tint`), 1.5px primary border, and a small check icon, so the state isn't carried by color alone; `aria-pressed`
Onboarding progress	Four 3px segments with 4px gaps, filled primary / unfilled border color, no numbers shown; screen-reader text "Step 2 of 4"
Severity pill	Icon + word, fully rounded, filled in the level's color with white text (light) or `#081725` text (dark); No data outline only
Motion	150ms fades, 250ms movement (sheets; onboarding steps as a short slide or a cross-fade), 400ms score ring. Reduced motion: instant, except very short opacity fades

Part B: Pages

B1. Onboarding

Purpose LOCKED

Collect the absolute minimum needed to produce a real result, and nothing more. No account creation, no email, no password, no permission prompt, the anonymous account already exists before this screen renders (§10.1). A household that finishes onboarding lands on real, personalized results; one that abandons it sees nothing. That's why this is the one page that stays deliberately short.

Address and entry

Built and approved (on `main`, commit `c84cbb3`, refined in `e3f2147`). The real entry is `/`, which redirects to `/onboarding?entry=1`; the first-visit rules are in A1. `/onboarding?change=1` starts address replacement (from Settings, HomeGuard's "no county" error). `/onboarding/preview` is a development-only review route with a scenario selector (loading, errors, missing data, wells, springs, outside NC); production hides it unless `HALO_ENABLE_FRONTEND_PREVIEW=1`. Full screen: no header, no tab bar, no assistant button (§51.1).

As built REFERENCE

> **ONBOARDING IS THE REFERENCE FOR EVERY OTHER PAGE.** Study these files before building anything: `app/OnboardingScreen.tsx` (the screen takes an `api` prop: the live adapter or a mock), `app/onboarding/page.tsx`, `app/onboarding/preview/`, `app/onboarding.css` (tokens and foundation), `lib/frontend/{api,copy,mock,onboarding,storage,types,useHaloData}.ts`, `components/ui/{Button,DataState,ErrorCallout,FieldError,SeverityPill}.tsx`, `test/ui/` (Playwright), and `docs/onboarding/` (`README`, `preservation`, `integration`, `requirements-checklist`, `verification`, 179 approved screenshots and a hash manifest). Verified on `main`: 354 of 354 node tests pass, and typecheck, lint and build succeed. Live-provider and real-phone acceptance are still to do (Vibhav's wrap-up list). The pattern each later page repeats: a screen component that takes an API object, a live adapter, a mock adapter with named scenarios, a preview route, a copy catalog, immutable approved screenshots.

> **WHAT ONBOARDING ALREADY DOES THAT PAGES RELY ON.** Session bootstrap and its two error messages; the GPS path (sends `"lng,lat"`, no fallback address); a fresh `request_id` (UUID) on every address submission so a timed-out write can be confirmed through the profile's `onboard_request_id`; bare five-digit ZIPs are constrained to the United States in the geocoder (without it `28025` resolves to Madrid); the identity-scoped storage and the `halo:identity-changed` event; `withDeadline` (8 seconds, 15 for onboard, daily-score and home-guard); after a location change the client clears old readings and calls `/api/daily-score?fresh=1`. The Today destination is a labeled stub (`app/today/page.tsx`) that renders `ProfileRecheck` for `?verify=1`; Today's owner replaces it and must keep that behavior.

Two follow-ups before Settings is built (Part P7): in change mode, Back on the address step should return to `/settings`, and the Household step (shown prefilled) must be described accurately in Settings.

Structure LOCKED

Four full-screen steps. A slim four-part progress indicator at the top (not shown on Welcome). A back arrow from step 2 onward returns to the previous step with every entered value intact. Each step is a history entry. Each step: the question, any supporting line, the input, then the primary button at the bottom (§10.2, §51).

Data calls, in order BIND

When	Call	Sends	Why here
Page load	`ensureAnonSession()`	n/a	Already runs app-wide; onboarding surfaces its failure
Step 2 "Continue"	`POST /api/onboard`	`{ address, request_id }` (a new UUID each submission)	So address problems surface on the address step, where the blueprint puts them (§10.4)
Step 3 "Continue" (not on Skip)	`PUT /api/household`	all seven booleans, every time	Saves composition. Skip sends nothing, leaving `household_set: false`
Step 4 "See my results"	`PATCH /api/profile`	`{ water_source, home_year }`	Saves the home details without re-sending the address, the address was used once at step 2 and never kept
Results reveal	In parallel: `GET /api/daily-score` and `GET /api/home-guard`	No parameters, both use what onboarding just stored. The onboard response's `state` decides the outside-NC note	Fills the four reveal lines and warms the cache for Today and HomeGuard

TRAP: `PUT /api/household` is a full replace: any of the seven not sent become `false`. Always send all seven, as real booleans.

Step 1: Welcome LOCKED

Element	Content	Notes
Mark	HALO logo, white, on the navy band	`public/halo-logo-mark.png` used as a mask (A5). The blue tile and `finallogo.png` are superseded
Headings	A handwritten "*Welcome*" that writes itself along the pen strokes, then a smaller, lighter "*To Halo*" subtitle that fades in beneath it. Both show immediately under reduced motion; never delay Get started	Approved; supersedes a HALO-only heading
Sentence	"*See what's in the air, water, and ground around your home, and what to do about it.*" then "*No account needed.*"	§10.3, punctuation per A5. Built in `copy.welcome`
Primary button	*Get started*	→ step 2
Supporting line	"*Takes about a minute.*"	Below the button
Session failure	"*Verification could not be completed. Check your connection and try again.*" or "*Could not start a session. Please reload and try again.*"	From `ensureAnonSession` error `.code`; Get started disabled until a session exists

Step 2: Address LOCKED

Element	Content / behavior	Binding / notes
Heading	"*Where do you live?*"	
Text field	Label "*Address or ZIP code*" visible; placeholder "*Street address or ZIP code*"; device address autofill; 16px+ text	Validation on blur: 5+ characters else "*Enter an address or ZIP code.*"
Secondary button	*Use my location*	`navigator.geolocation` → field shows "*Using your current location*"; sends `"lng,lat"` as `address`. Unsupported: "*This device can't share its location. Please type your address.*" Denied/failed: "*Couldn't get your location. Please type your address instead.*" Typing in the field discards the coordinates
Privacy line	"*We use your address once to find your local data, then throw it away. We keep only your approximate coordinates, your county, and which water utility serves you. We never keep the street address itself.*"	Directly under the field, visible before they type (§10.4)
Primary button	*Continue*; while processing: loading state + supporting text "*Finding your local data.*"	Calls `POST /api/onboard { address }`

Outcomes, by response BIND

Response	Show	Next
200	n/a	Step 3. Keep `service_area_status`, `county` for later
404	"*We couldn't find that address. Try adding your city or ZIP code.*" Field keeps its contents	Stay
429 (either message)	"*You've made a lot of requests. Please try again later.*" No blame, no detail	Stay
500 / network	"*Something went wrong on our end, not yours. Try again in a moment.*" + Retry	Stay
401 / 503	Handled by the auth layer (A2)	n/a

OUTSIDE NORTH CAROLINA: a non-NC address proceeds normally (§10.4). The onboard response carries `state`; when it isn't `NC`, the results reveal adds a note that daily air, UV, and pollen readings work everywhere while water and radon detail is strongest in North Carolina. PROPOSED wording: "*Daily air, UV, and pollen readings work anywhere. Water and radon detail is strongest in North Carolina.*"

NOTE: `service_area_status: "outside_known_area"` is a normal, common result in NC-08 (no utility boundary contains the point), not an error. `lookup_failed` means the boundary lookup itself failed; onboarding still continues. Either way, step 4's water-source question matters more for these households.

Step 3: Household LOCKED

Element	Content / behavior	Binding
Heading	*"Who lives here?"*	
Supporting text	*"This lets HALO tell you what today's readings mean for the people in your home. We never ask for ages, birthdays, or names, just which groups apply."*	
Seven toggle chips, wrapping, this order, all start unselected	Toddler · Child · Teen · Adult · Senior (65+) · Someone pregnant · Someone with asthma or a breathing condition	`has_toddler`, `has_child`, `has_teen`, `has_adult`, `has_senior`, `has_pregnant`, `has_respiratory`
Primary	*Continue*	`PUT /api/household` with all seven. On failure: keep selections, show the generic error, allow Continue again
Skip	*Skip this*, same row as Continue, equal visual weight so skipping never feels like failing	No call. General wording everywhere; never prompted again; editable in Settings

Step 4: Your home LOCKED

Element	Content / behavior	Binding
Heading	*"A little about your home."*	
Build year	Label *"Year built"*, optional, numeric keypad, digits only. Under it: *"Homes built before 1988 may have lead solder in their plumbing. If you're not sure, leave it blank. We'll tell you we don't know rather than guess."*	`home_year`: number or `null`. Validation per A7 (1700–current year)
Water source	Label *"Where does your water come from?"*; options: City or town water · Private well · Spring · Other · Not sure. Required	`water_source`. The server normalizes these exact labels: City or town water → `utility`, Private well → `well`, Spring → `spring`, Other and Not sure → `other`
Primary	*See my results*	`PATCH /api/profile { water_source, home_year }`, set the local onboarded flag, go to the results reveal. On failure: keep both values, show the generic error, allow retry

This one choice is the largest branch in the application: well or spring sends all of HomeGuard to a testing plan instead of measurements (§10.6).

The results reveal LOCKED

A transitional screen after step 4, before Today (§33, §51.5): the logo, then four lines appearing in sequence, each starting with a small working indicator that turns into a check as its result resolves, with the result on the right. Then a primary button *See my results* → `/today`. Under reduced motion all four appear at once. Purpose: makes the work visible, introduces the four things HALO tracks, and turns a redirect into proof the app did something real.

The four line labels and the example results come from §33. Result strings not in the blueprint are marked PROPOSED.

Line	Result shown	Binding
Finding your water utility...	Utility name (through `formatUtilityName`, A5), or PROPOSED *No utility matched your address* · *Private well (no utility)* / *Spring (no utility)*	`water.pws_name`; `water.status` = `no_pwsid_available` / `private_well` (+ `water.source_type`). For `no_data_yet` and `lookup_failed` there's no name in the response, show PROPOSED *Found your water system ([pwsid])* from the onboard response's `pwsid`
Checking federal testing results...	*PFOS found above the limit* (§33) · *No published water results yet for your utility.* (§33) · PROPOSED: *Nothing found above federal limits* · *Wells aren't tested by any agency* · *Found compounds with no federal limit* · *Couldn't check right now*	Measured: the worst entry in `water.scored_contaminants` with `exceeds_limit: true` (the server decides, the frontend never compares a value to a limit); none → "Nothing found above federal limits." Other states by `water.status`: `no_data_yet`, `private_well`, `no_pwsid_available` ("No utility to check"), `detected_unregulated`, `lookup_failed`
Looking up radon for [County]...	*Zone 2* (§33's example, every NC-08 county is actually Zone 3, A12) · or PROPOSED *Not available for this county* · for out-of-state addresses PROPOSED *Covers North Carolina only*	`radon.zone`; `radon.error` present → not available; `radon.out_of_state` → the out-of-state line
Getting today's air quality...	Severity label, e.g. *Good*	`air.severity` → label

If a lookup produced nothing, the line says so plainly rather than disappearing (§33). For an address outside North Carolina

(`state` isn't `NC`), a `no_data_yet` water result reads **PROPOSED** *HALO's water testing data covers North Carolina utilities* instead of the "not published yet" line, which would be false there. If a call fails outright, its line reads *Couldn't check right now* and the button still appears, Today and HomeGuard have their own error states.

> **VOICE CHECK:** "Nothing found above federal limits" is a statement about a measurement, which the rules allow. Never "Your water is safe" (§30.3).

> **ROUTING:** the local `halo.onboarded` flag is set only after step 4 succeeds. The server's `onboarding_complete` turns true at the same moment (coordinates and water source both stored); `onboarded` turns true earlier, at step 2, which is why routing never uses it (A1).

Proposed content PROPOSED

None beyond the outside-NC note. This page is deliberately minimal (see the note in section 0).

Visual decisions: made DECIDED

Onboarding's design pass is done and approved in light and dark. Everything below is now the foundation for every page.

Decision	Outcome
Palette, typefaces, tokens	A11 and P6. Appearance follows the phone; every color is a `light-dark()` token
Overall tone	"Something in between," quiet and immersive: the page background everywhere, a brand moment on Welcome only. Steps 2 to 4 and the reveal stay light and calm
Welcome band	A navy band (`--halo-band`, 220px) behind the white logo mark, on Welcome only. This band and Today's time-of-day hero are the only gradients or bands in the app
Base components	Built: buttons (primary, secondary, tertiary), inputs and dropdown, toggle chips, progress segments, severity pill, error callout, field error, type scale, spacing (gutter 20, radius 14), motion. These are the values in A13, and the built CSS wins if they ever differ
Step transition and reveal	As built and approved. Reduced motion: instant, all four reveal lines at once

> **REMOVED, AND MUST NOT COME BACK.** The fixed 390×844 phone shell, the aurora and plexus canvases, the frosted glass card, the "Proxima Soft" and "DM Sans" fonts, the old tagline "Know what surrounds your family.", the non-link "Terms of Service" footer, and the single-screen form.

B2. Today

Purpose LOCKED

Answer, in under two seconds of looking: *is it a good day to be outside, and is there anything I should do differently today?* This is the page a household opens habitually, legible at a glance, and it never requires scrolling to deliver its headline (§11.1).

Address and entry

`/today`: default landing for onboarded households, Tab 1. `/today?learn=air|uv|pollen|mold` opens that Learn sheet over it (any other value is ignored; on a deep link Today rebuilds the reading and context from its loaded data). **Today replaces the stub `app/today/page.tsx` and must keep its behavior:** render `<ProfileRecheck />` when `?verify=1`, which keeps its redirect to `/onboarding?entry=1` for an incomplete profile (A1). The shell renders the header, bell and assistant button; Today renders content only (Part P2).

Data calls BIND

Call	Params	Feeds
`GET /api/profile` through the shared `useProfile()` (Part P3)	None	`profile.county`, `household`, `profile.locale`
`GET /api/daily-score`	none, the server uses the household's stored location. `?fresh=1` on pull to refresh	Hero, four cards
`GET /api/history?days=7`	None	Trend lines (today plus the six days before) and yesterday's row for the proposed change line. Use `history.to` as "today" (A5)

Today makes no call to `/api/alerts`: the shell owns the alerts fetch and the badge (Part P3).

> **DON'T SEND COORDINATES:** call `/api/daily-score` without `lat` / `lng` so it uses the household's stored location. Readings for any other coordinates (a GPS fix, a rounded value) are fetched but never saved to the household's history or cache.

Layout, top to bottom LOCKED

1. Header (rendered by the shell): *Today* · alerts bell with unread badge · gear
2. Hero block (roughly the top third, no scroll needed): score ring → headline → notices → location/freshness line → contribution bar, on the time-of-day background
3. Four reading cards, fixed order: Air quality, UV index, Pollen, Mold risk. Order never changes, habitual users navigate by position (§11.4)
4. Floating assistant button (rendered by the shell)

Hero LOCKED

Element	Content	Binding	States
Score ring	The composite, 0–100, higher is better	`score.display_score` (never `score.score`, which is a two-decimal float; never `score.risk`, which runs the other way)	Partial when `score.is_partial`. No-data when `display_score` is null, and see the guard below
Ring color / headline level	n/a	`score.severity`	No data when null
Headline	By level: Good → *"Good day to be outside"* · Moderate → *"A mostly good day, with one thing to watch"* · Elevated → *"Worth taking some care outside today"* · High or Severe → *"A rough day out there. See what's driving it."* · No data → *"We couldn't get readings for your area today"*	From the level above	
Incomplete notice	*"Based on 3 of 4 readings. Pollen data isn't available right now."* Two or three missing: *"Based on 2 of 4 readings. Pollen and UV data aren't available right now."* (join with commas and "and"; singular "isn't", plural "aren't"; the count	Count = `score.included_inputs.length`; names from `score.missing_inputs` (air → "air quality", uv → "UV", pollen → "pollen", mold → "mold")	Only when `is_partial` **and** `display_score` is not null (with nothing available it would read "0 of

	is a data string, so it keeps its numeral). The score still shows; it's never presented as complete when it isn't		4")
Estimate notice	"*Includes an estimated value, not only measurements.*"	`score.includes_proxy === true`	Nearly always true (mold is always an estimate)
Location line	"*Union County · updated 14 minutes ago*"	`profile.county` + `retrieved_at` as relative time (A5)	Manual refresh that returns cached data still updates this line
Contribution bar	What's driving today's score, without opening a card	Segment size = each factor's share of the total of `score.inputs.air/uv/pollen/mold` (these are risks, higher = worse); segment color = that factor's `.severity`. Factors with `null` input are absent. Labels: Air quality, UV, Pollen, Mold. Edge cases per A10 (total 0 hides the bar; a zero-risk factor with data gets a sliver and a legend entry; segments under 10% are labeled only in the legend; mold's risk is capped at 30 by the server, which is correct)	Tap segment → scroll to + expand that card
Time-of-day background	Changes with the device clock. Fixed local-device-clock ranges (PROPOSED): 05:00 to 09:59 dawn, 10:00 to 16:59 midday, 17:00 to 19:59 dusk, 20:00 to 04:59 night. Light and dark mode change the gradient stops. Atmospheric only; carries no information; text contrast verified in all four looks in both modes (Phase 0 adds the stop tokens, P6)	Device clock (the one place the device clock is right: it is decoration)	Concept LOCKED (§6.7, §11.3, Today's signature element); the color stops are DESIGN, drawn from the A11 tokens, with text contrast verified in every state

> **WHEN EVERY SOURCE FAILS:** the server returns `display_score: null` with every input in `missing_inputs` and `reason` set. The hero shows its No data state and the headline "We couldn't get readings for your area today", never a number built from nothing, and the incomplete notice is hidden. Today's `isEmpty(data)` is `display_score === null` and every card's severity is `no_data`.

The four reading cards LOCKED

Collapsed (every card): severity icon, card name, measured value (tabular digits, A11) with unit, severity pill, any provenance/confidence pills, the one-sentence household-aware explanation beneath, and a chevron. **Expanded:** the detail below, and a "*What does this mean?*" link that opens Learn for that topic.

Card	Collapsed value	Pills	Sentence	Expanded contents
Air quality	`air.aqi`, whole number, unit "AQI". Null → No data	`air.severity`; *measured* if `is_measured` true, *modeled* if false, none if null	`air.sentence`	7-day trend (history `air` words + `values.aqi`) · provenance line: true → "*Measured at a monitoring station near you*"; false → "*Modeled estimate. The nearest monitoring station is more than 25 miles away.*"; null → PROPOSED "*The source of this reading wasn't recorded.*" · static note PROPOSED: "*The air quality index reflects the single worst pollutant, not an average.*" · the dominant pollutant with a one-line explanation, held by the frontend and hidden when null (`air.dominant_pollutant`), PROPOSED strings: `"PM2.5"` → "*PM2.5: fine particles from smoke, traffic and industry that can reach deep into the lungs.*"; `"PM10"` → "*PM10: coarser dust and particles.*"; `"O3"` → "*Ozone: a gas formed when sunlight reacts with vehicle and industrial emissions. It is highest on hot afternoons.*" · Learn: `topic=air` with its context (C1)
UV index	`uv.index`, one decimal. Null → No data	`uv.severity`	`uv.sentence`	7-day trend · the hours UV is elevated, e.g. "*UV is 3 or higher from 11am to 3pm. Highest today: 7.4.*" The window is the first hour at or above UV 3 to the end of the last such hour, and `max` is the day's highest value (`uv.peak_window.start`, `.end`, `.max`). Format `"HH:MM"` as "11am" (drop ":00"); an end of `"00:00"` reads "midnight". **When null, hide the row** (it means UV never reached 3 or data is missing; do not claim "UV stays low") · static note PROPOSED: "*UV passes through cloud cover, so a cloudy day can still burn.*" · Learn: `topic=uv` with its context (C1)
Pollen	The worst category named:	`pollen.severity` (severity of the	`pollen.sentence`	All three categories: `pollen.tree`, `.grass`, `.weed` each with its own value and severity pill

	`pollen.dominant` + its value, e.g. *Grass 3*. All null → No data	dominant category)		(`pollen.categories.tree / grass / weed` → `value` , `severity`) · 7-day trend of the dominant category · seasonal note by month, static, NC (**PROPOSED** , content owner to confirm): February to April *"Tree pollen is usually highest now."*; May to July *"Grass pollen is usually highest now."*; August to October *"Ragweed pollen is usually highest now."*; November to January *"Pollen is usually low this time of year."* (device month; hidden when `profile.state` is a two-letter code other than NC) · Learn: `topic=pollen` with its context (C1)
Mold risk	No numeric value exists and the value slot stays empty: the *estimate* pill and the severity pill carry the card. Do not display `mold.risk` ("low", "moderate", "high"): a "high" label beside an Elevated pill would contradict itself	`mold.severity` + *estimate* always	`mold.sentence`	Always, without exception: *"This is an estimate based on humidity and rainfall forecasts, not a sensor in your home."* · the humidity and chance of rain the estimate is built from (`mold.basis.humidity_pct` , `.precip_pct`) · static **PROPOSED** : *"Humid, rainy stretches raise the estimate. Dry, breezy days lower it."* · Learn: `topic=mold` with its context (C1)

NEVER BIND `status` : each reading object also carries a `status` field (`unhealthy` , `very_high` , `low` , `none` …) from older code with different thresholds. It disagrees with `severity` for many real values. Only `severity` exists as far as the UI is concerned.

MOLD NEVER READS HIGHER THAN ELEVATED: the server caps it on purpose because it's an estimate. When the weather forecast couldn't be retrieved, mold is `no_data` with `risk: null` , not a quiet "low."

Page-level states

State	Shows
First load, nothing stored	Skeleton: ring-shaped placeholder, headline bar, location bar, four card-shaped blocks, same sizes as the real content (§52)
Stored readings exist	Show them immediately with their original time; fetch fresh in the background
Fresh fetch fails, stored exists	Keep stored + *"Showing your last readings from [time]. We couldn't reach the data source."*
Request fails, nothing stored	Error state (A2): *"Something went wrong on our end, not yours. Try again in a moment."* + Retry
Request succeeds but every reading is missing (`display_score` null)	The page-level empty state: *"We couldn't get readings for your area right now. This is usually temporary. Pull down to try again."* plus each card's own No data sentence. Error and empty copy stay different (§24)
One reading missing	That card shows No data and its server sentence (e.g. *"We couldn't get a UV reading for your area right now."*); the other three render normally; the hero shows the incomplete notice
Trend with under 2 days of history	*"Your trend will appear here after a few days of readings."*
Offline	Banner + stored readings

Interactions LOCKED

- Pull down to refresh (disabled mid-request) → `refresh({ fresh: true })` , which sends `/api/daily-score?fresh=1` , bypassing the server's one-hour cache. The server allows one provider refresh per household every 5 minutes (a plain call that misses the cache is limited too); a repeat inside that window returns 429 `{ retry_after_seconds: 300 }` : keep the current readings on screen and show the short toast from A2, never the error state. Show stored readings immediately; the "couldn't reach the data source" notice appears only after the 15 second deadline (A2).
- A 400 *"No location yet..."* means there's no stored location, route to onboarding (A1).
- Auto-refresh on returning to the app after 15+ minutes away.
- Tap a card → expand/collapse in place (not a history entry). Tap a contribution-bar segment → scroll to and expand that card.
- Bell → Alerts sheet, gear → Settings, assistant button → Assistant with `page=today` : all three belong to the shell (Part P2). The badge shows the unread count and clears as rows are read, not when the sheet opens (decided in P3).

Proposed content PROPOSED

Addition	From	Why
Change since yesterday under the ring, e.g. *"6 lower than yesterday"*	`history` : yesterday's `score` vs today's `display_score` . Show it **only when** yesterday's row has no `no_data` factor and today's `score.is_partial` is false (a new household's backfilled "yesterday" is a partial, modeled score, and comparing it with a	Answers "is today worse than usual" without opening anything

	full one would be wrong). Otherwise hidden. Ask 4 in P9 makes this exact	
A "worded for your household" line, e.g. "*Readings explained for: a toddler, someone with asthma · Edit*"	`household` from profile; hidden when no household boolean is true (`household_set` alone is not enough, B6)	Makes the personalization visible, a judge sees immediately that the sentences are tailored, and the household can fix it if wrong
A week summary inside each expanded card, e.g. "*Air quality was Elevated or worse on 2 of the last 7 days.*" (the same sentence is the chart's caption; the screen-reader text lists each day and its value)	`history` severity words, counted	Turns the trend line into a sentence (every chart needs a takeaway caption anyway, §6.8)
Provenance explained once in the air card: measured vs modeled, and that roughly half the district has no monitor within 25 miles	Static, from §41.2	This disclosure is unusual for a consumer app and worth making visible (§11.4)

Visual decisions to make DESIGN

The shared pieces (score ring, contribution bar, pills, expandable card, callouts, skeletons, the shell) are decided in Phase 0 and frozen (Part P2). Today's own: the seven-day trend line (`TrendLine` , also used by Journal if needed), pull to refresh, the time-of-day background's color stops, and the hero's composition. Static strings marked PROPOSED above are built and listed for Yogi to keep or cut.

B3. HomeGuard

Purpose LOCKED

Answer *what is true about this specific building, and what should we do about it?* These facts change over years, not hours, so the page is read carefully rather than glanced at, it can carry more depth and more text than Today (§12.1). It should read as a building inspection, not a weather report.

Address and entry

`/home`, Tab 2. `/home?risk=water|radon|lead` opens with that card expanded; `/home?learn=pfas|radon|lead` opens that Learn sheet. One card is open at a time; expanding uses `router.replace` to keep `?risk=` in step (never a history entry); an unknown value is ignored. The shell renders the header and assistant button.

Data calls BIND

Call	Params	Feeds
`GET /api/profile` through `useProfile()`	None	`pwsid` (shown with the utility name), `home_year`, `renter_mode`, `state`, `water_source`, `locale`
`GET /api/home-guard`	none, the server uses the stored county, utility, water source, and build year	Everything on the page

> **DON'T PASS THE HOME'S DETAILS:** call `/api/home-guard` with no parameters so it always reflects what's stored. (The endpoint also accepts `county` / `pwsid` / `water_source` / `home_year` overrides; no page uses them.) Store the last response locally so it shows instantly on the next open (A2).

Response shape at a glance BIND

```
{ success, assembled_at, radon, water, score, lead, action_plan, breakdown }
  water  → five shapes, discriminate on water.status (table below)
  score  → four shapes: private well (no score) · neither input available · normal ·
           PARTIAL (one input only: is_partial true, missing_inputs ['water'] or ['radon'],
           display_score NOT null; happens when water is no_data_yet, lookup_failed,
           no_pwsid_available, or radon is out_of_state or errored)
  lead   → object for public water; null for well/spring
  action_plan → [] for well/spring
  breakdown → [water box, radon box, lead box (public only), one box per excluded substance (lithium)]
```

Layout, top to bottom LOCKED

1. Header (rendered by the shell): *Your home* · gear
2. Hero: score ring → headline naming the dominant contributor → basis line → contribution bar (water vs radon)
3. House diagram
4. Three risk cards, fixed order: Drinking water, Radon, Lead
5. Section heading *Your Action Plan* → numbered items (or the testing plan's summary for wells)
6. Floating assistant button (rendered by the shell)

Hero LOCKED

Element	Content	Binding / rule
Score ring	Home score 0–100, higher is better	`score.display_score`. Ring color: `score.severity`, water and radon only, never lead (A3). Null in two cases: private well/spring (`score.reason`: "No score can be calculated for a private well or spring...") and no inputs at all. Show the no-data ring with the reason text beneath, a missing score here is an expected state, not an error. Partial when `score.is_partial`
Floor	n/a	The score never goes below 6 (`score.floored`). A 6 means "as bad as it gets," not "6% good." No special copy needed; don't treat it as near-zero
Cap notice	When the water result couldn't be fully evaluated, the score is capped (70 or 85) and says why	Show `score.reason` when non-null for partial or unscoreable coverage (through `displayText`). Gate any wording that says "capped" on `score.capped === true` (the server writes "capped" even when no cap applied; ask 8 in P9). `score.ceiling` and `score.capped` are otherwise unused
Headline	Names the dominant contributor, e.g. "*Water is driving most of this*" /	**Name a driver only if that factor's `breakdown` severity is Elevated or worse**, choosing the larger of `score.water_risk` and `score.radon_risk` (two server numbers). Otherwise: `action_plan` non-empty → PROPOSED "*A few things here are worth doing.*"; `action_plan` empty and the score is not null → "*Nothing here needs urgent attention.*"; well or spring (score null) → PROPOSED "*Your well water hasn't been tested.*"; no score and not a well →

	"*Radon is driving most of this*"; or "*Nothing here needs urgent attention.*"	**PROPOSED** "*We couldn't check your home right now.*" (Naming the larger raw risk alone would call a Good radon zone "the driver" when lead is the only real item.)
Basis line	"*Based on EPA testing of your water utility and county radon data.*"	Well/spring variant **PROPOSED**: "*Based on county radon data. No agency has tested your well water.*" Partial variant (`score.is_partial`) **PROPOSED**, replaces the basis line: "*Based on 1 of 2 readings. Water data isn't available right now.*" (or "Radon data")
Contribution bar	Water vs radon share of the home risk	Sizes from `score.water_risk`, `score.radon_risk`, only for a box whose severity is not `no_data` (unscoreable water has `water_risk` 0, not null). A `no_data` box is a hatched "no data" segment, never zero width. Colors from each box's `severity`. Lead is never in this bar; it isn't scored. Tap → scroll to that card. Edge cases per A10. Bind cards by `breakdown[].key`, never by index; the lithium box (its `key` is the lowercase name) belongs to the water card's Excluded group, not a fourth card

House diagram LOCKED

Three zones, tap (water), foundation (radon), pipe (lead), each in its severity, grey with no data. Each zone taps to scroll to and expand its card. Purpose: shows three unrelated readings as facts about one building, and visually separates this page from Today (§12.2b). Zone severity: water and radon from their `breakdown[].severity`; lead from `lead.level`. Well households: lead zone grey with the lead card explaining why (below).

Card 1: Drinking water LOCKED

Collapsed: name, risk bar, severity pill, confidence pill, and one line naming the driving fact, by status:

`water.status`	One line (PROPOSED where marked)	Confidence pill
measured	"[worst scored contaminant] [value] ppt · [ratio]× the limit" from `scored_contaminants` sorted by ratio; when there are none: the no_detections sentence below	Shown, from the water box `confidence`
`detected_unregulated`	"Compounds detected, no federal limit"	Shown
`private_well`	"Not tested by any agency. See your testing plan."	Suppressed
`no_data_yet`	"No published results yet"	Suppressed
`lookup_failed`	"Couldn't check right now"	Suppressed
`no_pwsid_available`	"No utility matched"	Suppressed

Risk bar state keys on the box severity, not on `risk`: no-data when the box severity is `no_data` (its `risk` is 0, not null, and a zero-fill bar would look empty-good); "shown, not scored" (no fill) when `contributes_to_score` is false; otherwise fill = `risk` (0 to 100).

Expanded, the deepest single view in the app (§12.3). Contents depend entirely on `water.status`:

`water.status`	What it means	What the card shows
anything else (e.g. `measured`)	Real federal testing results exist	The full measured view, next table
`detected_unregulated`	Compounds were detected but none has a federal limit or benchmark	The measured view, but the threshold chart has no scored rows. Lead with `water.message` (ends "Detected is not the same as safe.") and the detected names from `water.detected_unregulated`. Score is capped, show `score.reason`
`private_well`	No agency has ever tested this water	The testing plan instead of any score (below)
`no_data_yet`	Valid utility, no published results	Neutral, not a warning: *"Testing results for your utility haven't been published yet. This is normal. Federal testing is still rolling out nationally."* (blueprint copy, not the server's message, which names the dataset). Show the utility's ID (`profile.pwsid`) so the household can check it's theirs; there's no name in this response. **Outside North Carolina** means `profile.state` is a two-letter code other than `NC` (use `isOutsideNC()`; a null or missing state counts as NC, as the server does). HALO's water data covers NC utilities only, so this copy would be false there. Show PROPOSED *"HALO's water testing data covers North Carolina utilities."*
`lookup_failed`	Something broke on our side	*"This is a problem on our end, not a finding about your water."* + Retry. Radon and lead still render. Never the same wording as `no_data_yet`: one means wait, the other means try again (§12.4)
`no_pwsid_available`	Address outside every mapped utility boundary	*"We couldn't match your address to a water utility. Are you on a private well?"* + choices (PROPOSED labels) *Private well* · *Spring* · *No, city or town water*. Well/Spring → save it with `PATCH /api/profile { water_source }`, then reload HomeGuard, which now shows the testing plan. "No" → no change; the card keeps the prompt's explanation

> **NEVER RENDER THESE SERVER FIELDS.** `breakdown[].note` (for radon it can be a raw error such as "County 'X' not found in North Carolina radon dataset."), `water.message` for `no_data_yet`, `lookup_failed` and `no_pwsid_available` (they name datasets), `water.hint`, `water.next_step`, and any `radon.error` text. Only `water.message` for `detected_unregulated` and `private_well`, and `score.reason`, are displayable (through `displayText`). A `water_source` of `other` is treated as public water (utility data if a `pwsid` exists); PROPOSED note under the utility name: *"You told HALO you're not sure where your water comes from. Change this in Settings."*

Measured sub-states BIND

The measured view depends on `water.coverage`: `complete`, `partial`, `unscoreable`, `no_detections`, `excluded_only`. Many utilities have no stored detections (the loaded EPA data kept only detections), so `contaminants` is empty.

coverage	Card shows
`complete`, `partial`	The full measured view (next table)
`no_detections`	**PROPOSED** *"No compounds were reported above detection levels in your utility's federal testing."* Never "safe" (non-detects aren't stored). No chart, no trend. Dates show "No data"
`excluded_only`	The excluded group (lithium) plus the same sentence
`unscoreable`	The `detected_unregulated` treatment above

The chart and the quarterly trend are hidden whenever `scored_contaminants` is empty. Freshness line under the hero (**PROPOSED**): *"Checked 14 minutes ago"* from `assembled_at`. Water data age comes from the sample dates, not from this line.

Measured view, in order BIND

Element	Content	Binding
Utility	Name and ID, so the household can check it's the right one	`water.pws_name` (through `formatUtilityName`, A5) + `profile.pwsid` (Plex Mono)
Threshold bar chart	One row per scored contaminant, worst first; unscored below a divider (A10)	Scored rows: `water.scored_contaminants[]` → `contaminant` (through `contaminantLabel`), `value_ppt`, `limit_ppt`; ratio = value/limit; sort by ratio. Row color per A3 (interim rule until ask 1 in P9). Below the divider: `water.excluded_from_score[]` (lithium, shown in µg/L, A5) and `water.detected_unregulated_latest[]` → `contaminant`, `value_ppt`, `date_iso`
Detail rows	Per scored contaminant: name, value with units, the limit, the ratio, date sampled. Worst first	`value_ppt`, `limit_ppt`, ratio, `date_iso` (→ "sampled July 2025")
Limit type	Per row: enforceable federal limit vs. a health benchmark that isn't enforceable	`is_enforceable`, `basis`, `source`. Rows scored against a benchmark: label "Health benchmark, not an enforceable limit". The words "above the federal limit" appear only when `exceeds_limit` and `is_enforceable` are both true. Print `limit_ppt` as the limit (not `mcl_ppt`)
Rescission marker	Per row whose limit is proposed for repeal	`proposed_for_rescission`
Quarterly trend	Worst contaminant across sampling dates with the limit as a reference line, "is it getting better?"	`water.contaminants[<worst name>]` = every reading as `{ date_iso, value_ppt }`. Draw only with 2+ **distinct** dated readings (one point per `date_iso`, the highest value that date); plot by `date_iso`. Caption **PROPOSED**: *"Only detections are shown."*
Unregulated group	Detected, no enforceable limit, shown, with the statement that these are not enforceable limits	`water.detected_unregulated`; `water.message` when present
Guidance-scored group	Scored against a health benchmark rather than a federal limit	`water.guidance_scored[]` → `contaminant`, `limit_ppt`, `basis`, `source`; `water.includes_guidance`
Excluded from score	Lithium: measured, shown, not scored, with the reason written out. Units are µg/L: *"EPA reference level 10 µg/L (non-regulatory)"* from `health_reference_level_ppt / 1000`; never ppt	`water.excluded_from_score[].reason` (through `displayText`), `value_ppt / 1000`, `date_iso`, `health_reference_source`; the lithium box adds `health_reference_level_ug_l` and `detail[].value_ug_l`
Regulatory notice	Callout when the score rests on a limit proposed for rescission	`water.regulatory_notice` (present only when it applies; through `displayText`). The server's wording names the specific compound ("The federal limit for PFHxS is currently the subject of a proposed EPA rescission..."); the blueprint's example is general ("Three of the five federal PFAS limits have been proposed for rescission. This score uses one of them.", §12.3). Recommendation: use the server's, it's more specific and stays correct as the rules change
Sample age	Caution callout when the latest sample is roughly 3+ years old: *"This sample is about 3 years old."*	Water box `confidence === "stale"` (server decides), `data_age_days`, `latest_sample_iso`
Learn link	*"What does this mean?"* → PFAS	`openLearn` with `topic: 'pfas'`, `reading` from the worst scored row, `context` `contaminant` (the raw key, e.g. `HFPO-DA`, not the display label), `value`, `limit`

Private well / spring: the testing plan LOCKED BIND

No score, ever, inventing a reading for untested water would be dishonest (§12.5).

Element	Binding
Framing	`water.message`: the well version ("Private wells are not covered by the Safe Drinking Water Act…") or the stronger spring version, chosen by `water.source_type`
Why these tests	`water.test_plan.reasons[]`, already tailored to the county's radon zone, the build year, and well vs spring
Tests, in priority order	`test_plan.tests[]` → `rank`, `name`, `priority` (words: *Critical*, *Recommended*, *Optional*), `cadence`, `why`, `where`, `estimated_cost.low` – `.high`. When `needs_local_context` is true (metals panel, PFAS): **PROPOSED** caveat "*Local conditions matter for this test.*"
Totals	`test_plan.start_here_cost` (all critical tests) and `test_plan.full_panel_cost`, as ranges

> **NOTE:** test costs are estimates (`estimated_cost.is_estimate: true`, same on both totals), label them as such.

Card 2: Radon LOCKED

Element	Content	Binding
Collapsed	Name, risk bar, severity pill, *estimate* pill, one line e.g. "*Zone 3 · a county average, not your home*"	Radon box `risk`; severity per A3; `radon.zone`
County and zone	*Union County · Zone 3* (every NC-08 county is Zone 3, A12)	`radon.county`, `radon.zone`
What a zone is and isn't	"*Zones predict the average indoor level across a whole county. They are not a measurement of your home. Homes in low-zone counties still test high, and neighboring houses can differ.*"	Static, §12.3
Action threshold, testing, cost	The federal action level (4 pCi/L), what a test involves, rough cost	Static; cost matches the radon action item (*$15–40 for a test kit*)
Seasonal note	October–February only: winter readings are the most accurate (sealed, heated homes draw in more soil gas)	Device month
County not found	**PROPOSED** "*Radon zone data isn't available for your county.*"	`radon.error` present, no `out_of_state`
Outside North Carolina	**PROPOSED** "*HALO's radon zone data covers North Carolina addresses. A home test is the only way to know your level anywhere.*" Card in its no-data state	`radon.out_of_state === true` (also carries `radon.state`, e.g. `"SC"`). Never show an NC zone for an out-of-state county with a matching name
Learn	"*What does this mean?*"	`topic=radon&county=&zone=`

Card 3: Lead LOCKED

Informational only. It has a severity for display and ordering and can produce an action item, but it never contributes to any score, settled, final (§12.3). The card header carries a permanent *shown, not scored* label, matching lithium.

Element	Content	Binding
Collapsed	Name, *shown, not scored*, severity pill, *estimate* pill when estimated, one line	`lead.level`, `lead.is_estimate`, `lead.contributes_to_score` (always false). Risk bar: no fill
Household sentence	e.g. "*…Young children absorb lead far more readily than adults, so this is worth testing sooner rather than later.*"	`lead.sentence`
Basis statement	Every age-based result says it's an estimate from the home's age, not a measurement of the plumbing	`lead.text` (includes that sentence)
By build year	Before 1950 → High · 1950–1987 → Elevated · 1988 or later → Good · unknown → No data	`lead.basis`: `built_before_1950`, `built_1950_1987`, `built_1988_or_later`, `unknown_year`
Unknown year	"*We don't know when this home was built, so we can't estimate lead risk. Testing is the only way to know.*" + link to add the build year in Settings. For an unknown year, `lead.sentence`, `lead.text` and this copy are the same sentence, show it **once**. In general, render `lead.text` only when it differs from `lead.sentence`	`lead.basis === "unknown_year"`. The link goes to `/settings?open=home` (Part P5), copy "*Add the year your home was built in Settings*". Saved via `PATCH /api/profile { home_year }` (B6)
Private well	**PROPOSED** "*For a private well, lead is part of your testing plan above.*" Show it only when `test_plan.tests` contains an entry with `id` `lead_copper` (the plan adds it only for homes built before 1988 or of unknown age). Otherwise show nothing for lead and keep the zone grey	`lead === null`
Learn	"*What does this mean?*"	`topic=lead&home_year=`

> **DECIDED (backend, recorded): THE SEVERE LEAD LEVEL IS KNOWINGLY UNREACHABLE:** the Severe level (`basis: "utility_lead_service_line"`) would need to know that this specific address has a lead service line. No public address-level inventory exists for NC-08 utilities, so the backend never produces it. Lead is always age-based (High, Elevated, Good, or No data), with the estimate pill whenever `lead.is_estimate` is true (A3). The mapping still handles `severe` in case that changes.

Element	Content	Binding
Your utility's service lines	**PROPOSED** Context under the lead estimate, only when the utility appears in the state's published inventory: *"[Utility] reported [needs_replacement] service lines that need replacing (lead or galvanized) and [unknown] of unknown material. This is for the whole utility, not your address."* + source line *"NC DEQ lead service line inventory, July 2025"*. It never changes the lead level	`lead.utility_inventory` : `{ utility, reported, lead, galvanized_requiring_replacement, non_lead, unknown, needs_replacement, address_level: false, source: { name, url, retrieved } }` or `null` (hide the row). Format counts with commas (A5)

Your Action Plan LOCKED BIND

Numbered, worst problem first, only items at Elevated or above. **Collapsed:** rank number, severity pill, title. **Expanded:** reason, cost, certification, action, and the tag. Each item expands to its full guidance (§12.6, §54).

Element	Binding
Order and number	`action_plan[].rank` (the server already moves water and lead up when a toddler or pregnancy is present)
Severity	`action_plan[].severity`
Title	`title` , e.g. "Filter your drinking water for PFOS"
Reason naming the finding	`reason` , e.g. "PFOS measured at 8.3 ppt, about 2.1× the federal limit of 4 ppt." Show as sent (through `displayText`); the value in it is already rounded to one decimal
Cost range	`cost` (already formatted, e.g. "$30–90"; lead and radon costs include words such as "$20–50 for a lead test"). Label it *"Estimated cost: [cost]"*, since every cost is an estimate
Certification to look for	`certification` (null for radon), shown inline, so it can be read standing in a store
What to do	`action` , already written for renter or owner by the server
Renter / owner tag	From `profile.renter_mode` : the text is already tailored; the tag says for whom. **PROPOSED** strings: *"For renters"* / *"For owners"*. `renter_can_self_serve` is unused
Empty	*"Nothing here needs action right now. We'll tell you if that changes."*, **but never for a well**: a well's action plan is always empty because the testing plan replaces it. Show a pointer to the testing plan instead

> **VERIFY BEFORE LAUNCH:** every certification number (NSF/ANSI 53, P473, NSF/ANSI 58, NSF/ANSI 42) against the certifying body's own documentation (§12.6, §66). A wrong filtration claim is materially harmful to a household acting on it.

Page-level states

State	Shows
Loading, nothing stored	Skeleton: ring, headline, house-diagram block, three card blocks
Request fails (500)	Stored response if any + notice; else *"Something went wrong on our end, not yours. Try again in a moment."* + Retry
Radon unavailable	`radon.error` present (county not matched, or `out_of_state`), the radon card shows its no-data state; water and lead render normally

Proposed content PROPOSED

Addition	From
A "what this is based on" line in the water card: *"Based on 6 readings from March 2023 to July 2025"*	`water.reading_count` , `earliest_sample_iso` , `latest_sample_iso`
Detected count: *"4 compounds detected · 3 scored · 1 with no federal limit"*	`water.detected_count` , `scored_count` , `detected_unregulated.length`
Renter/owner line above the action plan: *"Showing actions for renters · Change"*	`profile.renter_mode` . Change opens `/settings?open=renting` or, if Settings is cut, saves through `saveHousehold` (Part P3)
A small "what HomeGuard can't tell you" footer: what's at your tap depends on your own plumbing; zones aren't home measurements; build year isn't a pipe inspection	Static, from §69, the blueprint calls stating limits plainly an asset in judging

Visual decisions to make DESIGN

HomeGuard's own: the house diagram (the page's signature), risk bar, threshold bar chart, quarterly trend, testing-plan list, action-plan item anatomy, and how this page reads as "inspection" rather than "weather." The confidence pill, pills, cards, sheet and callouts are shared and frozen in Phase 0. Learn is this owner's sheet (C1).

B4. Journal

Purpose LOCKED

Connect environmental readings to how the people in the household actually feel. The only feature where the household puts information in and gets something specific back, what separates a tool from a display (§14.1).

> **THREE BINDING RULES FROM THE RETENTION RESEARCH (§14.2):** (1) The app must be fully useful to someone who never logs anything, Journal is optional depth, never a prerequisite. (2) No streaks, badges, points, or celebration animations. (3) The first insight must arrive immediately, not after two weeks.

Address and entry

`/journal` (Log mode), Tab 4. `/journal?mode=trends` opens Trends. Switching the segmented control uses `router.replace` (no history entry) and keeps the address in step. The page may already be open in Trends when an alert navigates to `/journal?mode=trends`; handle the parameter changing while mounted.

Data calls BIND

Call	Params	Feeds
`GET /api/profile` through `useProfile()`	None	`household` → group selector; `state`
`GET /api/journal`	default = last 90 days. For the "no entries at all" test use `from` = 365 days back, so a household whose entries are all older than 90 days does not see the first-visit prompt again	Entries: `entries[]` → `id, entry_date, band, symptoms[], severity, note, retrospective, possibly_illness`
`GET /api/history?days=90`	None	Heat grid, day detail, each entry's day condition, line chart. (The retrospective result is computed by the server, never here)
`POST /api/journal/retrospective`	`{ dates }` (at most 62; future dates are dropped silently)	The first-visit result: `{ ready, factor, statement, flagged_days_with_readings, flagged_avg, month_avg, unflagged_avg }`
`GET /api/journal/findings`	None	Trends findings or not-ready progress
`GET /api/journal/summary`	None (defaults to the season that most recently ended)	Season summary callout
`GET /api/home-guard`	Once, only in October to February and only when no stored home reading exists	The radon zone for the winter radon prompt
`POST /api/journal`	see the form table	Save / update / "Felt fine today" / retrospective days

Shape LOCKED

Header *Journal*. Directly beneath, a two-option segmented control: *Log* · *Trends*. Switching changes the content below without navigating (§14.3).

Log mode: first visit (no entries at all) LOCKED

Element	Content / behavior	Binding
Prompt	*"Think back over the last few weeks. Which days were rough? Think headaches, congestion, trouble sleeping. Tap them."*	Shown when `entries` is empty
Month calendar	Weekday initials header (Sunday first); past days selectable, future days dimmed and not selectable; month arrows, back no more than three months; at most 62 days selectable (the route's limit). A day with no readings in `/api/history` is still selectable but marked "no readings" (only about the last 28 days are backfilled, and pollen is never backfilled). Keyboard and screen-reader model per A8	n/a
Count	*"3 days selected."* (singular *"1 day selected."*, zero *"0 days selected."*)	n/a
Primary	*See what was in the air*: active at three or more days; below that it is disabled with the helper PROPOSED *"Pick at least 3 days to compare."* (an explicit exception to A7's "never disabled for validation", because the comparison has nothing to say below three)	Saves each selected day with `POST /api/journal` and `retrospective: true`, see the rule below
Result	The calendar collapses and a callout appears, e.g. *"On the 3 days you flagged, air quality averaged Elevated. The month's average was Moderate. Days you didn't flag averaged Moderate."* Then a tertiary *Start logging daily*. Show also PROPOSED	`POST /api/journal/retrospective` `{ dates }` → `statement`,

	"HALO has readings for 2 of your 3 days." from `flagged_days_with_readings` (the statement's count is the days that have readings, not the number selected). A multi-month selection reads "Those months averaged"; `factor: null` comes with a statement that none of the tracked readings were higher on flagged days: show it as returned	see below

HOW THE RETROSPECTIVE WORKS: after saving the flagged days, call `POST /api/journal/retrospective` with `{ dates: [...] }`. The server compares the flagged days' readings with the rest of that month and returns `{ ready, factor, statement }`, the frontend shows `statement` as-is and claims nothing more (§14.4). `ready: false` means too few of those days have readings; show its `statement` the same way. The flagged days are saved with `retrospective: true`, `band: "household"`, and `symptoms: ["unspecified"]`: the findings analysis counts a day as a symptom day only if `symptoms` is non-empty, and the person didn't say which symptoms. **Order and failure:** POST every selected day first (a POST is an upsert, so a retry is safe), then call the retrospective. If any POST fails, keep the selection, show the save-failure copy and retry only the failed dates. If the retrospective returns `flagged_days_with_readings` of 0 right after onboarding (backfill runs after the onboarding response), show its statement and a Retry.

WHY THIS WORKS ON A FIRST VISIT: the server records a day's readings for every household every day, and fills in recent weeks at onboarding where a historical source exists, so the flagged dates usually have readings before the person ever logs anything (§14.4). Dates without readings (older than the backfill, or a day a source had nothing) are left out of the comparison.

Log mode: ongoing LOCKED

Opens straight to today's form once any entry exists. The whole interaction must fit in under ten seconds (§14.5).

Element	Content / behavior	Binding (POST body)
Date header	*"Today, March 14"* with left/right arrows to log an earlier day (no future days; no earlier than 90 days back, the history window; arrows disabled at the limits). Last seven days use the weekday (*"Tuesday"*), older days *"March 4"* (A5)	`entry_date`, `YYYY-MM-DD`, the household's local calendar date (joined to history by the same string, see the date note in A5)
Who	Toggle chips, **single-select**, naming each group present; hidden entirely for a single-group or skipped household; last-used group preselected (stored at `halo.frontend.v1.<identity>.journal.lastBand`; if it no longer exists, the first group). Tag labels: Toddler, Child, Teen, Adult, Senior, Pregnant, Breathing, Household (`band: "household"`, also what retrospective entries show)	`band`, the group key (e.g. `has_toddler`), or `"household"` when there's one group or none. The server accepts only these values
Symptoms	Wrapping grid of seven chips, multi-select. Labels: Headache · Congestion · Cough · Itchy or watery eyes · Fatigue · Trouble sleeping · Wheezing or shortness of breath	`symptoms[]`, send stable keys, not display text: `headache`, `congestion`, `cough`, `itchy_eyes`, `fatigue`, `trouble_sleeping`, `wheezing` (so Spanish labels don't split the data)
Severity	Three-option control: Mild · Moderate · Bad. Appears only after a symptom is chosen; **no default** (null until chosen)	`severity`: `mild` / `moderate` / `bad`, or `null`. Not the same scale as environmental severity
Illness flag	Checkbox: *"I think this might be a cold or flu, not the environment."*	`possibly_illness`. The analysis drops these days entirely, which is the point: cold and flu season overlaps heating season (§14.5)
Note	Single line, expands; placeholder *"Anything else? (optional)"*; counter *"412/500"* past 400; max 500	`note` (the server truncates at 500); send `""` when empty
Save	Primary, full width: *Save*, or *Update* if an entry exists for this day and group. On save: brief toast, form collapses into a header row *Log another day* (opens the form for the header date) and the list shows the new entry on top. **Tapping a list row opens the form pre-filled for that date and group with Update.** A first visit also offers a tertiary *Log today instead* that opens today's form. Save with no symptom, no severity and no "Felt fine" shows PROPOSED *"Choose a symptom, or tap Felt fine today."*	Also `retrospective: false`. **While `GET /api/journal` is loading or has failed, Save is disabled** (*"Loading your entries..."* or Retry): a POST without the loaded entry would overwrite it with blanks
Felt fine today	Tertiary beneath: *Felt fine today*. Saves an entry with no symptoms, toasts. On a day that already has symptoms it replaces them (the tap is the intent)	`symptoms: []`, `severity: null`, and the form's `note` and `possibly_illness`. This matters: without good days recorded, the analysis only ever sees bad ones (Tier 1, §65)

Save failure	*"Couldn't save that entry. It's still here. Try again."*	Form keeps everything

> **TRAP, EVERY SAVE REWRITES EVERY FIELD.** Re-saving a day/group with a partial body resets whatever was left out (note → empty, symptoms → none, flags → false). Always send the whole entry, pre-filled from the existing one.

Recent entries list LOCKED

Below the form, up to seven recent entries. Each row: date, group as a small tag, symptoms as comma-separated display labels (*"Felt fine"* when empty; the retrospective key `unspecified` displays as PROPOSED *"Rough day"*), and that day's dominant condition as a severity pill. Note truncates to one line, full on tap (§47.4, §61).

BIND Day condition = worst of `history[date].air/uv/pollen/mold` (no_data if the date has no history). Offline entries show a pending marker until sent, only if the offline queue is built (Tier 3, A2); otherwise a failed save keeps the form filled and says so.

Delete: swipe left reveals delete, and every row also has a visible 44px Delete control (`SwipeRow`, A8). Deletion is immediate (`DELETE /api/journal?id=`; the id must be a UUID, and `{ deleted: 0 }` means it was already gone) with an undo toast. Undo re-saves the **full deleted entry** (keep it in memory: band, symptoms, severity, note, retrospective, possibly_illness) with `POST`, which gives it a new id (§47.4, §31.5).

Seasonal prompts: only on this page LOCKED

Seasonality appears in Journal and nowhere else (§14.7). As dismissible callouts at the top of Log mode, above the date header, never a modal, never interrupting the form. Dismissal lasts for that season (stored locally).

- **Winter radon prompt**: October to February, household in radon zone 1 or 2 (no NC-08 household qualifies, A12): *"Radon usually peaks in winter in homes like yours. This is the most accurate time of year to test, because a summer test can read low and miss the real number."* Zone from the stored HomeGuard response (`radon.zone`); if none is stored, fetch `GET /api/home-guard` once when the Journal opens in those months and store it; skip if still unknown.
- **Pollen season note**: NC only (`profile.state` is NC) and only when today's pollen severity is Elevated or worse. Months (PROPOSED, content owner to confirm): February to April tree, May to July grass, August to October ragweed. Copy: *"It's tree pollen season in your area. Elevated readings this month are normal."* (with "grass" or "ragweed").
- **Season summary**: *"This spring you logged 22 days. On the 9 high-pollen days, you flagged symptoms 7 times."* (the server's wording: "high-pollen", "poor-air", and so on). A milestone, not a nag. BIND `GET /api/journal/summary` → `statement`, `logged_days`, `season`, `year`. Request it on load; show it only when `logged_days` is above 0 and the season ended within the last 30 days. Shown at the top of **both** Log and Trends (D6), so the `season_summary` alert, which opens Trends, lands on it.

Order when several show: radon, pollen, summary; at most two visible. Dismissal is per season and stored at `halo.frontend.v1.<identity>.dismiss.<prompt>.<season>.<year>` (winter belongs to the year it starts).

Trends mode LOCKED

Element	Content / behavior	Binding
Calendar heat grid	The page's signature visual: one cell per day, colored by that day's worst severity, dot on days symptoms were logged. Answers Journal's whole question visually before any statistics, and shows gaps in logging without nagging about them. **Twelve weeks, Sunday first (seven columns), rows aligned to weekday, oldest at the top;** dates before the first history row and after today are not drawn	Cell = worst of the **available** factor words in `history[date]` (ignore `no_data`). All four `no_data`, or no history row → the no-data outline cell. Some factors missing (backfilled days always lack pollen) → the cell keeps its color plus a small hollow corner mark, and the day sheet lists the missing ones as No data (a missing factor is never read as green). Dot = any entry that date with non-empty `symptoms`. Today marked. Each cell's label: *"March 14, worst reading High, symptoms logged"*
Grid legend	Severity colors with labels + the dot labeled *you logged*	n/a
Day detail (tap a cell)	Standard sheet (Part A9): date as title; every reading that day with its severity pill; what was logged (every group's entry for that date), if anything; a button *Add or edit entry* that closes the sheet and switches to Log mode with that date	`history[date]` → `air, uv, pollen, mold` + `values`. Rows: Air quality "37" (whole number) + pill from `history.air`; UV "5.2" + pill; Pollen "Grass 3" for the highest of `values.pollen` + pill from `history.pollen` (the worst category); Mold "Low", "Moderate" or "High" with the estimate pill and the mold severity pill (the value here is the only place `mold_risk` is shown)
Findings	One card per finding: the statement, the underlying counts, a small pair of bars comparing the two rates. Factor labels: air = "Air quality", pollen = "Pollen", uv = "UV", mold = "Mold". Bar labels *"On elevated days"* and *"On other days"* (the rates are already percentages). Counts line *"12 elevated days, 40 other days"*. Caption: a descriptive one (the statement itself); screen-reader text is a small table	`findings[]` → `statement`, `elevated_rate`, `other_rate`, `counts.elevated_days`, `counts.other_days`, `factor`
Disclaimer	Permanent, never dismissible, under the findings	`disclaimer`, *"This is a pattern in your own logs, not a medical finding. Many things affect how you feel on a given day."*
Not ready	Replaces findings with progress: *"Keep logging. There isn't enough here yet to say anything reliable. You've logged 3 days; patterns need about 5."*	`ready === false`. The n is counted from `entries`: distinct dates with non-empty `symptoms` and `possibly_illness` false. **Not** `flaggedDays` (see below). When n already meets `needed.flagged` but `flaggedDays` is still below it (the logged days lack readings), the message is PROPOSED *"HALO needs readings for more of the days you logged. It has [flaggedDays] so far."*
Line chart	One line of `history[].score` (0 to 100, higher is better; a gap where the score is missing) with the severity bands behind it (A10 edges do not apply; the bands are the score's own severity words). A marker on each logged date; the marker's **shape** per group PROPOSED (design confirms in the wireframe): toddler circle, child square, teen triangle, adult diamond, senior star, pregnant cross, breathing hexagon, household ring. More than one entry on a date draws one marker per group, offset. Caption *"Each mark is a day you logged symptoms."* (neutral, no finding). Screen-reader text: the logged dates with that day's worst severity. Horizontally scrollable	`history` + `entries[].band`

DON'T SHOW `flaggedDays` AS "DAYS LOGGED": the server's `flaggedDays` counts only symptom days that also have a day of readings and aren't flagged as illness, it's "comparable days," not "days logged." Count logged days from `entries` for the message. Findings need two things: 5 comparable symptom days (`flaggedDays` vs `needed.flagged`) **and** 14 days of readings (`totalDays` vs `needed.total`). When the household has logged enough but there aren't 14 days of readings yet (a new household before its history fills in), use PROPOSED *"HALO needs about two weeks of readings for your area to compare against. It has [totalDays] so far."*

MISSING COPY, READY BUT NO FINDINGS. When there's enough data but no factor shows a big enough difference, the server returns `ready: true, findings: []`. The blueprint has no copy for this. PROPOSED: *"No clear pattern yet. Your symptom days don't line up with any one condition so far."*, the disclaimer still shows.

> **WHY THE RESTRAINT IS THE FEATURE:** findings are rates of co-occurrence, never a fitted model; illness-flagged days are excluded; factors with under three days on either side are skipped; differences under about 25 points aren't reported (§14.6). The frontend shows exactly what's returned and adds nothing. This is one of the strongest things to explain to a judge.

Page-level states

State	Shows
No entries at all	Log: the retrospective first visit. Trends: "*Nothing logged yet. Start by marking a few days that felt rough, and we'll show you what was in the air.*"
Loading	Trends: grid-shaped placeholder cells + two finding-card placeholders
Entries fail to load	Generic error + retry; the form still works for today
Offline	Default: the form stays filled and a failed save shows "*Couldn't save that entry. It's still here. Try again.*" (PROPOSED). With the Tier 3 queue built: entries queue with a pending marker (A2)

Connection to the Assistant LOCKED

Journal is the one feature that gives the Assistant information about the household itself: a question like "why does my son cough more on some days?" is answered from that household's own entries and environmental history, alongside cited guidance (§14.8). The server supplies this context to the Assistant; nothing for this page to do beyond saving entries, see C2.

Proposed content PROPOSED

Addition	From
Progress toward findings shown as two small counters in Trends: comparable symptom days 3 of 5, days of readings 9 of 14, so it's clear which one is holding things up. Note the readings counter excludes illness-flagged days, so it can drop when a day is flagged	`flaggedDays`, `totalDays`, `needed` (labeled as comparable days, not days logged)
In day detail, the household sentence for that day's worst reading isn't available from history, instead show the values with units	`history[date].values`
Per-group filter on the heat grid dots when more than one group logs	`entries[].band`

Visual decisions to make DESIGN

The Journal's own: the heat grid (signature), the month calendar for the retrospective step, the findings card with its bar pair, the scrollable line chart with its marker shapes. Segmented control, toggle chips, toast, empty state and swipe row are shared and frozen in Phase 0.

B5. Act

Purpose LOCKED

Everything a household can do beyond its own four walls, contributing, advocating, sharing. Distinct from Learn: Learn is personal protection (filters, sunscreen, closing windows); Act is acting on the problem itself. This is the layer that makes HALO civic rather than merely personal, and the clearest answer to why it belongs in a competition run by a member of Congress (§15.1, §17.2).

Address and entry

`/act`, Tab 5. Each of the four features opens a bottom sheet (history entries). **Each sheet fetches what it needs independently, on open, with its own skeleton; a failed part shows "No data" with a Retry for that part and never blanks the rest** (for example a failed `/api/sources` must not blank the health card). Act's sheets always fetch fresh rather than reading the stored copy, since Settings changes can make the stored copy stale (Part P3).

Shape LOCKED

Header *Act*. A framing line: "*Environmental problems aren't solved one household at a time. Here's what else you can do.*" Then four full-width cards, each with one strong icon, a title, a one-line description, and a chevron. Styled editorially, taller cards, larger type, more whitespace, no charts, because these are invitations, not readings. Same card component; only proportions and type scale differ (§15.2, §48.1).

Data calls BIND

Call	Feeds
`GET /api/profile` through `useProfile()`	County, water source, renter mode, coordinates (for the facility distance)
`GET /api/home-guard` (fresh on sheet open)	Which risks are elevated (volunteer causes, help sections), letter values, health card
`GET /api/daily-score` (fresh on sheet open, never `fresh=1`)	Health card daily block
`GET /api/volunteer?causes=…`	Volunteer sheet
`GET /api/district`	**PROPOSED** district context card

Card 1: Volunteer near you LOCKED

Description: "*Groups working on the problems HALO found in your area.*" The list is filtered by this household's own findings, a household with elevated water sees the riverkeeper working on exactly that; one with clean water and high radon doesn't (§15.3).

Element	Content	Binding
Request	None	`GET /api/volunteer?causes=`. The frontend derives causes; the server doesn't. From HomeGuard: water item in `action_plan` → `water,pfas`; radon item → `radon`; lead item → `water`. For a well or spring, use `wells` from `profile.water_source` (a well's `action_plan` is always empty, so the plan cannot supply it). Nothing else, adding a broad tag like `advocacy` would match every organization and undo the filtering. Nothing elevated → send no causes (every county-serving org is returned, unranked). County defaults to the profile's
Sheet title + basis line	*Volunteer near you*, then "*Groups working on [water quality / PFAS / radon / private wells] in [County].*" (join several with "and"); with no causes "*Groups serving [County].*"	`county`, `causes` from the response. Cause tag labels: water "Water quality", pfas "PFAS", radon "Radon", air "Air quality", wells "Private wells", advocacy "Advocacy"
Each organization	Name; description (2 lines max); cause tags; counties served (*Statewide* when `counties` includes `"statewide"`; else the household's county if listed, else the first two and "*and N more*"); *Visit site* opening externally; when it was last verified by a person	`orgs[]` → `name`, `description`, `causes[]`, `counties[]`, `url`, `last_verified`
Verification	`last_verified` set → "*Last verified March 2026*"; null → "*Not yet verified*"	Never claim verification that didn't happen. **Today all 13 seeded organizations have `last_verified` empty**, so every row says "Not yet verified" and the footer below never appears until a person verifies them (P10, D9)

Footer	*"Every group here was checked by hand. Last verified [month year]."*	Show only if every listed org has `last_verified`, otherwise the claim would be false
Empty	*"We don't have groups listed for your county yet."* (the list includes unverified groups, so the word "verified" would be wrong)	`orgs` empty

> **NOTE:** don't render `match_count` when no causes were sent (it's always 0 then). A household with nothing elevated sees every organization serving its county, unranked.

Card 2: How you can help LOCKED

Description: *"Ways to act on what's affecting your area."* Guidance tied to this household's actual readings, never generic (§15.4). Sections appear only for risks actually elevated here; each has a heading naming the risk and two to four numbered actions, each one line with an optional supporting sentence. A closing *Always useful* section applies to everyone (§48.3).

Section appears when	Covers	Trigger binding
Radon zone is elevated	How to test, and why it matters beyond one home, radon is geological, so a high result is information the neighborhood should have too	`radon.zone` 1 or 2
Water contamination is elevated	Finding the utility's public comment periods; who to contact; what a Consumer Confidence Report is and how to read your own	`action_plan` has a `water` item
Nearby facility violations	How enforcement works, looking up a facility's record, filing a complaint	A facility with open violations (`open_violations > 0`) within PROPOSED 5 miles of the home, from `/api/map?layer=facilities` (distance from the profile's `lat` / `lng`)
Always	How to read a water quality report; what to ask at a town or utility meeting	None

> **FACILITY SECTION HONESTY.** The facilities layer holds only active major facilities and repeat violators in NC-08, so the section appears only when the layer returns a facility with `open_violations` above 0 within 5 miles (haversine from the profile's rounded coordinates), and it is naturally absent outside NC-08. `open_violations` counts programs currently in violation (0 to 4), not violations. Tests need a fixture with such a facility.

> **CONTENT TO WRITE.** The blueprint names each section's topic but not its actual numbered actions. Draft actions are supplied below as PROPOSED so builders are not blocked; the content owner (Aadit, P10 D5) must verify every factual claim, and a claim that cannot be verified is cut, not softened. This is design/content-owner work (§26.3b), not engineering.

Section	PROPOSED draft actions (numbered in the sheet)
Radon	1. Test your home for radon. A short-term kit costs about $15 to $40 and takes a few days. 2. If the result is 4 pCi/L or higher, run a second test to confirm, then talk to a certified radon mitigation contractor. 3. Tell your neighbors. Radon comes from the ground, so a high result on one street is useful to the people around you. 4. Ask your state health department whether reduced-price test kits are offered.
Water	1. Find your utility's public comment periods. Utilities and state agencies post them on their websites. 2. Read your utility's Consumer Confidence Report. Utilities send one every year, by July 1; find the column that compares each result with its legal limit. 3. Ask your utility what its plan is to meet the federal limit, and when. 4. Contact your city council, county commission or the utility's board, which set water budgets.
Nearby facility	1. Look up the facility's record on EPA's ECHO site (Enforcement and Compliance History Online). 2. Report a concern to your state environmental agency (in North Carolina, the Department of Environmental Quality). 3. Keep notes: date, time, what you saw or smelled, and photos.
Always useful	1. Learn to read a water quality report. 2. At a town or utility meeting, ask: What are our most recent test results? What is the plan and timeline if a result is over a limit? How will households be told?

Card 3: Write a letter LOCKED

Description: *"A ready-to-send message about your water, filled in with your numbers."* The single strongest fit for a congressional competition: a tool that helps a constituent engage an institution, built from their own measured data (§15.5).

Element	Content / behavior	Binding
Recipient	Three-option control at the top: *Utility* · *Landlord* · *Representative*. Default: Landlord for a renter, Utility otherwise. Changing it shows that recipient's draft; each recipient keeps its own edited draft, and Start over resets only the current one	None

Data notice	*"Filled in with your utility's most recent test results."*	None
Draft	Editable multi-line text area filling most of a tall sheet	Utility template (verbatim below)
Actions	Primary *Copy* (toast confirms) · secondary *Share* (the device share sheet) · tertiary *Start over* restores the generated text after edits	`navigator.clipboard`, `navigator.share` (fall back to Copy where Share isn't supported; a cancelled share is not an error; if the clipboard is refused, select the text and say so)

```
I am a customer of [utility name], public water system [identifier]. Your most recent federal sampling,
dated [date], reported [contaminant] at [value] parts per trillion. The federal maximum contaminant level
is [limit] parts per trillion. This system serves approximately [population] people. I am writing to ask
what your compliance timeline is, and what interim steps are available to households in the meantime.
```

Placeholder	Binding
utility name / identifier	`water.pws_name` / `profile.pwsid`
contaminant, value, limit, date	The cited entry: from `water.scored_contaminants` keep entries with `is_enforceable === true` **and** `exceeds_limit === true`, take the highest `risk`; if none, show the explanation state. → `contaminant` (through `contaminantLabel`), `value_ppt` (1 decimal), `limit_ppt` (the limit; never say "federal maximum contaminant level" for a non-enforceable entry), `date_iso` as "July 2025"
population	`water.population_served` (formatted per A5, "serves approximately 120,000 people", no plus sign); drop the sentence if null. Utility name through `formatUtilityName`

> **CONTENT TO WRITE AND EDGE CASES:** (1) The blueprint gives the utility letter only; the landlord version ("asks about testing and tenant remedies rather than compliance schedules") and the representative version need writing (content owner). (2) HALO doesn't store who the household's representative is; the letter opens *"Dear Representative,"* for the person to complete. (3) When there's no measured water result to cite, show a short explanation instead of an empty template; for wells, hide the utility option.

PROPOSED draft	Text (the person completes bracketed parts; HALO never stores the address or a name)
Utility, greeting and closing	Opens "To whom it may concern," then the body above, then "Thank you," and "[Your name]".
Landlord	"Dear [landlord or property manager], I rent at [your address]. My water comes from [utility name], public water system [identifier]. That utility's most recent federal sampling, dated [date], reported [contaminant] at [value] parts per trillion. The federal maximum contaminant level is [limit] parts per trillion. I am writing to ask whether you have tested the water at this property, whether you plan to, and what steps you can take, such as providing a filter certified to reduce [contaminant]. Please reply in writing. Thank you, [Your name]"
Representative	"Dear Representative, I am a constituent in [county]. My water comes from [utility name], public water system [identifier]. Its most recent federal sampling, dated [date], reported [contaminant] at [value] parts per trillion; the federal maximum contaminant level is [limit] parts per trillion. The system serves approximately [population] people. I am writing to ask what is being done to help utilities like this one meet federal limits, and what help is available to households in the meantime. Thank you for your time, [Your name]"
No result: well	"Private wells aren't covered by the federal drinking water rules, so there is no utility result to cite. See your testing plan on the Home page."
No result: other cases	No data yet: "Your utility's federal results haven't been published yet, so there is nothing to cite." Lookup failed: "We couldn't check your utility's results right now. Try again in a moment." Nothing over a limit: "No enforceable federal limit was exceeded in your utility's most recent results, so this letter has no measurement to cite." Outside North Carolina: "HALO's water data covers North Carolina utilities, so it can't fill in this letter."

Card 4: Your home's health card LOCKED

Description: *"A one-page summary you can save or share."* Its practical purpose is to be handed to someone: a doctor asking about exposure, a landlord being asked to test, a relative being persuaded to test their own home. Formats data every other page already produces, plus the source list (§15.6, §48.5).

Section	Content	Binding
Header	HALO mark, county, generation date **with the year** (*"March 14, 2026"*, because it is printed)	`profile.county`, now
Two scores	Daily and home scores side by side with their rings and labels	`score.display_score` from each stored response
Category table	Every category with its reading, severity, and the date behind it: air, UV, pollen, mold, water, radon, lead	Today readings + time retrieved; water worst contaminant + `latest_sample_iso`; radon zone ("county estimate"); lead level + build year
Household notes	The household-aware sentences, if any	`*.sentence`, `lead.sentence`

Top actions	Top three action items. **For a well or spring** the action plan is always empty, so show the top three tests instead, never "Nothing here needs action"	`action_plan[0..2]` (title, cost, certification); well: `water.test_plan.tests[0..2]` (name, priority, estimated cost)
Sources	Every source with the date it was retrieved	`GET /api/sources` → `name`, `provides`, `last_retrieved`, rendered as returned (no hard-coded list); `last_retrieved` null → *"Retrieval date not recorded"*
Buttons	Primary *Save as image* · secondary *Print* (the card's description says "save or share": there is no separate Share button, so use "save or print" unless the image share below is built)	Client-side image render, always light, with a library approved by the team before use (the first-load budget is 500 KB; load it only when the button is first pressed); optionally `navigator.share` of the image. Print stylesheet per the box below

> **PRINT IS TIER 1.** Printing must produce a clean black-on-white page with no interface chrome, no backgrounds, severity conveyed by label (and pattern) rather than color, and it's always light even in dark mode (§48.5, §59.3, §65.1). Implementation: `@media print { body > *:not(.halo-health-card) { display: none } .halo-app { color-scheme: light } }`; severity shown as icon plus word; rings replaced by text such as "*Score: 78, High*". Missing values print "No data".

Page-level states

State	Shows
HomeGuard not yet loaded this session	Cards still render; each sheet fetches what it needs on open with its own skeleton
Loading (page)	Four card-shaped placeholders (the page itself needs almost no data)
A sheet's request fails	The sheet shows the parts it has, "No data" for the failed part, and a Retry for that part

Proposed content PROPOSED

Addition	From	Why
A district context card above the four features: e.g. "*[affected_population] people in NC-08 are served by a water system over a federal PFAS limit.*" ("at least" when `affected_population_complete` is false). **Show only** when `affected_population` is a number and the household's county appears in `counties[]` (in the district, wholly or partly); hide it otherwise (a household in Buncombe, Iredell or out of state is not in NC-08). Tap → `/map?panel=district`	`GET /api/district` → `affected_population`, `affected_population_complete`, `assembled_at`	Puts the household's result in civic context, "is this just my problem?", on the civic page. Strongest single line for a judge
In the letter sheet, a line showing which reading the letter cites, with a link to it in HomeGuard	The chosen `scored_contaminants` entry	Transparency: the person sees exactly what they're sending

Visual decisions to make DESIGN

Act's own: the editorial card treatment (taller, larger type, one strong icon each), organization list rows, the letter editor sheet, the health card as a formatted document (on screen and in print), the district context card. Bottom sheet, segmented control, toast and callouts are shared (Phase 0).

B6. Settings

Purpose LOCKED

Everything entered during onboarding must be changeable, and everything HALO knows must be inspectable. A household that can't correct a wrong address or remove a group will stop trusting the results (§16.1).

Owner: Yogi (Lane 1, Part P). Settings also owns `saveHousehold()`, which HomeGuard and Onboarding import, so it is built early in Lane 1.

Address and entry

`/settings`, from the gear in the shell header on every tabbed page. Header *Settings* with a back arrow (not a gear). No tab highlighted. Eight accordion sections (seven until reviewed Spanish exists: the Language section is hidden until then, see the Language box). Only one open at a time; each row shows its current value when closed. Every control saves immediately and toasts only on failure (§54). `/settings?open=household|home|renting|language|accessibility|notifications|privacy|about` opens that section on arrival (used by HomeGuard's "unknown year" link, `/settings?open=home`). Opening or closing a section does not change the URL after arrival.

Back arrow rule. Back goes to the page the person came from (`router.back()`). If the app was opened directly on `/settings` (no in-app history), back goes to `/today`.

Loading. Every control is disabled, with a skeleton for its summary, until `useProfile()` has resolved. A control that saves the household is never enabled with guessed values, because the save replaces all seven groups.

Data calls BIND

Call	Feeds
`GET /api/profile` (through `useProfile()`)	`profile.county`, `pwsid`, `water_source`, `home_year`, `renter_mode`, `locale`, `household`, `household_set`, `zip` (null after onboarding; never bind it), `onboard_request_id` (never displayed)
`PUT /api/household` (through `saveHousehold`)	Household groups, renting or owning, language
`PATCH /api/profile`	Build year, water source (send only the changed field)
Onboarding address step (`/onboarding?change=1`)	Change address (`POST /api/onboard`, made by the onboarding screen, not by Settings)
`GET` / `PUT /api/notifications`, `POST` / `DELETE /api/push/subscribe`	Notification switches and device notifications
`GET /api/journal?from=2000-01-01`	Entry count for the Clear journal confirmation (read `count`)
`DELETE /api/journal?all=true`	Clear journal (returns `{ deleted: n }`)
`DELETE /api/account`	Delete everything
`GET /api/sources`	About: data sources and dates
`GET /api/assistant`	Only `sends_household_context`, for the privacy statement (Part P9 ask 2; assume false until it exists)

> **ONE SAVE FUNCTION FOR `/api/household`:** `PUT /api/household` replaces all seven groups on every call. Send only `renter_mode` and the household's groups are wiped. Every Settings save that touches this endpoint goes through `saveHousehold({ household, renter_mode?, locale? })` (Part P3), which always sends all seven current booleans plus whatever changed. Its response only echoes `renter_mode` and `locale` if you sent them, so keep local state from the profile, never from the PUT response. Two quick taps on different chips must not race: queue saves so each one sends the latest full state, and after the last one settles refresh the profile once.

> **AFTER ANY SAVE THAT CHANGES WORDING:** household, renting or owning, water source, build year and language all change what Today, HomeGuard and Learn say. After a successful save Settings calls `clearReadings()` and dispatches `halo:settings-changed` (Part P3). The other pages refetch when they next show. Accessibility, notification and About changes do not dispatch it.

The eight sections LOCKED

Section	Collapsed summary	Expanded	Binding
Household	"*Toddler, adult*" or "*Not set*". "Not set" whenever no group is true. Don't rely on	The same seven chips as onboarding (the shared `ToggleChip`), same order; each tap saves; takes effect on the next reading refresh. Failure-only	Summary from `household` + `household_set`. Save: `saveHousehold`. Chip flips

	`household_set` alone: saving renting or language through `PUT /api/household` also sets it to true	toasts everywhere in Settings (the blueprint conflicts with itself: §49.2 toasts on save, §54 only on failure; failure-only is quieter and is the rule)	instantly, reverts on failure with the toast *Couldn't save that. Try again.*
Your home	"*Union County · City water*"	Build year field (A7 rules), water source dropdown (same five options as onboarding; "Not sure" is saved as `other`, so a stored `other` preselects "Other"), and secondary *Change address*	Build year and water source: `PATCH /api/profile`, saved on change (build year on blur). Change address: see the flow below
Renting or owning	*Owner* / *Renter*	Two-option control + one sentence that it changes which actions are recommended everywhere	`renter_mode` via `saveHousehold` (with all seven groups). Changes every action recommendation (§12.7)
Language	*English*	English · Español, applying immediately. Hidden unless `NEXT_PUBLIC_SPANISH_REVIEWED=1` (see the Language box)	`locale` via `saveHousehold`. Interface strings switch from the translation layer; Learn content is requested with the new `locale`
Accessibility	*Default*, or *Custom* if any of the three is changed	Text size slider with live preview sentence; higher-contrast switch; reduce-motion switch. Details in the Accessibility contract below	Device-local (stored in the browser under `halo.device.*`): these are about this device, not the household. Reduce motion defaults to the system setting
Notifications	"*4 of 5 on*"	One switch per alert type with its plain-language name and one-line description, all default on; plus one switch for device notifications on this browser (hidden until D8, see below)	Types: `GET` / `PUT /api/notifications` → `preferences.air_quality_change`, `weather_advisory`, `new_water_results`, `radon_season`, `season_summary`. Device notifications: see the push block below
Data and privacy	*Your data* (no value, so the row reads as a label)	"What HALO knows about you" (the stored values), the full privacy statement, then *Clear journal* and destructive *Delete everything*	`useProfile()`, `GET /api/journal`, `DELETE /api/journal?all=true`, `DELETE /api/account`
About	"*Version 1.0*"	Every data source with the date it was last retrieved; how scores are calculated; who built HALO; link to the source repository	`GET /api/sources` + static text (below)

Change address flow BIND

1. Tapping *Change address* opens a compact confirm sheet: "*This will replace your current results. Continue?*" with *Continue* and *Cancel*.
2. Continue goes to `/onboarding?change=1`, which starts at the address step (B1). The onboarding screen does the geocoding and the `POST /api/onboard` (with a fresh `request_id`), and discards the typed address afterward, exactly as in first-time onboarding.
3. Onboarding, not Settings, clears the stored Today and HomeGuard readings and requests the fresh reading. When the (rounded) location changes, the server clears that day's stored score and resets the daily-score limiter, so a one-hour-old cache can no longer serve the old address's readings. Re-entering the same address changes nothing and clears nothing. (If a `fresh=1` request still returns 429, keep the stored-readings notice and retry after `retry_after_seconds`.)
4. In change mode the flow continues to step 3 (Household), step 4 (Your home), then the results reveal, then Today. **Both steps are shown, prefilled** with the profile's current groups, build year and water source, because a new home may have a different water source and different people living in it. DECISION D2 (Part P10): default keep, as built. Yogi's call if he wants Household skipped.
5. Back on the address step in change mode returns to `/settings` (Part P7).

Accessibility contract LOCKED

Control	Stored key	What it does
Text size	`halo.device.textScale`	Slider with 5 steps: 100%, 115%, 130%, 150%, 175%. Sets `--halo-text-scale` on `.halo-app` and `data-text-scale` on it. All font sizes are rem-based and multiply by that variable (Part P2), so nothing is a fixed pixel size
Higher contrast	`halo.device.contrast`	Sets `data-contrast="high"`. Tokens swap to the high-contrast values (text and border colors only; status colors keep their hues, and the six severity words stay printed next to color)
Reduce motion	`halo.device.motion` (`reduce` | `full` |	Sets `data-motion="reduce"`. Absent means follow the system `prefers-reduced-motion`. When reduced, the ScoreRing draws its final state with no sweep, sheets appear without sliding, and

	absent)	skeleton shimmer stops

Live preview sentence under the slider, in the current size: "*Your air today is Moderate. Take it easy outside if you're sensitive.*" (Fixed sample, not real data.) The three values are read once at app start by the shell, before first paint where possible, so the page never flashes at the wrong size. These keys are device settings, never sent to the server, and never cleared by `clearReadings()`. They are cleared by Delete everything.

Notification switch copy PROPOSED

Preference	Name	Description	When it really fires (from the server)
`air_quality_change`	Air quality changes	When the air gets worse than yesterday. Earlier for homes with someone sensitive	Today is worse than yesterday (or yesterday has no reading), and today is Elevated or worse (High or worse for homes with no sensitive group). Message: "Air quality dropped to [label] today."
`weather_advisory`	Weather advisories	Flood and heat advisories for your county	Any National Weather Service alert with "flood" or "heat" in its name for the home's point. That includes watches and warnings, which the message still calls an advisory. Boil-water advisories are **not** sent: no national source exists, so the description must not promise them (the blueprint's "boil-water" wording is removed)
`new_water_results`	New water results	When new federal testing results are published for your utility	New testing results were published for the utility
`radon_season`	Radon testing season	A winter reminder when testing is most accurate	October to February, radon Zone 1 or 2, and no radon test recorded (the server has no record of radon tests yet and always treats it as "no test", so the reminder fires every winter in Zone 1 or 2). Zone 3 never gets it (every NC-08 county is Zone 3)
`season_summary`	Season summaries	A look back at your journal when a season ends	A season just ended and there are journal entries

The switch names and descriptions are PROPOSED for content-owner review (D5). The "when it fires" column is verified against the server and is for builders and the Alerts owner, not shown in the app. Turning a type off stops both inbox alerts and device notifications of that type. Alert titles and messages as the server sends them are in C3.

Device notifications (push) BIND DECISION D8

Device notifications are built **after the core pages** (Part P10, D8). Until then the switch is not rendered, and the Notifications section shows only the five type switches. The backend is ready: `POST /api/push/subscribe` and `DELETE`, and the daily job already sends pushes to every subscription once the four VAPID variables are set on Vercel.

Piece	What it is
Files	`public/sw.js` (service worker), `lib/frontend/push.ts` (subscribe, unsubscribe, permission state), the switch in `components/settings/NotificationsSection.tsx`
When the switch shows	Only when `'serviceWorker' in navigator`, `'PushManager' in window` and `NEXT_PUBLIC_VAPID_PUBLIC_KEY` is set. On iPhone this needs the app added to the home screen, so when push isn't available show nothing rather than a broken switch
Turning on	Ask browser permission only on this tap, and only after the person has used the app at least once (§19.5). Register `/sw.js`, convert `NEXT_PUBLIC_VAPID_PUBLIC_KEY` from base64url to a `Uint8Array` for `applicationServerKey`, call `pushManager.subscribe({ userVisibleOnly: true, applicationServerKey })`, then `POST /api/push/subscribe` with the subscription JSON (`endpoint`, `keys.p256dh`, `keys.auth`)
Permission denied	Switch returns to off, with the line *Notifications are blocked in your browser settings.* under it
Turning off	`DELETE /api/push/subscribe` with `{ endpoint }`, then `subscription.unsubscribe()`
Delete everything	Unsubscribe the browser subscription before calling `DELETE /api/account` (the server removes its own rows with the account)
Service worker handlers	`push`: parse the JSON `{ id, type, title, message, url }` and show a notification. `notificationclick`: focus an open HALO tab or open `url`. Landing pages by type: air and weather to `/today`; new water results to `/home?risk=water`; radon to `/home?risk=radon`; season summary to `/journal?mode=trends`
Rate limit	429 "Too many changes. Try again later." on subscribe or unsubscribe: switch returns to its previous state with the failure toast

Privacy statement LOCKED

What HALO stores: your approximate coordinates, your county, which water utility serves you, your home's build year, and which household groups you told us about.

What HALO never stores: your street address, anyone's name, anyone's age or birthday, and any medical record. Your journal notes stay on your account and are never shared with anyone.

The third paragraph depends on which Assistant mode is live (the server reports it as `sends_household_context` on `GET /api/assistant`; assume false until that exists). DECISION D4 default is off.

When false (default): *When you ask HALO's assistant a question, your question is sent to an AI service to write the answer. Your readings and symptom log are not sent. Don't include names or addresses in your question.*

When true: *When you ask HALO's assistant a question, the readings and logged symptoms that help answer it are sent to its AI service to write the answer. Your notes are never sent.*

Also stated here, per §37.4: HALO collects no behavioral analytics.

> **OPEN: THE "STORES" LINE IS INCOMPLETE.** §16.2 asks for exactly what is and isn't stored. The locked line omits water source, renting or owning, language, state, daily reading history, journal entries, alerts, device notification subscriptions, notification settings and the rounded coordinates' precision. PROPOSED addition for Yogi to approve (don't change the locked text without it): "*It also stores your water source, whether you rent or own, your language, your daily readings, the journal entries you write, the alerts it sends you, your notification settings, and, if you turn on device notifications, a code that lets your browser receive them.*" Coordinates are rounded to about 100 meters before saving, and street addresses are never stored.

> **DECIDED: THE ASSISTANT AND JOURNAL DATA.** The Assistant can send readings and journal symptom names and dates to its AI provider (never the free-text notes) to answer questions like "why does my son cough more on some days?" (§14.8, §18.2). The team approved that for this build, but the free model may log prompts, so it ships **off** until Yogi turns it on. The server variable `ASSISTANT_SEND_HOUSEHOLD_CONTEXT` must be exactly the string `false` for that: any other value, including unset, currently turns household data on (Part P9 ask 2 fixes the default). The statement above changes with the switch, so it can never claim more privacy than the server provides.

What HALO knows about you, and Clear journal BIND

- **"What HALO knows about you"** at the top of Data and privacy shows the stored values back: county, water utility ID, water source, build year, renting or owning, language, household groups. From `useProfile()`. A missing value reads *No data*. Makes "everything must be inspectable" (§16.1) literal. This is a required part of the page, not optional.
- **Clear journal** sits above Delete everything. Tapping it first calls `GET /api/journal?from=2000-01-01` and reads `count`. Confirm sheet: "*This deletes all [n] of your entries permanently.*" with a destructive confirm. At `count = 0` the row is disabled with *No entries to clear*. Confirm calls `DELETE /api/journal?all=true`; on success clear any stored journal data and toast *Journal cleared* (the one success toast in Settings, because the effect isn't visible on this page). On failure, stay on the sheet with the generic error.

Delete everything LOCKED BIND

Tapping *Delete everything* opens a compact sheet stating exactly what gets deleted: "*This permanently deletes everything HALO knows about your household. This cannot be undone.*" With a field that requires typing *DELETE* (exactly, capitals), and a destructive confirm button that stays disabled until it matches (§49.3). Steps on confirm:

1. Unsubscribe this browser from push if it is subscribed (see above). A failure here does not stop the deletion.
2. `DELETE /api/account` with body `{ "confirm": "DELETE" }`. On failure: stay on the sheet with the generic error *Couldn't delete your data. Nothing was deleted. Try again.* Do not clear anything locally.
3. On `{ deleted: true }`: remove every `halo.frontend.v1.*` key, every `halo.device.*` key and `halo.onboarded`; call `supabase.auth.signOut()` and discard the session; then `router.replace('/onboarding')`.
4. Onboarding opens on Welcome with the toast *Your data has been deleted.* (8 seconds, the shell's toast).

Tier 1 per the blueprint's own triage. Never test this against production data (Part P4).

About LOCKED

- **Data sources.** Render `GET /api/sources` exactly as returned: `sources: [{ name, provides, last_retrieved }]` (currently 10 entries). Show `name` as the row title, `provides` as the description, and `last_retrieved` as *Retrieved September 27, 2026* (date only, in the person's time zone). A `null` date reads *Retrieval date not recorded*, never a guess and never "No data" (this is a fact about the record, not a missing reading). Don't hard-code the list of sources in the app, so a new source appears without a release.
- **How scores are produced**, in plain language (static copy in `copy.settings`): daily conditions combine as independent risks; home conditions combine multiplicatively; missing data is excluded rather than counted as zero; one severe problem can't be averaged away by several mild ones.
- **Version.** The collapsed summary and the top of the expanded section read *Version 1.0*, a constant in `copy.settings` (changed by hand at release).
- **Who built HALO and the repository link.** PROPOSED text, content owner (Aadit, D5) to confirm names before submission: "*HALO was built by four students in North Carolina's 8th Congressional District for the 2026 Congressional App*

Challenge." and a link *View the source code* to the repository URL (a placeholder constant until the team publishes it; hide the link while the constant is empty). No names or personal contact details in the app itself.

- **Volunteer organizations.** No mention here. (All 13 organizations are still unverified, D9; verify before submission.)

Language and Spanish

> **DECISION:** the blueprint's rule is that machine-translated health guidance isn't acceptable; ship English-only rather than unreviewed Spanish (§68). The Language section appears only when a person has reviewed the Spanish interface strings and Learn content and the environment flag `NEXT_PUBLIC_SPANISH_REVIEWED=1` is set. Until then the profile's `locale` stays `en` and there are seven sections. Either way, every string goes through a translation layer from day one, no button's width is fixed to its English label, and Learn trusts the `locale` it gets back (it reports the language actually served).

Visual decisions DESIGN

Accordion rows with value summaries (`AccordionRow`, Part P2), switches (`Switch`), the text-size slider with live preview, the destructive button and delete sheet (built on `BottomSheet`), the formatted privacy statement. The wireframe for Settings shows: every section closed; each section open (eight screenshots); Household loading and save-failed states; Notifications with push available and unavailable; Clear journal with 0 and with several entries; the Delete sheet empty, matching and failed; About with a null date. Font sizes and colors come from the existing tokens (A11, Part P6); nothing new is invented here.

B7. Map and the District panel

Purpose LOCKED

Answer *is this just my problem, or is it everyone's?* Every other page is about one household; this one places it in its region, which is what turns a personal utility into something civic. The most visually striking screen, so it matters disproportionately in a demo (§13.1).

Owner: Vibhav (Lane 3, Part P), after Phase 0. The Map is the largest single page (four data layers, a slider, a district panel, a map library), so it starts early in Lane 3. It has no dependency on other pages except `openLearn` (Part P5) and the shell.

Address and entry

`/map` (Water layer, overall severity), Tab 3. The map library loads only when this tab is first opened, never on app start (§35.2). The header sits over the map. On wide screens the map column widens to 600px, the one exception to the 430px cap (§53.2); the shell does this by setting `data-wide` on `.halo-app` for `/map` (Part P2).

Parameter	Accepted values	History
`layer`	`water` | `radon` | `facilities` | `air`	replace
`contaminant` (water only)	`pfoa` | `pfos` | `pfhxs` | `pfna` | `genx` | `hazard` (means the combined hazard index; the default overall view has no parameter)	replace
`quarter` (water only)	Must match `^\d{4}Q[1-4]$` and be one of the response's `quarters[]`; otherwise ignored. Applies only when the slider exists	replace
`system`	A water system ID (`pwsid`), or a facility `id`, air station `id` or county name for the active layer. Opens that feature's sheet once its layer has loaded. A value not found in the loaded layer is ignored (no error)	push (a sheet, Part P3)
`panel`	`district`: opens the district panel	push (a sheet, Part P3)

Precedence when the page opens: address parameters, then the last state this session (kept in memory only: viewport, layer, contaminant, quarter, selection, per §46.3; the tab bar link is a plain `/map`, so returning to the tab restores it), then the defaults. Unrecognized or invalid values are ignored and the default is used, never an error page (A1). Changing layer, contaminant or quarter uses `router.replace` so the Back button never steps through every chip tap.

Map library, tiles and county shapes DECISION D7

Default: Mapbox GL JS on the team's existing Mapbox account, token `NEXT_PUBLIC_MAPBOX_TOKEN` (already in the environment table). Package `mapbox-gl`, loaded with a dynamic import inside the Map page only. A muted, low-saturation base style, and a dark variant of the same style when the phone is in dark mode (§59.3). The map's required attribution control (Mapbox and OpenStreetMap) stays visible in every state; it is never hidden to save space. Data layers are drawn as WebGL layers (GeoJSON sources with circle and fill layers), not DOM markers, because the water layer has hundreds of systems.

County shapes. Radon needs county polygons. The repository already has them: `lib/data/ncCounties.json` (U.S. Census Bureau TIGERweb, simplified to about 200 m, 100 counties, about 160 KB, each with `name` such as `"Union County"` and `geoid`). Phase 0 or the Map owner converts it once to `public/data/nc-counties.geojson` and loads it only when the Radon layer or the district panel needs it. The radon API's county names carry the same "County" suffix, so join by exact name; trim and compare case-insensitively as a guard. A county in the response with no shape is skipped, never guessed. Do not import the JSON under `lib/` into the browser bundle (it is server code and would add 160 KB to every page).

Cached map data. Layer responses are cached for the session and offline (§36) in an IndexedDB database named `halo-map`, keyed by identity + layer (+ quarter), holding the response and its `assembled_at`. It contains only public data, but it is keyed and cleared like household data (`clearReadings()`, Delete everything), per Part P2. Never use `storeReading` for map data: it strips coordinates. Offline, show the cached layer with *Showing saved map data from [date].* and disable the quarter slider if that quarter isn't cached.

Data calls BIND

Call	Feeds
`GET /api/profile` (`useProfile()`)	Home pin (`lat`, `lng`, rounded by the server to about 100 m, so the pin is approximate), own system (`pwsid`), `county`
`GET /api/map?layer=water`	`features[]`, `counts` (the live count line), `count` and `located`, `quarters[]` (oldest first), `assembled_at`, `population_source`, `prebuilt`
`GET /api/map?layer=water&quarter=2024Q3`	The same shape (except no `prebuilt`), only systems that have a reading in that quarter, `quarter` set. `assembled_at` here is the request time, not the data date (see Freshness)

`GET /api/map?layer=radon`	`features[]` → `county`, `zone`, `severity`; joined to the county shapes by name. One feature per county (100)
`GET /api/map?layer=facilities`	`features[]` → facility points (fields in the sheet table), plus `scope` ("NC-08") and `selection` (a server sentence, see the honesty line)
`GET /api/map?layer=air`	`features[]` → monitoring stations
`GET /api/district`	Comparison banner (`household_county`), district panel
Static asset `public/data/nc-counties.geojson`	County shapes for the radon fill

Each layer is fetched the first time it is selected and cached for the session (and offline). The daily job builds every layer once a day, so a normal request is fast; a cold rebuild can take up to about 20 seconds and up to 60 on the server, hence the 20-second client deadline (Part P3). The water layer is the default layer: request it at once; request `/api/district` only when the panel opens or the water banner needs it (after the water layer has drawn).

Layout LOCKED

The map fills everything between header and tab bar, edge to edge. Floating on top: a row of layer chips; beneath them (Water only) the contaminant selector, a live count line, and a conditional rescission notice; a locate button (crosshair icon; tapping it animates to the home pin at the default zoom, §54) and zoom buttons on the right edge (the blueprint hides zoom buttons on touch devices; that's Tier 3: **default: visible everywhere**, §65.3); a legend lower left; the time slider (Water only, when the data has quarters); a comparison banner; and a handle at the very bottom, *See the full NC-08 picture*, opening the district panel (§13.2, §46.2).

> **OVERLAP BUDGET PROPOSED:** on a small phone (375 x 667, so about 545 px of map after the header and tab bar) the floating controls must never cover more than 60% of the map height, or the map stops being a map. Suggested numbers: top stack (chips 44, selector 44, count 24, notice 24, gaps) about 160 px; bottom stack (handle 48, banner 56, slider 64, legend 28, gaps) about 210 px. That is already 68%, so the design owner must use these hiding rules in the wireframe: (1) below 600 px of map height the comparison banner hides while the slider is showing; (2) the legend collapses to a single *Legend* button that opens a small popover; (3) the rescission notice moves into the contaminant selector's popover as an info line. Design confirms the actual sizes; the 60% ceiling is the rule.

Drawing order (bottom to top): base map, radon fill, water circles (largest population first, so small systems on top of big ones stay tappable), facilities, air stations, the outline of the household's own system, the selected feature's ring, the home pin. **Tap targets:** a feature's drawn size can be smaller than 44 px, so a tap selects the nearest feature within a 22 px radius of the tap point (draw an invisible larger hit layer). Water circle radius is a square-root scale of population, from a 6 px minimum to a 22 px maximum PROPOSED.

Initial state and memory LOCKED

Centered on the household at a zoom showing roughly their county and its neighbors: close enough to identify their own system, wide enough to show others exist. Home pin visible; Water layer, overall severity. If the profile has no `lat` / `lng` (should not happen after onboarding) or `isOutsideNC(profile.county)` is true, center on NC-08 as a whole, hide the locate button, and hide the pin unless it falls inside the view. This is an NC-08 app, so an outside-NC household still sees the district; the comparison banner is hidden for them because `household_county` is null or unranked. On return within a session, restore the last viewport, layer, and selection (§46.3).

Layer chips LOCKED

Water · Radon · Facilities · Air. One active; active chip visibly filled. Switching re-renders data without moving the viewport, so the household's position stays anchored (§13.3). The row is a `radiogroup`: Left and Right arrow keys move between chips and select (roving tabindex), Tab moves on.

What each layer draws LOCKED BIND

Layer	Drawn as	Binding
Water	One circle per public water system at its service-area center. Size by population served (square-root scale so big systems don't swamp small ones); fill by severity. No reading for the selected compound: small grey marker, legend PROPOSED *No detection reported*, not "Not tested for this". The loaded EPA data kept only detections (non-detects were dropped), so a system with no reading may have been tested and found nothing. The same limit affects the district's `tested` counts until Vibhav reloads the data with non-detects or adds a per-compound tested flag (F4, still open). In a quarter view, systems with no reading in that quarter are absent from the response and are not drawn at all. Water and air layers never cluster (§46.5): the pattern is the point. Systems without a location are not drawn; they are listed in the district panel (see below)	`lat`, `lng`, `population` (may be null: draw at the minimum size), `overall_severity` or the selected compound's `severity`. `located` vs `count` tells you how many can be drawn
Radon	Filled county shapes by the county's radon-zone severity, low enough fill opacity that county names stay legible. No clustering, no points	`features[].severity`, `zone`. Zone 1 is severity `high`, Zone 2

		`elevated`, Zone 3 `good`
Facilities	Points colored by compliance; clusters that separate as you zoom in; cluster badge shows a count; tapping a cluster zooms to fit it rather than opening a sheet (§46.5). Use the library's built-in clustering (cluster radius about 40 px, stop clustering at about zoom 12)	`lat`, `lng`, `severity`, `compliance_status` (`no_violation` good · `violation` elevated · `significant_violation` high · `unknown` no_data)
Air	One point per monitoring station by current air level, visually distinct from water circles (a square or diamond marker, not a circle, so the shape carries the difference and color isn't the only cue)	`lat`, `lng`, `severity`, `aqi`
Home pin	Always on top of every layer, never recolored by severity: the household never loses itself	Profile `lat` / `lng`

> **HONESTY LINE (FACILITIES).** The facilities layer is not every regulated site. The server selects **active major facilities and repeat violators (more than 4 of the last 12 quarters in noncompliance), inside NC-08 only**. Show, under the legend when Facilities is active: *Shows major facilities and repeat violators in NC-08, not every regulated site.* (static copy; do not display the server's `selection` sentence). A facility outside NC-08 will never appear, even if it is near an out-of-district household.

Legends LOCKED

Layer	Legend entries, with the same colors, icons and words as everywhere else (A10)
Water, overall	Good · Moderate · Elevated · High · Severe, then grey *No detection reported*. When a system has `coverage: "unscoreable"`: PROPOSED *Detected, no federal limit to compare*
Water, one compound	The same, plus the limit for that compound: *Limit: 4 ppt* (from the count line's response, `contaminants[X].limit_ppt`)
Radon	*Zone 1, highest potential* (High color) · *Zone 2, moderate* (Elevated) · *Zone 3, lowest* (Good). EPA zone definitions: Zone 1 predicted county average above 4 pCi/L, Zone 2 between 2 and 4, Zone 3 below 2. The legend never uses the word "safe"
Facilities	*No violation* · *Violation* · *Significant violation* · *Unknown* (No data color)
Air	Good · Moderate · Elevated · High · Severe (by AQI: 50, 100, 150, 200 boundaries). Caption *Latest hourly reading*

The contaminant selector (Water) LOCKED

The single most interactive element in the app: it lets someone answer their own questions rather than read a fixed answer (§13.4). Choosing a compound recolors every system by that compound alone, against its own limit, animated over about 300ms (motion off: instant).

Option	Colors by	Binding
Overall water health (default)	Each system's combined severity across everything measured	`overall_severity`
PFOA · PFOS	That compound vs its 4 ppt limit	`contaminants.PFOA.severity`, `contaminants.PFOS.severity`
PFHxS · PFNA · GenX	vs its 10 ppt limit, **with the rescission notice**	`contaminants.PFHxS`, `.PFNA`, `["HFPO-DA"]` (short label *GenX* on the selector chip; `contaminantLabel()`, which reads *GenX (HFPO-DA)*, in sheets and tables); notice when `proposed_for_rescission`
Combined hazard index	The federal combined measure: 1.0 or above is a violation even when no single compound exceeds its own limit. Applies only to systems where two or more of four PFAS were detected, so systems without a hazard index (`hazard_index: null`) get the No detection reported marker	`hazard_index.severity` (`value` and `exceeds` shown in the sheet), with the rescission notice when `hazard_index.proposed_for_rescission`

- An untested entry is `{ severity: "no_data", limit_ppt, proposed_for_rescission }` with no value, ratio, or date; check before reading them. Absence of a measurement is never shown as safety.
- `overall_severity: "no_data"` with `coverage: "unscoreable"` means compounds were detected but none has a federal limit; see the water legend.
- **Count line** beneath the selector, updating with the selection: the most quotable line on the page. Bind the map response's own `counts`; they describe exactly the systems drawn, statewide. The district panel's figures are NC-08 only. Strings (PROPOSED, numbers with thousands separators, singular when 1):
  - Overall: *[counts.over_limit] of [counts.total] systems exceed a federal limit.*
  - One compound: *[by_contaminant[X].over] of [by_contaminant[X].tested] systems tested for [PFOS] exceed the limit.*

(denominator is systems tested for that compound, matching the district panel's wording)
    - Hazard index: *[counts.hazard_index_over] systems exceed the combined PFAS hazard index.* (no denominator exists)
    - Zero: *No systems exceed the limit for [PFOS].*
    - In a quarter view append *in Q3 2024*. The counts then describe only the systems that reported that quarter.
- Rescission-flagged selection (a compound's `proposed_for_rescission`, or `hazard_index.proposed_for_rescission` for the hazard index): one-line notice beneath the count that the standard itself is contested. **PROPOSED** *EPA has proposed removing this limit. It still applies today.* (Content owner verifies against current federal status before launch.)
- The household's own system (`pwsid` matches the profile) stays outlined and labeled regardless of selection.
- Any other layer active: the selector and count line are replaced by that layer's legend and the count line *[count] [stations | counties | facilities] shown.*

Tapping a feature LOCKED

Every feature opens a compact bottom sheet over the map (never navigates away); the feature shows as selected while its sheet is open and deselects when it closes. Height is the one exception to standard sheets: sized to its content, at most 50% of the screen, so the selected feature stays visible (A9). The blueprint pans the map just enough to keep it visible above the sheet (§13.6, §46.6 and §46.7); that is Tier 3: **default is no auto-pan**; the sheet may cover part of the map and people drag (§65.3). Each sheet: title row, severity pill, body, footer with the data date. Opening a sheet adds one history entry (address `?system=`); Learn opened from inside it replaces that entry (Part P3).

Feature	Sheet contents, in order	Binding
Water system	System name (title); ID beneath; severity pill; *Serves [n] people* (or *No data* when `population` is null); a bar row for each of PFOA, PFOS, PFHxS, PFNA, GenX using the threshold-row component (value, limit, ratio, severity; a row with no reading says *No detection reported*) and each row's own sample date *Sampled March 4, 2025* (from `date_iso`); the hazard index row, *Hazard index 1.4. 1.0 or above is a violation.* with *Based on the latest sample.* (hidden when `hazard_index` is null; show the rescission notice when flagged); footer *Latest sample: March 4, 2025* (from `latest_sample_iso`, or *No data*); *Learn about PFAS*	`name`, `pwsid`, `overall_severity`, `population`, `contaminants[X]` (`value_ppt`, `limit_ppt`, `ratio`, `severity`, `date_iso`), `hazard_index` (`value`, `exceeds`, `basis`, `components`), `latest_sample_iso`. `openLearn({ topic: 'pfas', reading, context })` uses the selected compound if it has a value, else the compound with the highest ratio; context is that compound, value and limit (C1)
County (radon)	County name; zone as a severity pill; two sentences on what the zone means; the sentence that zones predict county averages rather than measuring a home; *Learn about radon* (`openLearn({ topic: 'radon', context: { county, zone } })`)	`county`, `zone`, `severity`. **PROPOSED** copy: Zone 1 *Homes in this county average above 4 pCi/L, the level where EPA recommends fixing a home.* Zone 2 *Homes in this county average between 2 and 4 pCi/L.* Zone 3 *Homes in this county average below 2 pCi/L.* then always *Any home can have high radon. Only a test tells you about yours.* (content owner verifies against EPA)
Facility	Facility name; compliance status pill; which programs regulate it (small tags, the program names as the server sends them); *Quarters with a violation in the last 3 years: 5 (of 12)*; *Programs currently in violation: 2* (only when `open_violations` is more than 0); *Penalties in the last 5 years: $12,500* (only when `penalties` is more than 0); *Last inspection: June 3, 2026* (or *No data*); county	`name`, `compliance_status`, `severity`, `programs[]`, `recent_violations` (this is quarters in noncompliance out of the last 12, not a count of violations), `open_violations`, `penalties` (dollars), `last_inspection`, `county`
Air monitor	Station name; *AQI 87* and severity pill; dominant pollutant (through `displayText`); *[d] miles from your home*; last reading time (format below)	`name`, `aqi`, `severity`, `dominant_pollutant`, `last_reading_at`. Distance is the haversine distance from the profile `lat` / `lng` to the station, in miles to one decimal, minimum *0.1*; hidden when the profile has no coordinates
Home pin	*Your home*; county; today's score (only when a stored Today reading exists; never fetched here); primary *Go to Today* (closes the sheet by replace, then pushes `/today`)	`profile.county`, stored daily `display_score`

Reading time format (air stations): reading today *Today at 3:00 PM*, yesterday *Yesterday at 3:00 PM*, older *September 26 at 3:00 PM*, in the device's time zone. **Freshness format** (below) is the same rule, prefixed *Updated* .

Time slider (Water) LOCKED BIND

Federal water testing is quarterly, so the water layer can move through time: a horizontal slider with quarter labels above the comparison banner; dragging recolors every system to its severity at that quarter, showing whether contamination is spreading, receding, or steady (§13.7). **Shown only when `quarters` has two or more entries**; per the blueprint, if the data holds only the latest reading, the slider is cut and the layer is static (§68). Yogi confirmed the data has the history: about 12 quarters, 2023Q1 to 2025Q4 (F4).

- **Positions.** The response's `quarters[]` (oldest first) followed by one final position, *Latest*, which is the default and means "each system's most recent reading", i.e. the normal prebuilt water layer. Quarter labels read *Q3 2024* (from `2024Q3`). Show first, last and every third label if space is tight.
- **Loading a quarter.** Each quarter is a live request (`?layer=water&quarter=2024Q3`), sent when the person releases the thumb, or 250 ms after the last keyboard step. Abort any request still in flight when a new position is chosen. Keep the previous colors on screen with a small *Loading Q3 2024* label until the response lands; on failure, snap back to the previous position and show the layer error callout. Cache each quarter's response for the session (and in `halo-map`).
- **What the quarter shows.** Only systems that reported in that quarter; the rest are not drawn (not grey). The count line says *in Q3 2024* and counts only those systems. The contaminant selection persists across quarters.
- **Freshness while scrubbing.** Use the `assembled_at` of the default (Latest) payload. A quarter response's `assembled_at` is the moment it was requested and would say "Updated today" over years-old data.
- **Keyboard.** The slider is a native range input (or `role="slider"` with `aria-valuetext="Q3 2024"`): Left and Right step one quarter, Home and End jump to first and Latest.

Comparison banner LOCKED

A small floating card near the bottom with one sharp, specific comparison, updating with the layer; tap opens the district panel (§13.8). Deliberately a single number rather than a chart: specific rankings are what people repeat to each other.

Layer	Banner	Binding
Water	*Cabarrus County ranks 34th of 100 North Carolina counties for water quality.* with a second, smaller line *1st means the fewest people on water over a federal limit.* (§13.8, and the rank's meaning stated, because "34th" alone doesn't say which end is good)	`district.household_county` → `county`, `water_rank`, `of`. Bind `of`: it is the number of counties actually ranked (`ranked_counties`), which can be below 100. **Hide the banner** when `household_county` is null, or `water_rank` or `of` is null. Ordinals come from one shared `ordinal()` in `format.ts`. An unranked county comes back as an object with `water_rank: null` (not a null `household_county`), so test the rank, not the object. `share_over` is a 0 to 1 fraction of the *people served* on over-limit systems; it is not comparable with the district-versus-state pair below, which is a share of *systems tested*
Radon	PROPOSED *Union County is Zone 3. 8 of 100 North Carolina counties are Zone 1.* (counts from the radon features, never hard-coded)	Profile county + radon features (`count` is the county total)
Facilities / Air	PROPOSED *[n] facilities within 10 miles have open violations.* (`open_violations` more than 0, haversine from the profile point) · *The nearest air monitor is [d] miles away.* Zero case: *No facilities in this layer are within 10 miles of you.* Hidden when the profile has no coordinates	Layer features + profile location

What the ranking is (verified in the server code): counties are ranked by the **share of the people served by mapped water systems whose most recent sample exceeds a federal limit**; 1 is the lowest share; ties share a rank; counties with no mapped system and population figure are not ranked. The district panel repeats this in one plain line, *Ranked by the share of people served by water systems over a federal limit.* (do not display the server's `ranking_rule` string).

Freshness LOCKED

Every layer shows when its data was assembled, e.g. *Updated today at 7:00 AM*; undated environmental data invites the reasonable suspicion that it's stale (§13.9). BIND the active layer's `assembled_at` (for Water always the default payload's, see the slider). The line sits with the legend. The daily job builds layers at 11:00 UTC, so "today at 7:00 AM" (6:00 AM in winter, Eastern) is normal; a layer older than 36 hours is rebuilt by the server on request, so the client needs no staleness rule of its own.

The district panel LOCKED

An expandable panel inside Map, not a separate page. Opened by dragging the handle up or tapping the comparison banner or handle; address `/map?panel=district` (push, so Back closes it). It is a standard-height sheet (about 70%). Shows no household data: it's the view a community group or legislative office could open (§20). Not tied to the active layer.

Content, in order	Binding and rules
Headline figure: people affected across the district, at hero size, with a one-line caption PROPOSED *people are served by water systems over a federal*	**Data guard first:** if `total_systems` is 0, or `unlocated_systems` is 90% or more of `total_systems`, the server's geography lookup failed and every figure below is meaningless (the server would otherwise report 0 systems over the limit): show the district error state, never an all-clear (Part P9 ask 9 fixes it at the source). Then `affected_population`. Prefix *At least* when `affected_population_complete` is false. If `affected_population` is null (population data couldn't be joined), show `systems_over_limit` as the hero instead, with *water systems over a federal limit*. If it is 0 and `systems_over_limit` is 0: *No water system in NC-08 is over a federal limit.*

limit	
Comparison bar pair: district vs state, share of tested systems over a limit	`state_comparison.district_share_over` and `.state_share_over` are **fractions from 0 to 1**: multiply by 100 for the percent (0.214 is 21%). Either can be null: hide the pair. Caption *Share of tested water systems over a federal limit*. Denominator is systems tested for anything regulated (`systems_tested_any`)
County table: one row per NC-08 county, always nine rows: county, worst water severity, radon zone, systems over limit	`counties[]` : `county` , `in_district` (`"whole"` or `"partial"`), `systems` , `systems_over_limit` , `worst_water_severity` , `radon_zone` , `radon_severity` . Partial counties (Cabarrus, Mecklenburg, Robeson) carry a small *Partly in NC-08* tag and a footnote *Only water systems inside the district are counted for counties that are partly in it.* A county with 0 systems shows *No data* for water (the server sends `no_data`)
Per-compound detail: how many systems tested, how many over, how many people they serve	`by_contaminant[X]` for PFOA, PFOS, PFHxS, PFNA, HFPO-DA (label `contaminantLabel()` : GenX (HFPO-DA)): `tested` , `over` , `limit` , `population_over` , `over_population_unknown` . Compose the sentence in the app copy: *[over] of [tested] systems tested for [PFOS] exceed the limit.* (the same wording as the server's `exceedance_lines` , with GenX in place of HFPO-DA). When `over_population_unknown` is more than 0 prefix the people figure with *At least* . Same caveat as the map: "tested" counts systems with a detection on record
Named local specifics: the Concord treatment plant serving 120,000+ customers above the federal limit; Union County's reading against the federal maximum; compliance timelines extending to 2031	Static content in `lib/frontend/copy/district-facts.ts` : an array of `{ text, source, dateVerified }` , empty at first. Each item is added only after a person verifies the figure against its source (§66), and the section is **omitted entirely while the array is empty**. Nothing here is typed from memory. Content owner: Aadit (D5)
Systems not shown on the map	`unlocated_systems` is a **number**, not a list. Show *[n] water systems have no mapped service area and aren't shown on the map.* (hidden at 0). The number counts every located-less system statewide, the same as the water layer's `count` minus `located` . The names and severities come from the water layer's features whose `lat` or `lng` is null (statewide, filtered to unlocated); an expandable list of those, if the water layer is loaded, is **PROPOSED**
Footer: sources and date	*Data assembled September 28, 2026* (`assembled_at`). *District and county lines: [boundaries.district]; [boundaries.counties].* (server strings via `displayText`). Never show the response's `note` or `ranking_rule` text: `note` still claims there is no per-county breakdown, which is no longer true

> **OPEN (VIBHAV AND YOGI): WHICH DISTRICT MAP.** The district lines in the app are the 119th Congress NC-08 (the 2023 map). A newer 120th Congress map exists for the 2026 elections (`lib/data/nc08District.json` holds the one in use). The panel should say which map it uses. **PROPOSED** footer line: *District lines: 2023 map.* Decide before submission whether to switch maps; if switched, the counties list and the "partial county" tags change.

Loading and failure LOCKED

- Loading: neutral canvas with a centered spinner and *Loading map data.* Controls present but disabled (the one place a spinner is right, §46.9).
- **Layer error** (the map request returns 503 because a source is down, or 500, or times out at 20 seconds): base map + home pin + a callout *This layer's data couldn't load.* with *Try again*. Never show the server's message (a 503 carries one, but it names internal sources). The other layers stay usable. A 400 means the app sent a bad request; treat it as a bug: same callout, and log it in development.
- **District error** (`/api/district` 500 or timeout): the map keeps working; the comparison banner is hidden; opening the panel shows the panel chrome with an inline error and *Try again*.
- Map library fails entirely (no WebGL, missing or rejected token, or the import fails): show the district panel's content as a full page, so the tab is never empty, with a one-line notice *The map couldn't load on this device.*
- No data in view: *No data for this layer in the area you're viewing. Try zooming out.* (shown when the layer loaded but no feature is inside the viewport).
- Offline: last loaded datasets from `halo-map` , with the saved-data line. A layer never loaded offline shows the layer error callout without Try again until the app is back online.

Keyboard and screen reader model LOCKED

Focus order: header, layer chips (one tab stop, arrows inside), contaminant selector, count line (read-only, `aria-live="polite"` so a change is announced), locate, zoom in, zoom out, the map itself, time slider, comparison banner, district handle. The map canvas is one tab stop labeled *Map of North Carolina. Use the arrow keys to move and plus and minus to zoom. Open the district panel for a text summary.* Arrow keys pan and + and - zoom (the library's keyboard handler). Individual features on the canvas are not tab stops (hundreds of them would trap keyboard users). The equivalent content is available as text: the district panel (county table, per-compound counts, headline) for the region, HomeGuard for the household's own system, and an opened sheet through `?system=` . Severity is always a word or a shape beside the color (A10). Sheets follow the standard focus rules (A8): focus moves into the sheet on open and returns to the map on close.

Proposed content PROPOSED

Addition	Status
"Systems not shown on the map": a list with names and severities	In the district panel table above (the count is real; the list comes from water features with null coordinates)
In the household's own water-system sheet, a line comparing it to the district: "Your system is one of [n] in NC-08 over a federal limit"	Deferred, not in the first build: it needs to know whether the household's system is inside NC-08, which the current API does not say

Decisions to make

Technical: D7 above (Mapbox GL JS by default; MapLibre with free tiles only if the Mapbox account is a problem, because it changes the token, attribution and style files); the county shapes come from the existing `ncCounties.json` (first-load budget under 500 KB, §35.1: the map library and the 160 KB shapes load only when the Map tab opens). Performance targets: first render under 2s, layer switch under 300ms (§35.1). DESIGN: a muted, low-saturation base map where the data layer is the only saturated thing on screen (§57.1's principle), switching to a dark variant with the same muted treatment in dark mode (§59.3); floating control styling; chip row; legend; home pin; cluster badges; the overlap budget; district panel. The Map wireframe shows every layer; the Water layer with each contaminant option; the slider at two positions with the loading label; each feature sheet; the district panel full, with a null population, and with the error state; the layer error callout; the no-WebGL full-page fallback; offline with saved data.

Part C: Overlays

All three are bottom sheets over the current page: they add one history entry, trap focus, lock the page's scroll, dismiss by drag / backdrop / Escape / close control, and only one is ever open (A1). The opening functions, history rules and the sheet host are Phase 0 (Part P3): `openLearn`, `openAlerts`, `openAssistant` in `lib/frontend/sheets.ts`. Each overlay has one owner who builds the sheet's content; the entry points in other pages call the function and never build sheet content themselves.

C1. Learn

Owner: Naggi (Lane 2, with HomeGuard). Other pages (Today, HomeGuard, Map) call `openLearn`; until Learn merges, the sheet host shows a labeled placeholder in development only (Part P5).

Purpose and the rule that governs it LOCKED

Explain what a number means, at the moment someone is looking at it. Never reachable from the tab bar, no index, no browsable encyclopedia: content that can only be reached from a household's own reading, and references that reading in its second sentence, can't be generic by construction (§17.1).

The opening contract BIND

```
openLearn({
  topic: 'air' | 'uv' | 'pollen' | 'mold' | 'pfas' | 'radon' | 'lead',
  reading?: { severity: Severity; valueText: string },  // the pill under the title
  context?: Record<string, string | number>             // sent to the server as query parameters
}): void
```

Standard height. One history entry: `router.push` with `?learn=<topic>` (Part P3 gives the close rules, including deep links). A second sheet opened from inside a Learn sheet (none exist) would replace it. `valueText` per topic is in Part P3 (for example *AQI 37*, *8.3 ppt · 2.1×*, *Zone 3*, *Built 1975*).

Opened from, and the context to send BIND

The server builds the "why yours" sentence only from these exact parameter names. Send every one that the page has; a missing one is left out (never sent empty or as "undefined"). Always add `locale=en|es` (the profile's language) and `severity` as the canonical word when the reading has one.

Where	Topic	Context params BIND
Today, Air card	`air`	`source` (`airnow` or `open-meteo`, from `air.source`), `value` (`air.aqi`), `severity`, `pollutant` (the dominant pollutant as Today shows it)
Today, UV card	`uv`	`value`, `severity`, `peak_start` and `peak_end` (the same clock strings Today shows for the burn window; omit both when Today shows no window)
Today, Pollen card	`pollen`	`value` and `category` of the dominant pollen type (for example `Grass`), `severity`
Today, Mold card	`mold`	`risk`, `humidity` (percent), `precip` (percent chance of rain), `severity`. There is no numeric `value`. Omit humidity or rain when not shown
HomeGuard Water card or threshold row; Map water sheet	`pfas`	`contaminant` (send the label the person sees, from `contaminantLabel`), `value` (ppt), `limit` (ppt), `severity`. The tapped row, or the worst row when opened from the card
HomeGuard Radon card; Map county sheet	`radon`	`county`, `zone`
HomeGuard Lead card	`lead`	`home_year`. Omit it when unknown: the server then writes its "we don't know when this home was built" text

Call: `GET /api/learn?topic=<topic>&locale=<locale>&…context`. Deadline 8 seconds (Part P3). Topics are exactly those seven; anything else is a 400.

> **CACHING:** cache only the parts that don't depend on the reading or the household: `title`, `what_it_is`, `protect`, `sources`, `locale`, keyed by topic + locale, so the sheet can open offline (§36). Always request fresh `why_yours` and `household_note` while online (they depend on the reading passed and on the household's groups). Offline, show the cached parts and omit sections 2 and 3 without a notice. Clear this cache with `clearReadings()` and on `halo:settings-changed`.

Anatomy, top to bottom LOCKED BIND

Element	Content	Binding
Title	Topic name. While loading, the static topic name from `copy.learn`; then the server's title	`title`

Your reading	The household's own reading as a severity pill with its value, right under the title: what makes this contextual rather than generic. **Omitted** when the sheet was opened by a deep link the page couldn't rebuild a reading for (P3)	Passed in from the page that opened it, not from this endpoint
1. What it is	Two sentences, plain language	`what_it_is`
2. Why yours reads this way	References the household's actual value and situation. **Section omitted (heading too) when `why_yours` is null**. The server returns null when the context wasn't passed, except for Lead (always a sentence, using the "we don't know when this home was built" wording when `home_year` is missing) and Air (a sentence from `source` alone). Never filler	`why_yours`
3. What it means for your household	Branches on household composition	`household_note` : section omitted when null
4. How to protect yourself	A short checklist, most effective first; every item a concrete, checkable step; certifications named inline so they can be read standing in a store aisle: set text that matches `/NSF\/ANSI \d+|P473/` in the mono font (IBM Plex Mono, loaded in Phase 0). The section that makes Learn worth opening twice	`protect[]`
5. Sources	Every claim's origin, each with the date retrieved, each a working link, visually distinct. Format: *EPA, PFAS drinking water regulation* then *Retrieved September 27, 2026* on the next line. Links open in a new tab (`rel="noopener noreferrer"`) and have at least a 44 px target	`sources[]` → `label` , `url` , `retrieved` (an ISO date)

Every string from the response passes through `displayText()` : the content and the source labels still contain em dashes (Part P9 asks Vibhav to remove them at the source), and `displayText` turns them into commas. Render as plain text (list items as list items), never as HTML.

> **LANGUAGE:** the response's `locale` is the language actually served. If Spanish was requested but a topic hasn't been translated and reviewed yet, it comes back in English with `locale: "en"` : show it, don't pretend otherwise. (Until Spanish is reviewed the Language setting is hidden and only `en` is ever requested, B6.)

> **THE LEARN / ACT BOUNDARY:** Learn is personal protection (sunscreen, filters, flushing taps, closing windows, humidity). Act is collective action (volunteering, advocacy, sharing). Keeping that line clean keeps both coherent (§17.2).

States LOCKED

- **Loading:** skeleton (title bar, a pill shape, five paragraph blocks).
- **Error** (400, 404, 500, timeout, network): generic error *Couldn't load this explanation.* with *Try again*. Never the server's message (the 400 lists topic names). Reading pill and title still show.
- **Offline with a cached topic:** shown per the caching rule. Offline with no cache: the error state without Try again.

DESIGN: typographic treatment: strong headings, key facts given weight, sources as visible footnotes rather than hidden links (§6.7). The wireframe shows each of the seven topics with a reading, PFAS with a household note, a topic with `why_yours` null (section 2 absent), a deep link with no reading pill, loading and error.

C2. Assistant

Owner: Vibhav (Lane 3, after Map; the last build and first cut, §26.3, §68). The shell renders the entry button and hides it when `NEXT_PUBLIC_HALO_ASSISTANT=off` (Part P2), so it can be switched off without leaving a dead button.

Purpose LOCKED

Answer questions about the household's own readings and about environmental health generally, grounded entirely in cited sources, without ever practicing medicine (§18.1). It draws on curated source documents and, only when the server's household-context switch is on, the household's current readings and symptom entries; never the web, never general knowledge (§18.2).

> **HOW IT KNOWS THE HOUSEHOLD:** the server assembles any household context itself, from the verified session, and weighs it by the page the question came from. The frontend sends only `{ question, page }`; it never sends household data in the request. **Whether household data is used at all is a server switch** (`ASSISTANT_SEND_HOUSEHOLD_CONTEXT`), decided by Yogi (D4). The team default is **off**: free models may log prompts, so only the question and public agency sources are sent, and responses say `uses_household_data: false`. When on, the context is today's readings, the home assessment, and journal symptom names and dates, **never the free-text notes**. The Settings privacy statement changes with the switch (B6). When an answer uses household context, the response says so (`uses_household_data: true`).

Entry LOCKED

The floating button on all five tabbed pages. Opens as the tall sheet (about 90%; the blueprint says two-thirds, a Tier 3 difference, tall wins). Carries the current page as context. The conversation isn't kept in the URL.

Calls BIND

Call	Sends	Returns
`GET /api/assistant?page=…` (on open, once per app session per page)	`page`: Today → `today`, HomeGuard → `homeguard`, Journal → `journal`, Map → `map`, Act → `act`	`suggestions` (exactly 3), `disclaimer`. Pages with their own set: today, homeguard, journal, map. Any other page (Act) gets the default three (*What is PFAS?*, *How is my score calculated?*, *What does NSF/ANSI 53 mean?*)
`POST /api/assistant`	`{ question, page }` (question trimmed, 1 to 1,000 characters; `page` as for GET; the server accepts today, home, homeguard, map, journal, act, settings, learn and ignores anything else)	One of the branches below
`GET /api/assistant` field `sends_household_context`	Not built yet (Part P9 ask 2)	Read by Settings only. The assistant sheet does not depend on it

The chips and disclaimer are server text: pass them through `displayText`. Because chips are server-supplied, every chip's answer must be tested against the source corpus before a demo (each should return a grounded answer, not a decline). Known problem: the Map chip *What does 'not tested' mean on the map?* disagrees with the map's own wording, *No detection reported* (Part P9 ask 7).

Anatomy LOCKED

Element	Content / behavior
Title row	*Ask HALO* + close control
Suggested questions	On first open with an empty conversation only: three tappable chips chosen by the page it came from (§18.6). Tap sends it. Hidden after the first question
Conversation	The person's messages on one side, answers on the other. Answers never truncate. Rendered as **plain text**, never as HTML or Markdown; blank lines make paragraphs. In memory for the app session (survives closing the sheet, not a reload), never sent to the server as history (each question stands alone), cleared on Delete everything and on an identity change
Thinking state	An in-progress indicator in the answer position until the response arrives. Answers arrive complete, not streamed; the indicator carries the wait
Markers in the answer text	The model writes `[1]` style markers and `[H]` (household readings). Strip every `[H]`. Show each `[n]` as a small superscript number matching the citation with that `n` (use `n` as sent; don't renumber); a marker with no matching citation is removed
Citations	Under each answer: each cited source as a tappable name that opens it externally, with its retrieval date. The server's rule: a grounded answer cites at least one source **or** the household's own readings (`[H]`). So `citations` can be an empty array on an answer that used household data; then show no sources list, only the marker below. An answer with neither is never sent
"Uses your readings"	A small icon and the label *Uses your readings* above answers that used household data (§50.2):

marker	`uses_household_data === true`. Never shown when the switch is off
Declined answer	Shown in a caution callout, visually distinct from a normal answer
Disclaimer	Fixed above the input, always: *Not medical advice. Talk to a clinician about health decisions.* A small tertiary link under it, **PROPOSED** *How your data is used*, opens `/settings?open=privacy`
Input	Expanding text field + send button. The blueprint conflicts: §50.2 says send is disabled while empty; §34.1 says submit is never disabled for validation. Follow §34.1: send is always enabled; an empty or whitespace-only question (trimmed client-side) shows *Ask a question to get started.*; over 1,000 characters shows *That question is too long.* (the server's own two messages, so the client never causes a 400). Send is disabled only while a question is in flight (one at a time, no queue)

Response branches, how to tell them apart BIND

Check in this order:

Condition	Meaning	Show
HTTP 429	Rate limited: 20 questions per hour per household, or the app-wide daily cap of 450 (default; the cap protects the free model's quota and fails closed, meaning it also pauses the assistant if the rate-limit service is down). The body has `rateLimited: true` and a `message`	`message`: *You've reached the question limit for now. Please try again in a little while.* Same text for both causes. No Retry button
`declined === true`	Diagnostic refusal: a check that runs before any answer is composed. Has `message` and `citations: []`	`message` (through `displayText`, since it contains an em dash) in the caution callout
`configured === false`	The assistant isn't configured on this server (no model key, e.g. a development environment). The server's `message` reads as "no source for that", which would mislead	**PROPOSED** *The assistant isn't available right now.* Not the server's `message`, no Retry
`typeof answer === "string"`	Grounded answer (`grounded: true`, `reason: null`)	`answer` text (markers handled as above) + `citations[]` → `{ n, label, url, retrieved }` + `uses_household_data`. The `[n]` markers match each citation's `n`; they aren't guaranteed to be in order or consecutive
`answer === null` and `reason: "no_source"`	No source covers the question	`message`: *I don't have a reliable source for that. I can only answer from the health and environmental agencies I've been given.* (§31.3). No Retry
`answer === null` and `reason: "unavailable"`	The model or the source lookup failed. The only case with Retry	`message`: *I couldn't reach my sources just now. Please try again in a moment.* + *Try again*
Timeout, network error, HTTP 5xx (including a platform 504), or a body that isn't JSON	Infrastructure failure	Treated like `reason: "unavailable"`: the same message with Try again. The server has no explicit limit on this route yet (Part P9 ask 3 sets one at 30 seconds)

Retry re-sends the same question; it does not add a second copy of the person's message. HTTP 400 should never happen because of the client checks above; if it does, show the generic unavailable message.

Required behavior LOCKED

- Household data and general guidance always distinguished in wording: "your reading today is 68" and "federal guidance suggests..." are separate statements (§18.4). With the household switch off, answers use public sources only, and that is the correct behavior, not a bug.
- Diagnosis questions declined before any answer exists, reliably: a demo audience may try (§18.4). The refusal is code, not an instruction. It also declines a question that mixes an environmental word with anything about a person's body or care ("Should I take medication since my radon is high?").
- No source: says so rather than improvising.
- Answers can take up to about 20 seconds: the thinking state holds that long, and the client timeout for `POST /api/assistant` is 25s, not the 8s default. A client timeout is treated as unavailable.
- **Keyboard and focus.** Focus moves to the input on open. Enter sends; Shift+Enter makes a new line; Escape closes. The conversation list is `aria-live="polite"` so a new answer is announced once, and the list scrolls to the start of the new answer (not the end of a long one). Chips and citation links are 44 px targets.
- **Offline:** the input and chips are disabled with **PROPOSED** *You're offline. Reconnect to ask a question.*

> **WHAT THE REFUSAL COVERS:** questions asking for a diagnosis, whether a symptom means an illness, or what treatment or medication to take. Questions about how serious a *reading* is ("Is an AQI of 160 serious?") or how to fix an environmental problem ("What are the treatment options for radon?") are environmental questions and get answered.

> **BEFORE LAUNCH:** federal guidance on wellness and decision-support software was revised in January 2026; how the assistant is described publicly should be checked against the current text (§18.5, §66). Test it against ten deliberately adversarial questions before any demo (§66, Vibhav's checklist item).

DESIGN: message bubbles, citation chips, the thinking indicator, the declined-answer callout, the input row. The wireframe shows: empty with chips; a grounded answer with citations; a household-data answer with the marker and no source list; declined; no source; unavailable with Try again; rate limited; not configured; offline.

C3. Alerts

Owner: Naggi (Lane 2, after HomeGuard and Learn; with Act). Alerts also exports `useAlerts()` and `<BellBadge/>` (Part P3), which the shell header renders; until Alerts merges the bell slot renders nothing.

Purpose LOCKED

Tell a household when something meaningfully changes, without them opening the app to find out (§19.1). Opened only from the bell on Today's header.

Calls BIND

Call	Use
`GET /api/alerts`	`count`, `unread` (badge), `alerts[]` newest first (max 100, dismissed ones excluded). Each alert: `id`, `type`, `title`, `message`, `severity` (air only), `read`, `fired_at`
`POST /api/alerts { id }`	Mark one read (it's POST, not PATCH)
`POST /api/alerts { all: true }`	Mark all read
`POST /api/alerts { id, dismissed: true | false }`	Dismiss, and undo. Success returns `{ ok: true, updated: n }`; `updated: 0` on an id or dismiss call means the alert no longer exists, so refetch (Mark all read returns 0 when nothing was unread). A 501 means migration 0010 isn't applied: treat as a failure

The badge BIND DECIDED

`<BellBadge/>` shows the `unread` count as a small number on the bell (*9+* above nine), hidden at zero, with the accessible name *Alerts, 3 unread* (or *Alerts* at zero). The count comes down by **reading rows** (or Mark all read), not by opening the sheet, so a person who opens it and leaves still sees that something is unread. `useAlerts()` fetches once when the shell mounts, again when the sheet closes, and when the app returns to the foreground after more than five minutes. A failed fetch keeps the last known count and shows no error on the bell.

The inbox LOCKED

Element	Content / behavior	Binding
Title row	*Alerts* + tertiary *Mark all read*, **hidden when nothing is unread**	`{ all: true }`
Row	Reverse chronological. Unread: small colored marker + bolder title. Title, one-line message (cut with an ellipsis), relative time (*Just now*, *2 hours ago*, *Yesterday*, then *September 26*; one shared `relativeTime()` in `format.ts`)	`title`, `message`, `fired_at`, `read`; marker color from `severity` when set (air alerts only), neutral otherwise
Tap	Expands to the full message (season summaries can be several sentences) and marks it read	`POST { id }`. Optimistic; on failure the marker returns and the toast *Couldn't update that alert. Try again.* shows
Go to the related page	The blueprint conflicts: §19.4 says tapping an expanded row again navigates; §50.3 puts a tertiary button inside the expanded row instead. Use the button (§50.3): a second tap on the same row is easy to trigger by accident. Closing the sheet first (by replace), then pushing the route	By `type` (labels PROPOSED): `air_quality_change` → Today, *See today's air* · `weather_advisory` → Today, *See today* · `new_water_results` → `/home?risk=water`, *See water results* · `radon_season` → `/home?risk=radon`, *See radon* · `season_summary` → `/journal?mode=trends`, *See your summary* · unknown type → no button
Swipe to dismiss	Dismiss, immediate, with an undo toast (8 seconds, *Alert dismissed* + *Undo*). **A visible tertiary *Dismiss* button in the expanded row does the same** (swipe is never the only way: `SwipeRow`, Part P2). On failure the row returns with the toast *Couldn't dismiss that alert. Try again.*	`POST { id, dismissed: true }`; undo → `POST { id, dismissed: false }`, then refetch
Empty	*No alerts yet. We'll let you know when something changes.*	`alerts` empty
Loading / error	Three skeleton rows; an inline error *Couldn't load your alerts.* with *Try again* (no toast)	Request state

History is kept rather than cleared, so the alert system's behavior can be seen without waiting for a live trigger, which is useful for a household back from a week away and for anyone evaluating the app (§19.4). Dismissed alerts don't return.

What generates alerts LOCKED (verified against the server rules)

type	Fires when	Title and message as the server sends them	Opens
`air_quality_change`	Today's air is worse than yesterday's (or yesterday has no reading) **and** today is at least Elevated for a household with a sensitive group (a respiratory condition, a toddler, or a senior), or at least High for everyone else (§19.3). At most one per day	*Air quality dropped* / *Air quality dropped to Elevated today.* (the label is one of the six severity words; the blueprint's "Unhealthy for Sensitive Groups" is not what is sent)	Today
`weather_advisory`	The National Weather Service has an alert with **"flood" or "heat" in its name** for the home's point (watches and warnings too; the message still calls it an advisory). **Boil-water advisories are never sent**: no national source exists (the rule text for them is in the code but nothing feeds it). One alert per advisory, not one per day	*Weather advisory* / *Flood advisory for Union County. Flooding can affect well water, retest after it recedes.* or *Heat advisory for Union County. Limit strenuous outdoor activity, especially for older adults and young children.*	Today
`new_water_results`	New federal testing results were published for the household's utility	*New water results* / *New water testing results were published for your utility.*	HomeGuard, water card
`radon_season`	October through February, household in radon Zone 1 or 2, and no radon test on record (the server has no record of radon tests yet, so it always counts as none). Once per winter. (Never sent in Zone 3, which is every NC-08 county)	*Time to test for radon* / *Winter is the most accurate time to test for radon.*	HomeGuard, radon card
`season_summary`	A season ends and the household has journal entries. The `message` is the **full summary statement** (a sentence or more), not a short teaser	*Your spring summary is ready* / the summary statement	Journal, Trends mode (`/journal?mode=trends`). Journal shows the seasonal-summary callout in both Log and Trends (B4, D6)

Types the household switched off in Settings are never generated, for both this inbox and device notifications. Show any unrecognized type with its title and message and no button. Every title and message passes through `displayText`, which turns the server's remaining em dash (in the flood text) into a comma. Words like "Elevated" in a message are the same six severity words used everywhere.

Device notifications LOCKED

Built after the core pages (D8, and specified in B6, "Device notifications"). Use the browser's notification capability where permission was granted; otherwise alerts simply appear in the inbox on next open. Ask permission only after the household has used the app at least once, never on first load (§19.5): in practice from the device-notifications switch in Settings. A notification tap opens the related page from the `url` in its payload (same destinations as the table above). Needs a service worker (which can also serve the offline behavior, §36).

DESIGN: row anatomy, unread marker, the bell badge, the expanded row with both buttons. The wireframe shows: empty; several rows mixed read and unread; one expanded per type (five); a long season summary expanded; the dismissed toast; the failure toast; loading; error.

Part D: API Reference (the finished backend)

Every endpoint and field this document binds to, in one place. This is the contract between the frontend and the finished backend: the backend was built to exactly these names. Field names are verbatim, bind to them as written.

D0. Conventions that apply to every endpoint

Rule	Detail
Identity	Every route except the daily job requires `Authorization: Bearer <session token>`. No route accepts identity from a body or query parameter.
401 vs 503	`401` = the session is definitively invalid (the account no longer exists) → the client discards the session and starts a new one, silently. `503` = identity couldn't be verified right now → the client retries and **never** resets the session. This distinction is what protects a household's data during an auth-service outage.
Errors	`{ "error": string }`. Never shown to people verbatim; mapped to the copy in A2. This is a hard rule, not a preference: many routes still return raw database text in this field on a 500 (Part P9 ask 6), so the frontend never displays a server error string. The only server text ever shown is the Assistant's fixed `message` sentences (C2), and those pass through `displayText()` too.
Absent fields	Any field can legitimately be `null` or missing, and the interface handles that explicitly, the words No data, an explicit state, a hidden element, never a crash and never a zero (§62.4). This is also what lets a page be built before every field exists.
Severity	A closed set: `good`, `moderate`, `elevated`, `high`, `severe`, `no_data`. Anything else renders as No data. Always read the field named `severity` (or `level` for lead). Some objects also carry an older `status` field kept for compatibility; never bind it. Any server string shown to people passes through `displayText()` first (it still contains em dashes in places).
Dates	Daily readings and journal entries: `YYYY-MM-DD` in the household's local date. Timestamps: ISO 8601. Water sample dates: bind the `*_iso` fields (`YYYY-MM-DD`); the older `M/D/YYYY` strings remain for compatibility only.
Freshness	Every cached or pre-built response states when its data was retrieved or assembled (`retrieved_at`, `assembled_at`).
Change discipline	Additive only once frontend work begins: fields may be added, never renamed, removed, or re-typed (§62.4).

D1. Household and account

Endpoint	Request	Response
`GET /api/profile`	None	`{ onboarded, onboarding_complete, profile: { county, state, zip, pwsid, lat, lng, home_year, water_source, renter_mode, locale, onboard_request_id } | null, household: { has_toddler, has_child, has_teen, has_adult, has_senior, has_pregnant, has_respiratory }, household_set }`. `onboarding_complete` is true once coordinates *and* water source are stored, route on this, not `onboarded`. `profile` is null when no row exists yet. `zip` is null once a household has onboarded again (older rows may still hold one; never bind it). `onboard_request_id` is the id of the address request that stored this location (only onboarding uses it, to confirm a timed-out save; never displayed). `lat` / `lng` are rounded to about 100 m for locations saved after migration 0009. A database failure returns 500 with a fixed message, never an empty profile.
`PATCH /api/profile`	`{ water_source?, home_year? }`	Same shape as GET. Only the keys sent change, and at least one is required (400 otherwise). `home_year`: integer 1700–current year or `null` (null clears it); an invalid year is a 400 whose message the app never shows (validate first, A7). `water_source`: `utility` · `well` · `spring` · `other` (the onboarding labels are also accepted and normalized)
`POST /api/onboard`	`{ address, request_id?, water_source?, home_year? }`. `address` may be `"lng,lat"` from device location. `request_id` is a UUID the client makes for each address submit and reuses on a retry of the same submit (if a POST times	`{ profile_id, lat, lng, county, state, pwsid, service_area_status, water_source, home_year, onboard_request_id }`. `state` is the two-letter code, e.g. `"NC"`. `lat` / `lng` are rounded to about 100 m before storage. The response also echoes `zip`, which is no longer stored, don't bind it. `service_area_status`: `measured` (a utility boundary contains the point) · `outside_known_area` (none does) · `lookup_failed` (the boundary lookup failed; a previously stored utility is kept). When the (rounded) location changes, the server also

	out, the client re-checks `GET /api/profile` for a matching `onboard_request_id`, B1). Water source and build year are normally saved with `PATCH /api/profile`; onboard also accepts them	clears today's stored reading and resets the daily-score limiter, so the results reveal's fresh reading works for the new home. Errors: 400 no address (or an invalid `home_year`) · 404 not found · 429 (a per-device daily limit, or the app's daily geocoding budget; both say to try tomorrow) · 500. Onboarding maps these to its own copy (B1)
`GET /api/household`	None	`{ household: {7 booleans}, household_set, renter_mode, locale }`
`PUT /api/household`	`{ has_toddler … has_respiratory, renter_mode?, locale? }`	Full replace of the seven booleans, **always send all seven**. Echoes `renter_mode` / `locale` only when sent
`DELETE /api/account`	`{ confirm: "DELETE" }`	`{ deleted: true }`; 400 if the word is missing or not exactly `DELETE`. Deletes the account and everything tied to it. The client then unsubscribes push, clears its own storage and signs out (B6)

D2. Today

`GET /api/daily-score`, no parameters: uses the household's stored location. `?fresh=1` bypasses the one-hour cache (pull to refresh). Any request that has to call the providers (a fresh request, or no reading for today from the last hour) is limited to one per household per five minutes, and gets a 429 otherwise, even without `fresh`. Response:

```
{
  air:    { aqi, severity, sentence, source: "airnow"|"open-meteo"|null, is_measured: true|false|null,
            dominant_pollutant },                                  // "PM2.5" | "PM10" | "O3" | null
  uv:     { index, severity, sentence, peak_window: { start, end, max } | null },   // local "HH:MM"
  pollen: { tree, grass, weed, dominant: "tree"|"grass"|"weed"|null, severity, sentence,
            categories: { tree: { value, severity }, grass: { value, severity }, weed: { value, severity } } },
  mold:   { risk: "low"|"moderate"|"high"|null, is_proxy: true, severity, sentence,
            basis: { humidity_pct, precip_pct } | null },          // no_data + null risk when the forecast failed
  score:  { display_score, score, risk, severity, is_partial, missing_inputs[], included_inputs[],
            includes_proxy, inputs: { air, uv, pollen, mold }, reason },
  cached, retrieved_at
}
```

Errors specific to this route: `400 { error: "No location yet. Finish onboarding, or pass lat and lng." }` when the household has no stored location (route to onboarding) · `429 { error: "Readings were just refreshed. Try again in a few minutes.", retry_after_seconds: 300 }` when a provider call is repeated within 5 minutes (keep the readings on screen; A2). `retry_after_seconds` is always 300, not the time remaining. Onboarding a changed location clears today's stored reading and resets this limit. `lat` / `lng` are still accepted as optional overrides; the frontend doesn't send them.

`GET /api/history?days=N` (or `from` / `to`; `days` is clamped to 1 to 365, default 90; 0 or a non-number gives 90) → `{ from, to, count, history: [ { date, score, air, uv, pollen, mold, values: { aqi, uv_index, pollen: { tree, grass, weed }, mold_risk } } ] }`. One record per local date. Recorded every day for every household by the daily job (not only on days the app is opened), with recent weeks backfilled at onboarding where a historical source exists; a day with no data is simply absent.

D3. HomeGuard

`GET /api/home-guard`, no parameters: uses the stored county, utility, water source, and build year. Optional `county`, `pwsid`, `water_source`, `home_year` override them for this request only. Computed on every request: no server cache, no `fresh` parameter. The response also carries `success: true` and `assembled_at` (when it was put together; not a measurement time). A `state` override parameter works like the others (radon is looked up only when the state is `NC`). Error: `400 { error: "No county on file…" }` when no county is stored (A2). Water results exist for North Carolina systems only, an out-of-state utility comes back `no_data_yet` (B3 handles the wording).

Field	Shape
`radon`	`{ county, zone: 1|2|3, risk_level, severity }`, or `{ error }` when the county isn't matched, or `{ error, out_of_state: true, state }` for addresses outside North Carolina (radon zones are looked up only when the stored `state` is `NC`)
`water`	Discriminate on `water.status`: `private_well` · `no_data_yet` · `lookup_failed` · `no_pwsid_available` · `detected_unregulated` · anything else = measured. Measured fields: `pws_name`, `population_served`, `contaminants` (every reading: `{ name: [ { date, date_iso, value_ppt } ] }`), `detected_unregulated_latest[]` (`{ contaminant, value_ppt, date_iso }`), `coverage`, `latest_sample_iso`, `earliest_sample_iso`, `data_age_days`, `reading_count`, `scored_count`, `detected_count`, `detected_unregulated[]`, `excluded_from_score[]`, `scored_contaminants[]`, `includes_guidance`, `guidance_scored[]`, `regulatory_notice` (only when it applies), `message` (only when relevant). Private well: `source_type`, `message`, `test_plan`. Never display these fields raw: `message` and `regulatory_notice` are

	shown only where B3 says so; internal keys (`risk` , `ceiling` , `basis` codes) never appear in the interface
`scored_contaminants[]`	`{ contaminant, value_ppt, limit_ppt, mcl_ppt, basis, source, is_enforceable, proposed_for_rescission, exceeds_limit, date_iso, risk }`
`excluded_from_score[]`	`{ contaminant, value_ppt, date_iso, reason, health_reference_level_ppt, health_reference_basis, health_reference_source, needs_local_context }`
`test_plan`	`{ tests: [ { rank, id, name, priority: critical|recommended|optional, cadence, why, where, estimated_cost: { low, high, is_estimate }, needs_local_context? } ], reasons[], start_here_cost: { low, high, is_estimate }, full_panel_cost: { low, high, is_estimate } }`
`score`	`{ display_score, score, risk, severity, is_partial, floored, capped, ceiling, missing_inputs[], included_inputs[], water_risk, radon_risk, reason }` (`capped` , `ceiling` and `floored` are internal: never display them or the word "capped" in `reason` ; Part P9 ask 8). `display_score` and `severity` are both `null` whenever there's no score, a private well or spring, or nothing available (with `reason`). Same rule for Today's `score`
`lead`	`{ level, basis, is_estimate, shown_not_scored: true, contributes_to_score: false, text, sentence, utility_inventory }` ; `null` for well/spring. `level` : `high` (built before 1950) · `elevated` (1950–1987) · `good` (1988 or later) · `no_data` (year unknown); `severe` is knowingly unreachable (no address-level inventory exists). `utility_inventory` : `{ utility, reported, lead, galvanized_requiring_replacement, non_lead, unknown, needs_replacement, address_level: false, source: { name, url, retrieved } }` , the utility's published service-line counts (NC DEQ, July 2025), context only, or `null` when the utility isn't in the published results
`action_plan[]`	`{ rank, key: water|lead|radon, severity, title, reason, cost, certification, renter_can_self_serve, action }` , only items at Elevated or above; text already written for renter or owner
`breakdown[]`	In order: water, radon, lead (public water only), one per excluded substance. Each: `{ key, label, severity, contributes_to_score, risk, score, confidence: full|limited|stale|none, is_measured, is_estimate, detail, note }` (`risk` and `score` are numbers for the server's use; the app shows the severity word and the page's own sentences). Excluded-substance boxes add `health_reference_level_ug_l` , `detail[].value_ug_l`

D4. Journal

Endpoint	Request	Response
`GET /api/journal`	`from` , `to` (default last 90 days)	`{ from, to, count, entries: [ { id, entry_date, band, symptoms[], severity, note, retrospective, possibly_illness, created_at, updated_at } ] }`
`POST /api/journal`	`{ entry_date, band, symptoms[], severity, note, retrospective, possibly_illness }` , every field, every time (a save rewrites the whole entry). `band` : one of the seven group keys or `household` . Validation: `entry_date` must be `YYYY-MM-DD` ; `severity` is `mild` , `moderate` , `bad` or null; `symptoms` keeps at most 20 (each as text); `note` at most 500 characters. `entry_date` is checked for shape only: an impossible date like 2026-02-31 fails in the database as a 500, so the date picker must only offer real dates	`{ entry }` . One entry per date and group; saving again updates
`DELETE /api/journal`	`?id=<uuid>` one entry · `?all=true` every entry (400 for anything else, including a non-UUID id)	`{ deleted: n }`
`POST /api/journal/retrospective`	`{ dates: ["YYYY-MM-DD", …] }`	`{ ready, factor, statement, flagged_days_with_readings, flagged_avg?, month_avg?, unflagged_avg? }` . 400 when `dates` isn't an array or has more than 62 entries; future dates are silently dropped. Compares the flagged days' readings with the rest of that month, e.g. "On the 3 days you flagged, grass pollen averaged High. The month's average was Low." `ready: false` (with a `statement`) when too few of the dates have readings
`GET /api/journal/findings`	None	Not ready: `{ ready: false, flaggedDays, totalDays, needed: { flagged, total } }` . Ready: `{ ready: true, findings: [ { factor, statement, elevated_rate,`

		`other_rate, counts: { elevated_days, other_days } } ], disclaimer, flaggedDays, totalDays }` (note the camelCase)
`GET /api/journal/summary`	`season`, `year` (default: the season that most recently ended)	`{ season, year, logged_days, statement, factor?, from?, to? }`. 400 for an unknown season or a year outside 2000 to 2100. E.g. "This spring you logged 22 days. On the 9 highest-pollen days, you flagged symptoms 7 times."

D5. Learn and Assistant

Endpoint	Request	Response
`GET /api/learn`	`topic` (pfas · radon · lead · air · pollen · uv · mold), `locale`, and context (every one optional): pfas → `contaminant, value, limit, severity`; radon → `county, zone`; lead → `home_year`; air → `source, value, severity, pollutant`; uv → `value, severity, peak_start, peak_end`; pollen → `value, severity, category`; mold → `risk, humidity, precip, severity`. Errors: 400 unknown topic · 404 no content · 500	`{ topic, locale, title, what_it_is, why_yours, household_note, protect[], sources: [ { label, url, retrieved } ] }`. `locale` in the response is the language actually served. `why_yours` is null when no context was sent (except Lead and Air, which always have a sentence); `household_note` is null when the topic has no note for this household
`GET /api/assistant`	`page`: today · homeguard · journal · map · act. Anything else (or none) gets the default three questions	`{ suggestions: [3 strings], disclaimer }`
`POST /api/assistant`	`{ question, page }` (`page` is one of today, home, homeguard, map, journal, act, settings, learn; anything else is ignored)	Branches, see C2: `declined`; `configured: false`; grounded `{ answer, grounded: true, citations: [ { n, label, url, retrieved } ], uses_household_data, disclaimer }`; no answer `{ answer: null, reason: "no_source" | "unavailable", message, disclaimer }`; 429 `{ answer: null, configured: true, rateLimited: true, message }`. `citations` can be empty when `uses_household_data` is true. 400 `{ error }` for an empty question or one over 1,000 characters. Limits: 20 per household per hour, 450 per day app-wide

D6. Alerts and notifications

Endpoint	Request	Response
`GET /api/alerts`	None	`{ count, unread, alerts: [ { id, type, severity, title, message, fired_at, read } ] }`, newest first, dismissed alerts excluded. `type`: `air_quality_change` · `weather_advisory` · `new_water_results` · `radon_season` · `season_summary`
`POST /api/alerts`	`{ id }` mark read · `{ all: true }` mark all read · `{ id, dismissed: true }` dismiss · `{ id, dismissed: false }` undo	`{ ok: true, updated }` (`updated: 0` on an id or dismiss call means no such alert; Mark all read returns 0 when nothing was unread). 400 for an invalid id or an unrecognized body; 501 when migration 0010 is missing (dismiss only)
`GET /api/notifications` · `PUT`	PUT: `{ preferences: { air_quality_change, weather_advisory, new_water_results, radon_season, season_summary } }`	`{ preferences: {…five booleans} }`, all default true. The daily job skips types a household turned off. 400 for an unknown type or a value that isn't true or false
`POST /api/push/subscribe` · `DELETE`	POST: the browser's push subscription object (public key: `NEXT_PUBLIC_VAPID_PUBLIC_KEY`). DELETE: `{ endpoint }`	`{ ok: true }`. 400 for an endpoint that isn't https or missing keys; 429 when changed too often. New alerts are also pushed to subscribed browsers with the payload `{ id, type, title, message, url }` (`url` = the page to open)

D7. Act, Map, District, Sources

Endpoint	Request	Response

`GET /api/volunteer`	`causes` (comma list: water, pfas, radon, air, wells, advocacy), `county` (defaults to the household's)	`{ county, causes, count, orgs: [ { id, name, description, counties[], causes[], url, last_verified, link_checked, match_count } ] }`. `last_verified` is null on all 13 organizations today (D9)
`GET /api/map?layer=water`	`quarter` (optional, from `quarters`)	`{ layer, count, located, quarters[], counts: { total, over_limit, hazard_index_over, by_contaminant: { X: { tested, over } } }, features: [{ pwsid, name, lat, lng, population, overall_severity, coverage, latest_sample_iso, detected_unregulated[], hazard_index: { value, severity, proposed_for_rescission } | null, contaminants: { PFOA, PFOS, PFHxS, PFNA, "HFPO-DA": { value_ppt, limit_ppt, ratio, severity, date_iso, proposed_for_rescission } | { severity: "no_data", limit_ppt, proposed_for_rescission } } }], assembled_at }`. Systems with only unscoreable detections report `overall_severity: "no_data"` with `coverage: "unscoreable"`. `counts` describe exactly the features returned (statewide). `quarters` uses the form `"2024Q3"`, oldest first, and is empty when the data has no time series. Also on the root: `quarter` (null unless requested), `population_source`, `note`, and `prebuilt` (not present on a `?quarter=` response). Each feature also carries legacy `status` and `latest_sample_date` (never bind); `hazard_index` also has `exceeds`, `components`, `basis`. A `?quarter=` request returns only the systems that reported in that quarter, and its `assembled_at` is the request time. Errors: 400 (bad layer or quarter), 503 (a layer source is down), 500
`GET /api/map?layer=radon`	None	`{ layer, count, features: [ { county, zone, severity } ], assembled_at }` (county shapes are a static asset bundled with the frontend)
`GET /api/map?layer=facilities`	None	`{ layer, count, features: [ { id, name, lat, lng, programs[], compliance_status, severity, open_violations, recent_violations, penalties, last_inspection } ], assembled_at }`. `compliance_status`: `no_violation` · `violation` · `significant_violation` · `unknown`. `recent_violations` = quarters in noncompliance out of the last 12 (3 years), not a count of violations; `open_violations` = programs currently reporting a violation; `penalties` = dollars over the last 5 years. Each feature also has `county`. The root has `scope` ("NC-08") and `selection` (a server sentence; don't display, B7). Only active major facilities and repeat violators inside NC-08
`GET /api/map?layer=air`	None	`{ layer, count, features: [ { id, name, lat, lng, aqi, severity, dominant_pollutant, last_reading_at, agency } ], assembled_at }` (`dominant_pollutant` as in D2; the root also has `source`)
`GET /api/district`	None	`{ scope: "NC-08", total_systems, systems_over_limit, affected_population, affected_population_complete, systems_tested_any, over_limit_population_unknown, by_contaminant: { X: { tested, over, limit, population_over, over_population_unknown } }, exceedance_lines: { X: string }, counties: [{ county, in_district: "whole"|"partial", systems, systems_over_limit, worst_water_severity, radon_zone, radon_severity }] (always the nine NC-08 counties), state_comparison: { district_share_over, state_share_over } (fractions 0 to 1, either can be null), household_county: { county, water_rank, of, share_over } (an unranked county has water_rank and share_over null; share_over is a 0 to 1 fraction of people served), ranked_counties, nc_counties, ranking_rule, unlocated_systems (a number), boundaries: { district, counties, retrieved }, prebuilt, note, assembled_at }`. `affected_population` is a number in practice, but if the geography lookup failed it is 0 alongside `total_systems: 0` (the frontend treats that as no data, B7). Never display `note` or `ranking_rule` (B7). Exceedance lines count only systems tested for that compound. Most fields are NC-08-wide and pre-built daily (exceptions: `state_comparison.state_share_over`, `ranked_counties`, `nc_counties` and `unlocated_systems` are statewide); `household_county` is filled in per request from the caller's profile
`GET /api/sources`	None	`{ sources: [ { name, provides, last_retrieved } ] }` (currently 10). `last_retrieved` is null when unknown (shown as "Retrieval date not recorded"). Feeds Settings, About and the health card's sources footer
`GET /api/health`	None	`{ all_ok, checked_at, checks: [ { name, reachable, ok?, ms, status?, error? } ] }` (an unreachable check has `error` and no `ok`). For Vibhav's live checks only; the app never calls it

Map and district data are built once a day by the scheduled job and served from storage, so they load fast and can't break during a demo because a government service is slow (§13.10, §62.1).

Part E: Definition of Done and Build Order

E1. A page is done when LOCKED

Not when it renders (§42). Every box, every page:

☐	All four states, loading, error, empty, populated, with different copy for error and empty
☐	Every value formatted per A5; every string from the content inventory or approved as PROPOSED, no invented copy
☐	Every severity shown as color + icon + word; missing data as the words "No data" or an explicit state, never zero or green
☐	Every estimate labeled as an estimate; every data point's source and date available to the person
☐	Bound to the exact fields in Part D; nothing computed that the server decides; no legacy `status` fields
☐	Every field on the page tested absent/null without breaking
☐	Back button correct for any sheet it opens; scroll position kept on return
☐	Keyboard navigable with visible focus; sheets trap and return focus; 44px targets; reduced motion respected
☐	Checked at 375px on a real phone, in light and dark mode
☐	No console errors; meets the performance targets (first content under 1.5s, cached readings under 200ms, tab change under 100ms, map first render under 2s)
☐	Every string goes through the translation layer; both languages present once reviewed Spanish exists
☐	Built only from Phase 0's shared components and tokens (Part P2); no page-local copy of a shared component; no hard-coded color, size or string
☐	Bound to the live API from the first commit; the `/<page>/preview` route shows every state with mock data; every server string passes through `displayText()`; the U+2014 lint test passes
☐	Wireframe frozen and proven: the built page has zero visual difference from the approved wireframe screenshots, and the element list in the page's README is complete (A13)
☐	(Once, app-wide) Page titles, link-preview metadata, `theme-color`, favicon and home-screen icons set per A5's identity rules

E2. Build order

#	Build	Why here
0	**Onboarding (done)**, merged to main (c84cbb3). The reference implementation for everything below: adapter pattern, preview route, copy catalog, tokens, screenshots (B1)	It already set the button, input, chip and spacing styles and proved the design foundation
1	**Phase 0: the shared foundation** (Parts P2 and P3): shell, shared components, tokens moved to `:root`, sheets and toast host, data hooks and modules, copy files, fonts and icons, the U+2014 lint test. One owner, named by Yogi (D1). Merged to main before anything in row 2 starts	Three people building at once need one shared layer; otherwise each rebuilds it differently
2	**Three lanes in parallel** (Part P1). Lane 1 Yogi: Today, Journal, Settings, then the two Onboarding follow-ups. Lane 2 Naggi: HomeGuard + Learn, then Act + Alerts. Lane 3 Vibhav: Map + District, then Assistant. Every page follows the same four steps: wireframe (preview route, mock states), freeze (screenshots + README element list), build (live API, same component), prove (zero visual difference, live check, flow check)	Cross-page dependencies are covered by Phase 0's contracts (P5); anything not built yet is a labeled placeholder, never a stub that ships
3	Device notifications (B6, D8), Spanish (if reviewed), and polish	After the core pages

Cut order if time runs short, from the end: Assistant, Settings, Act, district panel (the blueprint's own order). Today is the first page built on a real phone with live data (the blueprint's own measurement, §26.4): if it takes hours the plan holds; if days, cut scope now.

Two finished pages beat five half-finished ones under every judging criterion, and an unfinished page reads as broken rather than scoped (§26.3).

Part F: Decision Log

Every decision that shapes the frontend, made during planning (September 2026). LOCKED = decided, build to it. PROPOSED = a recommendation awaiting Yogi's confirmation. A superseded decision is listed with what replaced it, so an old document or branch can't reintroduce it.

F1. Product and scope

#	Decision	Status	Why / notes
1	HALO is a web app with a mobile-shaped layout, running in a phone browser, not a native iOS or Android app	LOCKED	§3. The Capacitor config in the repo isn't part of this build
2	The blueprint's function, content, buttons, states, copy, and data rules are ground truth. Its visual and style spec is void and redesigned	LOCKED	Team decision. This document carries the function; A11 and A13 carry the new look
3	Vibhav's glassmorphic HomeGuard frontend (branch `vmag211/homeguard-frontend`) and its `docs/HALO-design-language.md` are discarded	LOCKED	With Vibhav's permission. Don't merge, copy, or follow them
4	Every page holds all the content it can; Onboarding is the deliberate exception and stays minimal	LOCKED	§10.1, every extra onboarding question costs completions
5	This specification targets the finished backend; Part D is the contract both sides build to	LOCKED	The absent-field rule (D0) covers anything not live yet
6	Build order: Onboarding (done) → Phase 0 foundation → three parallel lanes (Today, Journal, Settings / HomeGuard + Learn, Act + Alerts / Map + District, Assistant)	LOCKED	E2 and Part P. Cut order from the end: Assistant, Settings, Act, district panel
7	Frontend work is built with Codex (wireframes, visual design, code) by three page owners, after one shared Phase 0 foundation with a single owner named by Yogi. Every page follows wireframe, freeze, build, prove. Yogi supplies reference inspiration and approves each page	LOCKED	September 27 (Codex), September 29 (three owners). Process in A13; owners and lanes in Part P
8	Spanish ships only after a person reviews it; every string goes through a translation layer from day one	LOCKED	§68, machine-translated health guidance is not acceptable

F2. Design

#	Decision	Status	Why / notes
9	Six severity words, good, moderate, elevated, high, severe, no_data, exactly as the backend's `lib/severity.js` emits them	LOCKED	An earlier idea to drop "elevated" was superseded: the scale follows the existing code
10	Severity colors: Good green, Moderate orange, Elevated red, High a red-leaning warning purple, Severe a darker purple, No data black drawn as an outline only	LOCKED	Exact light and dark values in A11. The palette wasn't tuned for red-green color blindness, the team's choice, which is safe only because severity is always color + icon + word
11	Brand: dark blues, navy, dark teals, turquoise, with a sprinkle of light aqua; no saturated green in the brand	LOCKED	A11. Green always means Good
12	Typefaces, the "field journal" set: Fraunces (headings, hero scores), Instrument Sans (everything else, tabular digits for data), IBM Plex Mono (codes only)	LOCKED	Chosen over more generic candidates for personality without losing clarity
13	The app follows the phone's appearance setting (light or dark). There is no default mode and no in-app toggle	LOCKED	§59. Every color is a token defined with `light-dark()`. Whether to force dark only is open (D3, Part P10); the default is to keep following the phone
14	The palette swatch sheet and type specimen made during planning were previews of colors and fonts only, not page designs	LOCKED	Page design happens per page, in Codex
15	Onboarding tone: "something in between", light and calm, with a navy-to-teal brand band behind the logo on Welcome only	LOCKED	Details in B1

16	Base visual system, type scale, spacing, radius, buttons, inputs, chips, progress, pills, motion	LOCKED	Built and final in Onboarding (A13, Part P6). New pages reuse it and add nothing without a Phase 0 change
17	The blueprint's simplifications are the defaults: three button variants, two sheet heights, pill rather than colored card edge, two skeleton shapes, three motion durations	LOCKED	§65.4 (A9 note)

F3. Data, privacy, and backend behavior the frontend depends on

#	Decision	Status	Why / notes
18	The server decides every severity, score, and household sentence; the frontend never applies a threshold	LOCKED	§6.1. Bind `severity`, never the legacy `status`
19	401 = session definitively invalid → reset silently. 503 = couldn't verify now → retry, never reset	LOCKED	Protects a household's data during an auth outage. Merged to main (the session handling was rewritten in the Onboarding commit c84cbb3; Vibhav reviews it)
20	Route on `onboarding_complete` (not `onboarded`), backed by the local `halo.onboarded` flag	LOCKED	A1
21	The Assistant may send the relevant readings and journal symptom names and dates to its AI provider, never the free-text notes, but only when a server switch is on. It ships off by default	LOCKED	Approved September 27 for this build; off by default because free models may log prompts (D4, Yogi turns it on). The privacy statement changes with the switch (B6)
22	Stored coordinates are rounded to about 100 m; ZIP is no longer stored	LOCKED	Matches the privacy statement's "approximate coordinates"
23	Radon zones apply only to North Carolina addresses; others get `out_of_state`	LOCKED	County names repeat across states (Union County, SC)
24	The Severe lead level is knowingly unreachable, no address-level service-line inventory exists; the utility's published counts are shown as context	LOCKED	Recorded by the backend, September 27
25	Journal's first-visit comparison is computed server-side (`POST /api/journal/retrospective`)	LOCKED	The frontend never compares readings itself
26	Map layers and district figures are pre-built daily and served from storage	LOCKED	Fast, and a slow government service can't break a demo
27	Anonymous sign-in is enabled in the Supabase project	LOCKED	Confirmed by Yogi. Re-check if the project is ever recreated (§66 item 1)
28	Settings toasts only on failure (the blueprint contradicts itself: §49.2 vs §54). The one exception is Clear journal	LOCKED	Quieter; one rule for all of Settings (B6)

F4. Decisions made while building (September 27 to 29)

#	Decision	Status	Why / notes
29	Onboarding is built and merged to main (c84cbb3). It is the reference implementation: every other page copies its adapter pattern, preview route, copy catalog, tokens and screenshot workflow	LOCKED	B1, Part P
30	No em dash (U+2014) in anything the app shows. A missing value reads "No data". Page titles read `HALO | Page`. Server strings pass through `displayText()`; a lint test enforces the rule	LOCKED	A5, A6, Part P2. The backend still emits em dashes in some text, so `displayText()` is required until Vibhav removes them (P9 ask 5)
31	Bind to the live API from the first commit. Mock data exists only for wireframe previews and tests. Never run a destructive call against the production database	LOCKED	The backend is merged into main; Part P4
32	Fonts are self-hosted with `next/font/local` (Fraunces, Instrument Sans; IBM Plex Mono is added in Phase 0). Body weight 500. The logo is `public/halo-logo-mark.png`, a white mask on Welcome's navy band	LOCKED	A11, A13, B1
33	Alerts badge: shows the unread count and drops as rows are read, not when the sheet opens	LOCKED	C3. Settled September 29 (was open)
34	The map time slider is built: the water data has 12 quarters, 2023Q1 to 2025Q4. Its final position is "Latest"	LOCKED	B7. Yogi confirmed the history exists and the backend notes list these quarters; the slider hides itself if `quarters` comes back

			short
35	The daily cache no longer serves an old address's reading after Change address: the server clears today's row and resets the limiter	LOCKED	Backend done for a changed location (closes a former open item with Vibhav). B6
36	Alerts never include boil-water advisories (no national source exists); weather alerts are flood and heat from the National Weather Service	LOCKED	C3, B6 switch copy

F5. Still open

- **Nine decisions with default rules (Part P10, D1 to D9).** Builders proceed on the default; Yogi changes any of them. D1 Phase 0 owner, D2 Change address household step, D3 dark-only or follow the phone, D4 Assistant household context, D5 content owner for missing copy, D6 season summary target, D7 map library, D8 device notifications timing, D9 volunteer organization verification.
- **Nine backend asks for Vibhav (Part P9).** Each has a stated frontend fallback, so none blocks a page.
- Every PROPOSED item in Parts A to C: the page's owner keeps or cuts each as the page is built, and Yogi approves.
- The privacy statement's incomplete "stores" line: a proposed addition awaits approval (B6).
- With Vibhav: server timeouts per source (the blueprint's §62.2 targets aren't enforced) and reloading the water data with non-detects, which would let the map say "no detection" and "tested" precisely (B7).
- Which district map the app uses (the 119th Congress lines, 2023) versus the newer 120th Congress map (B7).
- Whether Spanish ships: depends on a reviewer (§68).
- Whether the Assistant ships: it is last in the build order and its household-data mode is off (§68, D4).
- Next.js 16.2.10 has security advisories (fix in 16.3.6, outside the pinned range). Upgrade separately, never inside a page branch (Part P4).
- Which evaluation criteria the NC-08 office uses this year: one email to ask (§66 item 12).